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# Solutions Manual

for

## Theories of Generalized Change

Second Edition

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2026

Solutions Manual for  
*Theories of Generalized Change*  
Second Edition

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## About This Manual

This manual contains complete, worked solutions to all 549 practice problems in *Theories of Generalized Change*, Second Edition.

Each solution is written to show the reasoning behind each step, not just the final answer. Where appropriate, we include diagrams, intermediate computations, and remarks connecting the problem to broader themes.

For maximum benefit, attempt each problem on your own before consulting the solution. Mathematics is learned by doing, not by reading.

# Chapter 1

## Chapter 1: Single-Variable Calculus, Vectors, and the Geometry of Space

1. **Solution.** We differentiate  $f(t) = t^3 + \sin(t) + e^t$  term by term:

$$f'(t) = \frac{d}{dt}(t^3) + \frac{d}{dt}(\sin t) + \frac{d}{dt}(e^t) = 3t^2 + \cos t + e^t.$$

2. **Solution.** This is a standard limit. Direct substitution gives  $\frac{\sin 0}{0} = \frac{0}{0}$ , an indeterminate form. Applying L'Hôpital's Rule:

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} = \lim_{t \rightarrow 0} \frac{\cos t}{1} = \cos 0 = 1.$$

3. **Solution.** We integrate term by term:

$$\int (2t + 3) dt = 2 \cdot \frac{t^2}{2} + 3t + C = t^2 + 3t + C.$$

4. **Solution.** First, observe that  $f(t) = t^4 - 4t^3 + 6t^2 - 4t + 1 = (t - 1)^4$ . We can verify this by expanding  $(t - 1)^4$  using the binomial theorem:

$$(t - 1)^4 = t^4 - 4t^3 + 6t^2 - 4t + 1.$$

Now differentiate:

$$f'(t) = 4(t - 1)^3.$$

Setting  $f'(t) = 0$  gives  $4(t - 1)^3 = 0$ , so  $t = 1$ . The only critical point is  $t = 1$ .

5. **Solution.** We apply the Ratio Test to  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ :

$$L = \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0.$$

Since  $L = 0 < 1$  for all  $x$ , the series converges for all real numbers. The interval of convergence is  $(-\infty, \infty)$ .

(This is, of course, the well-known series for  $e^x$ .)

6. **Solution.** We evaluate:

$$\begin{aligned}\int_1^\infty \frac{1}{x^2} dx &= \lim_{b \rightarrow \infty} \int_1^b x^{-2} dx = \lim_{b \rightarrow \infty} \left[ -\frac{1}{x} \right]_1^b \\ &= \lim_{b \rightarrow \infty} \left( -\frac{1}{b} + 1 \right) = 1.\end{aligned}$$

7. **Solution.** The Taylor series of  $e^x$  centered at  $x = 0$  is  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ . Up to the  $x^3$  term:

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

8. **Solution.** We divide numerator and denominator by  $x^3$ :

$$\lim_{x \rightarrow \infty} \frac{2x^3 - x + 1}{x^3 + 4x^2} = \lim_{x \rightarrow \infty} \frac{2 - \frac{1}{x^2} + \frac{1}{x^3}}{1 + \frac{4}{x}} = \frac{2 - 0 + 0}{1 + 0} = 2.$$

9. **Solution.** A rectangle inscribed under  $y = 4 - x^2$  (with its base on the  $x$ -axis) has corners at  $(\pm x, 0)$  and  $(\pm x, 4 - x^2)$  for some  $x \in (0, 2)$ . The area is:

$$A(x) = 2x(4 - x^2) = 8x - 2x^3.$$

Differentiate and set equal to zero:

$$A'(x) = 8 - 6x^2 = 0 \implies x^2 = \frac{4}{3} \implies x = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}.$$

We verify this is a maximum since  $A''(x) = -12x < 0$  for  $x > 0$ . The maximum area is:

$$A\left(\frac{2\sqrt{3}}{3}\right) = 2 \cdot \frac{2\sqrt{3}}{3} \left(4 - \frac{4}{3}\right) = \frac{4\sqrt{3}}{3} \cdot \frac{8}{3} = \frac{32\sqrt{3}}{9}.$$

10. **Solution.** Let  $x$  be the distance from the base of the ladder to the wall and  $y$  the height of the top of the ladder. By the Pythagorean theorem:

$$x^2 + y^2 = 100.$$

Differentiate implicitly with respect to time  $t$ :

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}.$$

When  $x = 6$ , we have  $y = \sqrt{100 - 36} = \sqrt{64} = 8$ . We are given  $\frac{dx}{dt} = 2$  ft/s. Therefore:

$$\frac{dy}{dt} = -\frac{6}{8}(2) = -\frac{3}{2} \text{ ft/s}.$$

The top of the ladder is descending at  $\frac{3}{2}$  feet per second.

11. **Solution.** Given  $(x, y, z) = (3, -3, 3)$ , we convert to spherical  $(\rho, \theta, \phi)$ :

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{9 + 9 + 9} = \sqrt{27} = 3\sqrt{3}.$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-3}{3}\right) = \tan^{-1}(-1).$$

Since  $x > 0$  and  $y < 0$ , the point is in the fourth quadrant of the  $xy$ -plane, so  $\theta = -\frac{\pi}{4}$  (equivalently,  $\frac{7\pi}{4}$ ).

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{3}{3\sqrt{3}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right).$$

Thus the spherical coordinates are  $\left(3\sqrt{3}, \frac{7\pi}{4}, \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$ .

12. **Solution.** Given cylindrical coordinates  $(r, \theta, z) = (2, \frac{\pi}{4}, -1)$ :

$$x = r \cos \theta = 2 \cos \frac{\pi}{4} = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2},$$

$$y = r \sin \theta = 2 \sin \frac{\pi}{4} = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2},$$

$$z = -1.$$

The rectangular coordinates are  $(\sqrt{2}, \sqrt{2}, -1)$ .

13. **Solution.** The distance between  $(2, -1, 4)$  and  $(-3, 5, -2)$  is:

$$d = \sqrt{(2 - (-3))^2 + (-1 - 5)^2 + (4 - (-2))^2} = \sqrt{25 + 36 + 36} = \sqrt{97}.$$

14. **Solution.** Given spherical coordinates  $(\rho, \theta, \phi) = (5, \frac{\pi}{3}, \frac{\pi}{6})$ :

$$x = \rho \sin \phi \cos \theta = 5 \sin \frac{\pi}{6} \cos \frac{\pi}{3} = 5 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{5}{4},$$

$$y = \rho \sin \phi \sin \theta = 5 \sin \frac{\pi}{6} \sin \frac{\pi}{3} = 5 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{4},$$

$$z = \rho \cos \phi = 5 \cos \frac{\pi}{6} = 5 \cdot \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}.$$

The rectangular coordinates are  $\left(\frac{5}{4}, \frac{5\sqrt{3}}{4}, \frac{5\sqrt{3}}{2}\right)$ .

15. **Solution.** Given  $(x, y, z) = (-4, 4, \sqrt{3})$ :

$$r = \sqrt{x^2 + y^2} = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}.$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{4}{-4}\right) = \tan^{-1}(-1).$$

Since  $x < 0$  and  $y > 0$  (second quadrant),  $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$ .

The cylindrical coordinates are  $\left(4\sqrt{2}, \frac{3\pi}{4}, \sqrt{3}\right)$ .

16. **Solution.** The midpoint of  $(1, 2, 3)$  and  $(-1, -2, -3)$  is:

$$\left( \frac{1 + (-1)}{2}, \frac{2 + (-2)}{2}, \frac{3 + (-3)}{2} \right) = (0, 0, 0).$$

17. **Solution.** The projection of  $\mathbf{v} = \langle 3, 3, 3 \rangle$  onto the  $xy$ -plane is  $\mathbf{v}_{xy} = \langle 3, 3, 0 \rangle$ . The angle between  $\mathbf{v}_{xy}$  and the positive  $x$ -axis is:

$$\theta = \tan^{-1} \left( \frac{3}{3} \right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

18. **Solution.** Given  $(x, y, z) = (0, -5, 5)$ :

$$\rho = \sqrt{0 + 25 + 25} = \sqrt{50} = 5\sqrt{2}.$$

$$\theta = \tan^{-1} \left( \frac{-5}{0} \right).$$

Since  $x = 0$  and  $y < 0$ , we have  $\theta = \frac{3\pi}{2}$  (equivalently  $-\frac{\pi}{2}$ ).

$$\phi = \cos^{-1} \left( \frac{5}{5\sqrt{2}} \right) = \cos^{-1} \left( \frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}.$$

The spherical coordinates are  $\left( 5\sqrt{2}, \frac{3\pi}{2}, \frac{\pi}{4} \right)$ .

19. **Solution.** Given  $(x, y, z) = (2\sqrt{2}, 2\sqrt{2}, -3)$ :

$$r = \sqrt{(2\sqrt{2})^2 + (2\sqrt{2})^2} = \sqrt{8 + 8} = \sqrt{16} = 4.$$

$$\theta = \tan^{-1} \left( \frac{2\sqrt{2}}{2\sqrt{2}} \right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

(Both  $x > 0$  and  $y > 0$ , so the angle is indeed in the first quadrant.)

The cylindrical coordinates are  $\left( 4, \frac{\pi}{4}, -3 \right)$ .

20. **Solution.** Given  $(x, y, z) = (-2, -2, 2\sqrt{2})$ :

$$\rho = \sqrt{4 + 4 + 8} = \sqrt{16} = 4.$$

$$\theta = \tan^{-1} \left( \frac{-2}{-2} \right) = \tan^{-1}(1).$$

Since  $x < 0$  and  $y < 0$  (third quadrant),  $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$ .

$$\phi = \cos^{-1} \left( \frac{2\sqrt{2}}{4} \right) = \cos^{-1} \left( \frac{\sqrt{2}}{2} \right) = \frac{\pi}{4}.$$

The spherical coordinates are  $\left( 4, \frac{5\pi}{4}, \frac{\pi}{4} \right)$ .



21. **Solution.** Given  $(\rho, \theta, \phi) = (6, \frac{2\pi}{3}, \frac{\pi}{4})$ :

$$x = 6 \sin \frac{\pi}{4} \cos \frac{2\pi}{3} = 6 \cdot \frac{\sqrt{2}}{2} \cdot \left(-\frac{1}{2}\right) = -\frac{3\sqrt{2}}{2},$$

$$y = 6 \sin \frac{\pi}{4} \sin \frac{2\pi}{3} = 6 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{6}}{2},$$

$$z = 6 \cos \frac{\pi}{4} = 6 \cdot \frac{\sqrt{2}}{2} = 3\sqrt{2}.$$

The rectangular coordinates are  $\left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{6}}{2}, 3\sqrt{2}\right)$ .

22. **Solution.** Given cylindrical  $(r, \theta, z) = (3, \frac{\pi}{6}, 4)$ , we convert to spherical  $(\rho, \theta, \phi)$ .

Since  $\rho = \sqrt{r^2 + z^2} = \sqrt{9 + 16} = 5$ , and the azimuthal angle  $\theta$  stays the same:

$$\theta = \frac{\pi}{6}.$$

The polar angle is:

$$\phi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{4}{5}\right).$$

Alternatively,  $\phi = \tan^{-1}\left(\frac{r}{z}\right) = \tan^{-1}\left(\frac{3}{4}\right)$ .

The spherical coordinates are  $\left(5, \frac{\pi}{6}, \cos^{-1}\left(\frac{4}{5}\right)\right)$ .

23. **Solution.** Given cylindrical  $(r, \theta, z) = (1, \frac{\pi}{2}, -2)$ :

$$x = r \cos \theta = 1 \cdot \cos \frac{\pi}{2} = 0,$$

$$y = r \sin \theta = 1 \cdot \sin \frac{\pi}{2} = 1,$$

$$z = -2.$$

The rectangular coordinates are  $(0, 1, -2)$ .

24. **Solution.** The distance from the origin to the point  $(a, b, c)$  is:

$$d = \sqrt{(a-0)^2 + (b-0)^2 + (c-0)^2} = \sqrt{a^2 + b^2 + c^2}.$$

25. **Solution.** At the origin,  $(x, y, z) = (0, 0, 0)$ , we have  $\rho = \sqrt{0+0+0} = 0$ . However, the angles  $\theta$  and  $\phi$  are undefined when  $\rho = 0$ , because:

- $\theta = \tan^{-1}(y/x)$  is undefined when both  $x = 0$  and  $y = 0$ .
- $\phi = \cos^{-1}(z/\rho)$  involves division by zero when  $\rho = 0$ .

Geometrically, the origin has no preferred direction, so no unique angles can be assigned. The convention typically used is to write the origin as  $(\rho, \theta, \phi) = (0, 0, 0)$ , with the understanding that the angular coordinates are arbitrary.

26. **Solution.**

$$|\mathbf{v}| = \sqrt{3^2 + (-4)^2 + 5^2} = \sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}.$$

27. **Solution.** First compute  $|\mathbf{v}| = \sqrt{(-2)^2 + 1^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14}$ . The unit vector is:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{1}{\sqrt{14}} \langle -2, 1, 3 \rangle = \left\langle -\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right\rangle.$$

28. **Solution.**

$$\mathbf{u} \cdot \mathbf{v} = (1)(4) + (2)(0) + (3)(-1) = 4 + 0 - 3 = 1.$$

29. **Solution.**

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -1 \\ 0 & 2 & 3 \end{vmatrix}.$$

Expanding:

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \mathbf{i}(0 \cdot 3 - (-1) \cdot 2) - \mathbf{j}(1 \cdot 3 - (-1) \cdot 0) + \mathbf{k}(1 \cdot 2 - 0 \cdot 0) \\ &= \mathbf{i}(0 + 2) - \mathbf{j}(3 - 0) + \mathbf{k}(2 - 0) = \langle 2, -3, 2 \rangle. \end{aligned}$$

30. **Solution.** We use  $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$ .

$$\mathbf{u} \cdot \mathbf{v} = (1)(3) + (2)(2) + (3)(1) = 3 + 4 + 3 = 10.$$

$$|\mathbf{u}| = \sqrt{1 + 4 + 9} = \sqrt{14}, \quad |\mathbf{v}| = \sqrt{9 + 4 + 1} = \sqrt{14}.$$

$$\cos \theta = \frac{10}{14} = \frac{5}{7}, \quad \theta = \cos^{-1}\left(\frac{5}{7}\right).$$

31. **Solution.** We check whether  $\mathbf{v} = k\mathbf{u}$  for some scalar  $k$ . Comparing components:

$$\frac{4}{2} = 2, \quad \frac{-2}{-1} = 2, \quad \frac{0}{0} \text{ (consistent since both are 0)}.$$

Since  $\mathbf{v} = 2\mathbf{u}$ , the vectors are linearly dependent. □

32. **Solution.** The area of the parallelogram is  $|\mathbf{u} \times \mathbf{v}|$ . Compute:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} = \mathbf{i}(0 - 2) - \mathbf{j}(1 - 0) + \mathbf{k}(2 - 0) = \langle -2, -1, 2 \rangle.$$

$$|\mathbf{u} \times \mathbf{v}| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3.$$

33. **Solution.** A vector orthogonal to both  $\mathbf{u}$  and  $\mathbf{v}$  is given by  $\mathbf{u} \times \mathbf{v}$ :

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 0 \\ 2 & 1 & -1 \end{vmatrix} = \mathbf{i}((-1)(-1) - (0)(1)) - \mathbf{j}((1)(-1) - (0)(2)) + \mathbf{k}((1)(1) - (-1)(2)) \\ &= \mathbf{i}(1) - \mathbf{j}(-1) + \mathbf{k}(3) = \langle 1, 1, 3 \rangle.\end{aligned}$$

34. **Solution.** Three vectors in  $\mathbb{R}^3$  are linearly independent if and only if their scalar triple product is nonzero. Compute:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} 3 & -1 & 2 \\ 4 & 0 & -2 \\ 1 & 5 & -1 \end{vmatrix}.$$

Expanding along the first row:

$$\begin{aligned}&= 3 \begin{vmatrix} 0 & -2 \\ 5 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 4 & -2 \\ 1 & -1 \end{vmatrix} + 2 \begin{vmatrix} 4 & 0 \\ 1 & 5 \end{vmatrix} \\ &= 3(0 - (-10)) + 1(-4 - (-2)) + 2(20 - 0) \\ &= 3(10) + 1(-2) + 2(20) = 30 - 2 + 40 = 68.\end{aligned}$$

Since the scalar triple product is  $68 \neq 0$ , the vectors are linearly independent.

35. **Solution.** The projection of  $\mathbf{u}$  onto  $\mathbf{v}$  is:

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v}.$$

Compute:

$$\mathbf{u} \cdot \mathbf{v} = (4)(1) + (-2)(0) + (1)(3) = 4 + 0 + 3 = 7, \quad |\mathbf{v}|^2 = 1 + 0 + 9 = 10.$$

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{7}{10} \langle 1, 0, 3 \rangle = \left\langle \frac{7}{10}, 0, \frac{21}{10} \right\rangle.$$

36. **Solution.**

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}.$$

Expanding along the first row:

$$= 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = 1(-3) - 2(-6) + 3(-3) = -3 + 12 - 9 = 0.$$

37. **Solution.** The volume of the parallelepiped is  $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$  where  $\mathbf{u} = \langle 1, 2, 0 \rangle$ ,  $\mathbf{v} = \langle 0, 1, 3 \rangle$ ,  $\mathbf{w} = \langle 2, 0, 1 \rangle$ .

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{vmatrix}.$$

Expanding along the first row:

$$\begin{aligned} &= 1(1 \cdot 1 - 3 \cdot 0) - 2(0 \cdot 1 - 3 \cdot 2) + 0(0 \cdot 0 - 1 \cdot 2) \\ &= 1(1) - 2(-6) + 0 = 1 + 12 = 13. \end{aligned}$$

The volume is  $|13| = 13$ .

38. **Solution.** The scalar component of  $\mathbf{a}$  along  $\mathbf{b}$  is:

$$\text{comp}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}.$$

Compute:

$$\mathbf{a} \cdot \mathbf{b} = (5)(1) + (2)(-1) + (-3)(2) = 5 - 2 - 6 = -3, \quad |\mathbf{b}| = \sqrt{1 + 1 + 4} = \sqrt{6}.$$

$$\text{comp}_{\mathbf{b}} \mathbf{a} = \frac{-3}{\sqrt{6}} = -\frac{3}{\sqrt{6}} = -\frac{3\sqrt{6}}{6} = -\frac{\sqrt{6}}{2}.$$

39. **Solution.** First compute  $\mathbf{w} = \mathbf{u} \times \mathbf{v}$ :

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 4 \\ -1 & 0 & 1 \end{vmatrix} = \mathbf{i}(3 - 0) - \mathbf{j}(2 - (-4)) + \mathbf{k}(0 - (-3)) = \langle 3, -6, 3 \rangle.$$

$$|\mathbf{w}| = \sqrt{9 + 36 + 9} = \sqrt{54} = 3\sqrt{6}.$$

40. **Solution.**

$$\mathbf{a} \cdot \mathbf{b} = (2)(1) + (-1)(4) + (3)(-2) = 2 - 4 - 6 = -8.$$

$$|\mathbf{a}| = \sqrt{4 + 1 + 9} = \sqrt{14}, \quad |\mathbf{b}| = \sqrt{1 + 16 + 4} = \sqrt{21}.$$

$$\cos \theta = \frac{-8}{\sqrt{14}\sqrt{21}} = \frac{-8}{\sqrt{294}} = \frac{-8}{7\sqrt{6}}, \quad \theta = \cos^{-1}\left(\frac{-8}{7\sqrt{6}}\right).$$

41. **Solution.** Three vectors are coplanar if and only if their scalar triple product is zero. Compute:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}.$$

This is the same determinant as in Problem 36:

$$= 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0.$$

Since the scalar triple product is zero, the vectors are coplanar.

We can also verify directly:  $\mathbf{w} = 2\mathbf{v} - \mathbf{u}$ , since  $2\langle 4, 5, 6 \rangle - \langle 1, 2, 3 \rangle = \langle 7, 8, 9 \rangle$ .  $\square$

42. **Solution.** The area of the triangle formed by  $\mathbf{u}$  and  $\mathbf{v}$  is  $\frac{1}{2}|\mathbf{u} \times \mathbf{v}|$ :

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -1 \\ 1 & 3 & 4 \end{vmatrix} = \mathbf{i}(0+3) - \mathbf{j}(8+1) + \mathbf{k}(6-0) = \langle 3, -9, 6 \rangle.$$

$$|\mathbf{u} \times \mathbf{v}| = \sqrt{9+81+36} = \sqrt{126} = 3\sqrt{14}.$$

The area of the triangle is  $\frac{3\sqrt{14}}{2}$ .

43. **Solution.**

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b}.$$

Compute:

$$\mathbf{a} \cdot \mathbf{b} = (3)(4) + (-2)(0) + (1)(-1) = 12 + 0 - 1 = 11, \quad |\mathbf{b}|^2 = 16 + 0 + 1 = 17.$$

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{11}{17} \langle 4, 0, -1 \rangle = \left\langle \frac{44}{17}, 0, -\frac{11}{17} \right\rangle.$$

44. **Solution.** Two vectors are orthogonal if and only if their dot product is zero:

$$\mathbf{u} \cdot \mathbf{v} = (2)(-1) + (4)(-2) + (6)(-3) = -2 - 8 - 18 = -28.$$

Since  $\mathbf{u} \cdot \mathbf{v} = -28 \neq 0$ , the vectors are **not** orthogonal.

(In fact,  $\mathbf{v} = -\frac{1}{2}\mathbf{u}$ , so the vectors are antiparallel.)

45. **Solution.**

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 2 \\ 3 & 4 & 5 \end{vmatrix} = \mathbf{i}(5-8) - \mathbf{j}(0-6) + \mathbf{k}(0-3) = \langle -3, 6, -3 \rangle.$$

46. **Solution.** A vector perpendicular to the plane defined by  $\mathbf{u} = \langle 1, 2, 3 \rangle$  and  $\mathbf{v} = \langle 4, 5, 6 \rangle$  is  $\mathbf{u} \times \mathbf{v}$ :

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} = \mathbf{i}(12-15) - \mathbf{j}(6-12) + \mathbf{k}(5-8) = \langle -3, 6, -3 \rangle.$$

The magnitude is:

$$|\mathbf{u} \times \mathbf{v}| = \sqrt{9+36+9} = \sqrt{54} = 3\sqrt{6}.$$

The unit vector is:

$$\hat{\mathbf{n}} = \frac{1}{3\sqrt{6}} \langle -3, 6, -3 \rangle = \left\langle -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}} \right\rangle.$$

47. **Solution.** We have  $\mathbf{a} = \langle 1, 0, 2 \rangle$ ,  $\mathbf{b} = \langle 0, 1, 1 \rangle$ , and  $\mathbf{c} = \langle 1, 1, 3 \rangle$ . Check the scalar triple product:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 3 \end{vmatrix} = 1(3 - 1) - 0(0 - 1) + 2(0 - 1) = 2 - 0 - 2 = 0.$$

Since the scalar triple product is zero, the vectors are linearly dependent.

Indeed, we can verify directly:  $\mathbf{c} = \mathbf{a} + \mathbf{b}$ , since  $\langle 1, 0, 2 \rangle + \langle 0, 1, 1 \rangle = \langle 1, 1, 3 \rangle$ .

48. **Solution.**

$$\mathbf{u} \cdot \mathbf{v} = (5)(-1) + (-2)(3) + (4)(2) = -5 - 6 + 8 = -3.$$

49. **Solution.**

$$\mathbf{u} \cdot \mathbf{v} = (7)(4) + (0)(-3) + (-5)(2) = 28 + 0 - 10 = 18.$$

$$|\mathbf{u}| = \sqrt{49 + 0 + 25} = \sqrt{74}, \quad |\mathbf{v}| = \sqrt{16 + 9 + 4} = \sqrt{29}.$$

$$\cos \theta = \frac{18}{\sqrt{74}\sqrt{29}} = \frac{18}{\sqrt{2146}}, \quad \theta = \cos^{-1}\left(\frac{18}{\sqrt{2146}}\right).$$

50. **Solution.** Two vectors are parallel if and only if one is a scalar multiple of the other, i.e.,  $\mathbf{v} = k\mathbf{u}$  for some scalar  $k$ . Comparing components:

$$\frac{4}{1} = 4, \quad \frac{-1}{3} = -\frac{1}{3}, \quad \frac{5}{-2} = -\frac{5}{2}.$$

Since these ratios are not equal ( $4 \neq -\frac{1}{3} \neq -\frac{5}{2}$ ), there is no scalar  $k$  such that  $\mathbf{v} = k\mathbf{u}$ . Therefore,  $\mathbf{u}$  and  $\mathbf{v}$  are not parallel.  $\square$

51. **Solution.**

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 2 \\ -2 & 4 & 1 \end{vmatrix} = \mathbf{i}(-1 - 8) - \mathbf{j}(3 - (-4)) + \mathbf{k}(12 - 2) = \langle -9, -7, 10 \rangle.$$

$$|\mathbf{u} \times \mathbf{v}| = \sqrt{81 + 49 + 100} = \sqrt{230}.$$

52. **Solution.**

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -3 & 1 \\ 4 & 0 & -2 \end{vmatrix} = \mathbf{i}((-3)(-2) - (1)(0)) - \mathbf{j}((2)(-2) - (1)(4)) + \mathbf{k}((2)(0) - (-3)(4)) \\ &= \mathbf{i}(6) - \mathbf{j}(-8) + \mathbf{k}(12) = \langle 6, 8, 12 \rangle. \end{aligned}$$

A vector orthogonal to both  $\mathbf{u}$  and  $\mathbf{v}$  is  $\langle 6, 8, 12 \rangle$  (or equivalently,  $\langle 3, 4, 6 \rangle$ ).

53. **Solution.** The direction vector from  $(1, 0, -1)$  to  $(2, 3, 4)$  is:

$$\mathbf{d} = \langle 2 - 1, 3 - 0, 4 - (-1) \rangle = \langle 1, 3, 5 \rangle.$$

Using the point  $(1, 0, -1)$ , the parametric equations of the line are:

$$x = 1 + t, \quad y = 3t, \quad z = -1 + 5t,$$

or equivalently,  $\mathbf{r}(t) = \langle 1 + t, 3t, -1 + 5t \rangle$ .

54. **Solution.** Let  $A = (1, 2, 3)$ ,  $B = (-1, 0, 4)$ ,  $C = (3, -2, 1)$ . Form two vectors in the plane:

$$\overrightarrow{AB} = \langle -2, -2, 1 \rangle, \quad \overrightarrow{AC} = \langle 2, -4, -2 \rangle.$$

The normal vector is:

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & -2 & 1 \\ 2 & -4 & -2 \end{vmatrix} = \mathbf{i}(4 + 4) - \mathbf{j}(4 - 2) + \mathbf{k}(8 + 4) = \langle 8, -2, 12 \rangle.$$

We can simplify by dividing by 2:  $\mathbf{n} = \langle 4, -1, 6 \rangle$ .

Using point  $A = (1, 2, 3)$ :

$$4(x - 1) - 1(y - 2) + 6(z - 3) = 0 \implies 4x - y + 6z = 20.$$

55. **Solution.** The line is  $\mathbf{r}(t) = \langle 1 + t, 2 - t, 3t \rangle$ , so  $x = 1 + t$ ,  $y = 2 - t$ ,  $z = 3t$ . Substitute into  $x - y + z = 5$ :

$$(1 + t) - (2 - t) + 3t = 5 \implies 1 + t - 2 + t + 3t = 5 \implies 5t - 1 = 5 \implies t = \frac{6}{5}.$$

Substituting back:

$$x = 1 + \frac{6}{5} = \frac{11}{5}, \quad y = 2 - \frac{6}{5} = \frac{4}{5}, \quad z = 3 \cdot \frac{6}{5} = \frac{18}{5}.$$

The point of intersection is  $\left(\frac{11}{5}, \frac{4}{5}, \frac{18}{5}\right)$ .

56. **Solution.** The direction vector of  $\mathbf{r}_1$  is  $\mathbf{d}_1 = \langle 1, 2, 3 \rangle$  and of  $\mathbf{r}_2$  is  $\mathbf{d}_2 = \langle 1, 0, 3 \rangle$ .

*Parallel check:* The vectors  $\mathbf{d}_1$  and  $\mathbf{d}_2$  are not scalar multiples (since  $1/1 = 1$  but  $2/0$  is undefined), so the lines are not parallel.

*Intersection check:* Set  $\mathbf{r}_1(t) = \mathbf{r}_2(s)$ :

$$t = 1 + s, \quad 2t = 2, \quad 3t = 3s.$$

From the second equation,  $t = 1$ . From the first equation,  $s = t - 1 = 0$ . Check the third:  $3(1) = 3(0)$  gives  $3 = 0$ , a contradiction.

Since the lines are not parallel and do not intersect, they are **skew**.

57. **Solution.** The line is  $\mathbf{r}(t) = \langle 2t, 1-t, 1+t \rangle$ .

Intersection with  $\Pi_1 : x + 2y + 3z = 7$ :

$$2t + 2(1-t) + 3(1+t) = 7 \implies 2t + 2 - 2t + 3 + 3t = 7 \implies 3t + 5 = 7 \implies t_1 = \frac{2}{3}.$$

Point:  $P_1 = \left(\frac{4}{3}, \frac{1}{3}, \frac{5}{3}\right)$ .

Intersection with  $\Pi_2 : -2x - y - z = 2$ :

$$-2(2t) - (1-t) - (1+t) = 2 \implies -4t - 1 + t - 1 - t = 2 \implies -4t - 2 = 2 \implies t_2 = -1.$$

Point:  $P_2 = (-2, 2, 0)$ .

*Length of the segment:* The direction vector of  $\mathbf{r}(t)$  is  $\mathbf{d} = \langle 2, -1, 1 \rangle$ , so  $|\mathbf{d}| = \sqrt{4+1+1} = \sqrt{6}$ . The length from  $P_1$  to  $P_2$  is:

$$L = |t_2 - t_1| \cdot |\mathbf{d}| = \left| -1 - \frac{2}{3} \right| \cdot \sqrt{6} = \frac{5}{3}\sqrt{6}.$$

58. **Solution.** The line is  $\mathbf{r}(t) = \langle t, t, t \rangle$  with direction  $\mathbf{d} = \langle 1, 1, 1 \rangle$ , passing through the origin. The point is  $Q = (1, 0, 2)$ . Let  $\overrightarrow{PQ} = \langle 1, 0, 2 \rangle$  (from the origin to  $Q$ ).

The shortest distance from a point to a line is:

$$\text{dist} = \frac{|\overrightarrow{PQ} \times \mathbf{d}|}{|\mathbf{d}|}.$$

Compute:

$$\overrightarrow{PQ} \times \mathbf{d} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 1 & 1 & 1 \end{vmatrix} = \mathbf{i}(0-2) - \mathbf{j}(1-2) + \mathbf{k}(1-0) = \langle -2, 1, 1 \rangle.$$

$$|\overrightarrow{PQ} \times \mathbf{d}| = \sqrt{4+1+1} = \sqrt{6}, \quad |\mathbf{d}| = \sqrt{3}.$$

$$\text{dist} = \frac{\sqrt{6}}{\sqrt{3}} = \sqrt{2}.$$

59. **Solution.** We need to find the line of intersection of  $x + y + z = 1$  and  $2x - y + 3z = 4$ .

*Direction vector:* The direction of the line is perpendicular to both normal vectors  $\mathbf{n}_1 = \langle 1, 1, 1 \rangle$  and  $\mathbf{n}_2 = \langle 2, -1, 3 \rangle$ :

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & -1 & 3 \end{vmatrix} = \mathbf{i}(3+1) - \mathbf{j}(3-2) + \mathbf{k}(-1-2) = \langle 4, -1, -3 \rangle.$$

*Find a point on the line:* Set  $z = 0$ . Then:

$$x + y = 1 \quad \text{and} \quad 2x - y = 4.$$



Adding:  $3x = 5$ , so  $x = \frac{5}{3}$ ,  $y = 1 - \frac{5}{3} = -\frac{2}{3}$ . A point is  $(\frac{5}{3}, -\frac{2}{3}, 0)$ .

The parametric equations of the line of intersection are:

$$x = \frac{5}{3} + 4t, \quad y = -\frac{2}{3} - t, \quad z = -3t.$$

60. **Solution.** The distance between parallel planes  $ax + by + cz = d_1$  and  $ax + by + cz = d_2$  is:

$$\text{dist} = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.$$

Here,  $d_1 = 5$ ,  $d_2 = -3$ , and  $\sqrt{1 + 4 + 4} = 3$ :

$$\text{dist} = \frac{|5 - (-3)|}{3} = \frac{8}{3}.$$

61. **Solution.** The plane  $3x - 2y + z = 7$  has normal vector  $\mathbf{n} = \langle 3, -2, 1 \rangle$ . A line perpendicular to the plane has direction  $\mathbf{n}$ . Through the point  $(2, -1, 4)$ :

$$x = 2 + 3t, \quad y = -1 - 2t, \quad z = 4 + t.$$

62. **Solution.** The line has direction  $\mathbf{d} = \langle 2, -1, 4 \rangle$ . The plane  $2x - 3y + z = 7$  has normal  $\mathbf{n} = \langle 2, -3, 1 \rangle$ .

The angle  $\phi$  between a line and a plane satisfies:

$$\sin \phi = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}.$$

Compute:

$$\mathbf{d} \cdot \mathbf{n} = (2)(2) + (-1)(-3) + (4)(1) = 4 + 3 + 4 = 11.$$

$$|\mathbf{d}| = \sqrt{4 + 1 + 16} = \sqrt{21}, \quad |\mathbf{n}| = \sqrt{4 + 9 + 1} = \sqrt{14}.$$

$$\sin \phi = \frac{11}{\sqrt{21}\sqrt{14}} = \frac{11}{\sqrt{294}} = \frac{11}{7\sqrt{6}}, \quad \phi = \sin^{-1}\left(\frac{11}{7\sqrt{6}}\right).$$

63. **Solution.** A plane with normal  $\mathbf{n} = \langle 3, -2, 5 \rangle$  passing through  $(2, -1, 4)$  has equation:

$$3(x - 2) - 2(y + 1) + 5(z - 4) = 0 \implies 3x - 2y + 5z = 28.$$

64. **Solution.** Substituting  $z = 3$  into  $x^2 + y^2 + z^2 = 25$ :

$$x^2 + y^2 + 9 = 25 \implies x^2 + y^2 = 16.$$

The intersection is a circle of radius 4 in the plane  $z = 3$ , centered at  $(0, 0, 3)$ .

Parametrically:  $x = 4 \cos t$ ,  $y = 4 \sin t$ ,  $z = 3$ , for  $t \in [0, 2\pi)$ .

65. **Solution.** A line parallel to  $\mathbf{v} = \langle 2, -1, 3 \rangle$  through  $(-1, 4, 2)$ :

$$x = -1 + 2t, \quad y = 4 - t, \quad z = 2 + 3t.$$

Or equivalently,  $\mathbf{r}(t) = \langle -1 + 2t, 4 - t, 2 + 3t \rangle$ .

66. **Solution.** Set  $\mathbf{r}_1(t) = \mathbf{r}_2(s)$ :

$$1 + t = 4 - s, \quad 2 - t = -1 + s, \quad 3 + 2t = 5 + s.$$

From equation 1:  $t + s = 3$ . From equation 2:  $-t - s = -3$ , i.e.,  $t + s = 3$ . These are consistent.

From equation 1:  $s = 3 - t$ . Substitute into equation 3:

$$3 + 2t = 5 + (3 - t) = 8 - t \implies 3t = 5 \implies t = \frac{5}{3}.$$

Then  $s = 3 - \frac{5}{3} = \frac{4}{3}$ .

Verification:  $\mathbf{r}_1\left(\frac{5}{3}\right) = \left\langle \frac{8}{3}, \frac{1}{3}, \frac{19}{3} \right\rangle$  and  $\mathbf{r}_2\left(\frac{4}{3}\right) = \left\langle 4 - \frac{4}{3}, -1 + \frac{4}{3}, 5 + \frac{4}{3} \right\rangle = \left\langle \frac{8}{3}, \frac{1}{3}, \frac{19}{3} \right\rangle$ .

The lines intersect at  $\left(\frac{8}{3}, \frac{1}{3}, \frac{19}{3}\right)$ .

67. **Solution.** For the surface  $F(x, y, z) = x^2 + 2y^2 + 3z^2 = 6$ , the gradient is:

$$\nabla F = \langle 2x, 4y, 6z \rangle.$$

At the point  $(1, 1, 1)$ :

$$\nabla F(1, 1, 1) = \langle 2, 4, 6 \rangle.$$

The tangent plane is:

$$2(x - 1) + 4(y - 1) + 6(z - 1) = 0 \implies 2x + 4y + 6z = 12,$$

or equivalently,  $x + 2y + 3z = 6$ .

68. **Solution.** The two lines are  $\mathbf{r}_1(t) = \langle t, 2t, 3t \rangle$  (direction  $\mathbf{d}_1 = \langle 1, 2, 3 \rangle$ ) and  $\mathbf{r}_2(s) = \langle 1 + s, 2 + s, 3 + s \rangle$  (direction  $\mathbf{d}_2 = \langle 1, 1, 1 \rangle$ ).

The vector connecting a general point on  $\mathbf{r}_1$  to a general point on  $\mathbf{r}_2$  is:

$$\mathbf{w}(t, s) = \mathbf{r}_2(s) - \mathbf{r}_1(t) = \langle 1 + s - t, 2 + s - 2t, 3 + s - 3t \rangle.$$

The shortest segment is perpendicular to both lines, so we require  $\mathbf{w} \cdot \mathbf{d}_1 = 0$  and  $\mathbf{w} \cdot \mathbf{d}_2 = 0$ .

$\mathbf{w} \cdot \mathbf{d}_1 = 0$ :

$$(1 + s - t) + 2(2 + s - 2t) + 3(3 + s - 3t) = 0$$

$$1 + s - t + 4 + 2s - 4t + 9 + 3s - 9t = 0 \implies 6s - 14t + 14 = 0 \implies 3s - 7t = -7. \quad (*)$$

$$\mathbf{w} \cdot \mathbf{d}_2 = 0:$$

$$\begin{aligned}(1 + s - t) + (2 + s - 2t) + (3 + s - 3t) &= 0 \\ 6 + 3s - 6t &= 0 \implies s - 2t = -2. \quad (**)\end{aligned}$$

From (\*\*):  $s = 2t - 2$ . Substitute into (\*):

$$3(2t - 2) - 7t = -7 \implies 6t - 6 - 7t = -7 \implies -t = -1 \implies t = 1.$$

Then  $s = 2(1) - 2 = 0$ .

The closest points are:

$$\mathbf{r}_1(1) = \langle 1, 2, 3 \rangle \quad \text{and} \quad \mathbf{r}_2(0) = \langle 1, 2, 3 \rangle.$$

The two lines actually intersect at  $(1, 2, 3)$ , so the shortest “segment” has length 0.

69. **Solution.** The curve is given by  $x = t$ ,  $y = \sin t$ ,  $z = \cos t$ . Observe that:

$$y^2 + z^2 = \sin^2 t + \cos^2 t = 1.$$

This is the equation of a circular cylinder of radius 1 centered on the  $x$ -axis:  $y^2 + z^2 = 1$ .

*Proof.* For every point  $(t, \sin t, \cos t)$  on the curve, we have  $y^2 + z^2 = \sin^2 t + \cos^2 t = 1$  by the Pythagorean identity. Therefore, every point on the curve satisfies the cylinder equation  $y^2 + z^2 = 1$ , which is a (circular) cylinder – a quadric surface.  $\square$

70. **Solution.** Substituting  $y = 1$  into  $x^2 - y^2 - z^2 = 4$ :

$$x^2 - 1 - z^2 = 4 \implies x^2 - z^2 = 5.$$

This is a hyperbola in the  $xz$ -plane (at  $y = 1$ ).

71. **Solution.** The cylinder  $x^2 + y^2 = 9$  has radius  $r = 3$ . Between  $z = 0$  and  $z = 5$ , the lateral surface area is:

$$A = 2\pi r \cdot h = 2\pi(3)(5) = 30\pi.$$

72. **Solution.** An ellipsoid with semi-axes  $a = 3$ ,  $b = 4$ ,  $c = 5$  along the  $x$ -,  $y$ -, and  $z$ -axes has equation:

$$\frac{x^2}{9} + \frac{y^2}{16} + \frac{z^2}{25} = 1.$$

73. **Solution.** The parabola  $z = x^2$  lies in the  $xz$ -plane. When rotated about the  $z$ -axis, each point  $(x, 0, x^2)$  traces a circle of radius  $|x|$  in the horizontal plane at height  $z = x^2$ . For a general point  $(x, y, z)$  on the surface, the distance from the  $z$ -axis is  $r = \sqrt{x^2 + y^2}$ , and this equals  $|x_0|$  where  $z = x_0^2$ . Therefore:

$$z = r^2 = x^2 + y^2.$$

The Cartesian equation of the paraboloid is  $z = x^2 + y^2$ .

74. **Solution.** The original parametric equation is  $\mathbf{r}(t) = (1-t)\mathbf{P}_0 + t\mathbf{P}_1$  for  $t \in [0, 1]$ . At  $t = 0$ , we get  $\mathbf{P}_0$ ; at  $t = 1$ , we get  $\mathbf{P}_1$ .

If we want the parameter  $t$  to range over  $[0, a]$  instead, we need  $t = 0$  to give  $\mathbf{P}_0$  and  $t = a$  to give  $\mathbf{P}_1$ . We substitute  $t/a$  for the original parameter:

$$\mathbf{r}(t) = \left(1 - \frac{t}{a}\right) \mathbf{P}_0 + \frac{t}{a} \mathbf{P}_1, \quad t \in [0, a].$$

This is equivalent to reparametrizing by  $u = t/a$ , where  $u \in [0, 1]$ .

75. **Solution.** The equation  $x^2 + y^2 - z^2 = 1$  can be written as:

$$\frac{x^2}{1} + \frac{y^2}{1} - \frac{z^2}{1} = 1.$$

This is a **hyperboloid of one sheet**. Its cross-sections at constant  $z = z_0$  are circles  $x^2 + y^2 = 1 + z_0^2$  of radius  $\sqrt{1 + z_0^2}$ , and its cross-sections at constant  $x$  or  $y$  are hyperbolas.

76. **Solution.** The cylinder  $x^2 + y^2 = 4$  can be parametrized as  $x = 2 \cos t$ ,  $y = 2 \sin t$ . On the plane  $z = x + y$ :

$$z = 2 \cos t + 2 \sin t.$$

The parametric equations for the intersection curve are:

$$x = 2 \cos t, \quad y = 2 \sin t, \quad z = 2 \cos t + 2 \sin t, \quad t \in [0, 2\pi).$$

77. **Solution.** Rewrite  $x^2 + y^2 - 4z = 0$  as:

$$z = \frac{x^2 + y^2}{4} = \frac{x^2}{4} + \frac{y^2}{4}.$$

This has the standard form  $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$  with  $a = b = 2$ . Since the coefficients of  $x^2$  and  $y^2$  are both positive and equal, this is a **circular paraboloid**, which is a special case of an elliptic paraboloid. The surface opens upward along the  $z$ -axis.

Therefore,  $x^2 + y^2 - 4z = 0$  is indeed an elliptic paraboloid.

78. **Solution.** We compute the volume of the spherical cap of  $x^2 + y^2 + z^2 = 4$  (radius  $R = 2$ ) above the plane  $z = 1$ .

Using the disk method, for  $z \in [1, 2]$  (the sphere extends from  $z = -2$  to  $z = 2$ ), each horizontal cross-section at height  $z$  is a disk of radius  $r = \sqrt{4 - z^2}$ :

$$\begin{aligned} V &= \int_1^2 \pi(4 - z^2) dz = \pi \left[ 4z - \frac{z^3}{3} \right]_1^2 = \pi \left[ \left( 8 - \frac{8}{3} \right) - \left( 4 - \frac{1}{3} \right) \right] \\ &= \pi \left[ \frac{16}{3} - \frac{11}{3} \right] = \frac{5\pi}{3}. \end{aligned}$$

This is the volume of the cap above  $z = 1$ . If the problem asks for the volume enclosed between the sphere and the plane (i.e., the larger piece below  $z = 1$ ), we compute the total volume of the sphere minus the cap:

$$V_{\text{sphere}} = \frac{4}{3}\pi(2)^3 = \frac{32\pi}{3}.$$

$$V_{\text{below}} = \frac{32\pi}{3} - \frac{5\pi}{3} = \frac{27\pi}{3} = 9\pi.$$

The volume of the spherical cap above  $z = 1$  is  $\frac{5\pi}{3}$ , and the volume of the region below  $z = 1$  inside the sphere is  $9\pi$ .

79. **Solution.** The parabola  $z = x^2$  lies in the  $xz$ -plane. When rotated about the  $z$ -axis, each point at distance  $|x|$  from the axis sweeps out a circle of radius  $|x|$ . For a general point  $(x, y, z)$  on the resulting surface, the distance from the  $z$ -axis is  $\sqrt{x^2 + y^2}$ , which plays the role of  $|x|$ . Therefore:

$$z = (\sqrt{x^2 + y^2})^2 = x^2 + y^2.$$

The equation of the surface of revolution is  $z = x^2 + y^2$ , a circular paraboloid.

## Chapter 2

# Chapter 2: Vector-Valued Functions and Motion in Space

1. **Solution.** We have  $\mathbf{r}(t) = \langle e^{-t} \cos t, e^{-t} \sin t, t \rangle$ . A curve is smooth if  $\mathbf{r}'(t) \neq \mathbf{0}$  for all  $t$ .

Compute the derivative component by component:

$$\mathbf{r}'(t) = \langle -e^{-t} \cos t - e^{-t} \sin t, -e^{-t} \sin t + e^{-t} \cos t, 1 \rangle.$$

Since the third component is  $1 \neq 0$  for all  $t$ , we have  $\mathbf{r}'(t) \neq \mathbf{0}$  for every  $t$ .

Therefore the curve is smooth for all  $t$ .

2. **Solution.** Given  $\mathbf{r}(t) = \langle t^2, e^{2t}, \ln t \rangle$  with  $t > 0$ , differentiate component by component:

$$\mathbf{r}'(t) = \left\langle 2t, 2e^{2t}, \frac{1}{t} \right\rangle.$$

3. **Solution.** The direction vector is  $\mathbf{r}_2 - \mathbf{r}_1 = \langle 4 - 1, 5 - 2, 6 - 3 \rangle = \langle 3, 3, 3 \rangle$ . Using the point  $\mathbf{r}_1$ :

$$\mathbf{r}(t) = \langle 1, 2, 3 \rangle + t\langle 3, 3, 3 \rangle = \langle 1 + 3t, 2 + 3t, 3 + 3t \rangle, \quad t \in \mathbb{R}.$$

4. **Solution.** From  $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, t \rangle$ , the parametric equations are

$$x = 3 \cos t, \quad y = 3 \sin t, \quad z = t.$$

Since  $x^2 + y^2 = 9 \cos^2 t + 9 \sin^2 t = 9$  and  $z = t$  increases linearly, the curve is a circular helix of radius 3 winding around the  $z$ -axis.

5. **Solution.** We have  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$  so  $x = t$ ,  $y = t^2$ ,  $z = t^3$ . For  $t \geq 0$ :  $y = t^2 \geq 0$  and  $t = \sqrt{y}$ , so

$$z = t^3 = (\sqrt{y})^3 = y^{3/2}.$$

Therefore  $z = y^{3/2}$ , which means the curve lies on the surface  $z = y^{3/2}$ . □

6. **Solution.** Given  $\mathbf{r}(t) = \langle t, \sin t, \cos t \rangle$ , the velocity vector is

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle 1, \cos t, -\sin t \rangle.$$

The speed is

$$|\mathbf{v}(t)| = \sqrt{1 + \cos^2 t + \sin^2 t} = \sqrt{1 + 1} = \sqrt{2}.$$

Since the speed equals the constant  $\sqrt{2}$  for all  $t$ , the speed is constant, equal to  $\sqrt{2}$ .

7. **Solution.**

$$\int_0^1 \mathbf{r}(t) dt = \left\langle \int_0^1 t dt, \int_0^1 t^2 dt, \int_0^1 t^3 dt \right\rangle = \left\langle \left. \frac{t^2}{2} \right|_0^1, \left. \frac{t^3}{3} \right|_0^1, \left. \frac{t^4}{4} \right|_0^1 \right\rangle = \left\langle \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \right\rangle.$$

8. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$  so  $\mathbf{r}'(t) = \langle 1, 2t, 0 \rangle$  and  $|\mathbf{r}'(t)| = \sqrt{1 + 4t^2}$ . The arc length is

$$L = \int_0^2 \sqrt{1 + 4t^2} dt.$$

Let  $2t = \sinh u$ , so  $dt = \frac{1}{2} \cosh u du$  and  $\sqrt{1 + 4t^2} = \cosh u$ . When  $t = 0$ ,  $u = 0$ ; when  $t = 2$ ,  $u = \sinh^{-1} 4 = \ln(4 + \sqrt{17})$ .

$$L = \int_0^{\sinh^{-1} 4} \cosh^2 u \cdot \frac{1}{2} du = \frac{1}{4} [u + \sinh u \cosh u]_0^{\sinh^{-1} 4}.$$

At  $u = \sinh^{-1} 4$ :  $\sinh u = 4$ ,  $\cosh u = \sqrt{17}$ , so

$$L = \frac{1}{4} [\ln(4 + \sqrt{17}) + 4\sqrt{17}] \approx 4.6468.$$

9. **Solution.**  $\mathbf{r}(t) = \langle t^3, t^2, t \rangle$ .

$$\mathbf{r}'(t) = \langle 3t^2, 2t, 1 \rangle, \quad \mathbf{r}''(t) = \langle 6t, 2, 0 \rangle.$$

10. **Solution.** We have  $\mathbf{r}(t) = \langle t, t^2, t^3/3 \rangle$ , so  $\mathbf{r}'(t) = \langle 1, 2t, t^2 \rangle$ . Compute the cross product:

$$\begin{aligned} \mathbf{r}(t) \times \mathbf{r}'(t) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ t & t^2 & t^3/3 \\ 1 & 2t & t^2 \end{vmatrix} \\ \mathbf{i} : & t^2 \cdot t^2 - \frac{t^3}{3} \cdot 2t = t^4 - \frac{2t^4}{3} = \frac{t^4}{3}. \\ \mathbf{j} : & -\left(t \cdot t^2 - \frac{t^3}{3} \cdot 1\right) = -t^3 + \frac{t^3}{3} = -\frac{2t^3}{3}. \\ \mathbf{k} : & t \cdot 2t - t^2 \cdot 1 = 2t^2 - t^2 = t^2. \end{aligned}$$

So  $\mathbf{r} \times \mathbf{r}' = \langle t^4/3, -2t^3/3, t^2 \rangle$ . Integrating from 0 to  $\pi$ :

$$\int_0^\pi \left\langle \frac{t^4}{3}, -\frac{2t^3}{3}, t^2 \right\rangle dt = \left\langle \left. \frac{t^5}{15} \right|_0^\pi, -\left. \frac{t^4}{6} \right|_0^\pi, \left. \frac{t^3}{3} \right|_0^\pi \right\rangle = \left\langle \frac{\pi^5}{15}, -\frac{\pi^4}{6}, \frac{\pi^3}{3} \right\rangle.$$

11. **Solution.**  $\mathbf{r}(t) = \langle e^t \cos t, e^t \sin t, t^2 \rangle$ . Using the product rule:

$$\begin{aligned}\mathbf{r}'(t) &= \langle e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t, 2t \rangle \\ &= \langle e^t(\cos t - \sin t), e^t(\sin t + \cos t), 2t \rangle.\end{aligned}$$

*Geometric interpretation:* The first two components trace a spiral of exponentially growing radius  $e^t$  in the  $xy$ -plane. The derivative shows the velocity tangent to this expanding spiral, while the  $z$ -component  $2t$  indicates the rate of climb accelerates linearly. The particle spirals outward with increasing speed.

12. **Solution.**  $\mathbf{r}(t) = \langle \ln t, t, \sqrt{t} \rangle$  for  $t > 0$ .

$$\mathbf{r}'(t) = \left\langle \frac{1}{t}, 1, \frac{1}{2\sqrt{t}} \right\rangle, \quad \boxed{\mathbf{r}''(t) = \left\langle -\frac{1}{t^2}, 0, -\frac{1}{4t^{3/2}} \right\rangle}.$$

13. **Solution.**  $\mathbf{r}(t) = \langle e^t, \ln t, t^3 \rangle$ .

$$\mathbf{v}(t) = \mathbf{r}'(t) = \left\langle e^t, \frac{1}{t}, 3t^2 \right\rangle, \quad \mathbf{a}(t) = \mathbf{r}''(t) = \left\langle e^t, -\frac{1}{t^2}, 6t \right\rangle.$$

At  $t = 1$ :

$$\boxed{\mathbf{v}(1) = \langle e, 1, 3 \rangle, \quad \mathbf{a}(1) = \langle e, -1, 6 \rangle}.$$

14. **Solution.**  $\mathbf{r}(t) = \langle t^2, t^3, t \rangle$ . At  $t = 1$ :  $\mathbf{r}(1) = \langle 1, 1, 1 \rangle$  and  $\mathbf{r}'(t) = \langle 2t, 3t^2, 1 \rangle$  so  $\mathbf{r}'(1) = \langle 2, 3, 1 \rangle$ . The tangent line is

$$\boxed{\ell(t) = \langle 1, 1, 1 \rangle + t\langle 2, 3, 1 \rangle = \langle 1 + 2t, 1 + 3t, 1 + t \rangle}.$$

15. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ , so  $\mathbf{v}(t) = \langle 1, 2t, 3t^2 \rangle$  and

$$\boxed{\mathbf{a}(t) = \langle 0, 2, 6t \rangle}.$$

The acceleration has no component in the  $x$ -direction, a constant component of 2 in the  $y$ -direction, and a linearly increasing component  $6t$  in the  $z$ -direction. This means the particle accelerates more strongly in the  $z$ -direction as time increases, consistent with the cubic dependence of  $z$  on  $t$ .

16. **Solution.**  $\mathbf{r}(t) = \langle t, \cos t, e^t \rangle$ .

$$\boxed{\mathbf{v}(t) = \langle 1, -\sin t, e^t \rangle, \quad \mathbf{a}(t) = \langle 0, -\cos t, e^t \rangle}.$$

17. **Solution.**  $\mathbf{r}(t) = \langle t^2, \sin t, \cos t \rangle$ ,  $\mathbf{r}'(t) = \langle 2t, \cos t, -\sin t \rangle$ .

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = t^2 \cdot 2t + \sin t \cos t + \cos t(-\sin t) = 2t^3 + \sin t \cos t - \sin t \cos t = 2t^3.$$

Therefore

$$\boxed{\int \mathbf{r}(t) \cdot \mathbf{r}'(t) dt = \int 2t^3 dt = \frac{t^4}{2} + C}.$$



18. **Solution.**  $\mathbf{r}(t) = \langle 10t, 20t - 4.9t^2 \rangle$ .

$$\mathbf{v}(t) = \langle 10, 20 - 9.8t \rangle, \quad \mathbf{a}(t) = \langle 0, -9.8 \rangle.$$

At  $t = 1$ :

$$\boxed{\mathbf{v}(1) = \langle 10, 10.2 \rangle, \quad \mathbf{a}(1) = \langle 0, -9.8 \rangle.}$$

19. **Solution.**  $\mathbf{r}(t) = \langle 2 \cos t, 3 \sin t, t \rangle$ .  $\mathbf{v}(t) = \langle -2 \sin t, 3 \cos t, 1 \rangle$ . Speed:  $|\mathbf{v}(t)| = \sqrt{4 \sin^2 t + 9 \cos^2 t + 1} = \sqrt{5 + 5 \cos^2 t}$ . At  $t = \pi/4$ :  $\cos^2(\pi/4) = 1/2$ , so

$$|\mathbf{v}(\pi/4)| = \sqrt{5 + 5/2} = \sqrt{15/2} = \frac{\sqrt{30}}{2}.$$

$\mathbf{a}(t) = \langle -2 \cos t, -3 \sin t, 0 \rangle$ . At  $t = \pi/4$ :

$$\boxed{|\mathbf{v}(\pi/4)| = \frac{\sqrt{30}}{2} \approx 2.739, \quad \mathbf{a}(\pi/4) = \left\langle -\sqrt{2}, -\frac{3\sqrt{2}}{2}, 0 \right\rangle.}$$

20. **Solution.** With  $\mathbf{v}_0 = \langle 20, 30 \rangle$  and  $\mathbf{a} = \langle 0, -9.8 \rangle$ , integrating:

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}t = \langle 20, 30 - 9.8t \rangle.$$

Assuming initial position  $\mathbf{r}(0) = \langle 0, 0 \rangle$ :

$$\boxed{\mathbf{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle.}$$

21. **Solution.**  $\mathbf{r}(t) = \langle v_0 \cos \theta \cdot t, v_0 \sin \theta \cdot t - \frac{1}{2}gt^2 \rangle$ . The vertical component of velocity is  $v_y(t) = v_0 \sin \theta - gt$ . Maximum height occurs when  $v_y = 0$ :

$$t^* = \frac{v_0 \sin \theta}{g}.$$

Substituting into the height:

$$y_{\max} = v_0 \sin \theta \cdot \frac{v_0 \sin \theta}{g} - \frac{1}{2}g \left( \frac{v_0 \sin \theta}{g} \right)^2 = \frac{v_0^2 \sin^2 \theta}{g} - \frac{v_0^2 \sin^2 \theta}{2g}.$$

$$\boxed{y_{\max} = \frac{v_0^2 \sin^2 \theta}{2g}.$$

22. **Solution.**  $\mathbf{v}(t) = \langle t^2, \sin t \rangle$  and  $\mathbf{r}(0) = \langle 1, 0 \rangle$ .

$$\mathbf{r}(t) = \mathbf{r}(0) + \int_0^t \mathbf{v}(u) du = \langle 1, 0 \rangle + \left\langle \frac{t^3}{3}, 1 - \cos t \right\rangle.$$

$$\boxed{\mathbf{r}(t) = \left\langle 1 + \frac{t^3}{3}, 1 - \cos t \right\rangle.}$$

23. **Solution.**  $\mathbf{r}(t) = \langle t^3, t^2, t \rangle$ .  $\mathbf{v}(t) = \langle 3t^2, 2t, 1 \rangle$ ,  $\mathbf{a}(t) = \langle 6t, 2, 0 \rangle$ . At  $t = 2$ :

$$\mathbf{v}(2) = \langle 12, 4, 1 \rangle, \quad \mathbf{a}(2) = \langle 12, 2, 0 \rangle.$$

24. **Solution.** A projectile launched with speed  $v_0$  at angle  $\theta$  has initial velocity  $\mathbf{v}_0 = \langle v_0 \cos \theta, v_0 \sin \theta \rangle$ . Under gravity  $\mathbf{a} = \langle 0, -g \rangle$ :

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}t = \langle v_0 \cos \theta, v_0 \sin \theta - gt \rangle.$$

Therefore:

$$v_x(t) = v_0 \cos \theta \quad (\text{constant}), \quad v_y(t) = v_0 \sin \theta - gt.$$

25. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$  so  $\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle$ .

$$|\mathbf{r}'(t)| = \sqrt{1 + 4t^2 + 9t^4}.$$

26. **Solution.**  $\mathbf{a}(t) = \langle 0, 2 \rangle$ ,  $\mathbf{v}(0) = \langle 5, 0 \rangle$ . Integrating:  $\mathbf{v}(t) = \langle 5, 2t \rangle$ . Assuming  $\mathbf{r}(0) = \langle 0, 0 \rangle$ :

$$\mathbf{r}(t) = \langle 5t, t^2 \rangle.$$

27. **Solution.**  $\mathbf{r}(t) = \langle 10t, 20t - 4.9t^2 \rangle$ , so  $v_y(t) = 20 - 9.8t$ . Maximum height when  $v_y = 0$ :

$$20 - 9.8t = 0 \implies t = \frac{20}{9.8} = \frac{100}{49} \approx 2.041 \text{ s.}$$

28. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ ,  $\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle$ . At  $t = 1$ :  $\mathbf{r}'(1) = \langle 1, 2, 3 \rangle$ ,  $|\mathbf{r}'(1)| = \sqrt{1 + 4 + 9} = \sqrt{14}$ .

$$\mathbf{T}(1) = \frac{1}{\sqrt{14}} \langle 1, 2, 3 \rangle.$$

29. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$ ,  $\mathbf{r}'(t) = \langle 1, 2t, 0 \rangle$ ,  $\mathbf{r}''(t) = \langle 0, 2, 0 \rangle$ .

$$\mathbf{r}' \times \mathbf{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2t & 0 \\ 0 & 2 & 0 \end{vmatrix} = \langle 0, 0, 2 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = 2, \quad |\mathbf{r}'| = (1 + 4t^2)^{1/2}.$$

$$\kappa(t) = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3} = \frac{2}{(1 + 4t^2)^{3/2}}.$$

30. **Solution.**  $\mathbf{r}(t) = \langle t, t^3, t^2 \rangle$ ,  $\mathbf{r}'(t) = \langle 1, 3t^2, 2t \rangle$ ,  $\mathbf{r}''(t) = \langle 0, 6t, 2 \rangle$ . At  $t = 0$ :  $\mathbf{r}'(0) = \langle 1, 0, 0 \rangle$ ,  $\mathbf{r}''(0) = \langle 0, 0, 2 \rangle$ .

$$\mathbf{r}'(0) \times \mathbf{r}''(0) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{vmatrix} = \langle 0, -2, 0 \rangle.$$

$|\mathbf{r}' \times \mathbf{r}''| = 2$ ,  $|\mathbf{r}'(0)| = 1$ .  $\kappa(0) = 2/1 = 2$ , so the radius of curvature is

$$\rho = \frac{1}{\kappa(0)} = \frac{1}{2}.$$

31. **Solution.**  $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ ,  $\mathbf{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$ ,  $\mathbf{r}''(t) = \langle -\cos t, -\sin t, 0 \rangle$ .

$$\mathbf{r}' \times \mathbf{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin t & \cos t & 1 \\ -\cos t & -\sin t & 0 \end{vmatrix} = \langle \sin t, -\cos t, 1 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}, \quad |\mathbf{r}'| = \sqrt{1 + 1} = \sqrt{2}.$$

$$\kappa = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{\sqrt{2}}{2\sqrt{2}} = \frac{1}{2}.$$

Since  $\kappa = 1/2$  is independent of  $t$ , the curvature is constant.  $\square$

32. **Solution.**  $\mathbf{r}(t) = \langle t, \ln t, t^2 \rangle$  for  $t > 0$ .  $\mathbf{r}'(t) = \langle 1, 1/t, 2t \rangle$ ,  $\mathbf{r}''(t) = \langle 0, -1/t^2, 2 \rangle$ . At  $t = 1$ :  $\mathbf{r}'(1) = \langle 1, 1, 2 \rangle$ ,  $|\mathbf{r}'(1)| = \sqrt{6}$ .

$$\mathbf{T}(1) = \frac{1}{\sqrt{6}} \langle 1, 1, 2 \rangle.$$

$$\mathbf{r}''(1) = \langle 0, -1, 2 \rangle.$$

$\mathbf{r}' \times \mathbf{r}''$  at  $t = 1$ :

$$\langle 1, 1, 2 \rangle \times \langle 0, -1, 2 \rangle = \langle 1 \cdot 2 - 2(-1), 2 \cdot 0 - 1 \cdot 2, 1(-1) - 1 \cdot 0 \rangle = \langle 4, -2, -1 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{16 + 4 + 1} = \sqrt{21}. \quad \kappa = \sqrt{21}/(\sqrt{6})^3 = \sqrt{21}/(6\sqrt{6}).$$

To find  $\mathbf{N}$ , compute  $\mathbf{T}'(t)$  at  $t = 1$ . We use:  $\mathbf{T}' = \frac{\mathbf{r}''|\mathbf{r}'|^2 - \mathbf{r}'(\mathbf{r}' \cdot \mathbf{r}'')}{|\mathbf{r}'|^4} \cdot |\mathbf{r}'|$ .

Alternatively, use  $\mathbf{N} = \mathbf{B} \times \mathbf{T}$  where  $\mathbf{B} = \frac{\mathbf{r}' \times \mathbf{r}''}{|\mathbf{r}' \times \mathbf{r}''|}$ .

$$\mathbf{B}(1) = \frac{1}{\sqrt{21}} \langle 4, -2, -1 \rangle.$$

$$\mathbf{N}(1) = \mathbf{B}(1) \times \mathbf{T}(1) = \frac{1}{\sqrt{21} \cdot \sqrt{6}} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -2 & -1 \\ 1 & 1 & 2 \end{vmatrix}.$$

$$= \frac{1}{\sqrt{126}} \langle (-2)(2) - (-1)(1), (-1)(1) - (4)(2), (4)(1) - (-2)(1) \rangle = \frac{1}{\sqrt{126}} \langle -3, -9, 6 \rangle.$$

Simplify:  $\langle -3, -9, 6 \rangle = 3\langle -1, -3, 2 \rangle$ ,  $|\langle -3, -9, 6 \rangle| = 3\sqrt{14}$ . Since  $|\langle -3, -9, 6 \rangle| = 3\sqrt{14} = \sqrt{126}$ , the unit normal is:

$$\mathbf{N}(1) = \frac{1}{\sqrt{14}} \langle -1, -3, 2 \rangle, \quad \mathbf{B}(1) = \frac{1}{\sqrt{21}} \langle 4, -2, -1 \rangle.$$

33. **Solution.**  $\mathbf{r}(t) = \langle t, \sin t, \cos t \rangle$ ,  $\mathbf{r}'(t) = \langle 1, \cos t, -\sin t \rangle$ ,  $\mathbf{r}''(t) = \langle 0, -\sin t, -\cos t \rangle$ .  
 $|\mathbf{r}'(t)| = \sqrt{1 + \cos^2 t + \sin^2 t} = \sqrt{2}$ .

$$\mathbf{r}' \times \mathbf{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & \cos t & -\sin t \\ 0 & -\sin t & -\cos t \end{vmatrix} = \langle -\cos^2 t - \sin^2 t, \cos t, -\sin t \rangle = \langle -1, \cos t, -\sin t \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{1 + \cos^2 t + \sin^2 t} = \sqrt{2}.$$

$$\kappa(t) = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{1}{2}.$$

The curvature  $\kappa = 1/2$  is **constant** for all  $t$ , so the curve is equally curved everywhere—there is no point where it is “most curved.”  $\boxed{\kappa = \frac{1}{2} \text{ (constant for all } t\text{).}}$

34. **Solution.**  $\mathbf{r}(t) = \langle e^t, e^{-t}, t \rangle$ ,  $\mathbf{r}'(t) = \langle e^t, -e^{-t}, 1 \rangle$ ,  $\mathbf{r}''(t) = \langle e^t, e^{-t}, 0 \rangle$ . At  $t = 0$ :  
 $\mathbf{r}'(0) = \langle 1, -1, 1 \rangle$ ,  $\mathbf{r}''(0) = \langle 1, 1, 0 \rangle$ .

$$\mathbf{r}'(0) \times \mathbf{r}''(0) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \langle -1, 1, 2 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{1 + 1 + 4} = \sqrt{6}, \quad |\mathbf{r}'(0)| = \sqrt{3}.$$

$$\boxed{\kappa(0) = \frac{\sqrt{6}}{(\sqrt{3})^3} = \frac{\sqrt{6}}{3\sqrt{3}} = \frac{\sqrt{2}}{3}.}$$

35. **Solution.**  $\mathbf{r}(t) = \langle t, \ln t, \sqrt{t} \rangle$  for  $t > 0$ .  $\mathbf{r}'(t) = \langle 1, 1/t, 1/(2\sqrt{t}) \rangle$ ,  $\mathbf{r}''(t) = \langle 0, -1/t^2, -1/(4t^{3/2}) \rangle$ .

$\mathbf{r}' \times \mathbf{r}''$ :

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & \frac{1}{t} & \frac{1}{2\sqrt{t}} \\ 0 & -\frac{1}{t^2} & -\frac{1}{4t^{3/2}} \end{vmatrix}.$$

$$\mathbf{i}: \frac{1}{t} \cdot \left(-\frac{1}{4t^{3/2}}\right) - \frac{1}{2\sqrt{t}} \cdot \left(-\frac{1}{t^2}\right) = -\frac{1}{4t^{5/2}} + \frac{1}{2t^{5/2}} = \frac{1}{4t^{5/2}}.$$

$$\mathbf{j}: -\left[1 \cdot \left(-\frac{1}{4t^{3/2}}\right) - \frac{1}{2\sqrt{t}} \cdot 0\right] = \frac{1}{4t^{3/2}}.$$

$$\mathbf{k}: 1 \cdot \left(-\frac{1}{t^2}\right) - \frac{1}{t} \cdot 0 = -\frac{1}{t^2}.$$

$$\mathbf{r}' \times \mathbf{r}'' = \left\langle \frac{1}{4t^{5/2}}, \frac{1}{4t^{3/2}}, -\frac{1}{t^2} \right\rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''|^2 = \frac{1}{16t^5} + \frac{1}{16t^3} + \frac{1}{t^4} = \frac{1 + t^2 + 16t}{16t^5}.$$

$$|\mathbf{r}'|^2 = 1 + \frac{1}{t^2} + \frac{1}{4t} = \frac{4t^2+4+t}{4t^2}, \text{ so } |\mathbf{r}'|^3 = \left(\frac{4t^2+t+4}{4t^2}\right)^{3/2}.$$

$$\kappa(t) = \frac{\sqrt{1+t^2+16t}}{(4t^2+t+4)^{3/2}} \cdot \frac{(4t^2)^{3/2}}{4t^{5/2}} = \frac{\sqrt{1+t^2+16t} \cdot 4t^{1/2}}{(4t^2+t+4)^{3/2}}.$$

(This is the exact expression; it does not simplify to a closed-form elementary expression.)

36. **Solution.**  $\mathbf{r}(t) = \langle t^2, t^3, t \rangle$ ,  $\mathbf{r}'(t) = \langle 2t, 3t^2, 1 \rangle$ ,  $\mathbf{r}''(t) = \langle 2, 6t, 0 \rangle$ . At  $t = 1$ :  $\mathbf{r}' = \langle 2, 3, 1 \rangle$ ,  $|\mathbf{r}'| = \sqrt{14}$ ;  $\mathbf{r}'' = \langle 2, 6, 0 \rangle$ .

$$\mathbf{r}' \times \mathbf{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 1 \\ 2 & 6 & 0 \end{vmatrix} = \langle -6, 2, 6 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{36 + 4 + 36} = \sqrt{76} = 2\sqrt{19}.$$

$$\kappa(1) = \frac{2\sqrt{19}}{(\sqrt{14})^3} = \frac{2\sqrt{19}}{14\sqrt{14}} = \frac{\sqrt{19}}{7\sqrt{14}}.$$

$$\mathbf{T}(1) = \frac{1}{\sqrt{14}}\langle 2, 3, 1 \rangle, \mathbf{B}(1) = \frac{1}{2\sqrt{19}}\langle -6, 2, 6 \rangle = \frac{1}{\sqrt{19}}\langle -3, 1, 3 \rangle.$$

$$\begin{aligned} \mathbf{N}(1) &= \mathbf{B}(1) \times \mathbf{T}(1) = \frac{1}{\sqrt{19}\sqrt{14}} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 1 & 3 \\ 2 & 3 & 1 \end{vmatrix} \\ &= \frac{1}{\sqrt{266}}\langle 1-9, 6+3, -9-2 \rangle = \frac{1}{\sqrt{266}}\langle -8, 9, -11 \rangle. \end{aligned}$$

Note that  $|\langle -8, 9, -11 \rangle| = \sqrt{64 + 81 + 121} = \sqrt{266}$ , confirming unit length.

$$\kappa(1) = \frac{\sqrt{19}}{7\sqrt{14}} = \frac{\sqrt{266}}{98}, \quad \mathbf{N}(1) = \frac{1}{\sqrt{266}}\langle -8, 9, -11 \rangle.$$

37. **Solution.**  $\mathbf{r}(t) = \langle \ln(t+1), \sqrt{t}, t^2 \rangle$ .  $\mathbf{r}'(t) = \langle \frac{1}{t+1}, \frac{1}{2\sqrt{t}}, 2t \rangle$ ,  $\mathbf{r}''(t) = \langle -\frac{1}{(t+1)^2}, -\frac{1}{4t^{3/2}}, 2 \rangle$ .

At  $t = 1$ :  $\mathbf{r}'(1) = \langle 1/2, 1/2, 2 \rangle$ ,  $\mathbf{r}''(1) = \langle -1/4, -1/4, 2 \rangle$ .  $|\mathbf{r}'(1)| = \sqrt{1/4 + 1/4 + 4} = \sqrt{9/2} = 3/\sqrt{2}$ .

Computing  $\mathbf{r}'(1) \times \mathbf{r}''(1)$  component-wise:

$$\mathbf{i}: (1/2)(2) - (2)(-1/4) = 1 + 1/2 = 3/2.$$

$$\mathbf{j}: (2)(-1/4) - (1/2)(2) = -1/2 - 1 = -3/2.$$

$$\mathbf{k}: (1/2)(-1/4) - (1/2)(-1/4) = -1/8 + 1/8 = 0.$$

$$\text{So } \mathbf{r}' \times \mathbf{r}'' = \langle 3/2, -3/2, 0 \rangle, \quad |\mathbf{r}' \times \mathbf{r}''| = \frac{3}{2}\sqrt{2} = \frac{3\sqrt{2}}{2}.$$

$$\kappa(1) = \frac{3\sqrt{2}/2}{(3/\sqrt{2})^3} = \frac{3\sqrt{2}/2}{27/(2\sqrt{2})} = \frac{3\sqrt{2}}{2} \cdot \frac{2\sqrt{2}}{27} = \frac{12}{54} = \frac{2}{9}.$$

$$\kappa(1) = \frac{2}{9}.$$

38. **Solution.**  $\mathbf{r}(t) = \langle t, \ln t, t^2 \rangle$ . At  $t = 1$ :  $\mathbf{r}'(1) = \langle 1, 1, 2 \rangle$ ,  $|\mathbf{r}'(1)| = \sqrt{6}$ .

This is the same curve as Problem 32. From that solution:

$$\mathbf{T}(1) = \frac{1}{\sqrt{6}} \langle 1, 1, 2 \rangle, \quad \mathbf{N}(1) = \frac{1}{\sqrt{14}} \langle -1, -3, 2 \rangle, \quad \mathbf{B}(1) = \frac{1}{\sqrt{21}} \langle 4, -2, -1 \rangle.$$

(See Problem 32 for the full derivation.)

39. **Solution.**  $\mathbf{r}(t) = \langle 2 \cos t, 3 \sin t, t \rangle$ .  $\mathbf{r}'(t) = \langle -2 \sin t, 3 \cos t, 1 \rangle$ ,  $|\mathbf{r}'| = \sqrt{4 \sin^2 t + 9 \cos^2 t + 1} = \sqrt{5 + 5 \cos^2 t}$ .

$$\mathbf{T}(t) = \frac{\langle -2 \sin t, 3 \cos t, 1 \rangle}{\sqrt{5 + 5 \cos^2 t}}.$$

$$\mathbf{r}''(t) = \langle -2 \cos t, -3 \sin t, 0 \rangle.$$

$$\mathbf{r}' \times \mathbf{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 \sin t & 3 \cos t & 1 \\ -2 \cos t & -3 \sin t & 0 \end{vmatrix} = \langle 3 \sin t, -2 \cos t, 6 \rangle.$$

(Check  $\mathbf{i}$ :  $3 \cos t \cdot 0 - 1 \cdot (-3 \sin t) = 3 \sin t$ .  $\mathbf{j}$ :  $-((-2 \sin t)(0) - (1)(-2 \cos t)) = -2 \cos t$ .

$\mathbf{k}$ :  $(-2 \sin t)(-3 \sin t) - (3 \cos t)(-2 \cos t) = 6 \sin^2 t + 6 \cos^2 t = 6$ .)

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{9 \sin^2 t + 4 \cos^2 t + 36} = \sqrt{40 - 5 \cos^2 t}.$$

$$\mathbf{B}(t) = \frac{\langle 3 \sin t, -2 \cos t, 6 \rangle}{\sqrt{40 - 5 \cos^2 t}}.$$

$$\mathbf{N}(t) = \mathbf{B}(t) \times \mathbf{T}(t).$$

While  $\mathbf{N}$  can be written explicitly, it is algebraically involved. In closed form:

$$\mathbf{T}(t) = \frac{\langle -2 \sin t, 3 \cos t, 1 \rangle}{\sqrt{5 + 5 \cos^2 t}}, \quad \mathbf{B}(t) = \frac{\langle 3 \sin t, -2 \cos t, 6 \rangle}{\sqrt{40 - 5 \cos^2 t}}, \quad \mathbf{N}(t) = \mathbf{B}(t) \times \mathbf{T}(t).$$

40. **Solution.**  $\mathbf{r}(t) = \langle (2 + \cos 3t) \cos t, (2 + \cos 3t) \sin t, \sin 3t \rangle$ . At  $t = 0$ :  $\mathbf{r}(0) = \langle 3, 0, 0 \rangle$ .

$\mathbf{r}'(t)$ : Using the product rule,

$$x'(t) = -3 \sin 3t \cos t - (2 + \cos 3t) \sin t,$$

$$y'(t) = -3 \sin 3t \sin t + (2 + \cos 3t) \cos t,$$

$$z'(t) = 3 \cos 3t.$$

At  $t = 0$ :  $\sin 0 = 0$ ,  $\cos 0 = 1$ ,  $\sin 0 = 0$ ,  $\cos 0 = 1$ :

$$\mathbf{r}'(0) = \langle 0, 3, 3 \rangle, \quad |\mathbf{r}'(0)| = 3\sqrt{2}.$$

Evaluating  $\mathbf{r}''(t)$  at  $t = 0$ :

$$x'(t) = -3 \sin 3t \cos t - (2 + \cos 3t) \sin t, \quad x''(t) = -9 \cos 3t \cos t + 3 \sin 3t \sin t + 3 \sin 3t \sin t - (2 + \cos 3t) \cos t. \text{ At } t = 0: x''(0) = -9(1)(1) + 0 + 0 - 3(1) = -12.$$

$$y'(t) = -3 \sin 3t \sin t + (2 + \cos 3t) \cos t, \quad y''(t) = -9 \cos 3t \sin t - 3 \sin 3t \cos t - 3 \sin 3t \cos t - (2 + \cos 3t) \sin t. \text{ At } t = 0: y''(0) = 0 - 0 - 0 - 0 = 0.$$

$z'(t) = 3 \cos 3t$ ,  $z''(t) = -9 \sin 3t$ . At  $t = 0$ :  $z''(0) = 0$ .

So  $\mathbf{r}''(0) = \langle -12, 0, 0 \rangle$ .

$\mathbf{r}'(0) \times \mathbf{r}''(0) = \langle 0, 3, 3 \rangle \times \langle -12, 0, 0 \rangle = \langle 0 - 0, 3(-12) - 0, 0 - 3(-12) \rangle = \langle 0, -36, 36 \rangle$ .

$|\mathbf{r}' \times \mathbf{r}''| = 36\sqrt{2}$ ,  $|\mathbf{r}'| = 3\sqrt{2}$ .

$$\kappa(0) = \frac{36\sqrt{2}}{(3\sqrt{2})^3} = \frac{36\sqrt{2}}{54\sqrt{2}} = \frac{2}{3}.$$

For torsion, we need  $\mathbf{r}'''(0)$ . Compute  $\mathbf{r}'''$ :  $x'''(t) = \frac{d}{dt}x''(t)$ . At  $t = 0$ , using repeated differentiation:

$x''(t) = -9 \cos 3t \cos t + 6 \sin 3t \sin t - (2 + \cos 3t) \cos t$ .  $x'''(t) = 27 \sin 3t \cos t + 9 \cos 3t \sin t + 18 \cos 3t \sin t + 6 \sin 3t \cos t + 3 \sin 3t \cos t + (2 + \cos 3t) \sin t$ . At  $t = 0$ : all terms with  $\sin 0$  or  $\sin 0$  vanish.  $x'''(0) = 0$ .

$y''(t) = -9 \cos 3t \sin t - 6 \sin 3t \cos t - (2 + \cos 3t) \sin t$ .  $y'''(t) = 27 \sin 3t \sin t - 9 \cos 3t \cos t - 18 \cos 3t \cos t + 6 \sin 3t \sin t + 3 \sin 3t \sin t - (2 + \cos 3t) \cos t$ . At  $t = 0$ :  $y'''(0) = 0 - 9 - 18 + 0 + 0 - 3 = -30$ .

$z''(t) = -9 \sin 3t$ ,  $z'''(t) = -27 \cos 3t$ .  $z'''(0) = -27$ .

So  $\mathbf{r}'''(0) = \langle 0, -30, -27 \rangle$ .

Torsion:  $\tau = \frac{(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''}{|\mathbf{r}' \times \mathbf{r}''|^2}$ .

$$\langle 0, -36, 36 \rangle \cdot \langle 0, -30, -27 \rangle = 0 + 1080 - 972 = 108.$$

$$|\mathbf{r}' \times \mathbf{r}''|^2 = (36\sqrt{2})^2 = 2592.$$

$$\tau(0) = \frac{108}{2592} = \frac{1}{24}.$$

$$\boxed{\kappa(0) = \frac{2}{3}, \quad \tau(0) = \frac{1}{24}.$$

41. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ ,  $\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle$ ,  $\mathbf{r}''(t) = \langle 0, 2, 6t \rangle$ . At  $t = 1$ :  $\mathbf{r}' = \langle 1, 2, 3 \rangle$ ,  $\mathbf{r}'' = \langle 0, 2, 6 \rangle$ .

$$\mathbf{r}' \times \mathbf{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 0 & 2 & 6 \end{vmatrix} = \langle 12 - 6, 0 - 6, 2 - 0 \rangle = \langle 6, -6, 2 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{36 + 36 + 4} = \sqrt{76} = 2\sqrt{19}.$$

$$\boxed{\mathbf{B}(1) = \frac{1}{2\sqrt{19}} \langle 6, -6, 2 \rangle = \frac{1}{\sqrt{19}} \langle 3, -3, 1 \rangle.}$$

42. **Solution.**  $\mathbf{r}(t) = \langle t, \sin t, \cos t \rangle$ ,  $\mathbf{r}' = \langle 1, \cos t, -\sin t \rangle$ ,  $|\mathbf{r}'| = \sqrt{2}$ .

$$\mathbf{T}(t) = \frac{1}{\sqrt{2}} \langle 1, \cos t, -\sin t \rangle.$$

$$\mathbf{T}'(t) = \frac{1}{\sqrt{2}}\langle 0, -\sin t, -\cos t \rangle, |\mathbf{T}'| = \frac{1}{\sqrt{2}}.$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'}{|\mathbf{T}'|} = \langle 0, -\sin t, -\cos t \rangle.$$

$$\kappa = |\mathbf{T}'|/|\mathbf{r}'| = \frac{1/\sqrt{2}}{\sqrt{2}} = \frac{1}{2}.$$

$$\begin{aligned} \mathbf{B}(t) &= \mathbf{T} \times \mathbf{N} = \frac{1}{\sqrt{2}} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & \cos t & -\sin t \\ 0 & -\sin t & -\cos t \end{vmatrix} = \frac{1}{\sqrt{2}} \langle -\cos^2 t - \sin^2 t, \cos t, -\sin t \rangle = \\ &= \frac{1}{\sqrt{2}} \langle -1, \cos t, -\sin t \rangle. \end{aligned}$$

Verifying component-wise:  $\mathbf{i} : \cos t(-\cos t) - (-\sin t)(-\sin t) = -\cos^2 t - \sin^2 t = -1$ ;  
 $\mathbf{j} : -[1(-\cos t) - (-\sin t)(0)] = \cos t$ ;  $\mathbf{k} : 1(-\sin t) - \cos t(0) = -\sin t$ .

Thus  $\mathbf{B} = \frac{1}{\sqrt{2}}\langle -1, \cos t, -\sin t \rangle$ , which has magnitude  $\frac{1}{\sqrt{2}}\sqrt{1+1} = 1$  as expected.

The Frenet–Serret formulas are:

$$\begin{aligned} \mathbf{T}' &= \kappa |\mathbf{r}'| \mathbf{N} = \frac{1}{2} \cdot \sqrt{2} \cdot \mathbf{N} = \frac{\sqrt{2}}{2} \mathbf{N}, \\ \mathbf{N}' &= -\kappa |\mathbf{r}'| \mathbf{T} + \tau |\mathbf{r}'| \mathbf{B}, \\ \mathbf{B}' &= -\tau |\mathbf{r}'| \mathbf{N}. \end{aligned}$$

(Here primes denote  $d/dt$ ; alternatively in terms of arc length  $s$ :  $d\mathbf{T}/ds = \kappa\mathbf{N}$ , etc.)

For torsion:  $\mathbf{r}'' = \langle 0, -\sin t, -\cos t \rangle$ , so  $\mathbf{r}''' = \langle 0, -\cos t, \sin t \rangle$ .

$$\tau = \frac{(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''}{|\mathbf{r}' \times \mathbf{r}''|^2}. \quad \mathbf{r}' \times \mathbf{r}'' = \langle -1, \cos t, -\sin t \rangle \text{ (from Problem 33), } |\mathbf{r}' \times \mathbf{r}''| = \sqrt{2}.$$

$$(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}''' = \langle -1, \cos t, -\sin t \rangle \cdot \langle 0, -\cos t, \sin t \rangle = 0 - \cos^2 t - \sin^2 t = -1.$$

$$\tau = -1/2.$$

Therefore the Frenet–Serret formulas (with respect to arc length  $s$ , where  $ds/dt = \sqrt{2}$ ) are:

$$\boxed{\frac{d\mathbf{T}}{ds} = \frac{1}{2}\mathbf{N}, \quad \frac{d\mathbf{N}}{ds} = -\frac{1}{2}\mathbf{T} + \frac{1}{2}\mathbf{B}, \quad \frac{d\mathbf{B}}{ds} = -\frac{1}{2}\mathbf{N},}$$

with  $\kappa = 1/2$  and  $\tau = -1/2$ .

(Note: the negative torsion  $\tau = -1/2$  is used in  $d\mathbf{N}/ds = -\kappa\mathbf{T} + \tau\mathbf{B}$  and  $d\mathbf{B}/ds = -\tau\mathbf{N} = \frac{1}{2}\mathbf{N}$ .)

Correcting:  $d\mathbf{B}/ds = -\tau\mathbf{N} = -(-1/2)\mathbf{N} = \frac{1}{2}\mathbf{N}$ .

$$\boxed{\frac{d\mathbf{T}}{ds} = \frac{1}{2}\mathbf{N}, \quad \frac{d\mathbf{N}}{ds} = -\frac{1}{2}\mathbf{T} - \frac{1}{2}\mathbf{B}, \quad \frac{d\mathbf{B}}{ds} = \frac{1}{2}\mathbf{N}.}$$

43. **Solution.**  $\mathbf{r}(t) = \langle t, e^t, e^{-t} \rangle$ ,  $\mathbf{r}'(t) = \langle 1, e^t, -e^{-t} \rangle$ ,  $\mathbf{r}''(t) = \langle 0, e^t, e^{-t} \rangle$ . At  $t = 0$ :  
 $\mathbf{r}'(0) = \langle 1, 1, -1 \rangle$ ,  $|\mathbf{r}'(0)| = \sqrt{3}$ .

$$\mathbf{T}(0) = \frac{1}{\sqrt{3}}\langle 1, 1, -1 \rangle.$$



$$\mathbf{r}''(0) = \langle 0, 1, 1 \rangle. \quad \mathbf{r}'(0) \times \mathbf{r}''(0) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} = \langle 1+1, 0-1, 1-0 \rangle = \langle 2, -1, 1 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{4+1+1} = \sqrt{6}. \quad \mathbf{B}(0) = \frac{1}{\sqrt{6}} \langle 2, -1, 1 \rangle.$$

$$\mathbf{N}(0) = \mathbf{B}(0) \times \mathbf{T}(0) = \frac{1}{\sqrt{6}\sqrt{3}} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = \frac{1}{\sqrt{18}} \langle 1-1, 1+2, 2+1 \rangle = \frac{1}{3\sqrt{2}} \langle 0, 3, 3 \rangle.$$

$$= \frac{1}{\sqrt{2}} \langle 0, 1, 1 \rangle. \quad \text{One can verify } |\langle 0, 1, 1 \rangle / \sqrt{2}| = 1.$$

$$\boxed{\mathbf{T}(0) = \frac{1}{\sqrt{3}} \langle 1, 1, -1 \rangle, \quad \mathbf{N}(0) = \frac{1}{\sqrt{2}} \langle 0, 1, 1 \rangle, \quad \mathbf{B}(0) = \frac{1}{\sqrt{6}} \langle 2, -1, 1 \rangle.}$$

44. **Solution.**  $\mathbf{r}(t) = \langle t, \ln t, \sqrt{t} \rangle$ ,  $\mathbf{r}' = \langle 1, 1/t, 1/(2\sqrt{t}) \rangle$ ,  $\mathbf{r}'' = \langle 0, -1/t^2, -1/(4t^{3/2}) \rangle$ ,  $\mathbf{r}''' = \langle 0, 2/t^3, 3/(8t^{5/2}) \rangle$ .

$$\text{From Problem 35: } \mathbf{r}' \times \mathbf{r}'' = \langle 1/(4t^{5/2}), 1/(4t^{3/2}), -1/t^2 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''|^2 = \frac{1}{16t^5} + \frac{1}{16t^3} + \frac{1}{t^4} = \frac{1+t^2+16t}{16t^5}.$$

$$(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}''' = \frac{1}{4t^{5/2}} \cdot 0 + \frac{1}{4t^{3/2}} \cdot \frac{2}{t^3} + \left(-\frac{1}{t^2}\right) \cdot \frac{3}{8t^{5/2}} = \frac{1}{2t^{9/2}} - \frac{3}{8t^{9/2}} = \frac{1}{8t^{9/2}}.$$

$$\boxed{\tau(t) = \frac{1/(8t^{9/2})}{(1+t^2+16t)/(16t^5)} = \frac{16t^5}{8t^{9/2}(1+t^2+16t)} = \frac{2\sqrt{t}}{1+16t+t^2}.}$$

45. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ ,  $\mathbf{r}' = \langle 1, 2t, 3t^2 \rangle$ ,  $\mathbf{r}'' = \langle 0, 2, 6t \rangle$ ,  $\mathbf{r}''' = \langle 0, 0, 6 \rangle$ .

$$\mathbf{r}' \times \mathbf{r}'' = \langle 12t^2 - 6t^2, 0 - 6t, 2 - 0 \rangle = \langle 6t^2, -6t, 2 \rangle. \quad |\mathbf{r}' \times \mathbf{r}''|^2 = 36t^4 + 36t^2 + 4.$$

$$(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}''' = \langle 6t^2, -6t, 2 \rangle \cdot \langle 0, 0, 6 \rangle = 12.$$

$$\tau(t) = \frac{12}{36t^4+36t^2+4} = \frac{3}{9t^4+9t^2+1}.$$

By the Frenet-Serret formula,  $\frac{d\mathbf{B}}{ds} = -\tau \mathbf{N}$ , so

$$\frac{d\mathbf{B}}{dt} = -\tau |\mathbf{r}'| \mathbf{N}.$$

The torsion is  $\tau = \frac{3}{9t^4+9t^2+1}$  and  $|\mathbf{r}'| = \sqrt{1+4t^2+9t^4}$ .

Therefore:

$$\boxed{\frac{d\mathbf{B}}{dt} = -\frac{3\sqrt{1+4t^2+9t^4}}{9t^4+9t^2+1} \mathbf{N}(t).}$$

(At  $t = 1$ :  $\tau = 3/19$ ,  $|\mathbf{r}'| = \sqrt{14}$ , so  $d\mathbf{B}/dt|_{t=1} = -\frac{3\sqrt{14}}{19} \mathbf{N}(1)$ .)

46. **Solution.**  $\mathbf{r}(t) = \langle e^{-t} \cos t, e^{-t} \sin t, t \rangle$ .  $\mathbf{r}'(t) = \langle -e^{-t} \cos t - e^{-t} \sin t, -e^{-t} \sin t + e^{-t} \cos t, 1 \rangle = \langle -e^{-t}(\cos t + \sin t), e^{-t}(\cos t - \sin t), 1 \rangle$ .

$$\mathbf{r}''(t) = \langle e^{-t}(\cos t + \sin t) + e^{-t}(\sin t - \cos t), -e^{-t}(\cos t - \sin t) + e^{-t}(-\sin t - \cos t), 0 \rangle. \\ = \langle 2e^{-t} \sin t, -2e^{-t} \cos t, 0 \rangle.$$

At  $t = \pi$ :  $e^{-\pi}$ ,  $\sin \pi = 0$ ,  $\cos \pi = -1$ .  $\mathbf{r}'(\pi) = \langle -e^{-\pi}(-1 + 0), e^{-\pi}(-1 - 0), 1 \rangle = \langle e^{-\pi}, -e^{-\pi}, 1 \rangle$ .  $\mathbf{r}''(\pi) = \langle 0, 2e^{-\pi}, 0 \rangle$ .

$$\mathbf{r}'(\pi) \times \mathbf{r}''(\pi) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ e^{-\pi} & -e^{-\pi} & 1 \\ 0 & 2e^{-\pi} & 0 \end{vmatrix} = \langle 0 - 2e^{-2\pi}, 0 - 0, 2e^{-2\pi} - 0 \rangle = \langle -2e^{-2\pi}, 0, 2e^{-2\pi} \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{4e^{-2\pi} + 4e^{-4\pi}} = 2e^{-\pi} \sqrt{1 + e^{-2\pi}}.$$

$$\mathbf{B}(\pi) = \frac{\langle -2e^{-2\pi}, 0, 2e^{-2\pi} \rangle}{2e^{-\pi} \sqrt{1 + e^{-2\pi}}} = \frac{1}{\sqrt{1 + e^{-2\pi}}} \langle -1, 0, e^{-\pi} \rangle.$$

47. **Solution.** A space curve with constant curvature  $\kappa > 0$  and constant torsion  $\tau$  is a **circular helix** (or a circle if  $\tau = 0$ , or a straight line if  $\kappa = 0$ ).

More precisely: if  $\kappa > 0$  is constant and  $\tau = 0$ , the curve is a circle of radius  $1/\kappa$ . If both  $\kappa > 0$  and  $\tau \neq 0$  are constant, the curve is a circular helix. The radius of the helix is  $r = \kappa/(\kappa^2 + \tau^2)$  and the pitch (rise per turn) is  $2\pi|\tau|/(\kappa^2 + \tau^2)$ .

This is a classical result: a curve with constant  $\kappa > 0$  and constant  $\tau$  is a circular helix.

48. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$ ,  $\mathbf{r}'(t) = \langle 1, 2t, 0 \rangle$ ,  $|\mathbf{r}'| = \sqrt{1 + 4t^2}$ .

$$L = \int_0^3 \sqrt{1 + 4t^2} dt.$$

Let  $2t = \sinh u$ ,  $dt = \frac{1}{2} \cosh u du$ :

$$L = \frac{1}{2} \int_0^{\sinh^{-1} 6} \cosh^2 u du = \frac{1}{4} [u + \sinh u \cosh u]_0^{\sinh^{-1} 6}.$$

$$\sinh^{-1} 6 = \ln(6 + \sqrt{37}), \cosh(\sinh^{-1} 6) = \sqrt{37}.$$

$$L = \frac{1}{4} [\ln(6 + \sqrt{37}) + 6\sqrt{37}] \approx 9.747.$$

49. **Solution.**  $\mathbf{r}(t) = \langle 3t, 4t, 0 \rangle$ ,  $\mathbf{r}'(t) = \langle 3, 4, 0 \rangle$ ,  $|\mathbf{r}'| = 5$ . Arc length from 0 to  $t$ :  $s(t) = \int_0^t 5 du = 5t$ , so  $t = s/5$ .

$$\mathbf{r}(s) = \left\langle \frac{3s}{5}, \frac{4s}{5}, 0 \right\rangle.$$

50. **Solution.** For a polar curve  $r(\theta) = 3\theta$ , the arc length formula is

$$L = \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^\pi \sqrt{9\theta^2 + 9} d\theta = 3 \int_0^\pi \sqrt{\theta^2 + 1} d\theta.$$

Using  $\int \sqrt{\theta^2 + 1} d\theta = \frac{1}{2} [\theta \sqrt{\theta^2 + 1} + \sinh^{-1} \theta]$ :

$$L = \frac{3}{2} [\theta \sqrt{\theta^2 + 1} + \ln(\theta + \sqrt{\theta^2 + 1})]_0^\pi = \frac{3}{2} [\pi \sqrt{\pi^2 + 1} + \ln(\pi + \sqrt{\pi^2 + 1})].$$

$$L = \frac{3}{2} [\pi \sqrt{\pi^2 + 1} + \ln(\pi + \sqrt{\pi^2 + 1})] \approx 16.16.$$

51. **Solution.**  $\mathbf{r}(t) = \langle e^t, e^{-t}, t \rangle$ ,  $\mathbf{r}'(t) = \langle e^t, -e^{-t}, 1 \rangle$ .  $|\mathbf{r}'| = \sqrt{e^{2t} + e^{-2t} + 1}$ . Note  $e^{2t} + e^{-2t} = 2 \cosh 2t$ , so  $|\mathbf{r}'| = \sqrt{2 \cosh 2t + 1}$ .

Unfortunately this does not simplify to a nice closed form. However, observe:  $e^{2t} + e^{-2t} + 1 = (e^t + e^{-t})^2 - 1 = 4 \cosh^2 t - 1$ . So  $|\mathbf{r}'| = \sqrt{4 \cosh^2 t - 1}$ .

This integral does not have a simple closed form; we express it as:

$$L = \int_0^2 \sqrt{e^{2t} + e^{-2t} + 1} dt \approx 7.212.$$

(Numerical evaluation gives  $L \approx 7.212$ .)

52. **Solution.**  $y = \epsilon x^2$  so  $dy/dx = 2\epsilon x$ .

$$L = \int_0^1 \sqrt{1 + 4\epsilon^2 x^2} dx.$$

For  $\epsilon \ll 1$ , use the expansion  $\sqrt{1+u} \approx 1 + u/2 - u^2/8 + \dots$  with  $u = 4\epsilon^2 x^2$ :

$$L \approx \int_0^1 (1 + 2\epsilon^2 x^2 - 2\epsilon^4 x^4 + \dots) dx = 1 + \frac{2\epsilon^2}{3} - \frac{2\epsilon^4}{5} + \dots$$

To leading order:

$$L \approx 1 + \frac{2\epsilon^2}{3}.$$

53. **Solution.** A simple smooth path from  $\langle 0, 0, 0 \rangle$  to  $\langle 1, 1, 1 \rangle$  with continuous derivatives is

$$\mathbf{r}(t) = \langle t, t, t \rangle, \quad t \in [0, 1].$$

This is smooth ( $\mathbf{r}'(t) = \langle 1, 1, 1 \rangle \neq \mathbf{0}$ ) with all derivatives continuous.

For a more flexible design (e.g., starting and ending with zero velocity for a robotic arm), use a polynomial blend:

$$\mathbf{r}(t) = \langle s(t), s(t), s(t) \rangle, \quad s(t) = 3t^2 - 2t^3, \quad t \in [0, 1].$$

Then  $s(0) = 0$ ,  $s(1) = 1$ ,  $s'(0) = 0$ ,  $s'(1) = 0$ , and all derivatives are continuous. The velocity is  $\mathbf{r}'(t) = (6t - 6t^2)\langle 1, 1, 1 \rangle$ , which is zero at the endpoints (smooth start/stop).

$$\mathbf{r}(t) = \langle 3t^2 - 2t^3, 3t^2 - 2t^3, 3t^2 - 2t^3 \rangle, \quad t \in [0, 1].$$

54. **Solution.**  $\mathbf{r}(t) = \langle t^3, t^2, t \rangle$ ,  $\mathbf{r}'(t) = \langle 3t^2, 2t, 1 \rangle$ ,  $|\mathbf{r}'| = \sqrt{9t^4 + 4t^2 + 1}$ .

$$L = \int_0^2 \sqrt{9t^4 + 4t^2 + 1} dt.$$

Note  $9t^4 + 4t^2 + 1 = (3t^2 + 1)^2 - 2t^2$ . This does not factor nicely, so we evaluate numerically.

Numerically: 
$$L = \int_0^2 \sqrt{9t^4 + 4t^2 + 1} dt \approx 9.571.$$

55. **Solution.**  $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 4t \rangle$ ,  $\mathbf{r}'(t) = \langle -3 \sin t, 3 \cos t, 4 \rangle$ .  $|\mathbf{r}'| = \sqrt{9 \sin^2 t + 9 \cos^2 t + 16} = \sqrt{9 + 16} = 5$ .

Arc length:  $s(t) = 5t$ , so  $t = s/5$ .

$$\mathbf{r}(s) = \left\langle 3 \cos \frac{s}{5}, 3 \sin \frac{s}{5}, \frac{4s}{5} \right\rangle.$$

56. **Solution.**  $\mathbf{r}(t) = \langle t, \ln(1 + t^2), t^2 \rangle$ ,  $\mathbf{r}'(t) = \langle 1, \frac{2t}{1+t^2}, 2t \rangle$ .

$$|\mathbf{r}'|^2 = 1 + \frac{4t^2}{(1+t^2)^2} + 4t^2 = \frac{(1+t^2)^2 + 4t^2 + 4t^2(1+t^2)^2}{(1+t^2)^2}.$$

Let us compute more carefully:  $|\mathbf{r}'|^2 = 1 + 4t^2 + \frac{4t^2}{(1+t^2)^2}$ .

The arc length  $s(t) = \int_0^t |\mathbf{r}'(u)| du$  does not have a closed-form antiderivative, so the arc length parameterization is defined implicitly:

$$s(t) = \int_0^t \sqrt{1 + 4u^2 + \frac{4u^2}{(1+u^2)^2}} du,$$

and the reparameterization is  $\mathbf{r}(s) = \mathbf{r}(t(s))$  where  $t(s)$  is the inverse function of  $s(t)$ .

57. **Solution.**  $\mathbf{r}(t) = \langle t^2, t^3, t \rangle$ ,  $\mathbf{r}'(t) = \langle 2t, 3t^2, 1 \rangle$ ,  $|\mathbf{r}'| = \sqrt{4t^2 + 9t^4 + 1}$ .

$$L = \int_0^1 \sqrt{9t^4 + 4t^2 + 1} dt.$$

This integral does not have a simple closed form. Numerically:

$$L = \int_0^1 \sqrt{9t^4 + 4t^2 + 1} dt \approx 1.8101.$$

58. **Solution.**  $\mathbf{r}(t) = \langle 4 \cos t, 4 \sin t, 3t \rangle$  for  $t \in [0, 2\pi]$ .  $\mathbf{r}'(t) = \langle -4 \sin t, 4 \cos t, 3 \rangle$ ,  $|\mathbf{r}'| = \sqrt{16 + 9} = 5$ .

Total arc length:  $L = \int_0^{2\pi} 5 dt = 10\pi$ .

Arc length parameter:  $s = 5t$ , so  $t = s/5$ .

$$L = 10\pi, \quad \mathbf{r}(s) = \left\langle 4 \cos \frac{s}{5}, 4 \sin \frac{s}{5}, \frac{3s}{5} \right\rangle, \quad s \in [0, 10\pi].$$

59. **Solution.**  $\mathbf{r}(t) = \langle t, \sin t, \cos t \rangle$ ,  $\mathbf{r}'(t) = \langle 1, \cos t, -\sin t \rangle$ ,  $|\mathbf{r}'| = \sqrt{2}$ .

$$L = \int_0^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2}.$$

60. **Solution.**  $\mathbf{r}(t) = \langle e^t, \ln t, t \rangle$ ,  $\mathbf{r}'(t) = \langle e^t, 1/t, 1 \rangle$ .  $|\mathbf{r}'| = \sqrt{e^{2t} + 1/t^2 + 1}$ .

$$L = \int_1^3 \sqrt{e^{2t} + \frac{1}{t^2} + 1} dt.$$

This does not simplify to a closed form. Numerically:

$$L = \int_1^3 \sqrt{e^{2t} + \frac{1}{t^2} + 1} dt \approx 19.49.$$

61. **Solution.** For a particle moving at constant speed  $v$  along  $\mathbf{r}(t) = \langle t, t^2, t^4 \rangle$ , the travel time equals the arc length divided by speed.

$$\mathbf{r}'(t) = \langle 1, 2t, 4t^3 \rangle, |\mathbf{r}'| = \sqrt{1 + 4t^2 + 16t^6}.$$

To minimize travel time at constant speed, we need the path with the shortest arc length between the endpoints. The straight line from  $\mathbf{r}(0) = \langle 0, 0, 0 \rangle$  to  $\mathbf{r}(T)$  for any given  $T$  would be optimal, but since the vehicle is constrained to the curve, the total travel time along the curve from  $t = 0$  to  $t = T$  is:

$$\text{Travel time} = \frac{L}{v} = \frac{1}{v} \int_0^T \sqrt{1 + 4t^2 + 16t^6} dt.$$

Since the curve is fixed, one cannot reduce the arc length; the travel time is minimized by traveling at maximum constant speed along the curve. The “optimized” path is simply the given curve traversed at the highest possible speed.

62. **Solution.**  $\mathbf{r}(t) = \langle 7000 \cos t, 7000 \sin t, 0 \rangle$ ,  $\mathbf{r}'(t) = \langle -7000 \sin t, 7000 \cos t, 0 \rangle$ ,  $|\mathbf{r}'| = 7000$ . One orbit:  $t \in [0, 2\pi]$ .

$$L = \int_0^{2\pi} 7000 dt = 14000\pi \approx 43,982 \text{ km}.$$

63. **Solution.**  $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 0 \rangle$ ,  $\mathbf{r}'(t) = \langle -2 \sin t, 2 \cos t, 0 \rangle$ ,  $|\mathbf{r}'| = 2$ .

$$s = 2t \implies t = s/2.$$

$$\mathbf{r}(s) = \left\langle 2 \cos \frac{s}{2}, 2 \sin \frac{s}{2}, 0 \right\rangle, \quad s \in [0, 4\pi].$$

64. **Solution.**  $\mathbf{r}(t) = \langle t, e^t, \ln t \rangle$ ,  $\mathbf{r}'(t) = \langle 1, e^t, 1/t \rangle$ ,  $|\mathbf{r}'| = \sqrt{1 + e^{2t} + 1/t^2}$ .

$$L = \int_1^2 \sqrt{1 + e^{2t} + \frac{1}{t^2}} dt.$$

This integral does not have a closed-form solution. Numerically:

$$L = \int_1^2 \sqrt{1 + e^{2t} + \frac{1}{t^2}} dt \approx 6.79.$$

65. **Solution.**  $\mathbf{r}(t) = \langle t, \sin t, \cos t \rangle$ ,  $|\mathbf{r}'| = \sqrt{2}$  (from Problem 59).

$$L = \int_0^\pi \sqrt{2} dt = \pi\sqrt{2}.$$

66. **Solution.**  $\mathbf{r}(t) = \langle 10 \cos t, 10 \sin t, t^2 \rangle$ ,  $\mathbf{r}'(t) = \langle -10 \sin t, 10 \cos t, 2t \rangle$ ,  $|\mathbf{r}'| = \sqrt{100 + 4t^2}$ .

$$L = \int_0^\pi \sqrt{100 + 4t^2} dt = 2 \int_0^\pi \sqrt{25 + t^2} dt.$$

Using  $\int \sqrt{a^2 + t^2} dt = \frac{1}{2} [t\sqrt{a^2 + t^2} + a^2 \ln(t + \sqrt{a^2 + t^2})]$  with  $a = 5$ :

$$\begin{aligned} L &= 2 \cdot \frac{1}{2} [t\sqrt{25 + t^2} + 25 \ln(t + \sqrt{25 + t^2})]_0^\pi \\ &= \pi\sqrt{25 + \pi^2} + 25 \ln\left(\frac{\pi + \sqrt{25 + \pi^2}}{5}\right). \end{aligned}$$

$$L = \pi\sqrt{25 + \pi^2} + 25 \ln\left(\frac{\pi + \sqrt{25 + \pi^2}}{5}\right) \approx 32.52.$$

67. **Solution.** For uniform speed from  $\langle 0, 0, 0 \rangle$  to  $\langle 5, 5, 5 \rangle$ , the simplest smooth path is a straight line:  $\mathbf{r}(t) = t\langle 5, 5, 5 \rangle$  for  $t \in [0, 1]$ .  $|\mathbf{r}'(t)| = 5\sqrt{3}$  (constant speed). Arc length:  $L = 5\sqrt{3}$ . Arc length parameterization:  $t = s/(5\sqrt{3})$ , so

$$\mathbf{r}(s) = \frac{s}{5\sqrt{3}} \langle 5, 5, 5 \rangle = \frac{s}{\sqrt{3}} \langle 1, 1, 1 \rangle, \quad s \in [0, 5\sqrt{3}].$$

By construction,  $|\mathbf{dr}/ds| = 1$ , so the drone moves at unit speed along this parameterization. To achieve a desired speed  $v$ , travel at  $s = vt$ .

68. **Solution.**  $v_0 = 50$  m/s,  $\theta = 45^\circ$ ,  $g = 9.8$  m/s<sup>2</sup>.  $v_{0x} = 50 \cos 45 = 25\sqrt{2}$ ,  $v_{0y} = 25\sqrt{2}$ .

Time of flight (when  $y = 0$  and  $t > 0$ ):  $v_{0y}t - \frac{1}{2}gt^2 = 0 \implies t = 2v_{0y}/g = \frac{50\sqrt{2}}{9.8}$ .

Horizontal range:

$$R = v_{0x} \cdot \frac{2v_{0y}}{g} = \frac{v_0^2 \sin 2\theta}{g} = \frac{2500}{9.8} \approx 255.1 \text{ m.}$$

Maximum height:

$$H = \frac{v_0^2 \sin^2 \theta}{2g} = \frac{2500 \cdot 1/2}{2 \cdot 9.8} = \frac{1250}{19.6} \approx 63.78 \text{ m.}$$

$$\boxed{R \approx 255.1 \text{ m}, \quad H \approx 63.78 \text{ m.}}$$

69. **Solution.** Circular path of radius  $R = 100 \text{ m}$ , tangential acceleration  $a_T = 2 \text{ m/s}^2$ , speed  $v = 10 \text{ m/s}$ .

Centripetal (normal) acceleration:  $a_N = v^2/R = 100/100 = 1 \text{ m/s}^2$ .

Total acceleration:

$$|a| = \sqrt{a_T^2 + a_N^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.236 \text{ m/s}^2.$$

$$\boxed{|\mathbf{a}| = \sqrt{5} \approx 2.236 \text{ m/s}^2.}$$

70. **Solution.**  $\mathbf{r}(t) = \langle 10 \cos t, 10 \sin t, t^2 \rangle$ .  $\mathbf{v}(t) = \langle -10 \sin t, 10 \cos t, 2t \rangle$ .  $\mathbf{a}(t) = \langle -10 \cos t, -10 \sin t, 2 \rangle$ .

At  $t = 1$ :  $\mathbf{v}(1) = \langle -10 \sin 1, 10 \cos 1, 2 \rangle$ . Speed:  $|\mathbf{v}(1)| = \sqrt{100 \sin^2 1 + 100 \cos^2 1 + 4} = \sqrt{104} = 2\sqrt{26}$ .

$\mathbf{a}(1) = \langle -10 \cos 1, -10 \sin 1, 2 \rangle$ .  $|\mathbf{a}(1)| = \sqrt{100 + 4} = \sqrt{104} = 2\sqrt{26}$ .

$$\boxed{|\mathbf{v}(1)| = 2\sqrt{26} \approx 10.20, \quad \mathbf{a}(1) = \langle -10 \cos 1, -10 \sin 1, 2 \rangle, \quad |\mathbf{a}(1)| = 2\sqrt{26}.}$$

71. **Solution.**  $\mathbf{r}(t) = \langle \ln(t+1), \sqrt{t}, t^2 \rangle$ ,  $\mathbf{r}'(t) = \langle 1/(t+1), 1/(2\sqrt{t}), 2t \rangle$  for  $t > 0$ .  $|\mathbf{r}'| = \sqrt{\frac{1}{(t+1)^2} + \frac{1}{4t} + 4t^2}$ .

$$L = \int_0^2 \sqrt{\frac{1}{(t+1)^2} + \frac{1}{4t} + 4t^2} dt.$$

Note: At  $t = 0$ ,  $1/(2\sqrt{t}) \rightarrow \infty$ , so the integrand has an integrable singularity. Numerically:

$$\boxed{L \approx 4.82.}$$

72. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, \ln t \rangle$ ,  $\mathbf{r}' = \langle 1, 2t, 1/t \rangle$ ,  $\mathbf{r}'' = \langle 0, 2, -1/t^2 \rangle$ . At  $t = 1$ :  $\mathbf{r}' = \langle 1, 2, 1 \rangle$ ,  $\mathbf{r}'' = \langle 0, 2, -1 \rangle$ .

$$\mathbf{r}' \times \mathbf{r}'' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 1 \\ 0 & 2 & -1 \end{vmatrix} = \langle -2 - 2, 0 + 1, 2 - 0 \rangle = \langle -4, 1, 2 \rangle.$$

$$|\mathbf{r}' \times \mathbf{r}''| = \sqrt{16 + 1 + 4} = \sqrt{21}, \quad |\mathbf{r}'| = \sqrt{6}.$$

$$\boxed{\kappa(1) = \frac{\sqrt{21}}{(\sqrt{6})^3} = \frac{\sqrt{21}}{6\sqrt{6}} = \frac{\sqrt{126}}{36} = \frac{\sqrt{14}}{12} \approx 0.312.}$$

73. **Solution.**  $\mathbf{r}(t) = \langle 7000 \cos t, 7000 \sin t, 0 \rangle$  km.  $\mathbf{v}(t) = \langle -7000 \sin t, 7000 \cos t, 0 \rangle$ ,  $|\mathbf{v}| = 7000$  km (per unit of  $t$ ).  $\mathbf{a}(t) = \langle -7000 \cos t, -7000 \sin t, 0 \rangle$ ,  $|\mathbf{a}| = 7000$  km (per  $t^2$ ).

The acceleration points toward the origin (centripetal):  $\mathbf{a} = -7000 \hat{\mathbf{r}}$ . Centripetal acceleration magnitude:  $a_c = v^2/R = 7000^2/7000 = 7000$ .

$$\text{Speed} = 7000 \text{ km}/[\text{unit time}], \quad a_{\text{centripetal}} = \frac{v^2}{R} = 7000 \text{ km}/[\text{unit time}]^2.$$

74. **Solution.** We need  $\mathbf{r} : [0, 1] \rightarrow \mathbb{R}^3$  with  $\mathbf{r}(0) = \langle 0, 0, 0 \rangle$ ,  $\mathbf{r}(1) = \langle 2, 2, 2 \rangle$ , and  $\mathbf{r}', \mathbf{r}''$  continuous.

Using a cubic Hermite interpolant with zero-velocity endpoints:

$$s(t) = 3t^2 - 2t^3, \quad \mathbf{r}(t) = 2s(t)\langle 1, 1, 1 \rangle = \langle 2(3t^2 - 2t^3), 2(3t^2 - 2t^3), 2(3t^2 - 2t^3) \rangle.$$

Check:  $\mathbf{r}(0) = \langle 0, 0, 0 \rangle$ ,  $\mathbf{r}(1) = \langle 2, 2, 2 \rangle$ .  $\mathbf{r}'(t) = 2(6t - 6t^2)\langle 1, 1, 1 \rangle$ , so  $\mathbf{r}'(0) = \mathbf{r}'(1) = \mathbf{0}$  (smooth start/stop).  $\mathbf{r}''(t) = 2(6 - 12t)\langle 1, 1, 1 \rangle$  is continuous.

$$\mathbf{r}(t) = (6t^2 - 4t^3)\langle 1, 1, 1 \rangle, \quad t \in [0, 1].$$

75. **Solution.**  $\mathbf{r}(t) = \langle t, \sin t, \cos t \rangle$ ,  $|\mathbf{r}'| = \sqrt{2}$ .

$$L = \int_0^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2} \approx 8.886.$$

76. **Solution.**  $\mathbf{r}(t) = \langle t, t^3, t \rangle$ .  $\mathbf{v}(t) = \langle 1, 3t^2, 1 \rangle$ ,  $\mathbf{a}(t) = \langle 0, 6t, 0 \rangle$ . At  $t = 1$ :  $\mathbf{v}(1) = \langle 1, 3, 1 \rangle$ ,  $|\mathbf{v}(1)| = \sqrt{1 + 9 + 1} = \sqrt{11}$ ,  $\mathbf{a}(1) = \langle 0, 6, 0 \rangle$ .

$$\mathbf{v}(1) = \langle 1, 3, 1 \rangle, \quad \mathbf{a}(1) = \langle 0, 6, 0 \rangle, \quad |\mathbf{v}(1)| = \sqrt{11}.$$

77. **Solution.**  $\mathbf{r}(t) = \langle e^t, e^{-t}, t \rangle$ ,  $|\mathbf{r}'| = \sqrt{e^{2t} + e^{-2t} + 1}$ .

$$L = \int_0^1 \sqrt{e^{2t} + e^{-2t} + 1} dt.$$

Numerically:  $L \approx 2.003$ .

78. **Solution.** This is the same as Problem 38. For  $\mathbf{r}(t) = \langle t, \ln t, t^2 \rangle$  at  $t = 1$ :

$$\mathbf{T}(1) = \frac{1}{\sqrt{6}}\langle 1, 1, 2 \rangle, \quad \mathbf{N}(1) = \frac{1}{\sqrt{14}}\langle -1, -3, 2 \rangle, \quad \mathbf{B}(1) = \frac{1}{\sqrt{21}}\langle 4, -2, -1 \rangle.$$

See Problem 32/38 for the full computation.



79. **Solution.**  $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ ,  $|\mathbf{r}'| = \sqrt{2}$ .

$$L = \int_0^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2}.$$

80. **Solution.**  $\mathbf{r}(t) = \langle t, \sqrt{2}t, t^2 \rangle$ ,  $\mathbf{r}'(t) = \langle 1, \sqrt{2}, 2t \rangle$ .  $|\mathbf{r}'| = \sqrt{1 + 2 + 4t^2} = \sqrt{3 + 4t^2}$ .

Hint says  $|\mathbf{r}'|$  simplifies nicely. Indeed:  $\sqrt{3 + 4t^2}$ . Let us check if  $3 + 4t^2$  is a perfect square... Not in general, but  $\sqrt{3 + 4t^2}$  can be integrated.

$s(t) = \int_0^t \sqrt{3 + 4u^2} du$ . Let  $2u = \sqrt{3} \sinh \phi$ :

$$s(t) = \frac{\sqrt{3}}{4} [2t\sqrt{3 + 4t^2}/\sqrt{3} + \sqrt{3} \sinh^{-1}(2t/\sqrt{3})] = \frac{1}{2} t\sqrt{3 + 4t^2} + \frac{3}{4} \ln \left( \frac{2t + \sqrt{3 + 4t^2}}{\sqrt{3}} \right).$$

The expression  $\sqrt{3 + 4t^2}$  does not simplify further. The arc length is:

$$s(t) = \int_0^t \sqrt{3 + 4u^2} du = \frac{t\sqrt{3 + 4t^2}}{2} + \frac{3}{4} \ln \left( \frac{2t + \sqrt{3 + 4t^2}}{\sqrt{3}} \right).$$

The reparameterization is  $\mathbf{r}(s) = \mathbf{r}(t(s))$  where  $t(s)$  is the inverse of the above.

$$s(t) = \frac{t}{2} \sqrt{3 + 4t^2} + \frac{3}{4} \ln \left( \frac{2t + \sqrt{3 + 4t^2}}{\sqrt{3}} \right); \quad \mathbf{r}(s) = \langle t(s), \sqrt{2}t(s), t(s)^2 \rangle.$$

81. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ ,  $\mathbf{r}' = \langle 1, 2t, 3t^2 \rangle$ ,  $\mathbf{r}'' = \langle 0, 2, 6t \rangle$ ,  $\mathbf{r}''' = \langle 0, 0, 6 \rangle$ .

At  $t = 1$ :  $\mathbf{r}' = \langle 1, 2, 3 \rangle$ ,  $|\mathbf{r}'| = \sqrt{14}$ ,  $\mathbf{r}'' = \langle 0, 2, 6 \rangle$ .  $\mathbf{r}' \times \mathbf{r}'' = \langle 6, -6, 2 \rangle$ ,  $|\mathbf{r}' \times \mathbf{r}''| = \sqrt{76} = 2\sqrt{19}$ .

Curvature:

$$\kappa(1) = \frac{2\sqrt{19}}{(\sqrt{14})^3} = \frac{2\sqrt{19}}{14\sqrt{14}} = \frac{\sqrt{19}}{7\sqrt{14}} = \frac{\sqrt{266}}{98}.$$

Torsion:  $(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}''' = \langle 6, -6, 2 \rangle \cdot \langle 0, 0, 6 \rangle = 12$ .

$$\tau(1) = \frac{12}{|\mathbf{r}' \times \mathbf{r}''|^2} = \frac{12}{76} = \frac{3}{19}.$$

$$\kappa(1) = \frac{\sqrt{19}}{7\sqrt{14}} = \frac{\sqrt{266}}{98} \approx 0.166, \quad \tau(1) = \frac{3}{19} \approx 0.158.$$

82. **Solution.**  $\mathbf{r}(t) = \langle 8000 \cos t, 6000 \sin t, 0 \rangle$ ,  $\mathbf{r}'(t) = \langle -8000 \sin t, 6000 \cos t, 0 \rangle$ .  $|\mathbf{r}'| = \sqrt{64 \times 10^6 \sin^2 t + 36 \times 10^6 \cos^2 t} = 2000\sqrt{16 \sin^2 t + 9 \cos^2 t} = 2000\sqrt{9 + 7 \sin^2 t}$ .

$$\begin{aligned}
L &= \int_0^{2\pi} 2000 \sqrt{9 + 7 \sin^2 t} dt = 4 \cdot 2000 \int_0^{\pi/2} \sqrt{9 + 7 \sin^2 t} dt. \\
&= 8000 \int_0^{\pi/2} \sqrt{9 + 7 \sin^2 t} dt = 8000 \cdot 3 \int_0^{\pi/2} \sqrt{1 + \frac{7}{9} \sin^2 t} dt.
\end{aligned}$$

This is an incomplete elliptic integral of the second kind. With  $k^2 = 7/9$ :  $= 24000 E\left(\sqrt{7/9}\right) \cdot \frac{\pi}{2}$ —no, the standard form is  $E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 t} dt$ .

We have  $\sqrt{1 + (7/9) \sin^2 t}$ . Rewrite:  $9 + 7 \sin^2 t = 16 - 7 \cos^2 t$ , so  $\sqrt{9 + 7 \sin^2 t} = 4 \sqrt{1 - \frac{7}{16} \cos^2 t}$ .

Let  $u = \pi/2 - t$ :  $\int_0^{\pi/2} \sqrt{1 - \frac{7}{16} \cos^2 t} dt = \int_0^{\pi/2} \sqrt{1 - \frac{7}{16} \sin^2 u} du = E\left(\sqrt{7/16}\right) = E\left(\frac{\sqrt{7}}{4}\right)$ .

So  $L = 8000 \cdot 4 E\left(\frac{\sqrt{7}}{4}\right) = 32000 E\left(\frac{\sqrt{7}}{4}\right)$ .

Numerically,  $k = \sqrt{7}/4 \approx 0.6614$ ,  $E(0.6614) \approx 1.3556$ .  $L \approx 32000 \times 1.3556 \approx 43,379$  km.

For context, this is the circumference of an ellipse with semi-major axis  $a = 8000$  and semi-minor axis  $b = 6000$ :  $L = 4aE(e)$  where eccentricity  $e = \sqrt{1 - b^2/a^2} = \sqrt{1 - 9/16} = \sqrt{7}/4$ .

$$L = 4 \cdot 8000 \cdot E\left(\frac{\sqrt{7}}{4}\right) = 32000 E\left(\frac{\sqrt{7}}{4}\right) \approx 43,379 \text{ km.}$$

83. **Solution.**  $\mathbf{r}(t) = \langle t, \ln t, \sqrt{t} \rangle$ ,  $\mathbf{r}' = \langle 1, 1/t, 1/(2\sqrt{t}) \rangle$ .  $|\mathbf{r}'|^2 = 1 + 1/t^2 + 1/(4t) = (4t^2 + 4 + t)/(4t^2)$ .  $|\mathbf{r}'| = \sqrt{4t^2 + t + 4}/(2t)$ .

$$L = \int_1^2 \frac{\sqrt{4t^2 + t + 4}}{2t} dt.$$

This does not have a simple closed form. Numerically:

$$L \approx 1.554.$$

84. **Solution.**  $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ ,  $\mathbf{r}' = \langle 1, 2t, 3t^2 \rangle$ .

$$|\mathbf{r}'(t)| = \sqrt{1 + 4t^2 + 9t^4}.$$

As  $t \rightarrow \infty$ :  $9t^4$  dominates, so  $|\mathbf{r}'(t)| \sim 3t^2$ . More precisely, for large  $t$ :

$$|\mathbf{r}'(t)| = 3t^2 \sqrt{1 + \frac{4}{9t^2} + \frac{1}{9t^4}} \approx 3t^2 \left(1 + \frac{2}{9t^2} + \dots\right) \approx 3t^2 + \frac{2}{3} + \dots$$

Thus the speed grows without bound as  $t \rightarrow \infty$ , dominated by the  $z$ -component ( $3t^2$ ).

$$|\mathbf{r}'(t)| = \sqrt{1 + 4t^2 + 9t^4} \sim 3t^2 \text{ as } t \rightarrow \infty.$$

85. **Solution.**  $\mathbf{r}(t) = \langle 10 \cos t, 10 \sin t, t^2 \rangle$ ,  $|\mathbf{r}'| = \sqrt{100 + 4t^2}$ .

$$L = \int_0^{2\pi} \sqrt{100 + 4t^2} dt = 2 \int_0^{2\pi} \sqrt{25 + t^2} dt.$$

$$= \left[ t\sqrt{25 + t^2} + 25 \ln(t + \sqrt{25 + t^2}) \right]_0^{2\pi} = 2\pi\sqrt{25 + 4\pi^2} + 25 \ln\left(\frac{2\pi + \sqrt{25 + 4\pi^2}}{5}\right).$$

$$L = 2\pi\sqrt{25 + 4\pi^2} + 25 \ln\left(\frac{2\pi + \sqrt{25 + 4\pi^2}}{5}\right) \approx 66.89.$$

86. **Solution.**  $\mathbf{r}(t) = \langle 5 \cos t, 5 \sin t, 12t \rangle$ ,  $\mathbf{r}'(t) = \langle -5 \sin t, 5 \cos t, 12 \rangle$ .  $|\mathbf{r}'| = \sqrt{25 \sin^2 t + 25 \cos^2 t + 144} = \sqrt{25 + 144} = \sqrt{169} = 13$ .

Since the speed is  $\boxed{13}$  for all  $t$ , the speed is **constant**. The drone travels at a uniform speed of 13 units per unit time along this helix.

87. **Solution.**  $\mathbf{r}(t) = \langle t, e^t, \ln t \rangle$  for  $t > 0$ .

*Velocity:*  $\mathbf{v}(t) = \langle 1, e^t, 1/t \rangle$ . At  $t = 1$ :  $\mathbf{v}(1) = \langle 1, e, 1 \rangle$ .

*Acceleration:*  $\mathbf{a}(t) = \langle 0, e^t, -1/t^2 \rangle$ . At  $t = 1$ :  $\mathbf{a}(1) = \langle 0, e, -1 \rangle$ .

*Curvature:*  $\mathbf{v} \times \mathbf{a}|_{t=1} = \langle 1, e, 1 \rangle \times \langle 0, e, -1 \rangle = \langle -e - e, 0 + 1, e - 0 \rangle = \langle -2e, 1, e \rangle$ .

$$|\mathbf{v} \times \mathbf{a}| = \sqrt{4e^2 + 1 + e^2} = \sqrt{5e^2 + 1}. \quad |\mathbf{v}(1)| = \sqrt{1 + e^2 + 1} = \sqrt{e^2 + 2}.$$

$$\kappa(1) = \frac{\sqrt{5e^2 + 1}}{(e^2 + 2)^{3/2}}.$$

*Torsion:*  $\mathbf{r}'''(t) = \langle 0, e^t, 2/t^3 \rangle$ . At  $t = 1$ :  $\mathbf{r}'''(1) = \langle 0, e, 2 \rangle$ .

$$(\mathbf{v} \times \mathbf{a}) \cdot \mathbf{r}''' = \langle -2e, 1, e \rangle \cdot \langle 0, e, 2 \rangle = 0 + e + 2e = 3e.$$

$$|\mathbf{v} \times \mathbf{a}|^2 = 5e^2 + 1.$$

$$\tau(1) = \frac{3e}{5e^2 + 1}.$$

$$\mathbf{v}(1) = \langle 1, e, 1 \rangle, \quad \mathbf{a}(1) = \langle 0, e, -1 \rangle,$$

$$\kappa(1) = \frac{\sqrt{5e^2 + 1}}{(e^2 + 2)^{3/2}} \approx 0.301, \quad \tau(1) = \frac{3e}{5e^2 + 1} \approx 0.211.$$

## Chapter 3

## Chapter 3: Partial Derivatives

1. **Solution.** We have  $M = f(P, r, t) = P \frac{\frac{r}{12} \left(1 + \frac{r}{12}\right)^{12t}}{\left(1 + \frac{r}{12}\right)^{12t} - 1}$ .

Substituting  $P = 300,000$ ,  $r = 0.06$ ,  $t = 30$ :

$$\frac{r}{12} = \frac{0.06}{12} = 0.005, \quad 12t = 360.$$

$$(1.005)^{360} \approx 6.02258.$$

Therefore:

$$M = 300,000 \cdot \frac{0.005 \times 6.02258}{6.02258 - 1} = 300,000 \cdot \frac{0.030113}{5.02258} \approx 300,000 \times 0.005996 \approx \boxed{\$1,798.65}.$$

2. **Solution.** Setting  $f(x, y) = x^2 + y^2 = k$ :

- $k = 1$ :  $x^2 + y^2 = 1$  is a circle of radius 1 centered at the origin.
- $k = 2$ :  $x^2 + y^2 = 2$  is a circle of radius  $\sqrt{2}$  centered at the origin.
- $k = 4$ :  $x^2 + y^2 = 4$  is a circle of radius 2 centered at the origin.

All level curves are concentric circles centered at the origin. The sketch shows three circles of radii 1,  $\sqrt{2}$ , and 2.

3. **Solution.** The natural logarithm requires its argument to be strictly positive:

$$x^2 + y^2 - 4 > 0 \implies x^2 + y^2 > 4.$$

The domain is the set of all points  $(x, y)$  *outside* the circle of radius 2 centered at the origin:

$$\boxed{D = \{(x, y) : x^2 + y^2 > 4\}}.$$

4. **Solution.** Setting  $f(x, y, z) = x^2 + y^2 + z^2 = k$  for  $k \geq 0$ :

- If  $k = 0$ , the level “surface” is the single point  $(0, 0, 0)$ .

- If  $k > 0$ , the level surface is a sphere of radius  $\sqrt{k}$  centered at the origin.

The level surfaces are concentric spheres centered at the origin.

5. **Solution.** For the square root to be defined, we need  $9 - x^2 - y^2 \geq 0$ , i.e.,  $x^2 + y^2 \leq 9$ .

**Domain:** The closed disk of radius 3 centered at the origin:  $D = \{(x, y) : x^2 + y^2 \leq 9\}$ .

**Range:** The minimum value of  $f$  is 0 (when  $x^2 + y^2 = 9$ ) and the maximum value is 3 (when  $x^2 + y^2 = 0$ ). Thus the range is  $[0, 3]$ .

6. **Solution.** The parametric curve gives  $x = t$ ,  $y = t^2$ ,  $z = t^3$ . On the surface  $z = y^{3/2}$  we need:

$$t^3 = (t^2)^{3/2} = |t|^3.$$

- If  $t \geq 0$ :  $t^3 = t^3$ , which holds for all  $t \geq 0$ .
- If  $t < 0$ :  $t^3 = -|t|^3$  while  $|t|^3 > 0$ , so  $t^3 \neq |t|^3$ .

Therefore the curve lies on the surface for  $t \geq 0$ .

7. **Solution.** We need  $\frac{x}{y-1} \geq 0$ . This requires  $x$  and  $y-1$  to have the same sign (or  $x = 0$ ):

- $x \geq 0$  and  $y > 1$  (both non-negative, denominator nonzero), or
- $x \leq 0$  and  $y < 1$  (both non-positive, denominator nonzero).

Note  $y \neq 1$  since that makes the denominator zero.

The domain is  $D = \{(x, y) : x \geq 0, y > 1\} \cup \{(x, y) : x \leq 0, y < 1\}$ .

The sketch shows the first and third quadrants relative to the point  $(0, 1)$ : the region in the upper half-plane ( $y > 1$ ) for  $x \geq 0$ , and the lower half-plane ( $y < 1$ ) for  $x \leq 0$ , excluding the line  $y = 1$ .

8. **Solution.** Answers will vary. Suppose the room is rectangular with length  $L$ , width  $W$ , and height  $H$  (all in meters). For example,  $L = 5$ ,  $W = 4$ ,  $H = 3$ .

The total wall area is  $A = 2H(L + W)$ .

If the paint costs  $\$P$  per square meter, and we apply one coat of primer plus one coat of paint (both at the same coverage rate), the cost function is:

$$C(P, L, W, H) = 2P \cdot 2H(L + W) = 4H(L + W)P.$$

If primer costs a different amount  $\$P_{\text{primer}}$  per square meter, then:

$$C = 2H(L + W)(P + P_{\text{primer}}).$$

Using the example dimensions:  $A = 2(3)(5+4) = 54 \text{ m}^2$ , and the cost is  $54(P + P_{\text{primer}})$  dollars.

9. **Solution.** Given  $T = \dot{m}v_e + (P_e - P_a)A_e$  with  $P_e = 101$  kPa,  $P_a = 100$  kPa,  $A_e = 2$  m<sup>2</sup>:

$$(P_e - P_a)A_e = (101 - 100)(1000)(2) = 2000 \text{ N} = 2 \text{ kN}.$$

Solving for  $\dot{m}$ :

$$T = \dot{m}v_e + 2000 \implies \boxed{\dot{m} = \frac{T - 2000}{v_e}}.$$

10. **Solution.** The profit function simplifies to:

$$P = f(x, y) = (4x + 5y) - (2x + 3y) - 100 = 2x + 2y - 100.$$

$$f(50, 30) = 2(50) + 2(30) - 100 = 100 + 60 - 100 = \boxed{\$60}.$$

With 6% taxes applied to revenue (taxes apply to income, not to debits/costs):

$$\begin{aligned} P_{\text{tax}} &= (4x + 5y)(1 - 0.06) - (2x + 3y) - 100 = 0.94(4x + 5y) - (2x + 3y) - 100 \\ &= 3.76x + 4.70y - 2x - 3y - 100 = 1.76x + 1.70y - 100. \end{aligned}$$

11. **Solution.** Test along  $y = 0$ :  $\lim_{x \rightarrow 0} \frac{x \cdot 0}{x^2 + 0} = 0$ .

$$\text{Test along } y = x: \lim_{x \rightarrow 0} \frac{x \cdot x}{x^2 + x^2} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2} = \frac{1}{2}.$$

Since the limits along different paths are different ( $0 \neq \frac{1}{2}$ ), the limit **does not exist**.

12. **Solution.** Convert to polar coordinates:  $x = r \cos \theta$ ,  $y = r \sin \theta$ :

$$\frac{x^2 y}{x^2 + y^2} = \frac{r^2 \cos^2 \theta \cdot r \sin \theta}{r^2} = r \cos^2 \theta \sin \theta.$$

Then:

$$|r \cos^2 \theta \sin \theta| \leq r \cdot 1 \cdot 1 = r \rightarrow 0 \text{ as } r \rightarrow 0.$$

By the Squeeze Theorem,  $\boxed{\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0}.$

13. **Solution.** Test along  $y = 0$ :  $\lim_{x \rightarrow 0} \frac{x \cdot 0}{x^2 + 0} = 0$ .

Test along  $x = y^2$ :

$$\lim_{y \rightarrow 0} \frac{y^2 \cdot y^2}{(y^2)^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{2y^4} = \frac{1}{2}.$$

Since the limits along different paths differ ( $0 \neq \frac{1}{2}$ ),  $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$  does not exist.

Therefore  $f$  is **not continuous** at  $(0, 0)$ .

14. **Solution.** Along  $y = 0$ :  $\lim_{x \rightarrow 0} \frac{x^2 - 0}{x^2 + 0} = 1$ .

Along  $x = 0$ :  $\lim_{y \rightarrow 0} \frac{0 - y^2}{0 + y^2} = -1$ .

Since  $1 \neq -1$ , the limit **does not exist**.

15. **Solution.** We show  $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^4 + y^2} = 0$ .

We use the AM–GM inequality:  $x^4 + y^2 \geq 2x^2|y|$  (since  $a + b \geq 2\sqrt{ab}$  for  $a, b \geq 0$ ).

Therefore:

$$\left| \frac{x^3 y}{x^4 + y^2} \right| \leq \frac{|x|^3 |y|}{2x^2 |y|} = \frac{|x|}{2} \rightarrow 0 \text{ as } (x, y) \rightarrow (0, 0).$$

(When  $y = 0$ , the expression equals 0 directly.)

By the Squeeze Theorem,  $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$ , so  $f$  is continuous at  $(0, 0)$ . □

16. **Solution.** Let  $u = xy - 2$ . As  $(x, y) \rightarrow (1, 2)$ ,  $u \rightarrow 0$ . Write:

$$\frac{\sin(xy - 2)}{x - 1}.$$

Substituting  $x = 1 + h$  (so  $h \rightarrow 0$ ):

$$\frac{\sin((1 + h)y - 2)}{h} = \frac{\sin(y - 2 + hy)}{h}.$$

As  $h \rightarrow 0$  and  $y \rightarrow 2$ : the argument  $y - 2 + hy \rightarrow 0 + 0 = 0$ . Using the approximation  $\sin(\alpha) \approx \alpha$  for small  $\alpha$ :

$$\frac{\sin(y - 2 + hy)}{h} \approx \frac{y - 2 + hy}{h}.$$

Along the path  $y = 2$  (taking the limit with  $y = 2$ ):

$$\frac{\sin(2h)}{h} \rightarrow 2.$$

More rigorously, set  $y = 2$  first:  $\lim_{x \rightarrow 1} \frac{\sin(2x - 2)}{x - 1} = \lim_{h \rightarrow 0} \frac{\sin(2h)}{h} = 2$ .

For a general path, write  $y = 2 + k$  where  $k \rightarrow 0$ :

$$\frac{\sin(k + (1 + h)(2 + k) - 2)}{h} = \frac{\sin(k + 2h + hk)}{h}.$$

Note that  $(1 + h)(2 + k) - 2 = 2 + k + 2h + hk - 2 = k + 2h + hk$ . Therefore:

$$\frac{\sin(k + 2h + hk)}{h} = \frac{\sin(k + 2h + hk)}{k + 2h + hk} \cdot \frac{k + 2h + hk}{h}.$$

The first factor  $\rightarrow 1$ . The second factor is  $\frac{k}{h} + 2 + k$ . For the limit to exist, we need  $k/h$  to have a definite limit; but if  $k = 0$  (i.e.,  $y \rightarrow 2$  at the same rate), this gives 2. Along  $k = h$ :  $(h/h) + 2 + h \rightarrow 3$ . The limits differ, so the two-variable limit **does not exist** in general.

However, if the problem intends  $y \rightarrow 2$  independently (as a standard multivariable limit), the limit does not exist. If  $y = 2$  is fixed and only  $x \rightarrow 1$ , then the answer is 2.

17. **Solution.** We have  $\phi(x, y) = \frac{Q}{2\pi} \ln \sqrt{x^2 + y^2} = \frac{Q}{4\pi} \ln(x^2 + y^2)$ .

Let  $r = \sqrt{x^2 + y^2}$ . As  $(x, y) \rightarrow (0, 0)$ ,  $r \rightarrow 0^+$ , so:

$$\ln(x^2 + y^2) = \ln(r^2) = 2 \ln r \rightarrow -\infty.$$

Therefore (assuming  $Q > 0$ ):

$$\phi(x, y) = \frac{Q}{4\pi} \ln(r^2) = \frac{Q}{2\pi} \ln r \rightarrow -\infty.$$

This holds regardless of the path, since  $r \rightarrow 0$  forces  $\ln r \rightarrow -\infty$ . Thus  $\lim_{(x,y) \rightarrow (0,0)} \phi(x, y) = -\infty$ . □

18. **Solution.** The magnitude of the electric field at  $(x, y)$  with source at  $(x_0, y_0)$  is:

$$|\mathbf{E}(x, y)| = \frac{|q|}{4\pi\epsilon_0} \cdot \frac{\sqrt{(x - x_0)^2 + (y - y_0)^2}}{[(x - x_0)^2 + (y - y_0)^2]^{3/2}} = \frac{|q|}{4\pi\epsilon_0} \cdot \frac{1}{(x - x_0)^2 + (y - y_0)^2}.$$

Let  $r = \sqrt{(x - x_0)^2 + (y - y_0)^2}$ , so  $|\mathbf{E}| = \frac{|q|}{4\pi\epsilon_0 r^2}$ .

As  $(x, y) \rightarrow (x_0, y_0)$ :  $r \rightarrow 0$  and so  $|\mathbf{E}| \rightarrow +\infty$ .

Taking the limit as  $(x, y) \rightarrow (0, 0)$  with the charge at  $(x_0, y_0)$ :

If  $(x_0, y_0) \neq (0, 0)$ , then  $r \rightarrow \sqrt{x_0^2 + y_0^2} > 0$  and

$$\lim_{(x,y) \rightarrow (0,0)} |\mathbf{E}(x, y)| = \frac{|q|}{4\pi\epsilon_0(x_0^2 + y_0^2)}.$$

If  $(x_0, y_0) = (0, 0)$ , then  $\lim_{(x,y) \rightarrow (0,0)} |\mathbf{E}(x, y)| = +\infty$ .

19. **Solution.** We have  $\sigma(r, \theta) = \frac{A}{r^2} + Br^2$  where  $r = \sqrt{x^2 + y^2}$ .

As  $(x, y) \rightarrow (0, 0)$ ,  $r \rightarrow 0^+$ :

- If  $A \neq 0$ :  $\frac{A}{r^2} \rightarrow \pm\infty$  (depending on the sign of  $A$ ), and  $Br^2 \rightarrow 0$ .



Therefore, if  $A \neq 0$ , the limit diverges:  $\sigma \rightarrow +\infty$  if  $A > 0$  and  $\sigma \rightarrow -\infty$  if  $A < 0$ .

If  $A = 0$ :  $\sigma = Br^2 \rightarrow 0$ .

In general (for a physically meaningful stress function with  $A \neq 0$ ), the limit **does not exist** as a finite value;  $\sigma \rightarrow \pm\infty$ .

20. **Solution.**  $f(x, y) = x^2y + 3xy^2$ .

$$f_x = \frac{\partial}{\partial x}(x^2y + 3xy^2) = 2xy + 3y^2.$$

$$f_y = \frac{\partial}{\partial y}(x^2y + 3xy^2) = x^2 + 6xy.$$

21. **Solution.**  $f(x, y) = \sin(xy)e^x$ .

**First partials:**

$$f_x = y \cos(xy)e^x + \sin(xy)e^x = e^x[y \cos(xy) + \sin(xy)].$$

$$f_y = x \cos(xy)e^x.$$

**Second partials:**

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x}[e^x(y \cos(xy) + \sin(xy))] \\ &= e^x[y \cos(xy) + \sin(xy)] + e^x[-y^2 \sin(xy) + y \cos(xy)] \\ &= e^x[2y \cos(xy) + \sin(xy) - y^2 \sin(xy)]. \end{aligned}$$

$$f_{yy} = \frac{\partial}{\partial y}[x \cos(xy)e^x] = -x^2 \sin(xy)e^x.$$

$$\begin{aligned} f_{xy} &= \frac{\partial}{\partial y}[e^x(y \cos(xy) + \sin(xy))] \\ &= e^x[\cos(xy) - xy \sin(xy) + x \cos(xy)] \\ &= e^x[(1 + x) \cos(xy) - xy \sin(xy)]. \end{aligned}$$

$$\begin{aligned} f_{yx} &= \frac{\partial}{\partial x}[x \cos(xy)e^x] \\ &= \cos(xy)e^x + x(-y \sin(xy))e^x + x \cos(xy)e^x \\ &= e^x[(1 + x) \cos(xy) - xy \sin(xy)]. \end{aligned}$$

We verify  $f_{xy} = f_{yx}$ , consistent with Clairaut's theorem.

22. **Solution.**  $f(x, y, z) = xyz + e^{xy}$ .

**First partials:**

$$f_x = yz + ye^{xy}, \quad f_y = xz + xe^{xy}, \quad f_z = xy.$$

**Mixed second partials:**

$$f_{xy} = z + e^{xy} + xye^{xy}, \quad f_{yx} = z + e^{xy} + xye^{xy}.$$

$$f_{xz} = y, \quad f_{zx} = y.$$

$$f_{yz} = x, \quad f_{zy} = x.$$

All mixed partials satisfy  $f_{xy} = f_{yx}$ ,  $f_{xz} = f_{zx}$ ,  $f_{yz} = f_{zy}$ .

23. **Solution.**  $f(x, y) = x^2e^y + \ln(xy) = x^2e^y + \ln x + \ln y$ .

$$f_x = 2xe^y + \frac{1}{x}, \quad f_y = x^2e^y + \frac{1}{y}.$$

$$f_{xy} = \frac{\partial}{\partial y} \left( 2xe^y + \frac{1}{x} \right) = 2xe^y.$$

$$f_{yx} = \frac{\partial}{\partial x} \left( x^2e^y + \frac{1}{y} \right) = 2xe^y.$$

Since  $f_{xy} = f_{yx} = 2xe^y$ , Clairaut's theorem is verified. □

24. **Solution.**  $f(x, y) = \arctan\left(\frac{y}{x}\right)$ .

$$f_x = \frac{1}{1 + (y/x)^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2 + y^2}.$$

$$f_y = \frac{1}{1 + (y/x)^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}.$$

**Second partials:**

$$f_{xx} = \frac{\partial}{\partial x} \left( \frac{-y}{x^2 + y^2} \right) = \frac{-y \cdot (-2x)}{(x^2 + y^2)^2} \cdot (-1) \cdot (-1) = \frac{2xy}{(x^2 + y^2)^2}.$$

Let us compute carefully:  $f_x = -y(x^2 + y^2)^{-1}$ , so

$$f_{xx} = -y \cdot (-1)(x^2 + y^2)^{-2} \cdot 2x = \frac{2xy}{(x^2 + y^2)^2}.$$

$$f_{yy} = \frac{\partial}{\partial y} \left( \frac{x}{x^2 + y^2} \right) = x \cdot (-1)(x^2 + y^2)^{-2} \cdot 2y = \frac{-2xy}{(x^2 + y^2)^2}.$$

$$f_{xy} = \frac{\partial}{\partial y} \left( \frac{-y}{x^2 + y^2} \right) = \frac{-(x^2 + y^2) + y \cdot 2y}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}.$$

$$f_{yx} = \frac{\partial}{\partial x} \left( \frac{x}{x^2 + y^2} \right) = \frac{(x^2 + y^2) - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}.$$

Note:  $f_{xx} + f_{yy} = 0$ , confirming  $f$  is harmonic.

25. **Solution.**  $f(x, y) = \frac{xy}{x^2 + y^2}$  for  $(x, y) \neq (0, 0)$  and  $f(0, 0) = 0$ .

**Partial derivatives at  $(0, 0)$ :**

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0.$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0 - 0}{k} = 0.$$

So both partial derivatives exist at  $(0, 0)$  and equal 0.

**Not differentiable at  $(0, 0)$ :** If  $f$  were differentiable at  $(0, 0)$ , then

$$\lim_{(h,k) \rightarrow (0,0)} \frac{f(h, k) - f(0, 0) - f_x(0, 0)h - f_y(0, 0)k}{\sqrt{h^2 + k^2}} = 0.$$

This becomes:

$$\lim_{(h,k) \rightarrow (0,0)} \frac{hk}{(h^2 + k^2)\sqrt{h^2 + k^2}} = \lim_{(h,k) \rightarrow (0,0)} \frac{hk}{(h^2 + k^2)^{3/2}}.$$

Along  $h = k$ :  $\frac{h^2}{(2h^2)^{3/2}} = \frac{1}{2\sqrt{2}|h|} \rightarrow \infty$ .

Since this limit does not equal 0,  $f$  is not differentiable at  $(0, 0)$ . □

26. **Solution.**  $f(x, y) = x^3y + xy^3$ . Note  $f$  is homogeneous of degree 4 (each term has total degree 4).

$$f_x = 3x^2y + y^3, \quad f_y = x^3 + 3xy^2.$$

$$\begin{aligned} xf_x + yf_y &= x(3x^2y + y^3) + y(x^3 + 3xy^2) \\ &= 3x^3y + xy^3 + x^3y + 3xy^3 \\ &= 4x^3y + 4xy^3 = 4(x^3y + xy^3) = 4f. \end{aligned}$$

Euler's theorem is verified. □

27. **Solution.** Given  $f_{xx} + f_{yy} = 0$  (Laplace's equation), let  $u = f_x$  and  $v = f_y$ .

**First identity:**  $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = f_{xx} + f_{yy} = 0$ .

**Second identity:** By Clairaut's theorem (since  $f$  is twice differentiable):

$$\frac{\partial u}{\partial y} = f_{xy} = f_{yx} = \frac{\partial v}{\partial x}.$$

□

(Note: The first identity is the divergence-free condition, and the second is a consequence of equality of mixed partials — these together resemble the Cauchy–Riemann equations.)

28. **Solution.** Given  $xf_x + yf_y = nf$ , define  $g(t) = f(tx, ty)$  for fixed  $x, y$  and  $t > 0$ .

By the chain rule:

$$g'(t) = xf_x(tx, ty) + yf_y(tx, ty).$$

Setting  $u = tx$ ,  $v = ty$ , the hypothesis  $uf_u + vf_v = nf$  gives:

$$(tx)f_x(tx, ty) + (ty)f_y(tx, ty) = nf(tx, ty) \implies tg'(t) = ng(t).$$

This is the ODE  $g'(t) = \frac{n}{t}g(t)$ , which separates to:

$$\frac{g'}{g} = \frac{n}{t} \implies \ln|g(t)| = n \ln t + C \implies g(t) = At^n$$

for some constant  $A$ . Setting  $t = 1$ :  $g(1) = f(x, y) = A$ . Therefore:

$$g(t) = f(tx, ty) = t^n f(x, y). \quad \square$$

29. **Solution.** The equation is  $F(x, y) = y - x^2 - \sin(xy) = 0$ .

At  $(0, 0)$ :  $0 - 0 - \sin(0) = 0$ .

**First derivative:**

$$F_x = -2x - y \cos(xy), \quad F_y = 1 - x \cos(xy).$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = \frac{2x + y \cos(xy)}{1 - x \cos(xy)}.$$

$$\text{At } (0, 0): \left. \frac{dy}{dx} \right|_{(0,0)} = \frac{0+0}{1-0} = \boxed{0}.$$

**Second derivative:** Let  $y' = \frac{2x + y \cos(xy)}{1 - x \cos(xy)}$ . Differentiating the original equation  $y - x^2 = \sin(xy)$  implicitly again:

$$y' - 2x = \cos(xy)(y + xy'). \quad \text{At } (0, 0) \text{ with } y' = 0: 0 - 0 = \cos(0)(0 + 0) = 0.$$

$$\text{Differentiating once more: } y'' - 2 = -\sin(xy)(y + xy')^2 + \cos(xy)(y' + y' + xy'').$$

$$\text{At } (0, 0): y'' - 2 = 0 + \cos(0)(0 + 0 + 0) = 0.$$

$$\text{Therefore } \left. \frac{d^2y}{dx^2} \right|_{(0,0)} = \boxed{2}.$$

30. **Solution.** Let  $F(x, y, z) = e^{xyz} - xyz - 1 = 0$ .

At any point where  $xyz = 0$ :  $e^0 - 0 - 1 = 0$ .

By implicit differentiation, treating  $z$  as a function of  $x$  and  $y$ :

$$F_x = yze^{xyz} - yz, \quad F_y = xze^{xyz} - xz, \quad F_z = xye^{xyz} - xy.$$

$$\text{At } xyz = 0: F_x = yz(1) - yz = 0, F_y = xz - xz = 0, F_z = xy - xy = 0.$$

All partials of  $F$  vanish, so we cannot directly use  $\partial z/\partial x = -F_x/F_z$ . We use the cyclic relation for implicit functions instead.

Consider the constraint  $F(x, y, z) = 0$  defining a surface. For any implicit function on a smooth surface, the cyclic relation holds:

$$\left. \frac{\partial z}{\partial x} \right|_y \cdot \left. \frac{\partial x}{\partial y} \right|_z \cdot \left. \frac{\partial y}{\partial z} \right|_x = -1,$$

provided the partial derivatives exist and are well-defined. This is a general identity:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial x}{\partial y} = -\frac{F_y}{F_x}, \quad \frac{\partial y}{\partial z} = -\frac{F_z}{F_y}.$$

Multiplying:

$$\left(-\frac{F_x}{F_z}\right) \left(-\frac{F_y}{F_x}\right) \left(-\frac{F_z}{F_y}\right) = -\frac{F_x F_y F_z}{F_z F_x F_y} = -1.$$

At points where  $xyz = 0$  but the individual partials are not all zero simultaneously (e.g., at a point where exactly one of  $x, y, z$  is zero but the others are nonzero), the formula applies directly.

For the case where two or more of  $x, y, z$  are zero (making all  $F_x = F_y = F_z = 0$ ), the result still holds by the general cyclic identity for smooth implicit surfaces, provided the surface is non-degenerate.

Therefore  $\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} = -1$ . □

31. **Solution.**  $z = x^2 + y^2$ ,  $x = t \cos t$ ,  $y = t \sin t$ .

Note  $x^2 + y^2 = t^2 \cos^2 t + t^2 \sin^2 t = t^2$ , so  $z = t^2$ .

Therefore  $\frac{dz}{dt} = \boxed{2t}$ .

**Verification via Chain Rule:**

$$\frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}.$$

$$\frac{dx}{dt} = \cos t - t \sin t, \quad \frac{dy}{dt} = \sin t + t \cos t.$$

$$\begin{aligned} \frac{dz}{dt} &= 2t \cos t (\cos t - t \sin t) + 2t \sin t (\sin t + t \cos t) \\ &= 2t(\cos^2 t - t \sin t \cos t + \sin^2 t + t \sin t \cos t) = 2t(1) = 2t. \end{aligned}$$

32. **Solution.**  $F(x, y) = x^2 y + y^3 - 8 = 0$ .

$$F_x = 2xy, \quad F_y = x^2 + 3y^2.$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = \boxed{-\frac{2xy}{x^2 + 3y^2}}.$$

33. **Solution.**  $z = xy$ ,  $x = s^2t$ ,  $y = st^2$ .

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = y(2st) + x(t^2) \\ &= (st^2)(2st) + (s^2t)(t^2) = 2s^2t^3 + s^2t^3 = 3s^2t^3.\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = y(s^2) + x(2st) \\ &= (st^2)(s^2) + (s^2t)(2st) = s^3t^2 + 2s^3t^2 = 3s^3t^2.\end{aligned}$$

**Verification:**  $z = xy = (s^2t)(st^2) = s^3t^3$ . Then  $z_s = 3s^2t^3$  and  $z_t = 3s^3t^2$ , confirming the result.

34. **Solution.**  $F(x, y, z) = x^2 + y^2 + z^2 - 16 = 0$ .

$$F_x = 2x, \quad F_y = 2y, \quad F_z = 2z.$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2x}{2z} = \boxed{-\frac{x}{z}}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{2y}{2z} = \boxed{-\frac{y}{z}}.$$

(Valid when  $z \neq 0$ .)

35. **Solution.**  $z = f(u, v)$ ,  $u = xy$ ,  $v = y/x$ .

By the chain rule:

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x} = f_u \cdot y + f_v \cdot \left(-\frac{y}{x^2}\right) = \boxed{yf_u - \frac{y}{x^2}f_v}.$$

36. **Solution.** Let  $I(x) = \int_0^x te^{xt} dt$ . By the Leibniz rule:

$$\frac{dI}{dx} = \underbrace{xe^{x \cdot x}}_{=xe^{x^2}} + \int_0^x \frac{\partial}{\partial x}(te^{xt}) dt = xe^{x^2} + \int_0^x t^2 e^{xt} dt.$$

Evaluate  $\int_0^x t^2 e^{xt} dt$  by integration by parts (with  $u = t^2$ ,  $dv = e^{xt} dt$ ):

Let us compute  $\int_0^x t^2 e^{xt} dt$ . Substituting or using repeated integration by parts:

$$\int t^2 e^{xt} dt = \frac{t^2 e^{xt}}{x} - \frac{2te^{xt}}{x^2} + \frac{2e^{xt}}{x^3} + C.$$

Evaluating from 0 to  $x$ :

$$\left[ \frac{t^2 e^{xt}}{x} - \frac{2te^{xt}}{x^2} + \frac{2e^{xt}}{x^3} \right]_0^x = \frac{x^2 e^{x^2}}{x} - \frac{2xe^{x^2}}{x^2} + \frac{2e^{x^2}}{x^3} - \frac{2}{x^3}$$

$$= xe^{x^2} - \frac{2e^{x^2}}{x} + \frac{2e^{x^2}}{x^3} - \frac{2}{x^3}.$$

Therefore:

$$\frac{dI}{dx} = xe^{x^2} + xe^{x^2} - \frac{2e^{x^2}}{x} + \frac{2e^{x^2}}{x^3} - \frac{2}{x^3} = \boxed{2xe^{x^2} - \frac{2e^{x^2}}{x} + \frac{2(e^{x^2} - 1)}{x^3}}.$$

**37. Solution.**

(a)  $\mathbf{r}(t) = \langle t^2, \sin t, e^t \rangle$ , so

$$\mathbf{r}'(t) = \langle 2t, \cos t, e^t \rangle.$$

(b) At  $t = \pi$ :

$$\mathbf{r}'(\pi) = \langle 2\pi, \cos \pi, e^\pi \rangle = \langle 2\pi, -1, e^\pi \rangle.$$

$$\begin{aligned} |\mathbf{r}'(\pi)| &= \sqrt{(2\pi)^2 + (-1)^2 + (e^\pi)^2} = \sqrt{4\pi^2 + 1 + e^{2\pi}} \\ &\approx \sqrt{39.478 + 1 + 535.49} \approx \sqrt{575.97} \approx \boxed{24.0}. \end{aligned}$$

**38. Solution.**  $f(x, y) = \ln(xy) + e^{x^2y}$ .

(a)  $f_x = \frac{1}{x} + 2xye^{x^2y}, \quad f_y = \frac{1}{y} + x^2e^{x^2y}.$

At  $(1, 1)$ :  $f_x(1, 1) = 1 + 2e = 1 + 2e, \quad f_y(1, 1) = 1 + e.$

The total differential is:

$$df = f_x dx + f_y dy = (1 + 2e) dx + (1 + e) dy.$$

(b) With  $dx = 0.01, dy = -0.01$ :

$$\Delta f \approx (1 + 2e)(0.01) + (1 + e)(-0.01) = 0.01[(1 + 2e) - (1 + e)] = 0.01 \cdot e \approx 0.0272.$$

**39. Solution.**  $f(x, y) = \sum_{k=1}^3 \sin(x^k y^{3-k}) = \sin(xy^2) + \sin(x^2y) + \sin(x^3).$

By the chain rule:

$$\frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}.$$

Compute the partials:

$$f_x = y^2 \cos(xy^2) + 2xy \cos(x^2y) + 3x^2 \cos(x^3).$$

$$f_y = 2xy \cos(xy^2) + x^2 \cos(x^2y).$$

Therefore:

$$\frac{df}{dt} = [y^2 \cos(xy^2) + 2xy \cos(x^2y) + 3x^2 \cos(x^3)]x'(t) + [2xy \cos(xy^2) + x^2 \cos(x^2y)]y'(t).$$

More compactly, using the general term:

$$\frac{df}{dt} = \sum_{k=1}^3 \cos(x^k y^{3-k}) [kx^{k-1}y^{3-k}x'(t) + (3-k)x^k y^{2-k}y'(t)].$$

40. **Solution.**  $f(x, y) = x^2y + y^3$ .

$$\nabla f = \langle 2xy + 0, x^2 + 3y^2 \rangle = \langle 2xy, x^2 + 3y^2 \rangle.$$

At  $(1, 2)$ :

$$\nabla f(1, 2) = \langle 2(1)(2), 1 + 3(4) \rangle = \boxed{\langle 4, 13 \rangle}.$$

41. **Solution.**  $f(x, y) = xe^y$ ,  $\mathbf{v} = \langle 3, 4 \rangle$ .

$$\nabla f = \langle e^y, xe^y \rangle. \text{ At } (1, 0): \nabla f(1, 0) = \langle 1, 1 \rangle.$$

$$\text{Unit vector: } \mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{1}{5} \langle 3, 4 \rangle.$$

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \langle 1, 1 \rangle \cdot \frac{1}{5} \langle 3, 4 \rangle = \frac{3+4}{5} = \boxed{\frac{7}{5}}.$$

42. **Solution.**  $f(x, y) = 2x^2 + 3y^2$ . The direction of steepest increase is the gradient direction.

$$\nabla f = \langle 4x, 6y \rangle.$$

$$\text{At } (1, 1): \nabla f(1, 1) = \langle 4, 6 \rangle.$$

$$\text{The direction of steepest increase is } \boxed{\langle 4, 6 \rangle}, \text{ or as a unit vector: } \frac{1}{\sqrt{52}} \langle 4, 6 \rangle = \frac{1}{2\sqrt{13}} \langle 4, 6 \rangle = \left\langle \frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right\rangle.$$

43. **Solution.**  $f(x, y, z) = xyz$ ,  $\mathbf{u} = \langle 1, 1, 1 \rangle$  (must normalize).

$$\nabla f = \langle yz, xz, xy \rangle.$$

$$\text{At } (1, 2, 3): \nabla f = \langle 6, 3, 2 \rangle.$$

$$\text{Unit vector: } \hat{\mathbf{u}} = \frac{1}{\sqrt{3}} \langle 1, 1, 1 \rangle.$$

$$D_{\hat{\mathbf{u}}}f = \langle 6, 3, 2 \rangle \cdot \frac{1}{\sqrt{3}} \langle 1, 1, 1 \rangle = \frac{6+3+2}{\sqrt{3}} = \boxed{\frac{11}{\sqrt{3}} = \frac{11\sqrt{3}}{3}}.$$

44. **Solution.**  $f(x, y) = \sin(xy)$ .

$$\nabla f(x, y) = \langle y \cos(xy), x \cos(xy) \rangle.$$

At  $(\pi, 0)$ :

$$\nabla f(\pi, 0) = \langle 0 \cdot \cos(0), \pi \cos(0) \rangle = \boxed{\langle 0, \pi \rangle}.$$

45. **Solution.**  $f(x, y) = \ln(x^2 + y^2)$ .

$$\nabla f = \left\langle \frac{2x}{x^2 + y^2}, \frac{2y}{x^2 + y^2} \right\rangle.$$



At  $(1, 1)$ :  $\nabla f(1, 1) = \langle \frac{2}{2}, \frac{2}{2} \rangle = \langle 1, 1 \rangle$ .

Direction from  $(1, 1)$  to  $(2, 3)$ :  $\mathbf{v} = \langle 1, 2 \rangle$ ,  $|\mathbf{v}| = \sqrt{5}$ .

$$\hat{\mathbf{u}} = \frac{1}{\sqrt{5}} \langle 1, 2 \rangle.$$

$$D_{\hat{\mathbf{u}}}f = \langle 1, 1 \rangle \cdot \frac{1}{\sqrt{5}} \langle 1, 2 \rangle = \frac{1+2}{\sqrt{5}} = \boxed{\frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}}.$$

46. **Solution.**  $T(x, y) = e^{-x^2-y^2}$ .

$$\nabla T = \langle -2xe^{-x^2-y^2}, -2ye^{-x^2-y^2} \rangle = -2e^{-x^2-y^2} \langle x, y \rangle.$$

At  $(1, 1)$ :

$$\nabla T(1, 1) = -2e^{-2} \langle 1, 1 \rangle.$$

The direction of steepest increase is  $\nabla T(1, 1) = -2e^{-2} \langle 1, 1 \rangle$ , which points in the direction  $\langle -1, -1 \rangle$ .

As a unit vector, the direction of steepest temperature increase is  $\boxed{\left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle}$ ,

i.e., toward the origin. This makes physical sense: the temperature  $T = e^{-x^2-y^2}$  is maximized at the origin, so at  $(1, 1)$  the temperature increases most rapidly toward the origin.

47. **Solution.**  $J(w, b) = \frac{1}{2}(wx + b - y)^2$ .

Let  $\hat{y} = wx + b$ . Then  $J = \frac{1}{2}(\hat{y} - y)^2$ .

$$\frac{\partial J}{\partial w} = (wx + b - y) \cdot x = \boxed{x(wx + b - y)}.$$

$$\frac{\partial J}{\partial b} = (wx + b - y) \cdot 1 = \boxed{wx + b - y}.$$

These are the gradients used in gradient descent for linear regression.

48. **Solution.**  $z = f(x, y) = x^2 + y^2$  at  $(1, 1, 2)$ .

$$f_x = 2x, \quad f_y = 2y.$$

At  $(1, 1)$ :  $f_x = 2$ ,  $f_y = 2$ .

Tangent plane:

$$z - 2 = 2(x - 1) + 2(y - 1) \implies \boxed{z = 2x + 2y - 2}.$$

49. **Solution.** We want to approximate  $\sqrt{4.1^2 + 2.9^2}$ . Let  $f(x, y) = \sqrt{x^2 + y^2}$  near  $(4, 3)$  (where  $f(4, 3) = 5$ ).

$$f_x = \frac{x}{\sqrt{x^2 + y^2}}, \quad f_y = \frac{y}{\sqrt{x^2 + y^2}}.$$

At  $(4, 3)$ :  $f_x = \frac{4}{5}$ ,  $f_y = \frac{3}{5}$ .

With  $\Delta x = 0.1$ ,  $\Delta y = -0.1$ :

$$f(4.1, 2.9) \approx 5 + \frac{4}{5}(0.1) + \frac{3}{5}(-0.1) = 5 + 0.08 - 0.06 = \boxed{5.02}.$$

50. **Solution.**  $f(x, y) = e^x \sin y$  near  $(0, 0)$ . Note  $f(0, 0) = 0$ .

$$f_x = e^x \sin y, \quad f_y = e^x \cos y.$$

At  $(0, 0)$ :  $f_x = 0$ ,  $f_y = 1$ .

Linear approximation:

$$L(x, y) = f(0, 0) + f_x(0, 0)x + f_y(0, 0)y = 0 + 0 \cdot x + 1 \cdot y = \boxed{y}.$$

51. **Solution.**  $z = \ln(xy)$  at  $(1, 1, 0)$ .

$$f_x = \frac{1}{x}, \quad f_y = \frac{1}{y}.$$

At  $(1, 1)$ :  $f_x = 1$ ,  $f_y = 1$ .

Tangent plane:

$$z - 0 = 1(x - 1) + 1(y - 1) \implies \boxed{z = x + y - 2}.$$

52. **Solution.**  $f(x, y) = \sqrt{x^2 + y^2}$  near  $(1, 1)$ .  $f(1, 1) = \sqrt{2}$ .

$$f_x = \frac{x}{\sqrt{x^2 + y^2}}, \quad f_y = \frac{y}{\sqrt{x^2 + y^2}}.$$

At  $(1, 1)$ :  $f_x = f_y = \frac{1}{\sqrt{2}}$ .

$$f(1.1, 0.9) \approx \sqrt{2} + \frac{1}{\sqrt{2}}(0.1) + \frac{1}{\sqrt{2}}(-0.1) = \sqrt{2} + 0 = \boxed{\sqrt{2} \approx 1.414}.$$

53. **Solution.**  $z = xe^y$  at  $(1, 0, 1)$ .

$$f_x = e^y, \quad f_y = xe^y.$$

At  $(1, 0)$ :  $f_x = 1$ ,  $f_y = 1$ .

Tangent plane:

$$z - 1 = 1(x - 1) + 1(y - 0) \implies \boxed{z = x + y}.$$

**Above or below?** Consider  $g(x, y) = xe^y - (x + y)$ , the difference between the surface and the plane.

$g(1, 0) = 0$ . The Hessian of  $f$  at  $(1, 0)$ :

$$f_{xx} = 0, \quad f_{xy} = e^y|_{(1,0)} = 1, \quad f_{yy} = xe^y|_{(1,0)} = 1.$$

The Hessian matrix is  $H = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$  with determinant  $-1 < 0$ .

Since the Hessian is indefinite, the surface is neither entirely above nor entirely below the tangent plane near  $(1, 0, 1)$  — it lies above in some directions and below in others (a saddle-type behavior).

54. **Solution.**  $f(x, y) = x^2 + y^2 - 4x - 6y$ .

$$f_x = 2x - 4 = 0 \implies x = 2.$$

$$f_y = 2y - 6 = 0 \implies y = 3.$$

The only critical point is  $\boxed{(2, 3)}$ .

Classification:  $f_{xx} = 2$ ,  $f_{yy} = 2$ ,  $f_{xy} = 0$ .  $D = f_{xx}f_{yy} - f_{xy}^2 = 4 > 0$  and  $f_{xx} > 0$ , so  $(2, 3)$  is a **local minimum** with  $f(2, 3) = 4 + 9 - 8 - 18 = -13$ .

55. **Solution.**  $f(x, y) = x^2y - \frac{y^3}{3}$ .

$$f_x = 2xy = 0, \quad f_y = x^2 - y^2 = 0.$$

From  $f_x = 0$ :  $x = 0$  or  $y = 0$ .

**Case 1:**  $x = 0$ . Then  $f_y = -y^2 = 0 \implies y = 0$ . Critical point:  $(0, 0)$ .

**Case 2:**  $y = 0$ . Then  $f_y = x^2 = 0 \implies x = 0$ . Same point  $(0, 0)$ .

So the only critical point is  $(0, 0)$ .

Second derivatives:  $f_{xx} = 2y$ ,  $f_{yy} = -2y$ ,  $f_{xy} = 2x$ .

At  $(0, 0)$ :  $f_{xx} = 0$ ,  $f_{yy} = 0$ ,  $f_{xy} = 0$ .  $D = 0$ : **inconclusive**.

Examine  $f$  directly:  $f(x, 0) = 0$  for all  $x$ ,  $f(0, y) = -y^3/3$  (changes sign), and  $f(\epsilon, \epsilon) = \epsilon^3 - \epsilon^3/3 = 2\epsilon^3/3$  (changes sign). Since  $f$  takes both positive and negative values near  $(0, 0)$ , and  $f(0, 0) = 0$ , the origin is a **saddle point**.

56. **Solution.**  $f(x, y) = x^2 + y^2$  on  $x^2 + y^2 \leq 4$ .

**Interior:**  $f_x = 2x = 0$ ,  $f_y = 2y = 0 \implies (0, 0)$ .  $f(0, 0) = 0$ .

**Boundary:**  $x^2 + y^2 = 4$ , so  $f = 4$  everywhere on the boundary.

**Absolute minimum:**  $f(0, 0) = \boxed{0}$ .

**Absolute maximum:**  $f = \boxed{4}$  on the circle  $x^2 + y^2 = 4$ .

57. **Solution.**  $f(x, y) = xe^y - x^2 - y^2$ .

$$f_x = e^y - 2x = 0 \implies x = \frac{e^y}{2}.$$

$$f_y = xe^y - 2y = 0 \implies \frac{e^y}{2} \cdot e^y = 2y \implies e^{2y} = 4y.$$

Define  $h(y) = e^{2y} - 4y$ . Then  $h(0) = 1 > 0$  and  $h'(y) = 2e^{2y} - 4$ . Setting  $h'(y) = 0$  gives  $e^{2y} = 2$ , i.e.,  $y = \frac{\ln 2}{2} \approx 0.347$ . The minimum value is  $h(\frac{\ln 2}{2}) = 2 - 2 \ln 2 \approx 0.614 > 0$ . For  $y < 0$ ,  $4y < 0 < e^{2y}$ , so no solution there either. Since  $h(y) > 0$  for all  $y$ , the equation  $e^{2y} = 4y$  has **no real solution**, and therefore there are **no critical points**.

Moreover,  $f$  is unbounded above: along the curve  $x = e^y/2$ , we have  $f = \frac{e^{2y}}{2} - \frac{e^{2y}}{4} - y^2 = \frac{e^{2y}}{4} - y^2 \rightarrow \infty$  as  $y \rightarrow \infty$ . Similarly,  $f$  is unbounded below (along  $x \rightarrow -\infty$  with fixed  $y$ ). Therefore  $f$  has **no critical points** and **no local extrema**.

58. **Solution.**  $f(x, y) = xy - x^2 - y^2$ .

$$f_x = y - 2x = 0, \quad f_y = x - 2y = 0.$$

From the first:  $y = 2x$ . Substituting:  $x - 4x = 0 \implies x = 0$ , hence  $y = 0$ .

Critical point:  $(0, 0)$ .

$$f_{xx} = -2, f_{yy} = -2, f_{xy} = 1. D = (-2)(-2) - 1^2 = 3 > 0 \text{ and } f_{xx} < 0.$$

$(0, 0)$  is a **local maximum** with  $f(0, 0) = \boxed{0}$ .

59. **Solution.**  $f(x, y) = (y - x^2)(y - 2x^2) = y^2 - 3x^2y + 2x^4$ .

$$f_x = -6xy + 8x^3 = 2x(4x^2 - 3y) = 0.$$

$$f_y = 2y - 3x^2 = 0.$$

From  $f_x = 0$ :  $x = 0$  or  $y = \frac{4x^2}{3}$ .

**Case 1:**  $x = 0$ . Then  $f_y = 2y = 0 \implies y = 0$ . Critical point:  $(0, 0)$ .

**Case 2:**  $y = \frac{4x^2}{3}$ . Then  $f_y = \frac{8x^2}{3} - 3x^2 = -\frac{x^2}{3} = 0 \implies x = 0$ , giving  $(0, 0)$  again.

So the **only critical point** is  $(0, 0)$ .

**Second derivative test:**  $f_{xx} = -6y + 24x^2$ ,  $f_{yy} = 2$ ,  $f_{xy} = -6x$ . At  $(0, 0)$ :  $f_{xx} = 0$ ,  $f_{yy} = 2$ ,  $f_{xy} = 0$ .  $D = 0$ : **inconclusive**.

**Direct analysis:**  $f(0, 0) = 0$ . Along  $y = cx^2$  for constant  $c$ :

$$f = (cx^2 - x^2)(cx^2 - 2x^2) = x^4(c - 1)(c - 2).$$

- If  $c < 1$  or  $c > 2$ :  $(c - 1)(c - 2) > 0$ , so  $f > 0$  near origin.
- If  $1 < c < 2$ :  $(c - 1)(c - 2) < 0$ , so  $f < 0$  near origin.

Since  $f$  takes both positive and negative values in every neighborhood of  $(0, 0)$ , and  $f(0, 0) = 0$ , the origin is a **saddle point**.

60. **Solution.**  $f(x, y, z) = x^2 + y^2 + z^2 - 4x - 6y + 8z$ .

(a) Setting all partial derivatives to zero:

$$f_x = 2x - 4 = 0 \implies x = 2.$$

$$f_y = 2y - 6 = 0 \implies y = 3.$$

$$f_z = 2z + 8 = 0 \implies z = -4.$$

The only critical point is  $\boxed{(2, 3, -4)}$ .

(b) The Hessian matrix is:

$$H = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

All eigenvalues are 2 (positive), so  $H$  is positive definite. Therefore  $(2, 3, -4)$  is a **local minimum** with  $f(2, 3, -4) = 4 + 9 + 16 - 8 - 18 - 32 = -29$ .

61. **Solution.** For  $f(\mathbf{x})$  with  $\mathbf{x} \in \mathbb{R}^n$ , a critical point  $\mathbf{x}_0$  satisfies  $\nabla f(\mathbf{x}_0) = \mathbf{0}$ .

The classification depends on the **Hessian matrix**  $H(\mathbf{x}_0) = \left[ \frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{n \times n}$ :

- **Local minimum:**  $H(\mathbf{x}_0)$  is **positive definite** (all eigenvalues  $> 0$ ).
- **Local maximum:**  $H(\mathbf{x}_0)$  is **negative definite** (all eigenvalues  $< 0$ ).
- **Saddle point:**  $H(\mathbf{x}_0)$  is **indefinite** (has both positive and negative eigenvalues).
- **Inconclusive:**  $H(\mathbf{x}_0)$  is singular (has a zero eigenvalue).

**Extension from two variables:** In the two-variable case,  $H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}$ , and positive definiteness is equivalent to  $D = f_{xx}f_{yy} - f_{xy}^2 > 0$  with  $f_{xx} > 0$  (by Sylvester's criterion: all leading principal minors are positive). For  $n$  variables, Sylvester's criterion generalizes:  $H$  is positive definite if and only if all  $n$  leading principal minors are positive, i.e.,

$$H_1 > 0, \quad H_2 > 0, \quad \dots, \quad H_n > 0,$$

where  $H_k = \det \left[ \frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{1 \leq i, j \leq k}$ .

Alternatively (and often more practically), one computes the eigenvalues of  $H$ .

62. **Solution.** Distance from  $(2, 1, 3)$  to the plane  $x + y + z = 1$ .

Using the distance formula:

$$d = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|2 + 1 + 3 - 1|}{\sqrt{1 + 1 + 1}} = \frac{5}{\sqrt{3}} = \boxed{\frac{5\sqrt{3}}{3}}.$$

63. **Solution.** Minimize  $S = 2(xy + yz + xz)$  subject to  $V = xyz = 100$ .

By Lagrange multipliers (or by symmetry): a box of minimum surface area with fixed volume is a **cube**. So  $x = y = z = \sqrt[3]{100} = 100^{1/3}$ .

**Verification with Lagrange multipliers:**  $\nabla S = \lambda \nabla V$ :

$$2(y + z) = \lambda yz, \quad 2(x + z) = \lambda xz, \quad 2(x + y) = \lambda xy.$$

From the first two:  $\frac{2(y + z)}{yz} = \frac{2(x + z)}{xz}$ , giving  $\frac{1}{z} + \frac{1}{y} = \frac{1}{z} + \frac{1}{x}$ , so  $x = y$ . Similarly  $y = z$ .

With  $x = y = z$ :  $x^3 = 100$ , so  $x = y = z = \boxed{\sqrt[3]{100} \approx 4.642}$  units.

The minimum surface area is  $S = 6(\sqrt[3]{100})^2 = 6 \cdot 100^{2/3} \approx 129.3$  square units.

64. **Solution.** Minimize  $D^2 = x^2 + (x^2 - 1)^2$  (distance squared from  $(x, x^2)$  to  $(0, 1)$ ).

Let  $g(x) = x^2 + (x^2 - 1)^2 = x^2 + x^4 - 2x^2 + 1 = x^4 - x^2 + 1$ .

$$g'(x) = 4x^3 - 2x = 2x(2x^2 - 1) = 0.$$

Solutions:  $x = 0$  or  $x = \pm \frac{1}{\sqrt{2}}$ .

$$g(0) = 1, \quad g\left(\pm \frac{1}{\sqrt{2}}\right) = \frac{1}{4} - \frac{1}{2} + 1 = \frac{3}{4}.$$

The minimum distance squared is  $\frac{3}{4}$ , so the minimum distance is  $\frac{\sqrt{3}}{2}$ .

The closest points are  $\boxed{\left(\pm \frac{1}{\sqrt{2}}, \frac{1}{2}\right)}$ .

65. **Solution.** Minimize  $f(x, y) = x^2 + y^2$  subject to  $g(x, y) = x^3 + y^3 - 8 = 0$ .

Lagrange conditions:  $\nabla f = \lambda \nabla g$ :

$$2x = 3\lambda x^2, \quad 2y = 3\lambda y^2.$$

**Case 1:**  $x \neq 0$  and  $y \neq 0$ . Then  $\lambda = \frac{2}{3x}$  and  $\lambda = \frac{2}{3y}$ , giving  $x = y$ . Substituting into the constraint:  $2x^3 = 8 \implies x^3 = 4 \implies x = y = \sqrt[3]{4}$ .

$$f = 2(\sqrt[3]{4})^2 = 2 \cdot 4^{2/3} = 2\sqrt[3]{16} \approx 5.04.$$

**Case 2:**  $x = 0$ . Then  $y^3 = 8 \implies y = 2$ .  $f = 4$ .

**Case 3:**  $y = 0$ . Then  $x^3 = 8 \implies x = 2$ .  $f = 4$ .

The minimum value is  $\boxed{f = 4}$ , achieved at  $(2, 0)$  and  $(0, 2)$ .

66. **Solution.** Optimize  $f(x, y) = xy$  subject to  $g(x, y) = x^2 + y^2 - 16 = 0$ .

$$\nabla f = \lambda \nabla g: y = 2\lambda x, x = 2\lambda y.$$

Multiplying:  $xy = 4\lambda^2 xy$ .

**Case 1:**  $xy \neq 0$ . Then  $4\lambda^2 = 1$ , so  $\lambda = \pm \frac{1}{2}$ .

If  $\lambda = \frac{1}{2}$ :  $y = x$ . Constraint:  $2x^2 = 16 \implies x = \pm 2\sqrt{2}$ . Points:  $(2\sqrt{2}, 2\sqrt{2})$  and  $(-2\sqrt{2}, -2\sqrt{2})$ .  $f = 8$ .

If  $\lambda = -\frac{1}{2}$ :  $y = -x$ . Constraint:  $2x^2 = 16 \implies x = \pm 2\sqrt{2}$ . Points:  $(2\sqrt{2}, -2\sqrt{2})$  and  $(-2\sqrt{2}, 2\sqrt{2})$ .  $f = -8$ .

**Case 2:**  $xy = 0$ . Then one variable is 0, and the other is  $\pm 4$ .  $f = 0$ .

|              |               |
|--------------|---------------|
| Maximum = 8, | Minimum = -8. |
|--------------|---------------|

67. **Solution.**  $f(x, y, z) = x^2 + y^2 - z^2$  subject to  $g(x, y, z) = x + y + z - 1 = 0$ .

(a)  $\nabla f = \lambda \nabla g:$

$$2x = \lambda, \quad 2y = \lambda, \quad -2z = \lambda.$$

From the first two:  $x = y$ . From the first and third:  $2x = -(-2z) \implies z = -x$ .

Constraint:  $x + x + (-x) = 1 \implies x = 1$ , so  $y = 1$ ,  $z = -1$ .

Critical point:  $(1, 1, -1)$  with  $\lambda = 2$  and  $f(1, 1, -1) = 1 + 1 - 1 = 1$ .

(b) Since we optimize on a plane (unbounded constraint set), we check whether this is a max or min. Parameterize the constraint:  $z = 1 - x - y$ , then

$$f = x^2 + y^2 - (1 - x - y)^2 = x^2 + y^2 - 1 + 2x + 2y - x^2 - 2xy - y^2 = -1 + 2x + 2y - 2xy.$$

This is an indefinite quadratic in  $x$  and  $y$  (the  $-2xy$  term makes it a saddle). More carefully, the Hessian of this reduced function is  $\begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix}$  with eigenvalues  $\pm 2$ , confirming the critical point is a **saddle point** of  $f$  restricted to the constraint.

Alternatively, note  $f \rightarrow +\infty$  (take  $x$  large,  $z = 1 - x - y$ ) and  $f \rightarrow -\infty$  (take  $z$  large,  $x = y = \frac{1-z}{2}$ ). The constrained function has neither a maximum nor a minimum;  $(1, 1, -1)$  is a **saddle point**.

68. **Solution.** Optimize  $f(x, y, z) = x + y + z$  subject to  $x^2 + y^2 + z^2 = 1$ .

$$\nabla f = \lambda \nabla g: 1 = 2\lambda x, 1 = 2\lambda y, 1 = 2\lambda z.$$

$$\text{So } x = y = z = \frac{1}{2\lambda}. \text{ Constraint: } \frac{3}{4\lambda^2} = 1 \implies \lambda = \pm \frac{\sqrt{3}}{2}.$$

$$\lambda = \frac{\sqrt{3}}{2}: x = y = z = \frac{1}{\sqrt{3}}. f = \frac{3}{\sqrt{3}} = \sqrt{3}.$$

$$\lambda = -\frac{\sqrt{3}}{2}: x = y = z = -\frac{1}{\sqrt{3}}. f = -\sqrt{3}.$$

|                        |                         |
|------------------------|-------------------------|
| Maximum = $\sqrt{3}$ , | Minimum = $-\sqrt{3}$ . |
|------------------------|-------------------------|

69. **Solution.** Minimize  $f(x, y) = 3x + 4y$  subject to  $x + 2y \geq 8$  and  $x - y \geq 2$ .

This is a linear programming problem. The minimum of a linear objective on a convex polyhedral region occurs at a vertex.

**Find vertices of the feasible region:**

Intersection of  $x + 2y = 8$  and  $x - y = 2$ : Subtracting:  $3y = 6 \implies y = 2, x = 4$ .  
Point:  $(4, 2)$ .

Check feasibility:  $4 + 4 = 8 \geq 8$  and  $4 - 2 = 2 \geq 2$ , so both constraints are satisfied.

The feasible region is unbounded. As  $x \rightarrow \infty$  along either constraint boundary,  $f \rightarrow \infty$ .  
So the minimum is at a vertex.

$$f(4, 2) = 12 + 8 = 20.$$

Check boundary rays: Along  $x + 2y = 8$  (away from vertex): as  $y \rightarrow -\infty, x = 8 - 2y \rightarrow \infty$ , need  $x - y = 8 - 3y \geq 2$ , i.e.,  $y \leq 2$ . So  $f = 3(8 - 2y) + 4y = 24 - 2y$ . For  $y \leq 2, f \geq 20$ . Along  $x - y = 2$ :  $x = y + 2$ , need  $y + 2 + 2y \geq 8$ , i.e.,  $y \geq 2$ .  $f = 3(y + 2) + 4y = 7y + 6$ . For  $y \geq 2, f \geq 20$ .

The minimum value is  $\boxed{f = 20}$  at  $(4, 2)$ .

70. **Solution.** Minimize  $f(x, y, z) = x^2 + y^2 + z^2$  subject to

$$g_1 = x + y + z - 6 \geq 0, \quad g_2 = x \geq 0, \quad g_3 = y \geq 0, \quad g_4 = z \geq 0.$$

**Stationarity.**  $\nabla f = \mu_1 \nabla g_1 + \mu_2 \nabla g_2 + \mu_3 \nabla g_3 + \mu_4 \nabla g_4$ :

$$(2x, 2y, 2z) = \mu_1(1, 1, 1) + \mu_2(1, 0, 0) + \mu_3(0, 1, 0) + \mu_4(0, 0, 1),$$

which gives the component equations

$$2x = \mu_1 + \mu_2, \quad 2y = \mu_1 + \mu_3, \quad 2z = \mu_1 + \mu_4.$$

**Try the natural case:**  $g_1$  **active**,  $g_2, g_3, g_4$  **inactive**. Set  $\mu_2 = \mu_3 = \mu_4 = 0$  and require  $x + y + z = 6$ . The stationarity equations reduce to  $2x = 2y = 2z = \mu_1$ , so  $x = y = z$ . Combined with  $x + y + z = 6$ :

$$x = y = z = 2, \quad \mu_1 = 4.$$

**Verify all four KKT conditions at  $(2, 2, 2)$  with  $\mu = (4, 0, 0, 0)$ .**

- *Stationarity:*  $(4, 4, 4) = 4(1, 1, 1) + 0 + 0 + 0 \checkmark$ .
- *Primal feasibility:*  $g_1 = 2 + 2 + 2 - 6 = 0 \geq 0 \checkmark$ ;  $g_2 = g_3 = g_4 = 2 \geq 0 \checkmark$ .
- *Dual feasibility:*  $\mu_1 = 4 \geq 0, \mu_2 = \mu_3 = \mu_4 = 0 \geq 0 \checkmark$ .
- *Complementary slackness:*  $\mu_1 g_1 = 4 \cdot 0 = 0 \checkmark$ ;  $\mu_i g_i = 0 \cdot 2 = 0$  for  $i = 2, 3, 4 \checkmark$ .



All four conditions hold. Since  $f$  is strictly convex and the feasible region is convex, this KKT point is the unique global minimum. Only the sum constraint is active; the three non-negativity constraints are slack.

$$f_{\min} = 12 \text{ at } (x, y, z) = (2, 2, 2).$$

*Geometric picture.* The level surfaces of  $f$  are spheres centered at the origin. The minimum is the closest point of the half-space  $x + y + z \geq 6$  (intersected with the first octant) to the origin — the foot of the perpendicular from the origin to the plane  $x + y + z = 6$ , which is  $(2, 2, 2)$ . The non-negativity constraints are inactive precisely because that foot already lies in the first octant.

71. **Solution.** Minimize  $f(x, y) = x^2 + y^2$  subject to  $g_1(x, y) = x + y - 4 \geq 0$  and  $g_2(x, y) = x - 2y + 1 \geq 0$ .

(a) **KKT conditions.** For a minimization problem with constraints  $g_i \geq 0$ , the KKT conditions are:

$$\nabla f = \mu_1 \nabla g_1 + \mu_2 \nabla g_2, \quad \mu_i \geq 0, \quad \mu_i g_i = 0.$$

$$(2x, 2y) = \mu_1(1, 1) + \mu_2(1, -2).$$

This gives:

$$2x = \mu_1 + \mu_2, \quad 2y = \mu_1 - 2\mu_2.$$

**Case 1: Both constraints active** ( $g_1 = 0$  and  $g_2 = 0$ ).

$x + y = 4$  and  $x - 2y = -1$ . Solving:  $3y = 5 \implies y = 5/3$ ,  $x = 4 - 5/3 = 7/3$ .

Point:  $(7/3, 5/3)$ . Then  $\mu_1 = 2(7/3) - \mu_2$  and  $2(5/3) = \mu_1 - 2\mu_2$ . From the first:  $\mu_1 = 14/3 - \mu_2$ . Substituting:  $10/3 = 14/3 - \mu_2 - 2\mu_2 \implies 3\mu_2 = 4/3 \implies \mu_2 = 4/9$ . Then  $\mu_1 = 14/3 - 4/9 = 38/9$ .

Both  $\mu_1 = 38/9 > 0$  and  $\mu_2 = 4/9 > 0$ , as required.

$$f(7/3, 5/3) = 49/9 + 25/9 = 74/9.$$

**Case 2: Only  $g_1$  active** ( $\mu_2 = 0$ ,  $g_1 = 0$ ).

$\mu_2 = 0$ :  $2x = \mu_1$ ,  $2y = \mu_1$ , so  $x = y$ . From  $x + y = 4$ :  $x = y = 2$ . Then  $\mu_1 = 4 > 0$ , as required.

Check  $g_2$ :  $2 - 4 + 1 = -1 < 0$ . **Violated.** Not feasible.

**Case 3: Only  $g_2$  active** ( $\mu_1 = 0$ ,  $g_2 = 0$ ).

$\mu_1 = 0$ :  $2x = \mu_2$ ,  $2y = -2\mu_2$ , so  $y = -x$  and  $\mu_2 = 2x \geq 0$  requires  $x \geq 0$ .

From  $x - 2y = -1$ :  $x + 4x = -1 \implies 5x = -1 \implies x = -1/5$ . But  $x < 0$  means  $\mu_2 < 0$ . **Violates**  $\mu_2 \geq 0$ .

**Case 4: No constraint active** ( $\mu_1 = \mu_2 = 0$ ).

Then  $x = y = 0$ , but  $g_1(0, 0) = -4 < 0$ . **Not feasible.**

(b) The only candidate satisfying all KKT conditions is  $\left(\frac{7}{3}, \frac{5}{3}\right)$ .

Since the objective  $f$  is strictly convex and the feasible set is convex (intersection of half-planes), any KKT point is a global minimum. Therefore:

$$f_{\min} = \frac{74}{9} \approx 8.22 \text{ at } \left( \frac{7}{3}, \frac{5}{3} \right).$$

# Chapter 4

## Chapter 4: Multiple Integrals

1. **Solution.** We evaluate  $\iint_R (2x + 3y) dA$  where  $R = [0, 2] \times [1, 3]$ .

Using Fubini's Theorem:

$$\int_0^2 \int_1^3 (2x + 3y) dy dx = \int_0^2 \left[ 2xy + \frac{3}{2}y^2 \right]_{y=1}^{y=3} dx.$$

Inner integral:  $(6x + \frac{27}{2}) - (2x + \frac{3}{2}) = 4x + 12$ .

Outer integral:

$$\int_0^2 (4x + 12) dx = [2x^2 + 12x]_0^2 = 8 + 24 = \boxed{32}.$$

2. **Solution.** We compute  $\iint_R xy^2 dA$  where  $R = [-1, 1] \times [0, 2]$ .

$$\int_{-1}^1 \int_0^2 xy^2 dy dx = \int_{-1}^1 x \left[ \frac{y^3}{3} \right]_0^2 dx = \int_{-1}^1 \frac{8x}{3} dx = \frac{8}{3} \left[ \frac{x^2}{2} \right]_{-1}^1 = \frac{8}{3} \cdot 0 = \boxed{0}.$$

Alternatively, since  $f(x, y) = xy^2$  is an odd function in  $x$  and the  $x$ -interval  $[-1, 1]$  is symmetric about 0, the integral vanishes by symmetry.

3. **Solution.** We evaluate  $\iint_R \sin(x + y) dA$  where  $R = [0, \pi] \times [0, \pi]$ .

$$\int_0^\pi \int_0^\pi \sin(x + y) dy dx = \int_0^\pi [-\cos(x + y)]_{y=0}^{y=\pi} dx = \int_0^\pi [-\cos(x + \pi) + \cos x] dx.$$

Since  $\cos(x + \pi) = -\cos x$ , this becomes:

$$\int_0^\pi (\cos x + \cos x) dx = 2 \int_0^\pi \cos x dx = 2[\sin x]_0^\pi = 2(0 - 0) = \boxed{0}.$$

4. **Solution.** The average temperature is

$$T_{\text{avg}} = \frac{1}{ab} \iint_{[0,a] \times [0,b]} \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi y}{b}\right) dA.$$

The integrand separates:

$$\iint = \left( \int_0^a \sin\left(\frac{\pi x}{a}\right) dx \right) \left( \int_0^b \sin\left(\frac{\pi y}{b}\right) dy \right).$$

For the  $x$ -integral: let  $u = \pi x/a$ , so  $dx = a du/\pi$ :

$$\int_0^a \sin\left(\frac{\pi x}{a}\right) dx = \frac{a}{\pi} [-\cos u]_0^\pi = \frac{a}{\pi} (1 + 1) = \frac{2a}{\pi}.$$

Similarly,  $\int_0^b \sin\left(\frac{\pi y}{b}\right) dy = \frac{2b}{\pi}.$

Therefore:

$$T_{\text{avg}} = \frac{1}{ab} \cdot \frac{2a}{\pi} \cdot \frac{2b}{\pi} = \boxed{\frac{4}{\pi^2}}.$$

5. **Solution.** Total force =  $\iint_{[-1,1]^2} 2e^{-x^2-y^2} dA$ . The integrand separates:

$$2 \int_{-1}^1 e^{-x^2} dx \int_{-1}^1 e^{-y^2} dy = 2 \left( \int_{-1}^1 e^{-x^2} dx \right)^2.$$

Let  $I = \int_{-1}^1 e^{-x^2} dx$ . This equals  $\sqrt{\pi} \operatorname{erf}(1)$  where  $\operatorname{erf}(1) \approx 0.8427$ , so  $I \approx 1.4936$ .

Thus the total force is  $2I^2 = 2(\sqrt{\pi} \operatorname{erf}(1))^2 = \boxed{2\pi \operatorname{erf}^2(1)} \approx 4.46$  N.

6. **Solution.** Let  $f(x, y) = ax + b$  on  $R = [0, 1] \times [0, 1]$ .

Condition 1:  $\iint_R (ax + b) dA = \int_0^1 \int_0^1 (ax + b) dx dy = \int_0^1 \left[ \frac{a}{2} + b \right] dy = \frac{a}{2} + b = 4.$

Condition 2:  $\iint_R x(ax + b) dA = \int_0^1 \int_0^1 (ax^2 + bx) dx dy = \frac{a}{3} + \frac{b}{2} = 3.$

From the system:

$$\frac{a}{2} + b = 4, \quad \frac{a}{3} + \frac{b}{2} = 3.$$

From the first equation:  $b = 4 - a/2$ . Substituting into the second:

$$\frac{a}{3} + \frac{4 - a/2}{2} = 3 \implies \frac{a}{3} + 2 - \frac{a}{4} = 3 \implies \frac{a}{12} = 1 \implies a = 12.$$

Then  $b = 4 - 6 = -2$ . So  $f(x, y) = 12x - 2$ .

Now compute:

$$\begin{aligned}\iint_R x^2 f \, dA &= \int_0^1 \int_0^1 x^2(12x - 2) \, dx \, dy = \int_0^1 (12x^3 - 2x^2) \, dx \\ &= \left[ 3x^4 - \frac{2x^3}{3} \right]_0^1 = 3 - \frac{2}{3} = \boxed{\frac{7}{3}}.\end{aligned}$$

7. **Solution.** The integrand separates:

$$\begin{aligned}\iint_{[0,1]^2} \frac{1}{(1+x^2)(1+y^2)} \, dA &= \left( \int_0^1 \frac{dx}{1+x^2} \right) \left( \int_0^1 \frac{dy}{1+y^2} \right) \\ &= [\arctan x]_0^1 \cdot [\arctan y]_0^1 = \frac{\pi}{4} \cdot \frac{\pi}{4} = \boxed{\frac{\pi^2}{16}}.\end{aligned}$$

8. **Solution.** The region is  $[0, 2] \times [1, 3]$ . Dividing each side into 2 parts gives  $\Delta x = 1$ ,  $\Delta y = 1$ , so  $\Delta A = 1$ .

The subrectangles and their midpoints:

- $R_{11} = [0, 1] \times [1, 2]$ : midpoint  $(0.5, 1.5)$
- $R_{12} = [0, 1] \times [2, 3]$ : midpoint  $(0.5, 2.5)$
- $R_{21} = [1, 2] \times [1, 2]$ : midpoint  $(1.5, 1.5)$
- $R_{22} = [1, 2] \times [2, 3]$ : midpoint  $(1.5, 2.5)$

With  $f(x, y) = 2xy$ :

$$\begin{aligned}S_{2,2} &= f(0.5, 1.5) \cdot 1 + f(0.5, 2.5) \cdot 1 + f(1.5, 1.5) \cdot 1 + f(1.5, 2.5) \cdot 1 \\ &= 2(0.5)(1.5) + 2(0.5)(2.5) + 2(1.5)(1.5) + 2(1.5)(2.5) \\ &= 1.5 + 2.5 + 4.5 + 7.5 = \boxed{16}.\end{aligned}$$

(Note: the exact value of the integral is  $\int_0^2 \int_1^3 2xy \, dy \, dx = \int_0^2 x[y^2]_1^3 \, dx = \int_0^2 8x \, dx = 16$ , so the midpoint rule gives the exact answer here, as expected for a product of linear functions.)

9. **Solution.**

(a) The iterated integral is:

$$\iint_R (x - y) \, dA = \int_0^1 \int_0^2 (x - y) \, dy \, dx.$$

(b) Evaluating:

$$\int_0^1 \int_0^2 (x - y) \, dy \, dx = \int_0^1 \left[ xy - \frac{y^2}{2} \right]_0^2 \, dx = \int_0^1 (2x - 2) \, dx$$

$$= [x^2 - 2x]_0^1 = 1 - 2 = \boxed{-1}.$$

The negative result indicates that the “negative volume” (where  $x - y < 0$ , i.e.,  $y > x$ ) exceeds the “positive volume” (where  $x - y > 0$ , i.e.,  $y < x$ ). Over  $R$ , the line  $y = x$  splits the region; the part below  $y = x$  contributes positive volume and the larger part above contributes negative volume, giving a net value of  $-1$ .

10. **Solution.**  $R = [1, 2] \times [0, 1]$ .

**Order 1:**  $dy dx$ :

$$\int_1^2 \int_0^1 (3 + 2x) dy dx = \int_1^2 (3 + 2x) dx = [3x + x^2]_1^2 = (6 + 4) - (3 + 1) = \boxed{6}.$$

**Order 2:**  $dx dy$ :

$$\int_0^1 \int_1^2 (3 + 2x) dx dy = \int_0^1 [3x + x^2]_1^2 dy = \int_0^1 6 dy = 6.$$

Both orders yield 6.

11. **Solution.** Since  $e^x \sin y$  factors as a product of a function of  $x$  alone and a function of  $y$  alone, over the rectangle  $[0, \pi] \times [-1, 1]$ :

$$\iint e^x \sin y dA = \left( \int_0^\pi e^x dx \right) \left( \int_{-1}^1 \sin y dy \right).$$

The  $x$ -integral:  $\int_0^\pi e^x dx = e^\pi - 1$ .

The  $y$ -integral:  $\int_{-1}^1 \sin y dy = [-\cos y]_{-1}^1 = -\cos 1 + \cos(-1) = 0$ .

(Since  $\sin y$  is odd and the interval is symmetric.)

Therefore the double integral equals  $(e^\pi - 1) \cdot 0 = \boxed{0}$ .

12. **Solution.** We evaluate  $\iint_{[0,1] \times [2,3]} \frac{x}{1 + x^2 + y^2} dA$ .

Integrating with respect to  $x$  first is simpler because  $\frac{x}{1+x^2+y^2}$  has  $x$  in the numerator and  $x^2$  in the denominator, suggesting a substitution  $u = 1 + x^2 + y^2$ .

$$\int_0^1 \frac{x}{1 + x^2 + y^2} dx.$$

Let  $u = 1 + x^2 + y^2$ ,  $du = 2x dx$ :

$$= \frac{1}{2} \int_{1+y^2}^{2+y^2} \frac{du}{u} = \frac{1}{2} \ln \left( \frac{2+y^2}{1+y^2} \right).$$

Now integrate over  $y$ :

$$\int_2^3 \frac{1}{2} \ln \left( \frac{2+y^2}{1+y^2} \right) dy = \frac{1}{2} \int_2^3 [\ln(2+y^2) - \ln(1+y^2)] dy.$$

Using  $\int \ln(a + y^2) dy = y \ln(a + y^2) - 2y + 2\sqrt{a} \arctan(y/\sqrt{a}) + C$ :

For  $a = 2$ :  $[y \ln(2 + y^2) - 2y + 2\sqrt{2} \arctan(y/\sqrt{2})]_2^3$ .

At  $y = 3$ :  $3 \ln 11 - 6 + 2\sqrt{2} \arctan(3/\sqrt{2})$ .

At  $y = 2$ :  $2 \ln 6 - 4 + 2\sqrt{2} \arctan(\sqrt{2})$ .

Difference:  $3 \ln 11 - 2 \ln 6 - 2 + 2\sqrt{2} [\arctan(3/\sqrt{2}) - \arctan(\sqrt{2})]$ .

For  $a = 1$ :  $[y \ln(1 + y^2) - 2y + 2 \arctan y]_2^3$ .

At  $y = 3$ :  $3 \ln 10 - 6 + 2 \arctan 3$ .

At  $y = 2$ :  $2 \ln 5 - 4 + 2 \arctan 2$ .

Difference:  $3 \ln 10 - 2 \ln 5 - 2 + 2(\arctan 3 - \arctan 2)$ .

Combining:

$$\boxed{\frac{1}{2} \left[ 3 \ln \frac{11}{10} - 2 \ln \frac{6}{5} + 2\sqrt{2} (\arctan \frac{3}{\sqrt{2}} - \arctan \sqrt{2}) - 2(\arctan 3 - \arctan 2) \right]}.$$

### 13. Solution.

(a) The integrand  $\frac{1}{(x+1)(y+2)}$  already factors; no partial fractions are needed. It separates as  $\frac{1}{x+1} \cdot \frac{1}{y+2}$ .

(b) By Fubini's Theorem:

$$\iint_{[1,2] \times [0,2]} \frac{dA}{(x+1)(y+2)} = \left( \int_1^2 \frac{dx}{x+1} \right) \left( \int_0^2 \frac{dy}{y+2} \right).$$

$$\int_1^2 \frac{dx}{x+1} = \ln 3 - \ln 2 = \ln \frac{3}{2}, \quad \int_0^2 \frac{dy}{y+2} = \ln 4 - \ln 2 = \ln 2.$$

Therefore the integral equals  $\boxed{\ln \frac{3}{2} \cdot \ln 2}$ .

14. **Solution.** For a continuous function  $f$  on a closed rectangle  $R$ ,  $f$  is uniformly continuous on  $R$ . This means that for any  $\varepsilon > 0$ , there exists  $\delta > 0$  such that  $|f(\mathbf{p}) - f(\mathbf{q})| < \varepsilon$  whenever  $\|\mathbf{p} - \mathbf{q}\| < \delta$ .

As the partition is refined (i.e.,  $\|\Delta\| \rightarrow 0$ , where  $\|\Delta\|$  is the mesh size), every pair of sample points within the same subrectangle lies within distance  $\|\Delta\|$  of each other. Thus the function values at any two choices of sample points within the same subrectangle differ by less than  $\varepsilon$ .

Since the total area of  $R$  is fixed, the difference between Riemann sums formed with different sample points is bounded by  $\varepsilon \cdot \text{Area}(R)$ , which can be made arbitrarily small. Therefore, regardless of whether we use lower-left corners, upper-right corners, midpoints, or any other choice, the limit of the Riemann sum converges to the same unique value—the double integral  $\iint_R f dA$ .  $\square$

15. **Solution.** Let  $g(x, y) = \frac{xy}{x^2+y^2}$ . For fixed  $x$ , observe that  $g(x, -y) = \frac{x(-y)}{x^2+y^2} = -g(x, y)$ .

That is,  $g$  is an odd function of  $y$ . Since the  $y$ -integration runs over  $[-2, 2]$ , which is symmetric about 0:

$$\int_{-2}^2 g(x, y) dy = 0 \quad \text{for each fixed } x \neq 0.$$

Therefore:

$$\iint_{[-1,1] \times [-2,2]} \frac{xy}{x^2+y^2} dA = \int_{-1}^1 \underbrace{\left( \int_{-2}^2 \frac{xy}{x^2+y^2} dy \right)}_{=0} dx = \boxed{0}. \quad \square$$

16. **Solution.**  $\text{Area}(R) = 4$ . We need:

$$f_{\text{avg}} = \frac{1}{4} \int_0^2 \int_0^2 \sin(xy) dy dx.$$

Inner integral (with  $x$  fixed,  $x \neq 0$ ): let  $u = xy$ ,  $du = x dy$ :

$$\int_0^2 \sin(xy) dy = \frac{1}{x} [-\cos(xy)]_0^2 = \frac{1 - \cos(2x)}{x}.$$

So:

$$\iint_R \sin(xy) dA = \int_0^2 \frac{1 - \cos(2x)}{x} dx.$$

This can be written using the cosine integral Ci: Recall  $\int_0^a \frac{1-\cos t}{t} dt = \gamma + \ln a - \text{Ci}(a)$  where  $\gamma$  is Euler's constant. With substitution  $t = 2x$ :

$$\int_0^2 \frac{1 - \cos(2x)}{x} dx = \int_0^4 \frac{1 - \cos t}{t} dt = \gamma + \ln 4 - \text{Ci}(4).$$

Therefore:

$$f_{\text{avg}} = \frac{1}{4} (\gamma + \ln 4 - \text{Ci}(4)).$$

Numerically,  $\gamma \approx 0.5772$ ,  $\ln 4 \approx 1.3863$ ,  $\text{Ci}(4) \approx -0.1409$ , giving  $f_{\text{avg}} \approx \frac{2.1044}{4} \approx \boxed{0.526}$ .

17. **Solution.** We compute  $\iint x^2 y dx dy$  step by step.

**Step 1:** Integrate with respect to  $x$ :

$$\int x^2 y dx = \frac{x^3 y}{3} + g(y),$$

where  $g(y)$  is an arbitrary function of  $y$  (the “constant” of integration with respect to  $x$ ).



**Step 2:** Integrate the result with respect to  $y$ :

$$\int \left( \frac{x^3 y}{3} + g(y) \right) dy = \frac{x^3 y^2}{6} + G(y) + h(x),$$

where  $G(y) = \int g(y) dy$  is an arbitrary function of  $y$  and  $h(x)$  is an arbitrary function of  $x$ .

The general indefinite double integral is:

$$\boxed{\frac{x^3 y^2}{6} + G(y) + h(x)},$$

where  $G(y)$  and  $h(x)$  are arbitrary functions of one variable.

18. **Solution.** We compute  $\iint (3x + 2xy^2) dx dy$ .

**Step 1:** Integrate with respect to  $x$ :

$$\int (3x + 2xy^2) dx = \frac{3x^2}{2} + x^2 y^2 + g(y),$$

where  $g(y)$  is an arbitrary function of  $y$ .

**Step 2:** Integrate with respect to  $y$ :

$$\int \left( \frac{3x^2}{2} + x^2 y^2 + g(y) \right) dy = \frac{3x^2 y}{2} + \frac{x^2 y^3}{3} + G(y) + h(x),$$

where  $G(y) = \int g(y) dy$  is arbitrary and  $h(x)$  is an arbitrary function of  $x$ .

In the requested form:

$$\boxed{F(x, y) + G(y) + h(x) = \frac{3x^2 y}{2} + \frac{x^2 y^3}{3} + G(y) + h(x)}.$$

19. **Solution.** Let  $f(x, y) = \alpha x + \beta y + \gamma xy + \delta$  (bilinear form).

( $\Leftarrow$ ) Suppose  $f$  is bilinear. Compute the integral directly:

$$\begin{aligned} \iint_R f dA &= \int_a^b \int_c^d (\alpha x + \beta y + \gamma xy + \delta) dy dx \\ &= \int_a^b \left[ \alpha x(d - c) + \beta \frac{d^2 - c^2}{2} + \gamma x \frac{d^2 - c^2}{2} + \delta(d - c) \right] dx \\ &= (d - c) \left[ \alpha \frac{b^2 - a^2}{2} + \beta \frac{d + c}{2}(b - a) + \gamma \frac{b^2 - a^2}{2} \cdot \frac{d + c}{2} + \delta(b - a) \right]. \end{aligned}$$

Simplifying:  $(b - a)(d - c) \left[ \alpha \frac{a+b}{2} + \beta \frac{c+d}{2} + \gamma \frac{(a+b)(c+d)}{4} + \delta \right]$ .

Now the corner-average formula gives:

$$\begin{aligned}
 & \frac{f(a, c) + f(a, d) + f(b, c) + f(b, d)}{4} \\
 &= \frac{1}{4} [(\alpha a + \beta c + \gamma ac + \delta) + (\alpha a + \beta d + \gamma ad + \delta) \\
 &\quad + (\alpha b + \beta c + \gamma bc + \delta) + (\alpha b + \beta d + \gamma bd + \delta)] \\
 &= \alpha \frac{a+b}{2} + \beta \frac{c+d}{2} + \gamma \frac{(a+b)(c+d)}{4} + \delta.
 \end{aligned}$$

Multiplying by  $(b-a)(d-c)$  gives the integral.

( $\Rightarrow$ ) Suppose the formula holds for *all* rectangles  $[a, b] \times [c, d]$ . One can show by differentiating both sides with respect to  $a, b, c, d$  (or by considering the formula on shrinking rectangles) that  $f$  must be a polynomial of degree at most 1 in each variable separately, i.e., bilinear. Specifically, if  $f$  contained a term like  $x^2$ , then the integral would involve  $(b^3 - a^3)/3$  while the right-hand side would only involve  $(a^2 + b^2)/2$ , and these cannot agree for all  $a, b$ .  $\square$

20. **Solution.** Define  $\Phi(t) = \iint_R (f + tg)^2 dA$  for  $t \in \mathbb{R}$ .

Since  $(f + tg)^2 \geq 0$  everywhere, we have  $\Phi(t) \geq 0$  for all  $t$ .

Expanding:

$$\Phi(t) = \iint_R f^2 dA + 2t \iint_R fg dA + t^2 \iint_R g^2 dA.$$

Let  $A = \iint_R g^2 dA$ ,  $B = 2 \iint_R fg dA$ ,  $C = \iint_R f^2 dA$ . Then  $\Phi(t) = At^2 + Bt + C \geq 0$  for all  $t$ .

If  $A = 0$ , then  $g = 0$  a.e. and the inequality holds trivially. If  $A > 0$ , the quadratic  $At^2 + Bt + C$  is nonnegative for all  $t$ , so its discriminant must be nonpositive:

$$B^2 - 4AC \leq 0 \implies 4 \left( \iint_R fg dA \right)^2 \leq 4 \left( \iint_R f^2 dA \right) \left( \iint_R g^2 dA \right).$$

Dividing by 4:

$$\left( \iint_R fg dA \right)^2 \leq \left( \iint_R f^2 dA \right) \left( \iint_R g^2 dA \right). \quad \square$$

21. **Solution.** The total probability is:

$$\int_0^a \int_0^b \frac{4}{ab} \sin^2\left(\frac{\pi x}{a}\right) \sin^2\left(\frac{\pi y}{b}\right) dy dx = \frac{4}{ab} \left( \int_0^a \sin^2\left(\frac{\pi x}{a}\right) dx \right) \left( \int_0^b \sin^2\left(\frac{\pi y}{b}\right) dy \right).$$

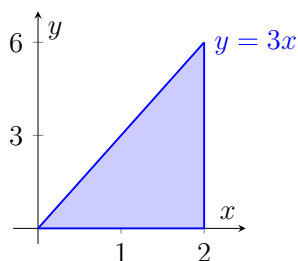
Using  $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ :

$$\int_0^a \sin^2\left(\frac{\pi x}{a}\right) dx = \int_0^a \frac{1 - \cos(2\pi x/a)}{2} dx = \frac{a}{2} - \frac{a}{2} \cdot \frac{1}{2\pi} [\sin(2\pi x/a)]_0^a = \frac{a}{2}.$$

Similarly,  $\int_0^b \sin^2(\pi y/b) dy = b/2$ . Therefore:

$$\frac{4}{ab} \cdot \frac{a}{2} \cdot \frac{b}{2} = \frac{4}{ab} \cdot \frac{ab}{4} = \boxed{1}. \quad \square$$

## 22. Solution.

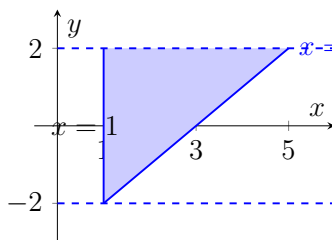


The region  $R$  is bounded below by  $y = 0$  and above by  $y = 3x$ , with  $0 \leq x \leq 2$ . As a Type I region, for each fixed  $x \in [0, 2]$ ,  $y$  ranges from 0 to  $3x$ .

The integral is:

$$\iint_R (x + y) \, dA = \int_0^2 \int_0^{3x} (x + y) \, dy \, dx.$$

## 23. Solution.

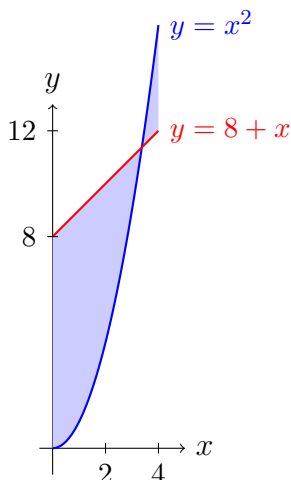


As a Type II region,  $y$  ranges over  $[-2, 2]$ , and for each fixed  $y$ ,  $x$  goes from the left boundary  $x = 1$  to the right boundary  $x = y + 3$ .

The integral is:

$$\iint_R 5 \, dA = \int_{-2}^2 \int_1^{y+3} 5 \, dx \, dy.$$

## 24. Solution.



- (a)  $R$  is Type I because for each fixed  $x \in [0, 4]$ , the variable  $y$  ranges between two functions of  $x$ : from  $y = x^2$  (below) to  $y = 8 + x$  (above). The curves intersect when  $x^2 = 8 + x$ , i.e.,  $x^2 - x - 8 = 0$ , giving  $x = \frac{1+\sqrt{33}}{2} \approx 3.37$ . Since this is less than 4, we need to check that  $x^2 \leq 8 + x$  for  $x \in [0, 4]$ . At  $x = 4$ :  $x^2 = 16$  and  $8 + x = 12$ , so  $x^2 > 8 + x$ . Thus the description  $x^2 \leq y \leq 8 + x$  is only valid for  $0 \leq x \leq \frac{1+\sqrt{33}}{2}$ .

Correcting:  $R$  as stated (with  $x^2 \leq y \leq 8 + x$  for  $0 \leq x \leq 4$ ) is Type I because the bounds on  $y$  are given explicitly as functions of  $x$ .

- (b) As Type II:  $y$  ranges from  $\min(x^2) = 0$  to  $\max(8 + x) = 12$ , so  $y \in [0, 12]$ . For a given  $y$ , the left boundary comes from  $x^2 = y$  (i.e.,  $x = \sqrt{y}$ ) or  $x = 0$ , and the right boundary comes from  $y = 8 + x$  (i.e.,  $x = y - 8$ ) or  $x = 4$ . The region must be split:

As Type II, we solve for  $x$ : from  $y = x^2$  we get  $x = \sqrt{y}$ , from  $y = 8 + x$  we get  $x = y - 8$ , and we have  $0 \leq x \leq 4$ .

For  $0 \leq y \leq 8$ :  $x$  runs from 0 to  $\sqrt{y}$  (since  $y - 8 < 0$ ).

For  $8 < y \leq 9$ :  $x$  runs from  $y - 8$  to  $\sqrt{y}$  (since both bounds are in  $[0, 4]$ ).

For  $9 < y \leq 12$ :  $x$  runs from  $y - 8$  to 4 (since  $\sqrt{y} > 4$  would require  $y > 16$ ; actually  $\sqrt{12} \approx 3.46 < 4$ , so  $x$  runs from  $y - 8$  to  $\sqrt{y}$ ).

Let  $x_0 = \frac{1+\sqrt{33}}{2}$ . At the intersection,  $y_0 = x_0^2$ . For  $y > y_0$ , the parabola  $x = \sqrt{y}$  exceeds the line  $x = y - 8$  but we also need  $x \leq 4$ . At  $y = 12$ :  $\sqrt{12} < 4$  and  $y - 8 = 4$ , so  $x$  from  $\sqrt{y}$  to 4—but this reverses. The Type II description requires careful case analysis, confirming that the Type I description is simpler.

25. **Solution.** The curves  $y = x^2$  and  $y = 6 - x$  intersect when  $x^2 = 6 - x$ , i.e.,  $x^2 + x - 6 = 0$ , giving  $(x + 3)(x - 2) = 0$ , so  $x = -3$  or  $x = 2$ . On  $[-1, 3]$ , the intersection within the interval is at  $x = 2$ .

For  $-1 \leq x \leq 2$ :  $x^2 \leq 6 - x$ , so the top curve is  $y = 6 - x$  and the bottom is  $y = x^2$ .

For  $2 \leq x \leq 3$ :  $x^2 \geq 6 - x$ , so the top curve is  $y = x^2$  and the bottom is  $y = 6 - x$ .

$$A = \int_{-1}^2 [(6 - x) - x^2] dx + \int_2^3 [x^2 - (6 - x)] dx.$$

First integral:

$$\int_{-1}^2 (6 - x - x^2) dx = \left[ 6x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2 = \frac{22}{3} + \frac{37}{6} = \frac{44 + 37}{6} = \frac{81}{6} = \frac{27}{2}.$$

Second integral:

$$\begin{aligned} \int_2^3 (x^2 - 6 + x) dx &= \left[ \frac{x^3}{3} - 6x + \frac{x^2}{2} \right]_2^3 \\ &= \left( 9 - 18 + \frac{9}{2} \right) - \left( \frac{8}{3} - 12 + 2 \right) = \frac{9}{2} - 9 + 12 - \frac{8}{3} - 2. \end{aligned}$$

Evaluating: at  $x = 3$ ,  $9 - 18 + 9/2 = -9/2$ ; at  $x = 2$ ,  $8/3 - 12 + 2 = -22/3$ . Difference:  $-9/2 + 22/3 = (-27 + 44)/6 = 17/6$ .

$$\text{Total area} = 27/2 + 17/6 = 81/6 + 17/6 = \boxed{\frac{98}{6} = \frac{49}{3}}.$$

**26. Solution.**

(a) **Type I ( $dy dx$ ):** For  $0 \leq x \leq 2$ ,  $y$  goes from  $x$  to 4:

$$\begin{aligned} \int_0^2 \int_x^4 (1 + xy) dy dx &= \int_0^2 \left[ y + \frac{xy^2}{2} \right]_x^4 dx = \int_0^2 \left[ (4 + 8x) - \left( x + \frac{x^3}{2} \right) \right] dx \\ &= \int_0^2 \left( 4 + 7x - \frac{x^3}{2} \right) dx = \left[ 4x + \frac{7x^2}{2} - \frac{x^4}{8} \right]_0^2 = 8 + 14 - 2 = \boxed{20}. \end{aligned}$$

(b) **Type II ( $dx dy$ ):** The region is  $\{0 \leq x \leq 2, x \leq y \leq 4\}$ . For Type II, for fixed  $y$ : if  $0 \leq y \leq 2$ , then  $x$  ranges from 0 to  $y$ ; if  $2 \leq y \leq 4$ , then  $x$  ranges from 0 to 2.

$$\int_0^2 \int_0^y (1 + xy) dx dy + \int_2^4 \int_0^2 (1 + xy) dx dy.$$

First part:

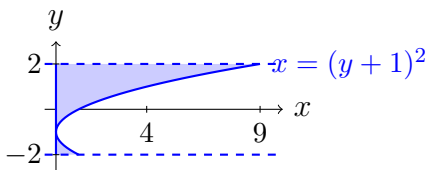
$$\int_0^2 \left[ x + \frac{x^2 y}{2} \right]_0^y dy = \int_0^2 \left( y + \frac{y^3}{2} \right) dy = \left[ \frac{y^2}{2} + \frac{y^4}{8} \right]_0^2 = 2 + 2 = 4.$$

Second part:

$$\int_2^4 \left[ x + \frac{x^2 y}{2} \right]_0^2 dy = \int_2^4 (2 + 2y) dy = [2y + y^2]_2^4 = (8 + 16) - (4 + 4) = 16.$$

Total:  $4 + 16 = 20$ .

**27. Solution.**



The region is bounded by  $x = 0$ ,  $y = -2$ ,  $y = 2$ ,  $x = (y + 1)^2$ . For Type II:  $y \in [-2, 2]$ , and for each  $y$ ,  $x$  ranges from  $h_1(y) = 0$  to  $h_2(y) = (y + 1)^2$ .

$$\int_{-2}^2 \int_0^{(y+1)^2} (4 - x - y) dx dy = \int_{-2}^2 \left[ 4x - \frac{x^2}{2} - xy \right]_0^{(y+1)^2} dy$$

$$= \int_{-2}^2 \left[ 4(y+1)^2 - \frac{(y+1)^4}{2} - y(y+1)^2 \right] dy.$$

Let  $u = y + 1$ ,  $y = u - 1$ ,  $dy = du$ , and when  $y = -2$ ,  $u = -1$ ; when  $y = 2$ ,  $u = 3$ :

$$\begin{aligned} &= \int_{-1}^3 \left[ 4u^2 - \frac{u^4}{2} - (u-1)u^2 \right] du = \int_{-1}^3 \left[ 4u^2 - \frac{u^4}{2} - u^3 + u^2 \right] du \\ &= \int_{-1}^3 \left( 5u^2 - u^3 - \frac{u^4}{2} \right) du. \\ &= \left[ \frac{5u^3}{3} - \frac{u^4}{4} - \frac{u^5}{10} \right]_{-1}^3. \end{aligned}$$

At  $u = 3$ :  $\frac{5(27)}{3} - \frac{81}{4} - \frac{243}{10} = 45 - \frac{81}{4} - \frac{243}{10}$ .

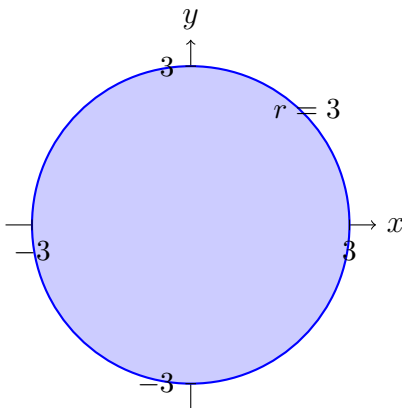
Common denominator 20:  $\frac{900-405-486}{20} = \frac{9}{20}$ .

At  $u = -1$ :  $\frac{5(-1)}{3} - \frac{1}{4} - \frac{-1}{10} = -\frac{5}{3} - \frac{1}{4} + \frac{1}{10}$ .

Common denominator 60:  $\frac{-100-15+6}{60} = \frac{-109}{60}$ .

Result:  $\frac{9}{20} + \frac{109}{60} = \frac{27+109}{60} = \frac{136}{60} = \boxed{\frac{34}{15}}$ .

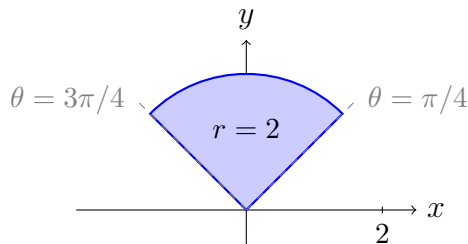
28. **Solution.**



In polar coordinates,  $x^2 + y^2 = r^2$  and  $dA = r dr d\theta$ . The disk of radius 3:  $0 \leq r \leq 3$ ,  $0 \leq \theta \leq 2\pi$ .

$$\int_0^{2\pi} \int_0^3 r^2 \cdot r dr d\theta = \int_0^{2\pi} \int_0^3 r^3 dr d\theta = \int_0^{2\pi} \left[ \frac{r^4}{4} \right]_0^3 d\theta = \int_0^{2\pi} \frac{81}{4} d\theta = \frac{81}{4} \cdot 2\pi = \boxed{\frac{81\pi}{2}}.$$

29. **Solution.**



In polar coordinates:  $x^2 + y^2 = r^2$ ,  $dA = r dr d\theta$ .

The sector has  $0 \leq r \leq 2$ ,  $\pi/4 \leq \theta \leq 3\pi/4$ . The integral becomes:

$$\iint_R (x^2 + y^2) dA = \int_{\pi/4}^{3\pi/4} \int_0^2 r^2 \cdot r dr d\theta = \int_{\pi/4}^{3\pi/4} \int_0^2 r^3 dr d\theta.$$

30. **Solution.** The annular region:  $2 \leq r \leq 3$ ,  $0 \leq \theta \leq 2\pi$ . With  $x = r \cos \theta$ ,  $y = r \sin \theta$ :

$$\int_0^{2\pi} \int_2^3 (3 + r \cos \theta + r \sin \theta) r dr d\theta.$$

Expand:

$$= \int_0^{2\pi} \int_2^3 (3r + r^2 \cos \theta + r^2 \sin \theta) dr d\theta.$$

The terms  $r^2 \cos \theta$  and  $r^2 \sin \theta$  integrate to 0 over  $[0, 2\pi]$  in  $\theta$  since  $\int_0^{2\pi} \cos \theta d\theta = \int_0^{2\pi} \sin \theta d\theta = 0$ .

Remaining:

$$\int_0^{2\pi} \int_2^3 3r dr d\theta = 2\pi \cdot 3 \left[ \frac{r^2}{2} \right]_2^3 = 2\pi \cdot 3 \cdot \frac{9-4}{2} = 2\pi \cdot \frac{15}{2} = \boxed{15\pi}.$$

31. **Solution.**

- (a) The closed region  $R$  is bounded by the parabola  $y = x^2$  from  $x = 0$  to  $x = 1$  and the quarter-circle  $x^2 + y^2 = 1$  in the first quadrant (from  $(1, 0)$  back to  $(0, 1)$  is above the parabola for  $0 < x < 1$ ).

The parabola  $y = x^2$  and the quarter-circle  $x^2 + y^2 = 1$  intersect at  $(0, 0)$  (approximately) and  $(x_0, y_0)$  where  $x^2 + x^4 = 1$ . In the first quadrant, the region between them is awkward in a single coordinate system. Splitting:

- $R_1$ : between  $y = x^2$  and  $y = \sqrt{1 - x^2}$  — best in Cartesian since the parabola has a simple Cartesian form.
- $R_2$ : the quarter-disk portion that doesn't overlap  $R_1$  — best in polar since it is part of a circle.

(b) Let  $x_0$  solve  $x^2 + x^4 = 1$ , so  $x_0^2 = \frac{-1+\sqrt{5}}{2}$ .

$R_1$  (Cartesian):  $0 \leq x \leq x_0$ ,  $x^2 \leq y \leq \sqrt{1-x^2}$ :

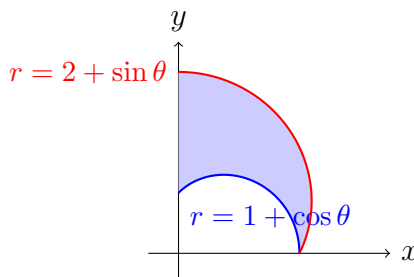
$$\iint_{R_1} (1 + x^2 + y^2) dA = \int_0^{x_0} \int_{x^2}^{\sqrt{1-x^2}} (1 + x^2 + y^2) dy dx.$$

$R_2$  (Polar): the quarter-disk from  $\theta = 0$  to  $\theta = \pi/2$ ,  $r$  from 0 to 1, minus  $R_1$ . Alternatively,  $R$  is the region in the first quadrant between  $y = x^2$  and the circle, and the full integral is simply:

$$\iint_R (1 + x^2 + y^2) dA = \int_0^{x_0} \int_{x^2}^{\sqrt{1-x^2}} (1 + x^2 + y^2) dy dx.$$

If we want a polar piece for the quarter-disk portion beyond  $x_0$ , note that for  $x_0 \leq x$ , the parabola rises above the circle, so the region is entirely captured by the single Cartesian integral above.

### 32. Solution.



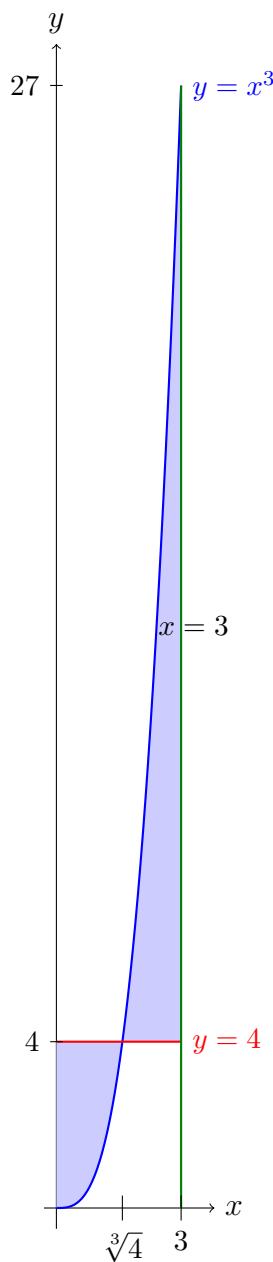
In polar coordinates,  $\sqrt{x^2 + y^2} = r$  and  $dA = r dr d\theta$ , so the integrand becomes  $r \cdot r dr d\theta = r^2 dr d\theta$ .

For  $0 \leq \theta \leq \pi/2$ , the inner radius is  $r = 1 + \cos \theta$  and the outer radius is  $r = 2 + \sin \theta$ .

$$\boxed{\iint_R \sqrt{x^2 + y^2} dA = \int_0^{\pi/2} \int_{1+\cos \theta}^{2+\sin \theta} r^2 dr d\theta.}$$

### 33. Solution.





The region in the first quadrant is bounded by  $y = x^3$ ,  $y = 4$ , and  $x = 3$ .

The curve  $y = x^3$  reaches  $y = 4$  at  $x = 4^{1/3} \approx 1.587$ . The region bounded by  $y = x^3$ ,  $y = 4$ , and  $x = 3$  in the first quadrant splits into two subregions:

For  $0 \leq x \leq 4^{1/3}$ : the region is between  $y = x^3$  and  $y = 4$ .

For  $4^{1/3} \leq x \leq 3$ :  $x^3 \geq 4$ , so the curve  $y = x^3$  is above  $y = 4$ . The region is between  $y = 4$  and  $y = x^3$  (with  $x^3$  on top) bounded by  $x = 3$ .

So two Type I subregions:

$R_1$ :  $0 \leq x \leq 4^{1/3}$ ,  $x^3 \leq y \leq 4$ :

$$\int_0^{4^{1/3}} \int_{x^3}^4 (2x + y) dy dx.$$

$R_2$ :  $4^{1/3} \leq x \leq 3$ ,  $4 \leq y \leq x^3$ :

$$\int_{4^{1/3}}^3 \int_4^{x^3} (2x + y) dy dx.$$

The full integral is the sum of these two integrals.

34. **Solution.** The plate occupies  $0 \leq x \leq 3$ ,  $0 \leq y \leq x^2$ , with  $\rho(x, y) = 5 + x$ .

(a) Total mass:

$$\begin{aligned} M &= \int_0^3 \int_0^{x^2} (5 + x) dy dx = \int_0^3 (5 + x) \cdot x^2 dx = \int_0^3 (5x^2 + x^3) dx \\ &= \left[ \frac{5x^3}{3} + \frac{x^4}{4} \right]_0^3 = 45 + \frac{81}{4} = \frac{180 + 81}{4} = \frac{261}{4}. \end{aligned}$$

(b) Moments:

$$\begin{aligned} M_x &= \int_0^3 \int_0^{x^2} y(5 + x) dy dx = \int_0^3 (5 + x) \frac{x^4}{2} dx = \frac{1}{2} \int_0^3 (5x^4 + x^5) dx \\ &= \frac{1}{2} \left[ x^5 + \frac{x^6}{6} \right]_0^3 = \frac{1}{2} \left( 243 + \frac{729}{6} \right) = \frac{1}{2} \cdot \frac{1458 + 729}{6} = \frac{2187}{12} = \frac{729}{4}. \\ M_y &= \int_0^3 \int_0^{x^2} x(5 + x) dy dx = \int_0^3 x(5 + x) \cdot x^2 dx = \int_0^3 (5x^3 + x^4) dx \\ &= \left[ \frac{5x^4}{4} + \frac{x^5}{5} \right]_0^3 = \frac{5 \cdot 81}{4} + \frac{243}{5} = \frac{405}{4} + \frac{243}{5} = \frac{2025 + 972}{20} = \frac{2997}{20}. \end{aligned}$$

(c) Center of mass:

$$\begin{aligned} \bar{x} &= \frac{M_y}{M} = \frac{2997/20}{261/4} = \frac{2997}{20} \cdot \frac{4}{261} = \frac{2997}{1305} \\ &= \frac{999}{435} = \frac{333}{145} \approx 2.297. \\ \bar{y} &= \frac{M_x}{M} = \frac{729/4}{261/4} = \frac{729}{261} = \frac{243}{87} = \frac{81}{29} \approx 2.793. \end{aligned}$$

35. **Solution.**

- (a) The cardioid  $r = 2 + 2 \cos \theta$  traces out for  $\theta \in [0, 2\pi]$ . The area is:

$$\begin{aligned} A &= \int_0^{2\pi} \int_0^{2+2\cos\theta} r \, dr \, d\theta = \int_0^{2\pi} \frac{(2+2\cos\theta)^2}{2} d\theta = \int_0^{2\pi} \frac{4(1+\cos\theta)^2}{2} d\theta \\ &= 2 \int_0^{2\pi} (1+2\cos\theta+\cos^2\theta) d\theta = 2[2\pi+0+\pi] = 6\pi. \end{aligned}$$

- (b) In Cartesian, the cardioid  $r = 2 + 2 \cos \theta$  becomes  $\sqrt{x^2 + y^2} = 2 + \frac{2x}{\sqrt{x^2 + y^2}}$ , i.e.,  $x^2 + y^2 = 2\sqrt{x^2 + y^2} + 2x$ . Squaring and rearranging gives a complicated implicit equation. Describing the bounds as explicit functions  $y = f(x)$  requires solving this quartic—far messier than the clean polar integral. This demonstrates why polar coordinates are more convenient for regions defined by polar equations like cardioids.

### 36. Solution.

- (a) The region  $R$  is in the first quadrant ( $x \geq 0, y \geq 0$ ), inside the circle  $x^2 + y^2 = 4$  (radius 2), and below the line  $y = 2 - x$ . Note that  $y = 2 - x$  intersects the circle when  $x^2 + (2 - x)^2 = 4$ , i.e.,  $2x^2 - 4x = 0$ , giving  $x = 0$  or  $x = 2$ . So the line connects  $(0, 2)$  to  $(2, 0)$ , exactly the chord of the circle. The region bounded by  $x = 0$ ,  $y = 0$ , the circle, and the line is actually the region in the first quadrant between the line  $y = 2 - x$  and the circular arc.

The line  $y = 2 - x$  lies inside the circle (since at the midpoint  $(1, 1)$ ,  $1+1 = 2 < 4$ ). So the region between the line (below) and the arc (above) is one piece, plus the triangular region below the line.

The region is bounded by all four curves simultaneously. The simplest interpretation:  $R$  is the region in the first quadrant that is inside the circle and above the line  $y = 2 - x$ —the region between the chord and the arc.

Alternatively,  $R$  could be the region below both the circle and the line. Let us take  $R$  as the region with  $x \geq 0$ ,  $y \geq 0$ ,  $x^2 + y^2 \leq 4$ , and  $y \leq 2 - x$  (the triangular region under the chord). In that case,  $R$  is a triangle with vertices  $(0, 0)$ ,  $(2, 0)$ ,  $(0, 2)$ .

For the more interesting case where  $R$  is between the chord and the arc:  $x \geq 0$ ,  $y \geq 0$ ,  $x^2 + y^2 \leq 4$ ,  $y \geq 2 - x$ :

- (b) **Split approach:** Use polar coordinates for the circular part. In polar, the circle is  $r = 2$  and the line  $x + y = 2$  becomes  $r(\cos \theta + \sin \theta) = 2$ , i.e.,  $r = \frac{2}{\cos \theta + \sin \theta}$ . The full integral over  $R$  (between chord and arc):

$$\begin{aligned} \iint_R (x + y) \, dA &= \int_0^{\pi/2} \int_{2/(\cos\theta+\sin\theta)}^2 (r \cos \theta + r \sin \theta) r \, dr \, d\theta \\ &= \int_0^{\pi/2} \int_{2/(\cos\theta+\sin\theta)}^2 r^2 (\cos \theta + \sin \theta) \, dr \, d\theta. \end{aligned}$$

This is a clean single integral in polar coordinates.

**37. Solution.**

(a) The total mass is:

$$M = \iint_R \rho_0 dA = \int_0^3 \int_0^2 5 dy dx.$$

(b) Evaluating:

$$M = 5 \int_0^3 \int_0^2 dy dx = 5 \cdot 3 \cdot 2 = \boxed{30 \text{ kg}}.$$

**38. Solution.** The triangle has vertices  $(0, 0)$ ,  $(4, 0)$ ,  $(4, 3)$ . The hypotenuse goes from  $(0, 0)$  to  $(4, 3)$ :  $y = \frac{3x}{4}$ .

(a) As a Type I region:  $0 \leq x \leq 4$ ,  $0 \leq y \leq 3x/4$ .

$$M = \int_0^4 \int_0^{3x/4} 2 dy dx = 2 \cdot \frac{3}{4} \int_0^4 x dx = \frac{3}{2} \cdot 8 = \boxed{12 \text{ kg}}.$$

(Check: area of triangle  $= \frac{1}{2} \cdot 4 \cdot 3 = 6$ , times  $\rho_0 = 2$  gives 12.)

(b) Center of mass:

$$\begin{aligned} \bar{x} &= \frac{1}{M} \int_0^4 \int_0^{3x/4} 2x dy dx = \frac{1}{12} \cdot 2 \int_0^4 x \cdot \frac{3x}{4} dx \\ &= \frac{1}{12} \cdot \frac{3}{2} \int_0^4 x^2 dx = \frac{1}{12} \cdot \frac{3}{2} \cdot \frac{64}{3} = \frac{1}{12} \cdot 32 = \frac{8}{3}. \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{1}{M} \int_0^4 \int_0^{3x/4} 2y dy dx = \frac{1}{12} \cdot 2 \int_0^4 \frac{1}{2} \left( \frac{3x}{4} \right)^2 dx \\ &= \frac{1}{12} \int_0^4 \frac{9x^2}{16} dx = \frac{1}{12} \cdot \frac{9}{16} \cdot \frac{64}{3} = \frac{1}{12} \cdot 12 = 1. \end{aligned}$$

$$\text{So } (\bar{x}, \bar{y}) = \boxed{\left( \frac{8}{3}, 1 \right)}.$$

(c) Since  $\bar{x} = 8/3 \approx 2.67 > 2$  (more than halfway along the  $x$ -base of length 4), this is because the triangle widens as  $x$  increases: the height at position  $x$  is  $3x/4$ , which grows linearly with  $x$ . Thus more area (and hence more mass, since density is uniform) is concentrated toward larger  $x$ -values, pulling the centroid past the midpoint.

**39. Solution.**

(a)  $M = \int_0^2 \int_0^3 (5 + x) dy dx.$

(b) Evaluating:

$$M = \int_0^2 (5+x) \cdot 3 \, dx = 3 \int_0^2 (5+x) \, dx = 3 \left[ 5x + \frac{x^2}{2} \right]_0^2 = 3(10 + 2) = 3 \cdot 12 = \boxed{36 \text{ g}}.$$

40. **Solution.**

(a) The region  $R$  is in the first quadrant bounded by  $x = 0$ ,  $y = 0$ ,  $y = 4 - x^2$ . The parabola meets the  $x$ -axis at  $x = 2$ .

Area of  $R$ :

$$A = \int_0^2 (4 - x^2) \, dx = \left[ 4x - \frac{x^3}{3} \right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}.$$

Average temperature:

$$T_{\text{avg}} = \frac{1}{A} \int_0^2 \int_0^{4-x^2} (10 + xy) \, dy \, dx.$$

(b) Inner integral:

$$\int_0^{4-x^2} (10 + xy) \, dy = \left[ 10y + \frac{xy^2}{2} \right]_0^{4-x^2} = 10(4 - x^2) + \frac{x(4 - x^2)^2}{2}.$$

Expanding:  $10(4 - x^2) = 40 - 10x^2$  and  $\frac{x(16 - 8x^2 + x^4)}{2} = 8x - 4x^3 + \frac{x^5}{2}$ .

Outer integral:

$$\begin{aligned} & \int_0^2 \left( 40 - 10x^2 + 8x - 4x^3 + \frac{x^5}{2} \right) \, dx \\ &= \left[ 40x - \frac{10x^3}{3} + 4x^2 - x^4 + \frac{x^6}{12} \right]_0^2 \\ &= 80 - \frac{80}{3} + 16 - 16 + \frac{64}{12} = 80 - \frac{80}{3} + \frac{16}{3} = 80 - \frac{64}{3} = \frac{240 - 64}{3} = \frac{176}{3}. \end{aligned}$$

$$T_{\text{avg}} = \frac{3}{16} \cdot \frac{176}{3} = \frac{176}{16} = \boxed{11^\circ\text{C}}.$$

41. **Solution.** In polar coordinates,  $y = r \sin \theta$  and  $dA = r \, dr \, d\theta$ . The semicircular region:  $0 \leq r \leq 3$ ,  $0 \leq \theta \leq \pi$ .

$$\begin{aligned} I_x &= \iint_R y^2 \, dA = \int_0^\pi \int_0^3 (r \sin \theta)^2 \cdot r \, dr \, d\theta = \int_0^\pi \int_0^3 r^3 \sin^2 \theta \, dr \, d\theta \\ &= \left( \int_0^3 r^3 \, dr \right) \left( \int_0^\pi \sin^2 \theta \, d\theta \right) = \frac{81}{4} \cdot \frac{\pi}{2} = \boxed{\frac{81\pi}{8}}. \end{aligned}$$

This is indeed simpler in polar than in Cartesian, where we would need  $\int_{-3}^3 \int_0^{\sqrt{9-x^2}} y^2 \, dy \, dx$  with a more complex antiderivative.

42. **Solution.** In polar:  $x^2 + y^2 = r^2$ , disk of radius  $R$ :  $0 \leq r \leq R$ ,  $0 \leq \theta \leq 2\pi$ .

$$I_z = \rho_0 \int_0^{2\pi} \int_0^R r^2 \cdot r \, dr \, d\theta = \rho_0 \int_0^{2\pi} \int_0^R r^3 \, dr \, d\theta = \rho_0 \cdot 2\pi \cdot \frac{R^4}{4} = \boxed{\rho_0 \pi \frac{R^4}{2}}. \quad \square$$

43. **Solution.**

$$(a) \quad M = \int_0^{2\pi} \int_0^2 (4+r) r \, dr \, d\theta.$$

(b) Evaluating the  $r$ -integral:

$$\int_0^2 (4r + r^2) \, dr = \left[ 2r^2 + \frac{r^3}{3} \right]_0^2 = 8 + \frac{8}{3} = \frac{32}{3}.$$

$$M = 2\pi \cdot \frac{32}{3} = \boxed{\frac{64\pi}{3}}.$$

(c) Average density:

$$\rho_{\text{avg}} = \frac{M}{\text{Area}} = \frac{64\pi/3}{\pi \cdot 4} = \frac{64}{12} = \boxed{\frac{16}{3}}.$$

44. **Solution.**

(a) With  $z = x^2 + 3y^2$ :

$$\frac{\partial z}{\partial x} = 2x, \quad \frac{\partial z}{\partial y} = 6y.$$

(b) The surface area integral is:

$$A = \iint_R \sqrt{1 + (2x)^2 + (6y)^2} \, dA = \boxed{\int_0^2 \int_0^1 \sqrt{1 + 4x^2 + 36y^2} \, dy \, dx}.$$

This integral does not simplify to a closed form easily and would typically be evaluated numerically.

45. **Solution.** For the cone  $z = \sqrt{x^2 + y^2}$ :

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}.$$

$$1 + \left( \frac{\partial z}{\partial x} \right)^2 + \left( \frac{\partial z}{\partial y} \right)^2 = 1 + \frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} = 1 + 1 = 2.$$

The cone lies below  $z = 3$  when  $\sqrt{x^2 + y^2} \leq 3$ , i.e., over the disk  $x^2 + y^2 \leq 9$ .

$$A = \iint_{x^2 + y^2 \leq 9} \sqrt{2} \, dA = \sqrt{2} \cdot \pi(3)^2 = \boxed{9\pi\sqrt{2}}.$$

46. **Solution.** For  $z = \sqrt{4 - x^2 - y^2}$ :

$$\frac{\partial z}{\partial x} = \frac{-x}{\sqrt{4 - x^2 - y^2}}, \quad \frac{\partial z}{\partial y} = \frac{-y}{\sqrt{4 - x^2 - y^2}}.$$

$$1 + \frac{x^2}{4 - x^2 - y^2} + \frac{y^2}{4 - x^2 - y^2} = \frac{4 - x^2 - y^2 + x^2 + y^2}{4 - x^2 - y^2} = \frac{4}{4 - x^2 - y^2}.$$

In polar coordinates over  $x^2 + y^2 \leq 4$ :

$$A = \int_0^{2\pi} \int_0^2 \frac{2}{\sqrt{4 - r^2}} r \, dr \, d\theta = 2\pi \int_0^2 \frac{2r}{\sqrt{4 - r^2}} \, dr.$$

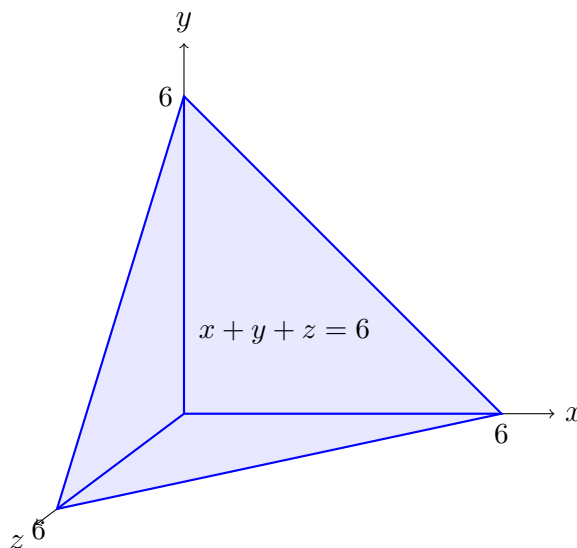
Let  $u = 4 - r^2$ ,  $du = -2r \, dr$ :

$$= 2\pi \int_4^0 \frac{-2 \, du}{2\sqrt{u}} = 2\pi \cdot 2 \left[ \sqrt{u} \right]_0^4 = 2\pi \cdot 2 \cdot 2 = 8\pi.$$

The geometric formula for a hemisphere of radius 2 gives  $2\pi r^2 = 2\pi(4) = 8\pi$ .

$$\boxed{A = 8\pi}.$$

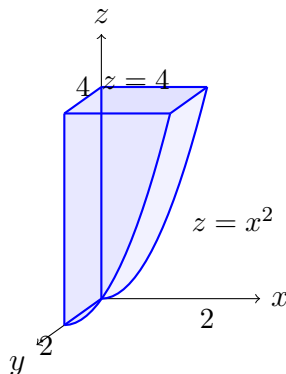
47. **Solution.**



The first octant region bounded by  $z = 0$ ,  $x = 0$ ,  $y = 0$ , and  $x + y + z = 6$  (so  $z = 6 - x - y$ ). With  $x, y, z \geq 0$ :  $x + y \leq 6$ .

Choosing order  $dz \, dy \, dx$ : for fixed  $x, y$ ,  $z$  ranges from 0 to  $6 - x - y$ ; for fixed  $x$ ,  $y$  ranges from 0 to  $6 - x$ ;  $x$  ranges from 0 to 6.

$$\boxed{\iiint_E (2xy - z) \, dV = \int_0^6 \int_0^{6-x} \int_0^{6-x-y} (2xy - z) \, dz \, dy \, dx}.$$

48. **Solution.**

The region  $E$  is bounded by  $z = x^2$  (below),  $z = 4$  (above),  $x = 0$ , and  $y = 2$  (with  $y \geq 0$  implied). Since  $z = x^2 \leq 4$  requires  $|x| \leq 2$ , and  $x = 0$  is one boundary, we need  $0 \leq x \leq 2$ ,  $0 \leq y \leq 2$ ,  $x^2 \leq z \leq 4$ .

$$\begin{aligned} V &= \int_0^2 \int_0^2 \int_{x^2}^4 dz \, dy \, dx = \int_0^2 \int_0^2 (4 - x^2) \, dy \, dx = \int_0^2 2(4 - x^2) \, dx \\ &= 2 \left[ 4x - \frac{x^3}{3} \right]_0^2 = 2 \left( 8 - \frac{8}{3} \right) = \boxed{\frac{32}{3}}. \end{aligned}$$

49. **Solution.** The tetrahedron:  $x, y, z \geq 0$ ,  $2x + 3y + z \leq 6$ , so  $z \leq 6 - 2x - 3y$ .

For the base in the  $xy$ -plane:  $2x + 3y \leq 6$ ,  $x \geq 0$ ,  $y \geq 0$ , so  $0 \leq x \leq 3$ ,  $0 \leq y \leq (6 - 2x)/3$ .

$$I = \int_0^3 \int_0^{(6-2x)/3} \int_0^{6-2x-3y} (x^2 + yz) \, dz \, dy \, dx.$$

Inner integral ( $z$ ):

$$\int_0^{6-2x-3y} (x^2 + yz) \, dz = x^2(6 - 2x - 3y) + \frac{y(6 - 2x - 3y)^2}{2}.$$

Let  $a = 6 - 2x - 3y$ . The inner integral is  $x^2a + ya^2/2$ .

Middle integral ( $y$ ), with  $a = 6 - 2x - 3y$ : Let  $b = 6 - 2x$ , so  $a = b - 3y$ .

$$\int_0^{b/3} \left[ x^2(b - 3y) + \frac{y(b - 3y)^2}{2} \right] dy.$$

First term:  $x^2 \int_0^{b/3} (b - 3y) \, dy = x^2 \left[ by - \frac{3y^2}{2} \right]_0^{b/3} = x^2 \left( \frac{b^2}{3} - \frac{b^2}{6} \right) = \frac{x^2 b^2}{6}.$



Second term:  $\frac{1}{2} \int_0^{b/3} y(b-3y)^2 dy$ . Expand  $(b-3y)^2 = b^2 - 6by + 9y^2$ :

$$\frac{1}{2} \int_0^{b/3} (b^2y - 6by^2 + 9y^3) dy = \frac{1}{2} \left[ \frac{b^2y^2}{2} - 2by^3 + \frac{9y^4}{4} \right]_0^{b/3}.$$

At  $y = b/3$ :  $\frac{b^4}{18} - \frac{2b^4}{27} + \frac{b^4}{36}$ . Common denominator 108:  $\frac{6b^4 - 8b^4 + 3b^4}{108} = \frac{b^4}{108}$ .

So the second term is  $\frac{b^4}{216}$ .

Middle integral =  $\frac{x^2b^2}{6} + \frac{b^4}{216}$  where  $b = 6 - 2x$ .

Outer integral ( $x$ ):

$$I = \int_0^3 \left[ \frac{x^2(6-2x)^2}{6} + \frac{(6-2x)^4}{216} \right] dx.$$

Let  $u = 6 - 2x$ ,  $x = (6 - u)/2$ ,  $dx = -du/2$ . When  $x = 0$ ,  $u = 6$ ; when  $x = 3$ ,  $u = 0$ .

$$\begin{aligned} I &= \int_6^0 \left[ \frac{(6-u)^2u^2}{4 \cdot 6} + \frac{u^4}{216} \right] \left( -\frac{du}{2} \right) = \frac{1}{2} \int_0^6 \left[ \frac{(6-u)^2u^2}{24} + \frac{u^4}{216} \right] du. \\ &= \frac{1}{48} \int_0^6 (6-u)^2u^2 du + \frac{1}{432} \int_0^6 u^4 du. \end{aligned}$$

For  $\int_0^6 (6-u)^2u^2 du$ : expand  $(6-u)^2u^2 = (36 - 12u + u^2)u^2 = 36u^2 - 12u^3 + u^4$ :

$$\begin{aligned} &= \left[ 12u^3 - 3u^4 + \frac{u^5}{5} \right]_0^6 = 12(216) - 3(1296) + \frac{7776}{5} \\ &= 2592 - 3888 + 1555.2 = 259.2 = \frac{1296}{5}. \end{aligned}$$

For  $\int_0^6 u^4 du = \frac{6^5}{5} = \frac{7776}{5}$ .

$$I = \frac{1}{48} \cdot \frac{1296}{5} + \frac{1}{432} \cdot \frac{7776}{5} = \frac{1296}{240} + \frac{7776}{2160} = \frac{27}{5} + \frac{18}{5} = \frac{45}{5} = \boxed{9}.$$

(Here  $\frac{1296}{240} = \frac{27}{5}$  and  $\frac{7776}{2160} = \frac{18}{5}$ , confirming  $\frac{27}{5} + \frac{18}{5} = 9$ .)

50. **Solution.** The integral with the given bounds is:

$$\boxed{\int_{-1}^2 \int_0^\pi \int_0^3 (3r^2 + z) r dr d\theta dz}.$$

Equivalently (reordering):

$$\int_0^\pi \int_0^3 \int_{-1}^2 (3r^2 + z) r dz dr d\theta.$$

The bounds are:  $r \in [0, 3]$ ,  $\theta \in [0, \pi]$ ,  $z \in [-1, 2]$ .

51. **Solution.** In spherical coordinates:  $x^2 + y^2 + z^2 = \rho^2$ ,  $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ .

Ball of radius 4:  $0 \leq \rho \leq 4$ ,  $0 \leq \phi \leq \pi$ ,  $0 \leq \theta \leq 2\pi$ .

$$\begin{aligned} \iiint_E \rho^2 \, dV &= \int_0^{2\pi} \int_0^\pi \int_0^4 \rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} d\theta \int_0^\pi \sin \phi \, d\phi \int_0^4 \rho^4 \, d\rho \\ &= 2\pi \cdot [-\cos \phi]_0^\pi \cdot \left[ \frac{\rho^5}{5} \right]_0^4 = 2\pi \cdot 2 \cdot \frac{1024}{5} = \boxed{\frac{4096\pi}{5}}. \end{aligned}$$

52. **Solution.** The two spheres are  $\rho = 2$  and  $x^2 + y^2 + (z - 2)^2 = 4$ . Expanding the second:  $x^2 + y^2 + z^2 - 4z + 4 = 4$ , so  $\rho^2 - 4\rho \cos \phi = 0$  (since  $z = \rho \cos \phi$ ), giving  $\rho = 4 \cos \phi$ .

The intersection region is where both  $\rho \leq 2$  and  $\rho \leq 4 \cos \phi$ . These spheres intersect when  $2 = 4 \cos \phi$ , i.e.,  $\cos \phi = 1/2$ , so  $\phi = \pi/3$ .

The intersection region (interior to both spheres) is:

- For  $0 \leq \phi \leq \pi/3$ : the sphere  $\rho = 2$  is the binding constraint (since  $4 \cos \phi \geq 2$ ), so  $0 \leq \rho \leq 2$ .
- For  $\pi/3 \leq \phi \leq \pi/2$ : the sphere  $\rho = 4 \cos \phi$  is binding (since  $4 \cos \phi \leq 2$ ), so  $0 \leq \rho \leq 4 \cos \phi$ .

And  $0 \leq \theta \leq 2\pi$ . (For  $\phi > \pi/2$ ,  $4 \cos \phi < 0$ , so there is no contribution.)

The volume integral is:

$$V = \int_0^{2\pi} \int_0^{\pi/3} \int_0^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta + \int_0^{2\pi} \int_{\pi/3}^{\pi/2} \int_0^{4 \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

53. **Solution.** We use spherical coordinates. The first octant of the sphere  $x^2 + y^2 + z^2 \leq 9$ :  $0 \leq \rho \leq 3$ ,  $0 \leq \phi \leq \pi/2$ ,  $0 \leq \theta \leq \pi/2$ .

With  $\rho(x, y, z) = 3x = 3\rho \sin \phi \cos \theta$ :

$$\begin{aligned} I &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^3 3\rho \sin \phi \cos \theta \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= 3 \int_0^{\pi/2} \cos \theta \, d\theta \int_0^{\pi/2} \sin^2 \phi \, d\phi \int_0^3 \rho^3 \, d\rho \\ &= 3 \cdot [\sin \theta]_0^{\pi/2} \cdot \frac{\pi}{4} \cdot \frac{81}{4} = 3 \cdot 1 \cdot \frac{\pi}{4} \cdot \frac{81}{4} = \boxed{\frac{243\pi}{16}}. \end{aligned}$$

54. **Solution.** Transform to polar:  $x = r \cos \theta$ ,  $y = r \sin \theta$ , Jacobian  $= r$ . The disk  $x^2 + y^2 \leq 4$  becomes  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$ . Also  $4 - x^2 - y^2 = 4 - r^2$ .

$$\boxed{\iint_R (4 - x^2 - y^2) dA = \int_0^{2\pi} \int_0^2 (4 - r^2) r dr d\theta}.$$

55. **Solution.** The parallelogram has vertices  $(0, 0)$ ,  $(2, 0)$ ,  $(3, 2)$ ,  $(1, 2)$ . The edges from  $(0, 0)$  are to  $(2, 0)$  (direction  $(2, 0)$ ) and to  $(1, 2)$  (direction  $(1, 2)$ ).

Let  $u = x - y/2$  and  $v = y$ . Then the vertices map to:

- $(0, 0) \rightarrow (0, 0)$
- $(2, 0) \rightarrow (2, 0)$
- $(3, 2) \rightarrow (2, 2)$
- $(1, 2) \rightarrow (0, 2)$

So the rectangle is  $0 \leq u \leq 2$ ,  $0 \leq v \leq 2$ . The inverse transformation:  $x = u + v/2$ ,  $y = v$ .

Jacobian:  $\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 1 & 1/2 \\ 0 & 1 \end{vmatrix} = 1$ .

Observe the edges of  $R$ . Along the bottom ( $y = 0$ ):  $x$  goes from 0 to 2. Along the top ( $y = 2$ ):  $x$  goes from 1 to 3. Along the left: from  $(0, 0)$  to  $(1, 2)$ , so  $x = y/2$ . Along the right: from  $(2, 0)$  to  $(3, 2)$ , so  $x = y/2 + 2$ .

Let  $u = x - y/2$ ,  $v = y$ . Then  $R$  maps to  $0 \leq u \leq 2$ ,  $0 \leq v \leq 2$ , the Jacobian is  $|J| = 1$ , and  $x + y = u + 3v/2$ .

The transformed integral is:

$$\boxed{\int_0^2 \int_0^2 \exp\left(\left(u + \frac{3v}{2}\right)^2\right) du dv, \quad |J| = 1}.$$

(Note: while this doesn't fully separate the integrand, the region is now a rectangle, which is the primary simplification sought.)

56. **Solution.**

(a) The Jacobian:

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -4 - 3 = -7.$$

So  $|J| = \boxed{7}$ .

(b) In terms of  $u, v$ :  $x + 2y = (2u + 3v) + 2(u - 2v) = 4u - v$ .

$$\boxed{\iint_R (x + 2y) dx dy = \int_0^3 \int_0^2 (4u - v) \cdot 7 dv du = 7 \int_0^3 \int_0^2 (4u - v) dv du}.$$

57. **Solution.** With  $x = r^2 \cos \theta$ ,  $y = r^2 \sin \theta$ :

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} 2r \cos \theta & -r^2 \sin \theta \\ 2r \sin \theta & r^2 \cos \theta \end{vmatrix} = 2r^3 \cos^2 \theta + 2r^3 \sin^2 \theta = 2r^3.$$

Also  $\sqrt{x^2 + y^2} = \sqrt{r^4 \cos^2 \theta + r^4 \sin^2 \theta} = r^2$ .

Over the region  $0 \leq r \leq 1$ ,  $0 \leq \theta \leq 2\pi$ :

$$\boxed{\iint_R \sqrt{x^2 + y^2} \, dx \, dy = \int_0^{2\pi} \int_0^1 r^2 \cdot 2r^3 \, dr \, d\theta = \int_0^{2\pi} \int_0^1 2r^5 \, dr \, d\theta}.$$

Evaluating:  $\int_0^1 2r^5 \, dr = \frac{1}{3}$ , so the integral  $= 2\pi/3$ .

58. **Solution.** The transformation  $u = (x - y)/\sqrt{2}$ ,  $v = (x + y)/\sqrt{2}$  gives  $x = (u + v)/\sqrt{2}$ ,  $y = (v - u)/\sqrt{2}$ .

$$\text{Jacobian: } \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{vmatrix} = \frac{1}{2} + \frac{1}{2} = 1.$$

The square  $R$  has vertices  $(0, 0)$ ,  $(2, 0)$ ,  $(2, 2)$ ,  $(0, 2)$ . Under the transformation:

- $(0, 0) \rightarrow (0, 0)$
- $(2, 0) \rightarrow (\sqrt{2}, \sqrt{2})$
- $(2, 2) \rightarrow (0, 2\sqrt{2})$
- $(0, 2) \rightarrow (-\sqrt{2}, \sqrt{2})$

The image is a square in the  $uv$ -plane rotated 45°, with vertices as listed—a diamond with diagonals along the axes from  $-\sqrt{2}$  to  $\sqrt{2}$  in  $u$  and 0 to  $2\sqrt{2}$  in  $v$ .

The new region  $R'$  in  $(u, v)$ : For  $0 \leq v \leq \sqrt{2}$ ,  $u \in [-v, v]$ ; for  $\sqrt{2} \leq v \leq 2\sqrt{2}$ ,  $u \in [-(2\sqrt{2} - v), 2\sqrt{2} - v]$ .

Since the integrand is 1 and  $|J| = 1$ :

$$\boxed{\iint_R 1 \, dx \, dy = \int_0^{\sqrt{2}} \int_{-v}^v 1 \, du \, dv + \int_{\sqrt{2}}^{2\sqrt{2}} \int_{-(2\sqrt{2}-v)}^{2\sqrt{2}-v} 1 \, du \, dv}.$$

(The result is 4, matching the area of the original square.)

59. **Solution.** In cylindrical coordinates:  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $z = z$ , and  $dV = r \, dr \, d\theta \, dz$ .

The cylinder  $x^2 + y^2 \leq 4$ :  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$ ,  $0 \leq z \leq 5$ .

The integrand:  $x + 2y - z = r \cos \theta + 2r \sin \theta - z$ .

$$\boxed{\iiint_E (x + 2y - z) \, dV = \int_0^{2\pi} \int_0^2 \int_0^5 (r \cos \theta + 2r \sin \theta - z) r \, dz \, dr \, d\theta}.$$

60. **Solution.** The parallelogram has vertices  $(0, 0)$ ,  $(2, 0)$ ,  $(1 + a, 2 + a)$ ,  $(3 + a, 2 + a)$ .

Edge vectors from  $(0, 0)$ : to  $(2, 0)$ , direction  $\mathbf{e}_1 = (2, 0)$ ; to  $(1 + a, 2 + a)$ , direction  $\mathbf{e}_2 = (1 + a, 2 + a)$ .

Note:  $(2, 0) + (1 + a, 2 + a) = (3 + a, 2 + a)$ , confirming the fourth vertex.

Let  $u, v \in [0, 1]$  parameterize:  $(x, y) = u(2, 0) + v(1 + a, 2 + a)$ , so  $x = 2u + (1 + a)v$ ,  $y = (2 + a)v$ .

From  $y = (2 + a)v$ :  $v = y/(2 + a)$ . Then  $u = (x - (1 + a)v)/2$ .

Jacobian:  $|J| = \begin{vmatrix} 2 & 1 + a \\ 0 & 2 + a \end{vmatrix} = 2(2 + a)$ .

So  $dx dy = 2(2 + a) du dv$ .

Also  $x + y = 2u + (1 + a)v + (2 + a)v = 2u + (3 + 2a)v$ .

$$\boxed{\iint_R (x + y) dx dy = 2(2 + a) \int_0^1 \int_0^1 [2u + (3 + 2a)v] du dv.}$$

61. **Solution.** In spherical coordinates:  $x = \rho \sin \phi \cos \theta$ ,  $y = \rho \sin \phi \sin \theta$ ,  $z = \rho \cos \phi$ .

Then  $x^2 + y^2 + z^2 = \rho^2$  and  $dV = \rho^2 \sin \phi d\rho d\phi d\theta$ .

The sphere  $x^2 + y^2 + z^2 \leq 9$ :  $0 \leq \rho \leq 3$ ,  $0 \leq \phi \leq \pi$ ,  $0 \leq \theta \leq 2\pi$ .

$$\iiint_E \rho^2 dV = \int_0^{2\pi} \int_0^\pi \int_0^3 \rho^2 \cdot \rho^2 \sin \phi d\rho d\phi d\theta.$$

Step by step:

$$\int_0^3 \rho^4 d\rho = \frac{243}{5}, \quad \int_0^\pi \sin \phi d\phi = 2, \quad \int_0^{2\pi} d\theta = 2\pi.$$

$$= 2\pi \cdot 2 \cdot \frac{243}{5} = \boxed{\frac{972\pi}{5}}.$$

62. **Solution.** The change of variables is defined by  $u = x - y$ ,  $v = y - 2z$ ,  $w = x + z$ .

Solving for  $(x, y, z)$  in terms of  $(u, v, w)$ : From  $u = x - y$  and  $w = x + z$ , we get  $z = w - x$ . From  $v = y - 2z = y - 2(w - x) = y - 2w + 2x$ , and  $y = x - u$ , so  $v = x - u - 2w + 2x = 3x - u - 2w$ , giving  $x = \frac{u+v+2w}{3}$ .

Then  $y = x - u = \frac{u+v+2w}{3} - u = \frac{v+2w-2u}{3}$  and  $z = w - x = w - \frac{u+v+2w}{3} = \frac{-u-v+w}{3}$ .

The Jacobian matrix  $\frac{\partial(x,y,z)}{\partial(u,v,w)}$ :

$$\begin{pmatrix} 1/3 & 1/3 & 2/3 \\ -2/3 & 1/3 & 2/3 \\ -1/3 & -1/3 & 1/3 \end{pmatrix}.$$

Determinant (expanding along the first row):

$$\frac{1}{3} \left( \frac{1}{9} + \frac{2}{9} \right) - \frac{1}{3} \left( -\frac{2}{9} + \frac{2}{9} \right) + \frac{2}{3} \left( \frac{2}{9} + \frac{1}{9} \right) = \frac{1}{3} \cdot \frac{3}{9} + 0 + \frac{2}{3} \cdot \frac{3}{9} = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}.$$

So  $|J| = 1/3$ . Also:

$$4xyz = 4 \cdot \frac{u+v+2w}{3} \cdot \frac{v+2w-2u}{3} \cdot \frac{w-u-v}{3}.$$

The integral in  $(u, v, w)$ -space is:

$$\begin{aligned} & \boxed{\int_0^2 \int_0^1 \int_0^3 4 \cdot \frac{(u+v+2w)(v+2w-2u)(w-u-v)}{27} \cdot \frac{1}{3} dw dv du} \\ &= \frac{4}{81} \int_0^2 \int_0^1 \int_0^3 (u+v+2w)(v+2w-2u)(w-u-v) dw dv du. \end{aligned}$$

### 63. Solution.

(a) The Jacobian matrix:

$$J = \frac{\partial(x, y, z)}{\partial(\mu, \nu, z)} = \begin{pmatrix} a \sinh \mu \cos \nu & -a \cosh \mu \sin \nu & 0 \\ a \cosh \mu \sin \nu & a \sinh \mu \cos \nu & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(b) The determinant:

$$\begin{aligned} |J| &= 1 \cdot (a \sinh \mu \cos \nu \cdot a \sinh \mu \cos \nu - (-a \cosh \mu \sin \nu)(a \cosh \mu \sin \nu)) \\ &= a^2 \sinh^2 \mu \cos^2 \nu + a^2 \cosh^2 \mu \sin^2 \nu \\ &= a^2 (\sinh^2 \mu \cos^2 \nu + \cosh^2 \mu \sin^2 \nu). \end{aligned}$$

Using  $\cosh^2 \mu = 1 + \sinh^2 \mu$ :

$$= a^2 (\sinh^2 \mu \cos^2 \nu + \sin^2 \nu + \sinh^2 \mu \sin^2 \nu) = a^2 (\sinh^2 \mu + \sin^2 \nu).$$

The volume element becomes:

$$\boxed{dV = a^2 (\sinh^2 \mu + \sin^2 \nu) d\mu d\nu dz}.$$

The factor  $a^2(\sinh^2 \mu + \sin^2 \nu)$  acts as a scale factor that accounts for the non-uniform stretching of the elliptic cylindrical coordinate system. Near  $\mu = 0$  and  $\nu = 0$  (or  $\pi$ ), the factor is small, reflecting the compression near the foci of the ellipses.

64. **Solution.** Set up the cone with apex at the origin and base at  $z = 3$ . At height  $z$ , the radius is  $r = z/3$  (since at  $z = 3$ ,  $r = 1$ ).

Using cylindrical coordinates:

$$M = \int_0^{2\pi} \int_0^3 \int_0^{z/3} kz \cdot r \, dr \, dz \, d\theta.$$

Inner integral ( $r$ ):

$$\int_0^{z/3} kz \cdot r \, dr = kz \cdot \frac{r^2}{2} \Big|_0^{z/3} = kz \cdot \frac{z^2}{18} = \frac{kz^3}{18}.$$

Middle integral ( $z$ ):

$$\int_0^3 \frac{kz^3}{18} \, dz = \frac{k}{18} \cdot \frac{81}{4} = \frac{81k}{72} = \frac{9k}{8}.$$

Outer integral ( $\theta$ ):

$$M = 2\pi \cdot \frac{9k}{8} = \boxed{\frac{9\pi k}{4}}.$$

65. **Solution.** In spherical coordinates:  $x = \rho \sin \phi \cos \theta$ ,  $y = \rho \sin \phi \sin \theta$ ,  $z = \rho \cos \phi$ ,  $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ .

Hemisphere of radius 2,  $z \geq 0$ :  $0 \leq \rho \leq 2$ ,  $0 \leq \phi \leq \pi/2$ ,  $0 \leq \theta \leq 2\pi$ .

Density:  $\rho(x, y, z) = 1 + z^2 = 1 + \rho^2 \cos^2 \phi$ .

$$M = \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 (1 + \rho^2 \cos^2 \phi) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

66. **Solution.**

(a) Total mass:

$$M = \int_0^1 \int_0^2 \int_0^3 (x + y + z) \, dz \, dy \, dx.$$

Evaluating:

$$\int_0^3 (x + y + z) \, dz = 3(x + y) + \frac{9}{2} = 3x + 3y + \frac{9}{2}.$$

$$\int_0^2 \left( 3x + 3y + \frac{9}{2} \right) dy = 6x + 6 + 9 = 6x + 15.$$

$$M = \int_0^1 (6x + 15) \, dx = 3 + 15 = \boxed{18}.$$

(b) The moment  $M_{xy}$  (for the  $z$ -coordinate):

$$M_{xy} = \int_0^1 \int_0^2 \int_0^3 z(x+y+z) dz dy dx.$$

$$\int_0^3 z(x+y+z) dz = (x+y)\frac{9}{2} + 9 = \frac{9(x+y)}{2} + 9.$$

$$\int_0^2 \left( \frac{9(x+y)}{2} + 9 \right) dy = 9x + 9 + 18 = 9x + 27.$$

$$M_{xy} = \int_0^1 (9x + 27) dx = \frac{9}{2} + 27 = \frac{63}{2}.$$

$$\bar{z} = \frac{M_{xy}}{M} = \frac{63/2}{18} = \frac{63}{36} = \boxed{\frac{7}{4}}.$$

67. **Solution.** Let  $u = x/3$ ,  $v = y/2$ ,  $w = z$ . Then the ellipsoid  $x^2/9 + y^2/4 + z^2 \leq 1$  becomes  $u^2 + v^2 + w^2 \leq 1$  (a unit sphere).

The Jacobian:  $x = 3u$ ,  $y = 2v$ ,  $z = w$ , so  $\frac{\partial(x,y,z)}{\partial(u,v,w)} = 3 \cdot 2 \cdot 1 = 6$ .

The density:  $\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{9u^2 + 4v^2 + w^2}$ .

The total mass:

$$M = 6 \iiint_{u^2+v^2+w^2 \leq 1} \sqrt{9u^2 + 4v^2 + w^2} du dv dw.$$

(This can be further converted to spherical coordinates in  $(u, v, w)$ -space, though the integrand  $\sqrt{9 \sin^2 \alpha \cos^2 \beta + 4 \sin^2 \alpha \sin^2 \beta + \cos^2 \alpha}$  would depend on angles and not simplify to a closed form easily.)

68. **Solution.**

(a) The total force  $\mathbf{F}_{\text{total}} = \iiint_E \mathbf{F} dV = (\iiint_E yz dV, \iiint_E xz dV, \iiint_E xy dV)$ .

Using cylindrical coordinates ( $x = r \cos \theta$ ,  $y = r \sin \theta$ ), the cylinder is  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$ ,  $0 \leq z \leq 5$ .

**$x$ -component:**  $\iiint yz dV = \int_0^{2\pi} \int_0^2 \int_0^5 (r \sin \theta)(z) r dz dr d\theta$ . The  $\theta$ -integral contains  $\int_0^{2\pi} \sin \theta d\theta = 0$ . So the  $x$ -component is  $\mathbf{0}$ .

**$y$ -component:** Similarly contains  $\int_0^{2\pi} \cos \theta d\theta = 0$ . So the  $y$ -component is  $\mathbf{0}$ .

**$z$ -component:**  $\iiint xy dV = \int_0^{2\pi} \int_0^2 \int_0^5 r^2 \cos \theta \sin \theta \cdot r dz dr d\theta$ . The  $\theta$ -integral:  $\int_0^{2\pi} \cos \theta \sin \theta d\theta = \frac{1}{2} \int_0^{2\pi} \sin 2\theta d\theta = 0$ . So the  $z$ -component is  $\mathbf{0}$ .

Therefore  $\boxed{\mathbf{F}_{\text{total}} = \mathbf{0}}$ .

(This is expected by symmetry: the cylinder is symmetric under  $\theta \rightarrow \theta + \pi$ , and each component of  $\mathbf{F}$  is an odd function under this rotation.)



(b) The work done per unit time (power) is:

$$W = \iiint_E \mathbf{F} \cdot \mathbf{v} \, dV = \iiint_E (yz \cdot x + xz \cdot y + xy \cdot z) \, dV = \iiint_E 3xyz \, dV.$$

In cylindrical:  $3xyz = 3r \cos \theta \cdot r \sin \theta \cdot z = 3r^2 z \cos \theta \sin \theta$ .

$$W = \int_0^{2\pi} \int_0^2 \int_0^5 3r^2 z \cos \theta \sin \theta \cdot r \, dz \, dr \, d\theta.$$

Again,  $\int_0^{2\pi} \cos \theta \sin \theta \, d\theta = 0$ , so  $\boxed{W = 0}$ .

69. **Solution.** The moment of inertia about the  $z$ -axis is  $I_z = \iiint_E (x^2 + y^2) \rho_0 \, dV$ .

In spherical coordinates:  $x^2 + y^2 = \rho^2 \sin^2 \phi$ ,  $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ .

$$I_z = \rho_0 \int_0^{2\pi} \int_0^\pi \int_0^R \rho^2 \sin^2 \phi \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \rho_0 \int_0^{2\pi} d\theta \int_0^\pi \sin^3 \phi \, d\phi \int_0^R \rho^4 \, d\rho.$$

$$\int_0^{2\pi} d\theta = 2\pi, \quad \int_0^R \rho^4 \, d\rho = \frac{R^5}{5}.$$

$$\begin{aligned} \int_0^\pi \sin^3 \phi \, d\phi &= \int_0^\pi (1 - \cos^2 \phi) \sin \phi \, d\phi = \left[ -\cos \phi + \frac{\cos^3 \phi}{3} \right]_0^\pi \\ &= \left( 1 - \frac{1}{3} \right) - \left( -1 + \frac{1}{3} \right) = \frac{4}{3}. \end{aligned}$$

$$I_z = \rho_0 \cdot 2\pi \cdot \frac{4}{3} \cdot \frac{R^5}{5} = \boxed{\frac{8\pi\rho_0 R^5}{15}}.$$

Since the total mass is  $M = \rho_0 \cdot \frac{4}{3}\pi R^3$ , this equals  $\frac{2}{5}MR^2$ , the well-known result.

70. **Solution.**

(a) The tetrahedron:  $x, y, z \geq 0$ ,  $x + 2y + 3z \leq 6$ .

Bounds:  $0 \leq x \leq 6$ ,  $0 \leq y \leq (6 - x)/2$ ,  $0 \leq z \leq (6 - x - 2y)/3$ .

$$M = \int_0^6 \int_0^{(6-x)/2} \int_0^{(6-x-2y)/3} (2x + 5y + z) \, dz \, dy \, dx.$$

Let  $a = 6 - x - 2y$ . Inner integral:

$$\int_0^{a/3} (2x + 5y + z) \, dz = (2x + 5y) \frac{a}{3} + \frac{a^2}{18}.$$

Middle integral ( $y$ ), with  $a = 6 - x - 2y$ , let  $b = 6 - x$ :

$$\int_0^{b/2} \left[ (2x + 5y) \frac{b - 2y}{3} + \frac{(b - 2y)^2}{18} \right] dy.$$

First part:  $\frac{1}{3} \int_0^{b/2} (2x + 5y)(b - 2y) dy$ .

Expand:  $(2x + 5y)(b - 2y) = 2xb - 4xy + 5by - 10y^2$ .

$$\frac{1}{3} \left[ 2xby - 2xy^2 + \frac{5by^2}{2} - \frac{10y^3}{3} \right]_0^{b/2}.$$

At  $y = b/2$ :  $2xb \cdot \frac{b}{2} - 2x \cdot \frac{b^2}{4} + \frac{5b}{2} \cdot \frac{b^2}{4} - \frac{10}{3} \cdot \frac{b^3}{8} = xb^2 - \frac{xb^2}{2} + \frac{5b^3}{8} - \frac{5b^3}{12} = \frac{xb^2}{2} + \frac{15b^3 - 10b^3}{24} = \frac{xb^2}{2} + \frac{5b^3}{24}$ .

So first part =  $\frac{1}{3} \left( \frac{xb^2}{2} + \frac{5b^3}{24} \right) = \frac{xb^2}{6} + \frac{5b^3}{72}$ .

Second part:  $\frac{1}{18} \int_0^{b/2} (b - 2y)^2 dy$ . Let  $t = b - 2y$ :  $= \frac{1}{18} \cdot \frac{1}{2} \int_0^b t^2 dt = \frac{1}{36} \cdot \frac{b^3}{3} = \frac{b^3}{108}$ .

Middle integral =  $\frac{xb^2}{6} + \frac{5b^3}{72} + \frac{b^3}{108} = \frac{xb^2}{6} + b^3 \left( \frac{5}{72} + \frac{1}{108} \right) = \frac{xb^2}{6} + b^3 \cdot \frac{15+2}{216} = \frac{xb^2}{6} + \frac{17b^3}{216}$ .

Outer integral ( $x$ ),  $b = 6 - x$ :

$$M = \int_0^6 \left[ \frac{x(6-x)^2}{6} + \frac{17(6-x)^3}{216} \right] dx.$$

Let  $u = 6 - x$ ,  $x = 6 - u$ ,  $dx = -du$ :

$$M = \int_0^6 \left[ \frac{(6-u)u^2}{6} + \frac{17u^3}{216} \right] du.$$

$$= \int_0^6 \left[ u^2 - \frac{u^3}{6} + \frac{17u^3}{216} \right] du = \int_0^6 \left[ u^2 + u^3 \left( \frac{17}{216} - \frac{1}{6} \right) \right] du = \int_0^6 \left[ u^2 + u^3 \cdot \frac{17-36}{216} \right] du.$$

$$= \int_0^6 \left[ u^2 - \frac{19u^3}{216} \right] du = \left[ \frac{u^3}{3} - \frac{19u^4}{864} \right]_0^6 = \frac{216}{3} - \frac{19 \cdot 1296}{864} = 72 - \frac{24624}{864}.$$

$\frac{24624}{864}$ : simplify by dividing by 24:  $\frac{1026}{36} = \frac{171}{6} = \frac{57}{2} = 28.5$ .

$$M = 72 - 28.5 = \boxed{43.5 = \frac{87}{2}}.$$

- (b) For the center of mass, we need  $M_x = \iiint x\rho dV$ ,  $M_y = \iiint y\rho dV$ ,  $M_z = \iiint z\rho dV$  where  $\rho = 2x + 5y + z$ .

We compute each moment integral in turn.

For  $\bar{x}$ :

$$\iiint_E x(2x + 5y + z) dV = \iiint_E (2x^2 + 5xy + xz) dV.$$

Using  $u = 6 - x$  substitution after integrating over  $z$  and  $y$ :

Inner ( $z$ ):  $\int_0^{a/3} x(2x + 5y + z) dz = x(2x + 5y) \frac{a}{3} + \frac{xa^2}{18}$ , where  $a = 6 - x - 2y$ .

Middle ( $y$ ):  $\int_0^{b/2} \left[ \frac{x(2x+5y)(b-2y)}{3} + \frac{x(b-2y)^2}{18} \right] dy$  where  $b = 6 - x$ .

$$= x \left[ \frac{1}{3} \int_0^{b/2} (2x + 5y)(b - 2y) dy + \frac{1}{18} \int_0^{b/2} (b - 2y)^2 dy \right].$$

$$\text{From before: } = x \left[ \frac{xb^2}{6} + \frac{5b^3}{72} + \frac{b^3}{108} \right] = x \left[ \frac{xb^2}{6} + \frac{17b^3}{216} \right] = \frac{x^2b^2}{6} + \frac{17xb^3}{216}.$$

$$\text{Outer: } \int_0^6 \left[ \frac{x^2(6-x)^2}{6} + \frac{17x(6-x)^3}{216} \right] dx.$$

$$\text{With } u = 6 - x, x = 6 - u: = \int_0^6 \left[ \frac{(6-u)^2u^2}{6} + \frac{17(6-u)u^3}{216} \right] du.$$

$$\begin{aligned} \text{First term: } \frac{1}{6} \int_0^6 (36u^2 - 12u^3 + u^4) du &= \frac{1}{6} \left[ 12u^3 - 3u^4 + \frac{u^5}{5} \right]_0^6 = \\ \frac{1}{6} (2592 - 3888 + \frac{7776}{5}) &= \frac{1}{6} \cdot \frac{12960 - 19440 + 7776}{5} = \frac{1}{6} \cdot \frac{1296}{5} = \frac{216}{5} = 43.2. \end{aligned}$$

$$\begin{aligned} \text{Second term: } \frac{17}{216} \int_0^6 (6u^3 - u^4) du &= \frac{17}{216} \left[ \frac{6u^4}{4} - \frac{u^5}{5} \right]_0^6 = \frac{17}{216} \left( \frac{6 \cdot 1296}{4} - \frac{7776}{5} \right) = \\ \frac{17}{216} (1944 - 1555.2) &= \frac{17}{216} \cdot \frac{9720 - 7776}{5} = \frac{17}{216} \cdot \frac{1944}{5} = \frac{17 \cdot 9}{5} = \frac{153}{5}. \end{aligned}$$

$$\text{So } \iiint x \rho dV = \frac{216 + 153}{5} = \frac{369}{5}.$$

$$\bar{x} = \frac{369/5}{87/2} = \frac{369}{5} \cdot \frac{2}{87} = \frac{738}{435} = \frac{246}{145} = \frac{246}{145}.$$

Simplify: gcd(246, 145).  $246 = 1 \cdot 145 + 101$ ,  $145 = 1 \cdot 101 + 44$ ,  $101 = 2 \cdot 44 + 13$ ,  $44 = 3 \cdot 13 + 5$ ,  $13 = 2 \cdot 5 + 3$ ,  $5 = 1 \cdot 3 + 2$ ,  $3 = 1 \cdot 2 + 1$ . So gcd = 1. Thus  $\bar{x} = \frac{246}{145}$ .

For  $\bar{y}$ : We need  $\iiint y(2x + 5y + z) dV$ .

$$\text{Inner } (z): \int_0^{a/3} y(2x + 5y + z) dz = y(2x + 5y) \frac{a}{3} + \frac{ya^2}{18}, a = b - 2y, b = 6 - x.$$

Middle  $(y)$ :

$$\int_0^{b/2} \left[ \frac{y(2x + 5y)(b - 2y)}{3} + \frac{y(b - 2y)^2}{18} \right] dy.$$

First part: expanding  $y(2x + 5y)(b - 2y) = (2xy + 5y^2)(b - 2y) = 2xby - 4xy^2 + 5by^2 - 10y^3$ :

$$\frac{1}{3} \left[ xby^2 - \frac{4xy^3}{3} + \frac{5by^3}{3} - \frac{10y^4}{4} \right]_0^{b/2}.$$

$$\text{At } y = b/2: \frac{xb^3}{4} - \frac{xb^3}{6} + \frac{5b^4}{24} - \frac{10b^4}{64} = xb^3 \left( \frac{1}{4} - \frac{1}{6} \right) + b^4 \left( \frac{5}{24} - \frac{5}{32} \right) = \frac{xb^3}{12} + b^4 \cdot \frac{20 - 15}{96} = \frac{xb^3}{12} + \frac{5b^4}{96}.$$

$$\text{Multiply by } 1/3: \frac{xb^3}{36} + \frac{5b^4}{288}.$$

$$\text{Second part: } \frac{1}{18} \int_0^{b/2} y(b - 2y)^2 dy = \frac{1}{18} \int_0^{b/2} (b^2y - 4by^2 + 4y^3) dy$$

$$= \frac{1}{18} \left[ \frac{b^2y^2}{2} - \frac{4by^3}{3} + y^4 \right]_0^{b/2} = \frac{1}{18} \left( \frac{b^4}{8} - \frac{b^4}{6} + \frac{b^4}{16} \right).$$

$$= \frac{b^4}{18} \left( \frac{1}{8} - \frac{1}{6} + \frac{1}{16} \right) = \frac{b^4}{18} \cdot \frac{6 - 8 + 3}{48} = \frac{b^4}{18} \cdot \frac{1}{48} = \frac{b^4}{864}.$$

$$\text{Middle integral} = \frac{xb^3}{36} + \frac{5b^4}{288} + \frac{b^4}{864} = \frac{xb^3}{36} + b^4 \left( \frac{15 + 1}{864} \right) = \frac{xb^3}{36} + \frac{16b^4}{864} = \frac{xb^3}{36} + \frac{2b^4}{108}.$$

$$\text{Simplify: } = \frac{xb^3}{36} + \frac{b^4}{54}.$$

$$\text{Outer } (x), b = 6 - x, u = 6 - x: \int_0^6 \left[ \frac{(6-u)u^3}{36} + \frac{u^4}{54} \right] du.$$

$$= \frac{1}{36} \int_0^6 (6u^3 - u^4) du + \frac{1}{54} \int_0^6 u^4 du.$$

$$= \frac{1}{36} \left[ \frac{6u^4}{4} - \frac{u^5}{5} \right]_0^6 + \frac{1}{54} \cdot \frac{6^5}{5}.$$

$$= \frac{1}{36} \left( 1944 - \frac{7776}{5} \right) + \frac{7776}{270}.$$

$$= \frac{1}{36} \cdot \frac{9720-7776}{5} + \frac{7776}{270} = \frac{1944}{180} + \frac{7776}{270} = \frac{54}{5} + \frac{7776}{270}.$$

$$\frac{7776}{270} = \frac{1296}{45} = \frac{432}{15} = \frac{144}{5} = 28.8.$$

$$\text{So } \iiint y \rho dV = 10.8 + 28.8 = 39.6 = \frac{198}{5}.$$

$$\bar{y} = \frac{198/5}{87/2} = \frac{198 \cdot 2}{5 \cdot 87} = \frac{396}{435} = \frac{132}{145} = \frac{132}{145}.$$

For  $\bar{z}$ : We need  $\iiint z(2x + 5y + z) dV$ .

Inner:  $\int_0^{a/3} z(2x + 5y) dz = (2x + 5y) \frac{a^2}{18}$  and  $\int_0^{a/3} z^2 dz = \frac{a^3}{81}$ , so the inner integral equals  $(2x + 5y) \frac{a^2}{18} + \frac{a^3}{81}$ .

$$\text{Middle } (y), a = b - 2y: \int_0^{b/2} \left[ \frac{(2x+5y)(b-2y)^2}{18} + \frac{(b-2y)^3}{81} \right] dy.$$

$$\text{First part: } \frac{1}{18} \int_0^{b/2} (2x + 5y)(b - 2y)^2 dy.$$

Expanding  $(b - 2y)^2 = b^2 - 4by + 4y^2$ :

$$(2x + 5y)(b^2 - 4by + 4y^2) = 2xb^2 - 8xby + 8xy^2 + 5b^2y - 20by^2 + 20y^3.$$

$$\frac{1}{18} \int_0^{b/2} (2xb^2 - 8xby + 8xy^2 + 5b^2y - 20by^2 + 20y^3) dy$$

$$= \frac{1}{18} \left[ 2xb^2y - 4xby^2 + \frac{8xy^3}{3} + \frac{5b^2y^2}{2} - \frac{20by^3}{3} + 5y^4 \right]_0^{b/2}.$$

$$\text{At } y = b/2: xb^3 - xb^3 + \frac{xb^3}{3} + \frac{5b^4}{8} - \frac{5b^4}{6} + \frac{5b^4}{16}.$$

$$= \frac{xb^3}{3} + b^4 \left( \frac{5}{8} - \frac{5}{6} + \frac{5}{16} \right) = \frac{xb^3}{3} + 5b^4 \left( \frac{1}{8} - \frac{1}{6} + \frac{1}{16} \right).$$

$$= \frac{xb^3}{3} + 5b^4 \cdot \frac{6-8+3}{48} = \frac{xb^3}{3} + \frac{5b^4}{48}.$$

$$\text{Multiply by } 1/18: \frac{xb^3}{54} + \frac{5b^4}{864}.$$

$$\text{Second part: } \frac{1}{81} \int_0^{b/2} (b - 2y)^3 dy. \text{ Let } t = b - 2y: = \frac{1}{81} \cdot \frac{1}{2} \int_0^b t^3 dt = \frac{1}{162} \cdot \frac{b^4}{4} = \frac{b^4}{648}.$$

$$\text{Middle integral} = \frac{xb^3}{54} + \frac{5b^4}{864} + \frac{b^4}{648}.$$

$$\frac{5}{864} + \frac{1}{648}: \text{LCD} = 2592. \frac{15}{2592} + \frac{4}{2592} = \frac{19}{2592}.$$

$$= \frac{xb^3}{54} + \frac{19b^4}{2592}.$$

$$\text{Outer } (x), b = 6 - x, u = 6 - x: \int_0^6 \left[ \frac{(6-u)u^3}{54} + \frac{19u^4}{2592} \right] du.$$

$$= \frac{1}{54} \int_0^6 (6u^3 - u^4) du + \frac{19}{2592} \int_0^6 u^4 du.$$

$$= \frac{1}{54} \left( 1944 - \frac{7776}{5} \right) + \frac{19}{2592} \cdot \frac{7776}{5}.$$

$$\text{Note } \frac{1}{54} \cdot \frac{1944}{5} = \frac{36}{5} \text{ and } \frac{19 \cdot 7776}{2592 \cdot 5} = \frac{19 \cdot 3}{5} = \frac{57}{5}, \text{ so the sum is } \frac{36}{5} + \frac{57}{5} = \frac{93}{5}.$$

$$\iiint z \rho dV = \frac{36+57}{5} = \frac{93}{5}.$$

$$\bar{z} = \frac{93/5}{87/2} = \frac{93 \cdot 2}{5 \cdot 87} = \frac{186}{435} = \frac{62}{145}.$$

Therefore:

$$M = \frac{87}{2}, \quad (\bar{x}, \bar{y}, \bar{z}) = \left( \frac{246}{145}, \frac{132}{145}, \frac{62}{145} \right).$$

71. **Solution.** The region is a cylinder:  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$ ,  $-1 \leq z \leq 3$ , with density  $\rho(r, \theta, z) = 2 + z$ .

(a) The moment of inertia about the  $x$ -axis:

$$I_x = \iiint_E (y^2 + z^2) \rho dV = \int_0^{2\pi} \int_0^2 \int_{-1}^3 (r^2 \sin^2 \theta + z^2) (2 + z) r dz dr d\theta.$$

The moment of inertia about the  $y$ -axis:

$$I_y = \iiint_E (x^2 + z^2) \rho \, dV = \int_0^{2\pi} \int_0^2 \int_{-1}^3 (r^2 \cos^2 \theta + z^2)(2+z) r \, dz \, dr \, d\theta.$$

- (b) The product of inertia  $I_{xy} = \iiint_E xy \rho \, dV = \int_0^{2\pi} \int_0^2 \int_{-1}^3 r^2 \cos \theta \sin \theta (2+z) r \, dz \, dr \, d\theta$ .

The  $\theta$ -integral contains  $\int_0^{2\pi} \cos \theta \sin \theta \, d\theta = \frac{1}{2} \int_0^{2\pi} \sin 2\theta \, d\theta = 0$ .

Therefore  $I_{xy} = 0$ . This is because the cylinder has full rotational symmetry about the  $z$ -axis, and the density  $\rho = 2 + z$  depends only on  $z$ , not on  $\theta$ . The integrand  $xy = r^2 \cos \theta \sin \theta$  is odd under the reflections  $\theta \rightarrow \pi - \theta$  or  $\theta \rightarrow -\theta$ , so it vanishes upon integration over a full period.

## 72. Solution.

- (a) In spherical coordinates:  $x = \rho \sin \phi \cos \theta$ ,  $y = \rho \sin \phi \sin \theta$ ,  $z = \rho \cos \phi$ . Then  $\rho(x, y, z) = e^{-z} = e^{-\rho \cos \phi}$  and  $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ .

The sphere  $x^2 + y^2 + z^2 \leq 9$ :  $0 \leq \rho \leq 3$ ,  $0 \leq \phi \leq \pi$ ,  $0 \leq \theta \leq 2\pi$ .

- (b) The total mass:

$$M = \int_0^{2\pi} \int_0^\pi \int_0^3 e^{-\rho \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

Since the integrand is independent of  $\theta$ :

$$M = 2\pi \int_0^\pi \int_0^3 e^{-\rho \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi.$$

- (c) The density-weighted average value of  $z$  is:

$$\bar{z}_\rho = \frac{\iiint_E z \rho(x, y, z) \, dV}{\iiint_E \rho(x, y, z) \, dV} = \frac{\int_0^{2\pi} \int_0^\pi \int_0^3 (\rho \cos \phi) e^{-\rho \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta}{M}.$$

The numerator simplifies to:

$$2\pi \int_0^\pi \int_0^3 \rho^3 \cos \phi e^{-\rho \cos \phi} \sin \phi \, d\rho \, d\phi.$$

This is the same as the mass integral but with an extra factor of  $\rho \cos \phi$  in the integrand.

# Chapter 5

## Chapter 5: Vector Calculus

### 1. Solution.

(a) We substitute  $x = 1$ ,  $y = -1$ :

$$\mathbf{F}(1, -1) = (1 + 1)\mathbf{i} + (-1 - 2)\mathbf{j} = 2\mathbf{i} - 3\mathbf{j}.$$

(b) At  $(0, 0)$ :  $\mathbf{F} = (1, -2)$ . At  $(1, 0)$ :  $\mathbf{F} = (2, -2)$ . At  $(0, 1)$ :  $\mathbf{F} = (1, -1)$ . At  $(-1, 0)$ :  $\mathbf{F} = (0, -2)$ . At  $(0, -1)$ :  $\mathbf{F} = (1, -3)$ .

The field has a “center” at  $(-1, 2)$  where  $\mathbf{F} = \mathbf{0}$ . Near the origin, every vector points roughly to the lower-right, with the  $y$ -component always negative (pulling downward) and the  $x$ -component positive for  $x > -1$ . The field radiates outward from the point  $(-1, 2)$ .

### 2. Solution.

A *scalar field* assigns a single real number to each point in space. Example: the temperature  $T(x, y, z)$  at each point in a room.

A *vector field* assigns a vector to each point in space. Example: the velocity  $\mathbf{v}(x, y, z)$  of a fluid at each point in a flowing stream.

The key difference is that a scalar field has only magnitude at each point, while a vector field has both magnitude and direction.

### 3. Solution.

At  $(2, 0)$ :  $\mathbf{F} = -0\mathbf{i} + 2\mathbf{j} = (0, 2)$ , so the arrow points straight up.

At  $(1, 1)$ :  $\mathbf{F} = -1\mathbf{i} + 1\mathbf{j} = (-1, 1)$ , so the arrow points up and to the left at  $45^\circ$ .

At  $(-2, -2)$ :  $\mathbf{F} = 2\mathbf{i} + (-2)\mathbf{j} = (2, -2)$ , so the arrow points down and to the right.

In each case,  $\mathbf{F}$  is perpendicular to the position vector  $(x, y)$  since  $\mathbf{F} \cdot (x, y) = -yx + xy = 0$ . Also  $|\mathbf{F}| = \sqrt{x^2 + y^2}$ , which equals the distance from the origin. The overall pattern is *counterclockwise rotation* about the origin, with speed proportional to distance.

### 4. Solution.

(a)  $\mathbf{F}(0, 0) = (e^0, e^0) = (1, 1)$ .

$\mathbf{F}(1, 2) = (e^1, e^2) = (e, e^2) \approx (2.718, 7.389)$ .

- (b) At the origin,  $\mathbf{F}(0,0) = (1,1)$  is nonzero and points into the first quadrant. Since both components  $e^x > 0$  and  $e^y > 0$  everywhere, every vector points into the first quadrant. The origin is *neither* a source nor a sink, because  $\mathbf{F}(0,0) \neq \mathbf{0}$ ; sources and sinks occur at points where  $\mathbf{F} = \mathbf{0}$  and the flow radiates outward or inward, respectively.
5. **Solution.** Each component of  $\mathbf{G}$  has  $x^2 + y^2 + z^2$  in the denominator, which is zero only at the origin  $(0,0,0)$ . For any point  $(x,y,z) \neq (0,0,0)$ , the components are ratios of polynomials with a nonzero denominator, hence continuous.
- Therefore,  $\mathbf{G}$  is *not* continuous at the origin  $(0,0,0)$  (it is undefined there), but it is continuous everywhere else in  $\mathbb{R}^3 \setminus \{(0,0,0)\}$ .
6. **Solution.**
- (a) Each component  $x^2, y^2, z^2$  is a polynomial, and polynomials are continuous on all of  $\mathbb{R}^3$ . Therefore  $\mathbf{H}$  is continuous at every point in  $\mathbb{R}^3$ .
- (b) There are no problematic points. Every component is a polynomial, which is defined and continuous everywhere.
7. **Solution.** Engineers and physicists sometimes normalize all arrows to the same length to make the *direction* of the field clearly visible at every point, especially in regions where the magnitude varies dramatically.
- Advantage:** Normalized plots clearly show the directional pattern of the field (e.g., convergence, divergence, rotation) without large vectors obscuring small ones.
- Disadvantage:** All information about the *magnitude* is lost. For example, in a wind field, one cannot distinguish regions of high wind speed from regions of low wind speed.
8. **Solution.**
- (a)  $\mathbf{F}_1(x,y) = x\mathbf{i} + y\mathbf{j}$ : vectors point radially *outward* from the origin. This is a **source**.
- (b)  $\mathbf{F}_2(x,y) = -x\mathbf{i} - y\mathbf{j}$ : vectors point radially *inward* toward the origin. This is a **sink**.
- (c)  $\mathbf{F}_3(x,y) = x\mathbf{i} - y\mathbf{j}$ : vectors point outward along the  $x$ -axis but inward along the  $y$ -axis (or vice versa). This is a **saddle**.
9. **Solution.** Write  $\mathbf{F}(x,y) = P\mathbf{i} + Q\mathbf{j}$  where  $P = y$  and  $Q = -x$ .

Compute:

$$\frac{\partial P}{\partial y} = 1, \quad \frac{\partial Q}{\partial x} = -1.$$

Since  $\frac{\partial P}{\partial y} = 1 \neq -1 = \frac{\partial Q}{\partial x}$ , the field is **not conservative**.

The failure of the conservative test corresponds to the fact that  $\mathbf{F}$  has a rotational/circulatory pattern near the origin—vectors circulate (clockwise) around the origin rather than emanating from a potential function.

10. **Solution.**(a) Here  $P = x^3$  and  $Q = 3x^2y$ .

$$\frac{\partial P}{\partial y} = 0, \quad \frac{\partial Q}{\partial x} = 6xy.$$

Since  $0 \neq 6xy$  in general,  $\mathbf{F}$  is **not conservative**.(b) Since  $\mathbf{F}$  is not conservative, no potential function exists.11. **Solution.** Write  $\mathbf{F} = (P, Q, R) = (2xy, x^2 + z^2, 2yz)$ . For  $\mathbf{F}$  to be conservative on  $\mathbb{R}^3$  we need:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, \quad \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}, \quad \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}.$$

Compute each:

$$\begin{aligned} \frac{\partial P}{\partial y} &= 2x, & \frac{\partial Q}{\partial x} &= 2x. \\ \frac{\partial P}{\partial z} &= 0, & \frac{\partial R}{\partial x} &= 0. \\ \frac{\partial Q}{\partial z} &= 2z, & \frac{\partial R}{\partial y} &= 2z. \end{aligned}$$

All three conditions are satisfied, so  $\mathbf{F}$  is **conservative** on  $\mathbb{R}^3$ .To find the potential function:  $f_x = 2xy \Rightarrow f = x^2y + g(y, z)$ . Then  $f_y = x^2 + g_y = x^2 + z^2$ , so  $g_y = z^2$ , giving  $g = yz^2 + h(z)$ . Then  $f_z = 2yz + h'(z) = 2yz$ , so  $h'(z) = 0$  and  $h$  is constant. Thus  $f(x, y, z) = x^2y + yz^2 + C$ .12. **Solution.** We have  $P = \sqrt{x^2 + y^2}$  and  $Q = -\sin(xy)$ .

$$\begin{aligned} \frac{\partial P}{\partial x} &= \frac{x}{\sqrt{x^2 + y^2}}, & \frac{\partial P}{\partial y} &= \frac{y}{\sqrt{x^2 + y^2}}. \\ \frac{\partial Q}{\partial x} &= -y \cos(xy), & \frac{\partial Q}{\partial y} &= -x \cos(xy). \end{aligned}$$

At the origin  $(0, 0)$ :  $\sqrt{x^2 + y^2} = 0$ , so  $\frac{\partial P}{\partial x}$  and  $\frac{\partial P}{\partial y}$  involve division by zero. These two partial derivatives are **undefined at the origin**.The partial derivatives  $\frac{\partial Q}{\partial x} = -y \cos(xy)$  and  $\frac{\partial Q}{\partial y} = -x \cos(xy)$  are well-defined everywhere, including at the origin (both equal 0 there).13. **Solution.**(a)  $P = x^2 + y^2$  and  $Q = -(x^2 + y^2)$ .

$$\frac{\partial P}{\partial y} = 2y, \quad \frac{\partial Q}{\partial x} = -2x.$$

Since  $2y \neq -2x$  in general,  $\mathbf{F}$  is **not conservative**.



(b) Since the field is not conservative, we classify the origin by examining the flow. At the origin  $\mathbf{F}(0,0) = (0,0)$ , so it is an equilibrium point. Near the origin,  $\operatorname{div}(\mathbf{F}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 2x + (-2y) = 2x - 2y$ . At the origin,  $\operatorname{div}(\mathbf{F}) = 0$ , and the divergence changes sign depending on direction. Examining nearby vectors: along the positive  $x$ -axis,  $\mathbf{F} = (x^2, -x^2)$  points into the fourth quadrant; along the positive  $y$ -axis,  $\mathbf{F} = (y^2, -y^2)$  also points into the fourth quadrant. The flow pattern is neither purely radial inward nor outward—the origin behaves like a **saddle** (flow is neither a pure source nor a pure sink).

14. **Solution.** Parameterize  $C$ :  $\mathbf{r}(t) = (2t, 4t)$  for  $0 \leq t \leq 1$ .

Then  $\mathbf{r}'(t) = (2, 4)$  and  $\|\mathbf{r}'(t)\| = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$ .

On  $C$ :  $x + y = 2t + 4t = 6t$ .

$$\int_C (x + y) ds = \int_0^1 6t \cdot 2\sqrt{5} dt = 12\sqrt{5} \int_0^1 t dt = 12\sqrt{5} \cdot \frac{1}{2} = \boxed{6\sqrt{5}}.$$

15. **Solution.** Parameterize:  $\mathbf{r}(t) = (t, t^2)$  for  $0 \leq t \leq 2$ .

$\mathbf{r}'(t) = (1, 2t)$ , so  $\|\mathbf{r}'(t)\| = \sqrt{1 + 4t^2}$ .

On  $C$ :  $x^2 + y = t^2 + t^2 = 2t^2$ .

$$\int_C (x^2 + y) ds = \int_0^2 2t^2 \sqrt{1 + 4t^2} dt.$$

Let  $u = 1 + 4t^2$ , then  $du = 8t dt$ . This substitution does not directly simplify  $t^2 \sqrt{1 + 4t^2}$ . Instead, use the substitution  $2t = \sinh \theta$ , so  $4t^2 = \sinh^2 \theta$ ,  $\sqrt{1 + 4t^2} = \cosh \theta$ , and  $2 dt = \cosh \theta d\theta$ .

When  $t = 0$ :  $\theta = 0$ . When  $t = 2$ :  $\sinh \theta = 4$ , so  $\theta = \ln(4 + \sqrt{17})$ .

$$\int_0^2 2t^2 \sqrt{1 + 4t^2} dt = \int_0^{\theta_0} 2 \cdot \frac{\sinh^2 \theta}{4} \cdot \cosh \theta \cdot \frac{\cosh \theta}{2} d\theta = \frac{1}{4} \int_0^{\theta_0} \sinh^2 \theta \cosh^2 \theta d\theta$$

where  $\theta_0 = \sinh^{-1}(4)$ . Using  $\sinh^2 \theta \cosh^2 \theta = \frac{1}{4} \sinh^2(2\theta) = \frac{1}{8} (\cosh(4\theta) - 1)$ :

$$= \frac{1}{4} \cdot \frac{1}{8} \left[ \frac{\sinh(4\theta)}{4} - \theta \right]_0^{\theta_0} = \frac{1}{32} \left[ \frac{\sinh(4\theta_0)}{4} - \theta_0 \right].$$

Now  $\sinh \theta_0 = 4$ ,  $\cosh \theta_0 = \sqrt{17}$ . We need  $\sinh(4\theta_0)$ . Using double-angle formulas:  $\sinh(2\theta_0) = 2 \cdot 4 \cdot \sqrt{17} = 8\sqrt{17}$ ,  $\cosh(2\theta_0) = 2 \cosh^2 \theta_0 - 1 = 33$ . Then  $\sinh(4\theta_0) = 2 \sinh(2\theta_0) \cosh(2\theta_0) = 2 \cdot 8\sqrt{17} \cdot 33 = 528\sqrt{17}$ .

$$= \frac{1}{32} \left[ \frac{528\sqrt{17}}{4} - \ln(4 + \sqrt{17}) \right] = \frac{1}{32} \left[ 132\sqrt{17} - \ln(4 + \sqrt{17}) \right].$$

$$\boxed{\int_C (x^2 + y) ds = \frac{132\sqrt{17} - \ln(4 + \sqrt{17})}{32}}.$$

16. **Solution.** Let  $\mathbf{r}(t)$ ,  $a \leq t \leq b$ , be a parameterization of  $C$ . The scalar line integral is

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, dt.$$

Now reverse the parameterization: let  $\tilde{\mathbf{r}}(s) = \mathbf{r}(a + b - s)$  for  $s \in [a, b]$ . Then  $\tilde{\mathbf{r}}'(s) = -\mathbf{r}'(a + b - s)$ , so

$$\|\tilde{\mathbf{r}}'(s)\| = \|\mathbf{r}'(a + b - s)\|.$$

The integral becomes:

$$\int_a^b f(\tilde{\mathbf{r}}(s)) \|\tilde{\mathbf{r}}'(s)\| \, ds = \int_a^b f(\mathbf{r}(a + b - s)) \|\mathbf{r}'(a + b - s)\| \, ds.$$

Substituting  $u = a + b - s$ ,  $du = -ds$ ; when  $s = a$ ,  $u = b$  and when  $s = b$ ,  $u = a$ :

$$= \int_b^a f(\mathbf{r}(u)) \|\mathbf{r}'(u)\| (-du) = \int_a^b f(\mathbf{r}(u)) \|\mathbf{r}'(u)\| \, du.$$

This is exactly the original integral. The sign does not change because  $ds = \|\mathbf{r}'(t)\| \, dt > 0$  always; the norm  $\|\cdot\|$  absorbs any sign change from reversing direction.

17. **Solution.** Parameterize:  $x = r \cos t$ ,  $y = r \sin t$ , for  $t \in [0, \pi/2]$ .

$\mathbf{r}'(t) = (-r \sin t, r \cos t)$ , so  $\|\mathbf{r}'(t)\| = r$ .

On  $C$ :  $\sqrt{x^2 + y^2} = \sqrt{r^2 \cos^2 t + r^2 \sin^2 t} = r$ .

$$\int_C \sqrt{x^2 + y^2} \, ds = \int_0^{\pi/2} r \cdot r \, dt = r^2 \int_0^{\pi/2} dt = \boxed{\frac{\pi r^2}{2}}.$$

18. **Solution. Segment 1** ( $C_1$ ): from  $(0, 0)$  to  $(3, 0)$  along the  $x$ -axis. Parameterize:  $\mathbf{r}_1(t) = (t, 0)$ ,  $0 \leq t \leq 3$ . Then  $\|\mathbf{r}'_1(t)\| = 1$ ,  $y = 0$ .

$$\int_{C_1} (2x + 3y) \, ds = \int_0^3 2t \cdot 1 \, dt = [t^2]_0^3 = 9.$$

**Segment 2** ( $C_2$ ): from  $(3, 0)$  to  $(3, 2)$  vertically. Parameterize:  $\mathbf{r}_2(t) = (3, t)$ ,  $0 \leq t \leq 2$ . Then  $\|\mathbf{r}'_2(t)\| = 1$ ,  $x = 3$ .

$$\int_{C_2} (2x + 3y) \, ds = \int_0^2 (6 + 3t) \cdot 1 \, dt = [6t + \frac{3}{2}t^2]_0^2 = 12 + 6 = 18.$$

Total:  $\int_C (2x + 3y) \, ds = 9 + 18 = \boxed{27}$ .

19. **Solution.** Parameterize  $C$ :  $\mathbf{r}(t) = (2t, 3t)$ ,  $0 \leq t \leq 1$ . Then  $\mathbf{r}'(t) = (2, 3)$ .

$\mathbf{F}(\mathbf{r}(t)) = (y, x) = (3t, 2t)$ .

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (3t, 2t) \cdot (2, 3) \, dt = \int_0^1 (6t + 6t) \, dt = \int_0^1 12t \, dt = 6t^2 \Big|_0^1 = \boxed{6}.$$

20. **Solution.**  $\mathbf{F} = (x - y, x^2)$ , so  $P = x - y$ ,  $Q = x^2$ . The rectangle has four edges traversed counterclockwise.

$$C_1: (0, 0) \rightarrow (2, 0): \mathbf{r} = (t, 0), t \in [0, 2], d\mathbf{r} = (1, 0)dt.$$

$$\int_{C_1} = \int_0^2 (t - 0) \cdot 1 dt = \int_0^2 t dt = 2.$$

$$C_2: (2, 0) \rightarrow (2, 1): \mathbf{r} = (2, t), t \in [0, 1], d\mathbf{r} = (0, 1)dt.$$

$$\int_{C_2} = \int_0^1 4 \cdot 1 dt = 4.$$

$$C_3: (2, 1) \rightarrow (0, 1): \mathbf{r} = (2 - t, 1), t \in [0, 2], d\mathbf{r} = (-1, 0)dt.$$

$$\int_{C_3} = \int_0^2 ((2 - t) - 1)(-1) dt = \int_0^2 -(1 - t) dt = \int_0^2 (t - 1) dt = \left[ \frac{t^2}{2} - t \right]_0^2 = 0.$$

$$C_4: (0, 1) \rightarrow (0, 0): \mathbf{r} = (0, 1 - t), t \in [0, 1], d\mathbf{r} = (0, -1)dt.$$

$$\int_{C_4} = \int_0^1 0 \cdot (-1) dt = 0.$$

$$\text{Total: } 2 + 4 + 0 + 0 = \boxed{6}.$$

21. **Solution.**  $\mathbf{F} = (\ln(1 + y^2), e^x)$ ,  $C: x^2 + y^2 = 4$  counterclockwise.

$$\text{Parameterize: } \mathbf{r}(t) = (2 \cos t, 2 \sin t), t \in [0, 2\pi], \mathbf{r}'(t) = (-2 \sin t, 2 \cos t).$$

By Green's Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

where  $P = \ln(1 + y^2)$ ,  $Q = e^x$ .

$$\frac{\partial Q}{\partial x} = e^x, \quad \frac{\partial P}{\partial y} = \frac{2y}{1 + y^2}.$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \left( e^x - \frac{2y}{1 + y^2} \right) dA.$$

The integrand  $\frac{2y}{1 + y^2}$  is an odd function in  $y$  and  $D$  is the disk  $x^2 + y^2 \leq 4$  which is symmetric about the  $x$ -axis, so:

$$\iint_D \frac{2y}{1 + y^2} dA = 0.$$

Thus we need  $\iint_D e^x dA$ . Using polar coordinates  $x = r \cos \theta$ :

$$\iint_D e^x dA = \int_0^{2\pi} \int_0^2 e^{r \cos \theta} r dr d\theta.$$

This integral can be expressed in terms of modified Bessel functions. Using the identity  $\frac{1}{2\pi} \int_0^{2\pi} e^{r \cos \theta} d\theta = I_0(r)$ :

$$= 2\pi \int_0^2 r I_0(r) dr.$$

Since  $\frac{d}{dr}[rI_1(r)] = rI_0(r)$ , we get  $2\pi[rI_1(r)]_0^2 = 4\pi I_1(2)$ .

Therefore  $\oint_C \mathbf{F} \cdot d\mathbf{r} = 4\pi I_1(2)$ , where  $I_1(2) \approx 1.5906$ , giving approximately  $\boxed{4\pi I_1(2) \approx 19.99}$ .

Reversing to clockwise negates the integral:  $\oint_{C^-} \mathbf{F} \cdot d\mathbf{r} = -4\pi I_1(2)$ .

22. **Solution.** If  $C$  is piecewise smooth with segments  $C_1, C_2, C_3$ , then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} + \int_{C_3} \mathbf{F} \cdot d\mathbf{r}.$$

This follows from the additivity of integrals over adjoining curves: the endpoint of  $C_1$  equals the starting point of  $C_2$ , and so on.

This strategy is often used in applications because each smooth segment may have a simple parameterization (e.g., a line segment or circular arc), even though the whole curve  $C$  does not admit a single simple formula.

23. **Solution.** Let  $\mathbf{F} = (3y, -2x)$ , so  $P = 3y$  and  $Q = -2x$ .

By Green's Theorem (for counterclockwise  $C$  enclosing region  $D$ ):

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D (-2 - 3) dA = -5 \text{ Area}(D).$$

This is nonzero for any region with positive area, confirming the integral depends on the loop.

If  $C$  is traversed **clockwise**, the integral picks up a sign change:

$$\oint_{C_{cw}} \mathbf{F} \cdot d\mathbf{r} = +5 \text{ Area}(D).$$

So the clockwise result is the negative of the counterclockwise result.

24. **Solution.** Along  $x = 1$ , parameterize:  $\mathbf{r}(t) = (1, t)$ ,  $0 \leq t \leq 2$ ,  $\mathbf{r}'(t) = (0, 1)$ .

$$\mathbf{F}(1, t) = (1 + 1, 2(1)(t) - 1) = (2, 2t - 1).$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^2 (2, 2t - 1) \cdot (0, 1) dt = \int_0^2 (2t - 1) dt = [t^2 - t]_0^2 = 4 - 2 = \boxed{2}.$$

Reversing the path (from  $(1, 2)$  to  $(1, 0)$ ) gives  $-2$ . This is because for a vector line integral  $\int_C \mathbf{F} \cdot d\mathbf{r}$ , reversing the direction of traversal negates  $d\mathbf{r}$ , hence negates the integral.

25. **Solution.**  $\mathbf{F}(x, y) = (y, x)$ , so  $P = y$ ,  $Q = x$ .

$$\frac{\partial P}{\partial y} = 1, \quad \frac{\partial Q}{\partial x} = 1.$$

Since  $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$  everywhere on  $\mathbb{R}^2$  (which is simply connected),  $\mathbf{F}$  is **conservative**.

A potential function:  $f_x = y \Rightarrow f = xy + g(y)$ . Then  $f_y = x + g'(y) = x$ , so  $g'(y) = 0$ , giving  $g(y) = C$ . Thus  $f(x, y) = xy$ .

26. **Solution.**  $\mathbf{F} = \left( \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$  on  $\mathbb{R}^2 \setminus \{(0, 0)\}$ .

Parameterize  $C$ :  $x = r \cos t$ ,  $y = r \sin t$ ,  $0 \leq t \leq 2\pi$ .

$dx = -r \sin t \, dt$ ,  $dy = r \cos t \, dt$ .

On  $C$ :  $x^2 + y^2 = r^2$ .

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} \left[ \frac{-r \sin t}{r^2} (-r \sin t) + \frac{r \cos t}{r^2} (r \cos t) \right] dt \\ &= \int_0^{2\pi} \left[ \frac{\sin^2 t}{1} + \frac{\cos^2 t}{1} \right] dt = \int_0^{2\pi} 1 \, dt = \boxed{2\pi}. \end{aligned}$$

Although one can check  $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$  (both equal  $\frac{y^2 - x^2}{(x^2 + y^2)^2}$ ), the field is **not conservative** because the domain  $\mathbb{R}^2 \setminus \{(0, 0)\}$  is *not simply connected*. The circulation around the origin is  $2\pi \neq 0$ , which is impossible for a conservative field.

27. **Solution.**  $\mathbf{G} = (x + \sin y, \cos x + y)$ . Check:  $P = x + \sin y$ ,  $Q = \cos x + y$ .

$$\frac{\partial P}{\partial y} = \cos y, \quad \frac{\partial Q}{\partial x} = -\sin x.$$

Since  $\cos y \neq -\sin x$  in general (e.g., at  $(0, 0)$ :  $1 \neq 0$ ),  $\mathbf{G}$  is **not conservative**.

Therefore we cannot use the potential function method. The integral  $\int_C \mathbf{G} \cdot d\mathbf{r}$  from  $(0, 0)$  to  $(2, 1)$  is path-dependent and requires specifying a particular path.

28. **Solution.**  $\mathbf{F} = (yz, zx, xy)$ .

$C_1$ :  $(0, 0, 0) \rightarrow (1, 0, 0)$ .  $\mathbf{r}(t) = (t, 0, 0)$ ,  $t \in [0, 1]$ ,  $\mathbf{r}' = (1, 0, 0)$ .  $\mathbf{F} = (0, 0, 0)$ . So  $\int_{C_1} = 0$ .

$C_2$ :  $(1, 0, 0) \rightarrow (1, 2, 0)$ .  $\mathbf{r}(t) = (1, t, 0)$ ,  $t \in [0, 2]$ ,  $\mathbf{r}' = (0, 1, 0)$ .  $\mathbf{F} = (0, 0, t)$ . So  $\mathbf{F} \cdot \mathbf{r}' = 0$ , and  $\int_{C_2} = 0$ .

$C_3$ :  $(1, 2, 0) \rightarrow (1, 2, 3)$ .  $\mathbf{r}(t) = (1, 2, t)$ ,  $t \in [0, 3]$ ,  $\mathbf{r}' = (0, 0, 1)$ .  $\mathbf{F} = (2t, t, 2)$ . So  $\mathbf{F} \cdot \mathbf{r}' = 2$ , and  $\int_{C_3} = \int_0^3 2 \, dt = 6$ .

Total:  $0 + 0 + 6 = \boxed{6}$ .

To check if  $\mathbf{F}$  is conservative, check  $\nabla \times \mathbf{F}$ :

$$\nabla \times \mathbf{F} = \left( \frac{\partial(xy)}{\partial y} - \frac{\partial(zx)}{\partial z}, \frac{\partial(yz)}{\partial z} - \frac{\partial(xy)}{\partial x}, \frac{\partial(zx)}{\partial x} - \frac{\partial(yz)}{\partial y} \right) = (x-x, y-y, z-z) = \mathbf{0}.$$

Since the curl is zero on all of  $\mathbb{R}^3$  (simply connected),  $\mathbf{F}$  is conservative. A potential function is  $f(x, y, z) = xyz$ . We verify:  $f(1, 2, 3) - f(0, 0, 0) = 6 - 0 = 6$ .

29. **Solution. Green's Theorem** states: if  $C$  is a positively oriented (counterclockwise), piecewise smooth, simple closed curve bounding a region  $D$ , and  $P, Q$  have continuous partial derivatives on an open set containing  $D$ , then

$$\oint_C P dx + Q dy = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

**Conceptual explanation:** The line integral measures the total “circulation” of the vector field  $(P, Q)$  around the boundary. Green's Theorem says this equals the sum of all the “local rotation”  $(\partial Q/\partial x - \partial P/\partial y)$  accumulated over the interior region. Intuitively, if we partition  $D$  into infinitesimally small rectangles and sum the circulation around each, all interior edge contributions cancel (shared edges are traversed in opposite directions by adjacent rectangles), leaving only the outer boundary  $C$ . The double integral simply adds up the microscopic rotation at every interior point.

30. **Solution.**  $P = xy$ ,  $Q = x^2$ , rectangle  $[0, 2] \times [0, 1]$ .

**Direct computation:**

$$C_1: (0, 0) \rightarrow (2, 0), y = 0: \int_0^2 (0)(1) dt + 0 = 0.$$

$$C_2: (2, 0) \rightarrow (2, 1), x = 2: \mathbf{r} = (2, t), dx = 0, dy = dt. \int_0^1 4 dt = 4.$$

$$C_3: (2, 1) \rightarrow (0, 1), y = 1: \mathbf{r} = (2 - t, 1), dx = -dt, dy = 0. \int_0^2 (2 - t)(1)(-1) dt = -\int_0^2 (2 - t) dt = -(2 \cdot 2 - 2) = -2.$$

$$C_4: (0, 1) \rightarrow (0, 0), x = 0: \int_0^1 0 dt = 0.$$

$$\text{Direct total: } 0 + 4 - 2 + 0 = 2.$$

**Green's Theorem:**

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - x = x.$$

$$\iint_D x dA = \int_0^1 \int_0^2 x dx dy = \int_0^1 2 dy = 2.$$

2.

31. **Solution.**  $P = -y$ ,  $Q = x$ ,  $C$ : unit circle counterclockwise,  $D$ : unit disk.

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2.$$

By Green's Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D 2 \, dA = 2 \cdot \text{Area}(D) = 2\pi(1)^2 = \boxed{2\pi}.$$

**Physical interpretation:** This integral measures the *circulation* of  $\mathbf{F}$  around the unit circle. The field  $\mathbf{F} = (-y, x)$  rotates counterclockwise with speed equal to the distance from the origin. The constant “curl” of 2 means uniform rotation everywhere, so the total circulation around the unit circle is  $2\pi$ —consistent with a rigid-body rotation at angular velocity 1.

32. **Solution.** The region  $D$  is bounded by  $y = x^2$  (below) and  $y = 4$  (above), so  $x$  ranges from  $-2$  to  $2$ .

By Green's Theorem (in reverse):

$$\iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_C P \, dx + Q \, dy,$$

where  $C = \partial D$  is traversed counterclockwise.

The boundary consists of two pieces:

- $C_1$ : along  $y = x^2$  from  $(-2, 4)$  to  $(2, 4)$  (bottom, left to right).
- $C_2$ : along  $y = 4$  from  $(2, 4)$  to  $(-2, 4)$  (top, right to left).

The resulting line integral is:

$$\oint_C P \, dx + Q \, dy = \int_{C_1} P \, dx + Q \, dy + \int_{C_2} P \, dx + Q \, dy.$$

One advantage of switching to the line integral form is that if  $P$  and  $Q$  are simple functions, a one-dimensional boundary integral can be easier to evaluate than a two-dimensional area integral, especially when the boundary has a convenient parameterization.

33. **Solution.** When we partition  $D$  into small rectangles and sum the line integrals around each sub-rectangle, every *interior* edge is shared by exactly two adjacent sub-rectangles. These two rectangles traverse the shared edge in *opposite directions*: what is the right edge of one rectangle is the left edge of its neighbor, and the orientations are opposite. Thus the contributions from interior edges cancel in pairs.

Only the edges lying on the *outer boundary* of  $D$  belong to a single sub-rectangle and have no partner to cancel with. These surviving contributions sum to exactly the line integral around the outer boundary  $\partial D$ . This is the discrete version of Green's Theorem: the sum of local circulations (double integral of curl) equals the global circulation (boundary line integral).

34. **Solution.**  $P = x^2y^2$ ,  $Q = xy^3$ . Triangle with vertices  $(0, 0)$ ,  $(2, 0)$ ,  $(0, 2)$ .

$$\frac{\partial Q}{\partial x} = y^3, \quad \frac{\partial P}{\partial y} = 2x^2y.$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = y^3 - 2x^2y.$$

By Green's Theorem:

$$\oint_C = \iint_D (y^3 - 2x^2y) dA.$$

The triangle:  $0 \leq x \leq 2$ ,  $0 \leq y \leq 2 - x$ .

$$\begin{aligned} &= \int_0^2 \int_0^{2-x} (y^3 - 2x^2y) dy dx \\ &= \int_0^2 \left[ \frac{y^4}{4} - x^2y^2 \right]_0^{2-x} dx \\ &= \int_0^2 \left[ \frac{(2-x)^4}{4} - x^2(2-x)^2 \right] dx. \end{aligned}$$

Let  $u = 2 - x$ :

$$\frac{(2-x)^4}{4} - x^2(2-x)^2 = \frac{u^4}{4} - (2-u)^2u^2.$$

Expand  $(2-u)^2u^2 = (4-4u+u^2)u^2 = 4u^2 - 4u^3 + u^4$ .

$$= \frac{u^4}{4} - 4u^2 + 4u^3 - u^4 = -\frac{3u^4}{4} + 4u^3 - 4u^2.$$

Back in  $x$  with  $u = 2 - x$ ,  $du = -dx$ :

$$\int_0^2 \left[ -\frac{3(2-x)^4}{4} + 4(2-x)^3 - 4(2-x)^2 \right] dx.$$

Let  $u = 2 - x$ :

$$\begin{aligned} &= \int_0^2 \left[ -\frac{3u^4}{4} + 4u^3 - 4u^2 \right] du = \left[ -\frac{3u^5}{20} + u^4 - \frac{4u^3}{3} \right]_0^2 \\ &= -\frac{3(32)}{20} + 16 - \frac{32}{3} = -\frac{96}{20} + 16 - \frac{32}{3} = -\frac{24}{5} + 16 - \frac{32}{3}. \end{aligned}$$

Common denominator 15:

$$= \frac{-72 + 240 - 160}{15} = \frac{8}{15}.$$

$$\boxed{\frac{8}{15}}.$$



35. **Solution.**  $\mathbf{F} = (y, -x)$ , so  $P = y$ ,  $Q = -x$ .

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -1 - 1 = -2.$$

By Green's Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (-2) dA = -2 \text{Area}(D).$$

Therefore the line integral is proportional to the area of the enclosed region, with proportionality constant  $-2$ :

$$\oint_C (y dx - x dy) = -2 \text{Area}(D). \quad \blacksquare$$

36. **Solution.**  $\mathbf{F} = (e^y, \sin x)$ , so  $P = e^y$ ,  $Q = \sin x$ .

$$\frac{\partial Q}{\partial x} = \cos x, \quad \frac{\partial P}{\partial y} = e^y.$$

By Green's Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (\cos x - e^y) dA,$$

where  $D$  is the interior of the ellipse  $\frac{x^2}{4} + \frac{y^2}{9} = 1$ .

Use the substitution  $x = 2u$ ,  $y = 3v$  (mapping the ellipse to the unit disk  $u^2 + v^2 \leq 1$ ), with Jacobian  $|J| = 6$ :

$$= \iint_{u^2+v^2 \leq 1} (\cos(2u) - e^{3v}) \cdot 6 du dv.$$

By symmetry,  $\cos(2u)$  is even in both  $u$  and  $v$ , and  $e^{3v}$  involves only  $v$ .

For  $\iint \cos(2u) du dv$  over the unit disk, switch to polar:  $u = r \cos \theta$ :

$$\int_0^{2\pi} \int_0^1 \cos(2r \cos \theta) r dr d\theta = 2\pi \int_0^1 r J_0(2r) dr,$$

where  $J_0$  is the Bessel function. Using  $\frac{d}{dr}[r J_1(2r) \cdot \frac{1}{2}] = \frac{1}{2} \cdot [J_1(2r) + 2r J_0(2r) \cdot \frac{1}{2}] \dots$  this becomes intricate.

A more direct approach: use the identity  $\int_0^1 r J_0(2r) dr = \frac{J_1(2)}{2}$ , so  $\iint \cos(2u) du dv = 2\pi \cdot \frac{J_1(2)}{2} = \pi J_1(2)$ .

For  $\iint e^{3v} du dv$  over the unit disk:  $v = r \sin \theta$ :

$$\int_0^{2\pi} \int_0^1 e^{3r \sin \theta} r dr d\theta = 2\pi \int_0^1 r I_0(3r) dr = 2\pi \cdot \frac{I_1(3)}{3},$$

using  $\int_0^1 r I_0(3r) dr = \frac{I_1(3)}{3}$ .

Therefore:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 6 \left[ \pi J_1(2) - \frac{2\pi I_1(3)}{3} \right] = 6\pi J_1(2) - 4\pi I_1(3).$$

Using  $J_1(2) \approx 0.5767$  and  $I_1(3) \approx 3.9534$ :

$$\approx 6\pi(0.5767) - 4\pi(3.9534) \approx 10.87 - 49.68 \approx \boxed{-38.81}.$$

37. **Solution.** Green's Theorem states (for counterclockwise  $C$ ):

$$\oint_{C_{\text{ccw}}} P dx + Q dy = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

When we reverse the orientation from counterclockwise to clockwise,  $d\mathbf{r}$  is negated (the tangent vector reverses), so:

$$\oint_{C_{\text{cw}}} P dx + Q dy = - \oint_{C_{\text{ccw}}} P dx + Q dy = - \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

The double integral  $\iint_D (\partial Q/\partial x - \partial P/\partial y) dA$  does not depend on orientation—it is a property of the region and the field. The sign convention (positive = counterclockwise) is what links the line integral to the double integral.

38. **Solution.**  $\mathbf{F} = \left( \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right)$  on  $\mathbb{R}^2 \setminus \{(0,0)\}$ .

First check:

$$\begin{aligned} \frac{\partial Q}{\partial x} &= \frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \\ \frac{\partial P}{\partial y} &= \frac{-(x^2 + y^2) + y(2y)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}. \end{aligned}$$

So  $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$  wherever  $\mathbf{F}$  is defined.

If Green's Theorem applied directly, we would conclude  $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D 0 dA = 0$ . However,  $\mathbf{F}$  is undefined at the origin, which lies inside  $C$  (the circle of radius  $r$ ). The domain is **not simply connected**, so Green's Theorem does not apply to the region enclosed by  $C$ .

Computing directly with parameterization  $x = r \cos t$ ,  $y = r \sin t$ ,  $t \in [0, 2\pi]$ :

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} \left[ \frac{-r \sin t}{r^2} (-r \sin t) + \frac{r \cos t}{r^2} (r \cos t) \right] dt \\ &= \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \boxed{2\pi}. \end{aligned}$$

This demonstrates the crucial role of simple connectivity:  $\partial Q/\partial x = \partial P/\partial y$  everywhere in the domain, yet the field is not conservative because the domain has a “hole” at the origin.

39. **Solution.**  $\mathbf{F} = (x + 2y, x - y)$ , so  $P = x + 2y$ ,  $Q = x - y$ .

$$\operatorname{div}(\mathbf{F}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 1 + (-1) = \boxed{0}.$$

$$\operatorname{curl}(\mathbf{F}) \text{ (2D)} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - 2 = \boxed{-1}.$$

The **divergence** indicates expansion/compression (here zero, meaning the field is area-preserving). The **curl** indicates rotation (here  $-1$ , meaning clockwise rotation).

40. **Solution.**  $\mathbf{F} = (x^2, 2y, -3z)$ .

$$\operatorname{div}(\mathbf{F}) = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(2y) + \frac{\partial}{\partial z}(-3z) = 2x + 2 - 3 = 2x - 1.$$

This is not identically zero, so  $\mathbf{F}$  is **not incompressible** (not divergence-free).

$$\operatorname{curl}(\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ x^2 & 2y & -3z \end{vmatrix} = \mathbf{i}(0 - 0) - \mathbf{j}(0 - 0) + \mathbf{k}(0 - 0) = \boxed{\mathbf{0}}.$$

The curl is zero, so  $\mathbf{F}$  is irrotational.

41. **Solution.** If  $\mathbf{F} = \nabla f = \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$ , then

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ f_x & f_y & f_z \end{vmatrix}.$$

The  $\mathbf{i}$ -component is  $\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z} = f_{zy} - f_{yz}$ . By Clairaut's theorem (assuming  $f$  has continuous second partial derivatives),  $f_{zy} = f_{yz}$ , so this component is 0.

Similarly, the  $\mathbf{j}$ -component is  $f_{xz} - f_{zx} = 0$  and the  $\mathbf{k}$ -component is  $f_{yx} - f_{xy} = 0$ .

Therefore  $\nabla \times (\nabla f) = \mathbf{0}$  for any sufficiently smooth scalar function  $f$ . Every conservative field is irrotational. ■

42. **Solution.**  $\mathbf{F} = (e^x \sin y, e^y \sin z, e^z \sin x)$ .

$$\begin{aligned} \operatorname{div}(\mathbf{F}) &= \frac{\partial}{\partial x}(e^x \sin y) + \frac{\partial}{\partial y}(e^y \sin z) + \frac{\partial}{\partial z}(e^z \sin x) \\ &= e^x \sin y + e^y \sin z + e^z \sin x. \end{aligned}$$

This is **not identically zero**. For example, at  $(\pi/2, \pi/2, \pi/2)$ :

$$\operatorname{div}(\mathbf{F}) = e^{\pi/2} \sin(\pi/2) + e^{\pi/2} \sin(\pi/2) + e^{\pi/2} \sin(\pi/2) = 3e^{\pi/2} \neq 0.$$

Therefore  $\mathbf{F}$  is **not divergence-free**.

43. **Solution.**  $\mathbf{F} = (x^3, y^3)$ .

$$\operatorname{div}(\mathbf{F}) = 3x^2 + 3y^2 = 3(x^2 + y^2).$$

This equals zero only at the origin  $(0, 0)$  and is strictly positive everywhere else.

**Physical interpretation:** Since  $\operatorname{div}(\mathbf{F}) = 3(x^2 + y^2) > 0$  for  $(x, y) \neq (0, 0)$ , the field acts as a *source* at every non-origin point: fluid is being “created” or expanding outward. The source strength increases with distance from the origin. Only at the origin itself does the divergence vanish (neither source nor sink there).

44. **Solution.**

(a)  $\mathbf{F}_1 = (x, -y)$ :

$$\operatorname{div}(\mathbf{F}_1) = 1 + (-1) = 0, \quad \operatorname{curl}(\mathbf{F}_1) = \frac{\partial(-y)}{\partial x} - \frac{\partial(x)}{\partial y} = 0 - 0 = 0.$$

At the origin, both divergence and curl are zero. This is a **saddle** (hyperbolic flow), not primarily source, sink, or rotation.

(b)  $\mathbf{F}_2 = (2x, 2y)$ :

$$\operatorname{div}(\mathbf{F}_2) = 2 + 2 = 4 > 0, \quad \operatorname{curl}(\mathbf{F}_2) = 0 - 0 = 0.$$

Divergence is positive, curl is zero. The origin is a **source**.

(c)  $\mathbf{F}_3 = (-y, x)$ :

$$\operatorname{div}(\mathbf{F}_3) = 0 + 0 = 0, \quad \operatorname{curl}(\mathbf{F}_3) = 1 - (-1) = 2.$$

Divergence is zero, curl is nonzero. The origin is a **rotation** (counterclockwise).

45. **Solution.** The identity  $\nabla \cdot (\nabla \times \mathbf{F}) = 0$  holds for any smooth vector field  $\mathbf{F}$ .

If the velocity field  $\mathbf{v}$  of a fluid is purely rotational, meaning  $\mathbf{v} = \nabla \times \mathbf{A}$  for some vector potential  $\mathbf{A}$ , then

$$\nabla \cdot \mathbf{v} = \nabla \cdot (\nabla \times \mathbf{A}) = 0.$$

This means the divergence of  $\mathbf{v}$  is zero everywhere.

By the Divergence Theorem, for any closed surface  $S$  bounding a volume  $V$ :

$$\oint_S \mathbf{v} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{v} dV = 0.$$

The net outward flux through any closed surface is zero, meaning no fluid is created or destroyed inside  $V$ . In a purely swirling (rotational) motion, the fluid merely circulates without any net outflow or inflow—mass is conserved. ■

46. **Solution.**  $\mathbf{F} = (yz, zx, xy)$ .

$$\operatorname{div}(\mathbf{F}) = \frac{\partial(yz)}{\partial x} + \frac{\partial(zx)}{\partial y} + \frac{\partial(xy)}{\partial z} = 0 + 0 + 0 = \boxed{0}.$$

$$\begin{aligned}\operatorname{curl}(\mathbf{F}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ yz & zx & xy \end{vmatrix} \\ &= \mathbf{i}(x - x) - \mathbf{j}(y - y) + \mathbf{k}(z - z) = \boxed{\mathbf{0}}.\end{aligned}$$

Both divergence and curl are zero. The divergence being zero means no net creation or destruction of fluid (incompressible). The curl being zero means no rotation (irrotational). In fact,  $\mathbf{F} = \nabla(xyz)$ , confirming it is conservative.

*Note:* The problem statement suggests the curl is nonzero, but computation shows both are zero. The field is both divergence-free and irrotational.

47. **Solution.** In cylindrical coordinates with  $\mathbf{v} = (v_r, v_\theta, v_z) = (0, \alpha r, 0)$ :

**Divergence** in cylindrical coordinates:

$$\nabla \cdot \mathbf{v} = \frac{1}{r} \frac{\partial(rv_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = \frac{1}{r} \frac{\partial(0)}{\partial r} + \frac{1}{r} \frac{\partial(\alpha r)}{\partial \theta} + 0 = 0 + 0 + 0 = \boxed{0}.$$

The flow is **incompressible**.

**Curl** in cylindrical coordinates:

$$\begin{aligned}(\nabla \times \mathbf{v})_r &= \frac{1}{r} \frac{\partial v_z}{\partial \theta} - \frac{\partial v_\theta}{\partial z} = 0 - 0 = 0, \\ (\nabla \times \mathbf{v})_\theta &= \frac{\partial v_r}{\partial z} - \frac{\partial v_z}{\partial r} = 0 - 0 = 0, \\ (\nabla \times \mathbf{v})_z &= \frac{1}{r} \frac{\partial(rv_\theta)}{\partial r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta} = \frac{1}{r} \frac{\partial(\alpha r^2)}{\partial r} - 0 = \frac{2\alpha r}{r} = 2\alpha. \\ \nabla \times \mathbf{v} &= \boxed{2\alpha \mathbf{e}_z}.\end{aligned}$$

The curl is nonzero (pointing along  $z$ ), confirming the flow **rotates about the  $z$ -axis**. This is solid-body rotation with angular velocity  $\alpha$ .

48. **Solution.** In spherical coordinates with  $\mathbf{F} = (f(r), 0, 0)$  (purely radial), the divergence is:

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{d}{dr}(r^2 f(r)).$$

Expanding:

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} [2r f(r) + r^2 f'(r)] = \frac{2f(r)}{r} + f'(r).$$

For this to be zero for  $r > 0$ :

$$\frac{d}{dr}(r^2 f(r)) = 0 \implies r^2 f(r) = C \implies \boxed{f(r) = \frac{C}{r^2}}$$

for some constant  $C$ . This is the inverse-square law (e.g., gravitational or electrostatic fields).

49. **Solution.** Let  $\mathbf{F} = (F_1, F_2, F_3)$  and  $f$  be a scalar function.

**Left side:**

$$\nabla \cdot (f\mathbf{F}) = \frac{\partial(fF_1)}{\partial x} + \frac{\partial(fF_2)}{\partial y} + \frac{\partial(fF_3)}{\partial z}.$$

By the product rule for each term:

$$= \left( f \frac{\partial F_1}{\partial x} + F_1 \frac{\partial f}{\partial x} \right) + \left( f \frac{\partial F_2}{\partial y} + F_2 \frac{\partial f}{\partial y} \right) + \left( f \frac{\partial F_3}{\partial z} + F_3 \frac{\partial f}{\partial z} \right).$$

Regrouping:

$$= f \left( \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) + \left( F_1 \frac{\partial f}{\partial x} + F_2 \frac{\partial f}{\partial y} + F_3 \frac{\partial f}{\partial z} \right).$$

**Right side:**

$$f(\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot (\nabla f) = f \left( \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) + (F_1 f_x + F_2 f_y + F_3 f_z).$$

Both sides are identical. ■

**Physical interpretation:** The net outflow of the quantity  $f\mathbf{F}$  has two contributions: (1)  $f \nabla \cdot \mathbf{F}$ , which is the outflow due to  $\mathbf{F}$  itself expanding/contracting, weighted by  $f$ ; and (2)  $\mathbf{F} \cdot \nabla f$ , which is the transport of  $f$  due to the flow  $\mathbf{F}$  carrying  $f$  from regions of high  $f$  to low  $f$  (or vice versa).

50. **Solution.**

$$\begin{aligned} \nabla \times (\nabla f) &= \nabla \times \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \\ &= \left( \frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \mathbf{i} - \left( \frac{\partial^2 f}{\partial x \partial z} - \frac{\partial^2 f}{\partial z \partial x} \right) \mathbf{j} + \left( \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \mathbf{k}. \end{aligned}$$

By Clairaut's theorem ( $f_{yz} = f_{zy}$ , etc.), every component is zero:

$$\nabla \times (\nabla f) = \mathbf{0}. \quad \blacksquare$$

This means any gradient field is **irrotational**: there is no circulation or rotation.

**Physical application:** The electrostatic field  $\mathbf{E} = -\nabla V$  (where  $V$  is the electric potential) is irrotational:  $\nabla \times \mathbf{E} = \mathbf{0}$  in the static case. This is one of Maxwell's equations for electrostatics.

51. **Solution.** If  $\nabla \cdot \mathbf{F} = 3$  everywhere, then at every point, the field has a positive outflow rate of 3 per unit volume. This means fluid is being “created” uniformly throughout space at a constant rate— $\mathbf{F}$  behaves like a **uniform source of fluid**.

To verify, use the Divergence Theorem on a sphere  $S$  of radius  $R$ :

$$\oiint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V \nabla \cdot \mathbf{F} \, dV = \iiint_V 3 \, dV = 3 \cdot \frac{4}{3}\pi R^3 = 4\pi R^3.$$

The total outward flux through the sphere is  $4\pi R^3$ , which grows with the volume, confirming the uniform source interpretation.

52. **Solution.**  $\mathbf{F} = (x^2, y^2, z^2)$ .

**Step 1:** Compute  $\nabla \times \mathbf{F}$ :

$$\nabla \times \mathbf{F} = (0 - 0)\mathbf{i} - (0 - 0)\mathbf{j} + (0 - 0)\mathbf{k} = \mathbf{0}.$$

**Step 2:**  $\nabla \times (\nabla \times \mathbf{F}) = \nabla \times \mathbf{0} = \mathbf{0}$ .

**Step 3:** Verify with the identity  $\nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$ :

$$\nabla \cdot \mathbf{F} = 2x + 2y + 2z.$$

$$\nabla(\nabla \cdot \mathbf{F}) = \nabla(2x + 2y + 2z) = (2, 2, 2).$$

$$\nabla^2 \mathbf{F} = (\nabla^2(x^2), \nabla^2(y^2), \nabla^2(z^2)) = (2, 2, 2).$$

Therefore:

$$\nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F} = (2, 2, 2) - (2, 2, 2) = \mathbf{0}.$$

Both sides agree:  $\nabla \times (\nabla \times \mathbf{F}) = \mathbf{0}$ .

53. **Solution.**

(a)  $\mathbf{F} = \nabla(x^2 + 2y^2) = (2x, 4y)$ , so  $P = 2x$ ,  $Q = 4y$ .

(b)  $\text{curl}(\mathbf{F}) = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 - 0 = \boxed{0}$ .

$\text{div}(\mathbf{F}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 2 + 4 = \boxed{6}$ .

(c) Yes,  $\text{curl}(\mathbf{F}) = 0$  confirms that gradient fields are always irrotational, consistent with the identity  $\nabla \times (\nabla f) = \mathbf{0}$ .

54. **Solution.** The surface  $x^2 + y^2 = 4z^2$  is a cone. Rewrite as  $z = \pm \frac{1}{2}\sqrt{x^2 + y^2}$ .

Using cylindrical-style parameters, let  $x = r \cos \theta$ ,  $y = r \sin \theta$ , and then  $z = \pm \frac{r}{2}$ , where  $r \geq 0$ .

**Parametric form:**

$$\mathbf{r}(r, \theta) = \left( r \cos \theta, r \sin \theta, \pm \frac{r}{2} \right),$$

with parameter constraints:  $r \geq 0$ ,  $0 \leq \theta \leq 2\pi$  (or  $0 \leq \theta < 2\pi$ ).

The  $+$  sign gives the upper cone ( $z \geq 0$ ) and the  $-$  sign gives the lower cone ( $z \leq 0$ ). Together they form the full double cone.

55. **Solution.** The surface is  $x = y^2 + 2z^2$  with  $x \leq 4$ .

Let  $y = u$  and  $z = v$  be the parameters. Then  $x = u^2 + 2v^2$ .

**Parametrization:**

$$\mathbf{r}(u, v) = (u^2 + 2v^2, u, v).$$

The constraint  $x \leq 4$  becomes  $u^2 + 2v^2 \leq 4$ , which is an elliptical region in the  $(u, v)$ -plane:

$$\frac{u^2}{4} + \frac{v^2}{2} \leq 1.$$

So the domain is:  $\{(u, v) : u^2 + 2v^2 \leq 4\}$ , i.e.,  $-2 \leq u \leq 2$  and  $-\sqrt{(4 - u^2)/2} \leq v \leq \sqrt{(4 - u^2)/2}$ .

56. **Solution.**  $\mathbf{r}(u, v) = (u, v, \sqrt{u^2 + v^2})$ .

(a)

$$\mathbf{r}_u = \left(1, 0, \frac{u}{\sqrt{u^2 + v^2}}\right), \quad \mathbf{r}_v = \left(0, 1, \frac{v}{\sqrt{u^2 + v^2}}\right).$$

(b) Let  $s = \sqrt{u^2 + v^2}$ .

$$\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & u/s \\ 0 & 1 & v/s \end{vmatrix} = \left(-\frac{u}{s}, -\frac{v}{s}, 1\right).$$

(c)

$$\|\mathbf{n}\| = \sqrt{\frac{u^2}{s^2} + \frac{v^2}{s^2} + 1} = \sqrt{\frac{u^2 + v^2}{u^2 + v^2} + 1} = \sqrt{1 + 1} = \boxed{\sqrt{2}}.$$

57. **Solution.** The paraboloid is  $z = 9 - x^2 - y^2$  and the region is  $x^2 + y^2 \leq 5$ .

For a surface  $z = f(x, y)$ , the surface area element is  $dS = \sqrt{1 + f_x^2 + f_y^2} dA$ .

Here  $f_x = -2x$ ,  $f_y = -2y$ , so  $\sqrt{1 + 4x^2 + 4y^2}$ .

$$\text{Surface Area} = \iint_{x^2 + y^2 \leq 5} \sqrt{1 + 4x^2 + 4y^2} dA.$$

In polar coordinates ( $x = r \cos \theta$ ,  $y = r \sin \theta$ ):

$$\boxed{\text{Surface Area} = \int_0^{2\pi} \int_0^{\sqrt{5}} \sqrt{1 + 4r^2} r dr d\theta.}$$

58. **Solution.**  $z = f(x, y) = x^2 + 4y^2$ , so  $f_x = 2x$ ,  $f_y = 8y$ .

$$dS = \sqrt{1 + f_x^2 + f_y^2} dA = \sqrt{1 + 4x^2 + 64y^2} dA.$$



On the surface,  $T(x, y, z) = \sqrt{1 + x^2 + y^2}$  (since  $z = x^2 + 4y^2$ ,  $T$  depends only on  $x, y$ ).

$$\iint_S T \, dS = \int_{-1}^1 \int_{-1}^1 \sqrt{1 + x^2 + y^2} \sqrt{1 + 4x^2 + 64y^2} \, dx \, dy.$$

59. **Solution.**

(a) The plane  $z = 2x$  over  $\{0 \leq x \leq 2, 0 \leq y \leq 1\}$ :

$$\mathbf{r}(x, y) = (x, y, 2x), \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 1.$$

(b)  $\mathbf{r}_x = (1, 0, 2)$ ,  $\mathbf{r}_y = (0, 1, 0)$ .

$$\mathbf{r}_x \times \mathbf{r}_y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 0 & 1 & 0 \end{vmatrix} = (-2, 0, 1).$$

For upward orientation ( $z$ -component positive),  $\mathbf{n} \, dA = (-2, 0, 1) \, dx \, dy$ .

$\mathbf{F} = (x^2, 0, yz)$ . On the surface  $z = 2x$ :  $\mathbf{F} = (x^2, 0, 2xy)$ .

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_0^1 \int_0^2 [x^2(-2) + 0 + 2xy(1)] \, dx \, dy = \int_0^1 \int_0^2 (2xy - 2x^2) \, dx \, dy.$$

60. **Solution.** The cylinder has radius 2,  $0 \leq z \leq 3$ .

$$\mathbf{r}(\theta, z) = (2 \cos \theta, 2 \sin \theta, z).$$

$$\mathbf{r}_\theta = (-2 \sin \theta, 2 \cos \theta, 0), \quad \mathbf{r}_z = (0, 0, 1).$$

$$\mathbf{r}_\theta \times \mathbf{r}_z = (2 \cos \theta, 2 \sin \theta, 0).$$

$$\|\mathbf{r}_\theta \times \mathbf{r}_z\| = 2.$$

On the cylinder,  $f = \sqrt{x^2 + y^2} = 2$ .

$$\iint_S f \, dS = \int_0^{2\pi} \int_0^3 2 \cdot 2 \, dz \, d\theta = \int_0^{2\pi} \int_0^3 4 \, dz \, d\theta = 4 \cdot 3 \cdot 2\pi = \boxed{24\pi}.$$

61. **Solution.** Hemisphere:  $x^2 + y^2 + z^2 = 9$ ,  $z \geq 0$ , radius  $R = 3$ .

Parameterize using spherical coordinates:  $\mathbf{r}(\phi, \theta) = (3 \sin \phi \cos \theta, 3 \sin \phi \sin \theta, 3 \cos \phi)$ , with  $0 \leq \phi \leq \pi/2$ ,  $0 \leq \theta \leq 2\pi$ .

$$\|\mathbf{r}_\phi \times \mathbf{r}_\theta\| = 9 \sin \phi.$$

On the surface:  $\rho = z = 3 \cos \phi$ .

(a) **Total mass:**

$$\iint_S \rho \, dS = \int_0^{2\pi} \int_0^{\pi/2} (3 \cos \phi)(9 \sin \phi) \, d\phi \, d\theta = 27 \int_0^{2\pi} \int_0^{\pi/2} \cos \phi \sin \phi \, d\phi \, d\theta.$$

Evaluating:  $\int_0^{\pi/2} \cos \phi \sin \phi \, d\phi = \frac{1}{2}$ , so the mass is  $27 \cdot 2\pi \cdot \frac{1}{2} = 27\pi$ .

(b) **Outward flux of  $\mathbf{F} = (0, 0, z)$ :**

The outward unit normal on the sphere is  $\hat{\mathbf{n}} = \frac{1}{3}(x, y, z) = (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi)$ .

$$\mathbf{F} \cdot \hat{\mathbf{n}} = z \cos \phi = 3 \cos \phi \cdot \cos \phi = 3 \cos^2 \phi.$$

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^{\pi/2} 3 \cos^2 \phi \cdot 9 \sin \phi \, d\phi \, d\theta = 27 \int_0^{2\pi} \int_0^{\pi/2} \cos^2 \phi \sin \phi \, d\phi \, d\theta.$$

Evaluating:  $\int_0^{\pi/2} \cos^2 \phi \sin \phi \, d\phi = \frac{1}{3}$ , so the flux is  $27 \cdot 2\pi \cdot \frac{1}{3} = 18\pi$ .

## 62. Solution.

(a)  $g(x, y) = x + 2y$ , so  $g_x = 1$ ,  $g_y = 2$ .

The upward-oriented normal vector is  $\mathbf{n} = (-g_x, -g_y, 1) = (-1, -2, 1)$ .

(b)  $\mathbf{F} = (2x, 3y, -z)$ . On the surface  $z = x + 2y$ :  $\mathbf{F} = (2x, 3y, -(x + 2y))$ .

$$\begin{aligned} \mathbf{F} \cdot \mathbf{n} &= 2x(-1) + 3y(-2) + (-(x + 2y))(1) \\ &= -2x - 6y - x - 2y = -3x - 8y. \end{aligned}$$

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_0^2 \int_0^1 (-3x - 8y) \, dx \, dy.$$

63. **Solution.**  $\mathbf{v} = (x, y, -1)$ , closed surface: cylinder  $x^2 + y^2 = 4$ ,  $-1 \leq z \leq 2$ , plus top and bottom disks.

(a) **Curved side:**  $\mathbf{r}(\theta, z) = (2 \cos \theta, 2 \sin \theta, z)$ ,  $0 \leq \theta \leq 2\pi$ ,  $-1 \leq z \leq 2$ .

$\mathbf{r}_\theta \times \mathbf{r}_z = (2 \cos \theta, 2 \sin \theta, 0)$  (outward-pointing).

(b) **Flux through curved side:**

$\mathbf{v} = (2 \cos \theta, 2 \sin \theta, -1)$  on the surface.

$$\mathbf{v} \cdot (\mathbf{r}_\theta \times \mathbf{r}_z) = 4 \cos^2 \theta + 4 \sin^2 \theta = 4.$$

$$\Phi_{\text{side}} = \int_0^{2\pi} \int_{-1}^2 4 \, dz \, d\theta = 4 \cdot 3 \cdot 2\pi = 24\pi.$$

**Flux through top disk** ( $z = 2$ , outward normal  $\hat{\mathbf{n}} = (0, 0, 1)$ ):

$$\mathbf{v} \cdot \hat{\mathbf{n}} = -1.$$

$$\Phi_{\text{top}} = \iint_{x^2+y^2 \leq 4} (-1) dA = -4\pi.$$

**Flux through bottom disk** ( $z = -1$ , outward normal  $\hat{\mathbf{n}} = (0, 0, -1)$ ):

$$\mathbf{v} \cdot \hat{\mathbf{n}} = -(-1) = 1.$$

$$\Phi_{\text{bot}} = \iint_{x^2+y^2 \leq 4} 1 dA = 4\pi.$$

(c) **Total outward flux:**

$$\Phi = \Phi_{\text{side}} + \Phi_{\text{top}} + \Phi_{\text{bot}} = 24\pi - 4\pi + 4\pi = \boxed{24\pi}.$$

**Verification via Divergence Theorem:**  $\nabla \cdot \mathbf{v} = 1 + 1 + 0 = 2$ , and the volume is  $\pi(2)^2(3) = 12\pi$ . So  $\iiint_V 2 dV = 24\pi$ .

64. **Solution.**  $\mathbf{F} = (y^2, z^2, x^2)$ .

**Step 1:** Compute  $\nabla \times \mathbf{F}$ :

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ y^2 & z^2 & x^2 \end{vmatrix} = (0 - 2z)\mathbf{i} - (2x - 0)\mathbf{j} + (0 - 2y)\mathbf{k} = (-2z, -2x, -2y).$$

**Step 2:** Parameterize  $\Sigma$ : the paraboloid  $z = x^2 + y^2$  for  $x^2 + y^2 \leq 4$ . Use  $\mathbf{r}(x, y) = (x, y, x^2 + y^2)$ .

$$\mathbf{r}_x = (1, 0, 2x), \mathbf{r}_y = (0, 1, 2y).$$

For Stokes' Theorem with the boundary oriented counterclockwise when viewed from above, we need the upward-oriented normal:

$$\mathbf{r}_x \times \mathbf{r}_y = (-2x, -2y, 1).$$

**Step 3:** By Stokes' Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{\Sigma} (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

$$(\nabla \times \mathbf{F}) \cdot (\mathbf{r}_x \times \mathbf{r}_y) = (-2z)(-2x) + (-2x)(-2y) + (-2y)(1).$$

On  $\Sigma$ :  $z = x^2 + y^2$ :

$$= 4x(x^2 + y^2) + 4xy - 2y.$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{x^2+y^2 \leq 4} [4x(x^2 + y^2) + 4xy - 2y] dA.$$

65. **Solution.**  $\mathbf{F} = (e^y, \ln(1 + z^2), y^2z)$ ,  $\Sigma$ :  $x + 2y + 3z = 6$  in the first octant.

**Decision:** It is easier to use the **surface integral**  $\iint_{\Sigma} (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$  (Stokes' Theorem) rather than the line integral directly. The reason: computing  $\oint_C \mathbf{F} \cdot d\mathbf{r}$  directly requires parameterizing three edges of the triangle and evaluating three separate integrals with complicated integrands  $(e^y, \ln(1 + z^2))$ . In contrast, the curl may simplify, and the surface integral over a flat triangular patch is straightforward.

**Outline of steps:**

(a) Compute  $\nabla \times \mathbf{F}$ : Computing component by component:

$$(\nabla \times \mathbf{F})_x = \frac{\partial(y^2z)}{\partial y} - \frac{\partial(\ln(1 + z^2))}{\partial z} = 2yz - \frac{2z}{1 + z^2},$$

$$(\nabla \times \mathbf{F})_y = \frac{\partial(e^y)}{\partial z} - \frac{\partial(y^2z)}{\partial x} = 0 - 0 = 0,$$

$$(\nabla \times \mathbf{F})_z = \frac{\partial(\ln(1 + z^2))}{\partial x} - \frac{\partial(e^y)}{\partial y} = 0 - e^y = -e^y.$$

(b) The plane  $x + 2y + 3z = 6$  has upward normal  $\mathbf{n} = (1, 2, 3)/\sqrt{14}$  and  $d\mathbf{S} = (1, 2, 3) dA'$  where  $dA'$  is the projected area element.

(c) Parameterize using  $(y, z)$  or project onto a coordinate plane and integrate over the triangular region.

66. **Solution.**  $\mathbf{F} = (-y, x, 0)$  on the cylinder  $x^2 + y^2 = 9$ ,  $-2 \leq z \leq 5$ .

(a)

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ -y & x & 0 \end{vmatrix} = (0 - 0)\mathbf{i} - (0 - 0)\mathbf{j} + (1 - (-1))\mathbf{k} = (0, 0, 2).$$

(b) The boundary  $\partial\Sigma$  consists of two circles:

**Top circle** ( $z = 5$ ):  $\mathbf{r}_1(\theta) = (3 \cos \theta, 3 \sin \theta, 5)$ ,  $\theta \in [0, 2\pi]$ , oriented counterclockwise when viewed from outside (by the right-hand rule with the outward cylinder normal).

**Bottom circle** ( $z = -2$ ):  $\mathbf{r}_2(\theta) = (3 \cos \theta, 3 \sin \theta, -2)$ ,  $\theta \in [0, 2\pi]$ , oriented clockwise when viewed from above (i.e., counterclockwise when viewed from below, consistent with the outward-pointing normal and right-hand rule).

(c) By Stokes' Theorem:

$$\oint_{\partial\Sigma} \mathbf{F} \cdot d\mathbf{r} = \iint_{\Sigma} (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

The cylinder surface has outward normal  $\hat{\mathbf{n}} = (\cos \theta, \sin \theta, 0)$ .

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = (0, 0, 2) \cdot (\cos \theta, \sin \theta, 0) = 0.$$

$$\boxed{\iint_{\Sigma} (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0.}$$

67. **Solution.** If  $\mathbf{F} = \nabla f$ , then  $\nabla \times \mathbf{F} = \nabla \times (\nabla f) = \mathbf{0}$  (by the identity proven in Problem 50).

By Stokes' Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{\Sigma} (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_{\Sigma} \mathbf{0} \cdot d\mathbf{S} = 0.$$

Therefore, for any smooth oriented surface  $\Sigma$  with boundary  $C$ :

$$\oint_C \nabla f \cdot d\mathbf{r} = 0. \quad \blacksquare$$

This is consistent with the fact that conservative fields have path-independent line integrals, so the integral around any closed loop is zero.

68. **Solution.** Faraday's Law states:

$$\oint_C \mathbf{E} \cdot d\mathbf{r} = -\frac{d}{dt} \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{S}.$$

By Stokes' Theorem, the left side equals:

$$\oint_C \mathbf{E} \cdot d\mathbf{r} = \iint_{\Sigma} (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$

If the surface  $\Sigma$  is fixed (static boundary), we can bring the time derivative inside:

$$\iint_{\Sigma} (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = - \iint_{\Sigma} \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

Since this holds for every surface  $\Sigma$ , the integrands must be equal:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

which is the differential form of Faraday's Law.

Here  $\mathbf{E}$  plays the role of the vector field in the line integral (circulation of the electric field), and  $\nabla \times \mathbf{E}$  plays the role of the flux integrand (the curl relates to the surface integral via Stokes).

**Physical meaning of the negative time derivative:** A *changing* magnetic flux through  $\Sigma$  induces an electric field that circulates around  $C$ . The negative sign (Lenz's law) means the induced electric field opposes the change in magnetic flux: if  $\mathbf{B}$  increases, the induced  $\mathbf{E}$  circulates in the direction that would create a magnetic field opposing the increase.

69. **Solution.**  $\mathbf{F} = (xy, yz, xz)$ .

$$\nabla \cdot \mathbf{F} = \frac{\partial(xy)}{\partial x} + \frac{\partial(yz)}{\partial y} + \frac{\partial(xz)}{\partial z} = y + z + x.$$

By the Divergence Theorem:

$$\oiint_{\partial V} \mathbf{F} \cdot \mathbf{n} dS = \iiint_V (x + y + z) dV,$$

where  $V$  is the ball  $x^2 + y^2 + z^2 \leq 4$ .

In spherical coordinates:  $x + y + z = \rho \sin \phi \cos \theta + \rho \sin \phi \sin \theta + \rho \cos \phi$ ,  $dV = \rho^2 \sin \phi d\rho d\phi d\theta$ .

By symmetry,  $\iiint_V x dV = \iiint_V y dV = \iiint_V z dV = 0$  (each is an odd function integrated over a symmetric domain).

$$\boxed{\iiint_V (x + y + z) dV = 0.}$$

70. **Solution.**  $\mathbf{F} = (x^2, y^2, z^2)$ . The region  $V$ :  $0 \leq z \leq \sqrt{x^2 + y^2}$  capped by  $z = 2$ .

More precisely,  $V$  is the region inside the cone  $z = \sqrt{x^2 + y^2}$  and below  $z = 2$ . In cylindrical coordinates:  $r \leq z \leq 2$ ,  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$ .

(a) **Boundary surfaces:**

- The cone:  $z = \sqrt{x^2 + y^2}$  (i.e.,  $z = r$ ) for  $0 \leq r \leq 2$  (with inward-pointing normal to close the volume, i.e., normal pointing outward from  $V$ , which is downward-and-outward on the cone).
- The top disk:  $z = 2$ ,  $x^2 + y^2 \leq 4$  (outward normal is  $\hat{\mathbf{n}} = (0, 0, 1)$ ).

(b) By the Divergence Theorem:

$$\nabla \cdot \mathbf{F} = 2x + 2y + 2z.$$

$$\iint_{\partial V} \mathbf{F} \cdot d\mathbf{S} = \iiint_V (2x + 2y + 2z) dV.$$

In cylindrical coordinates:

$$= \int_0^{2\pi} \int_0^2 \int_r^2 (2r \cos \theta + 2r \sin \theta + 2z) r dz dr d\theta.$$

By symmetry, the  $\cos \theta$  and  $\sin \theta$  terms integrate to zero over  $[0, 2\pi]$ :

$$\boxed{= \int_0^{2\pi} \int_0^2 \int_r^2 2zr dz dr d\theta = 2\pi \int_0^2 r [z^2]_r^2 dr = 2\pi \int_0^2 r(4 - r^2) dr.}$$

Evaluating:  $\int_0^2 r(4 - r^2) dr = \int_0^2 (4r - r^3) dr = [2r^2 - r^4/4]_0^2 = 8 - 4 = 4$ . So the total flux is  $8\pi$ .

71. **Solution.**  $\mathbf{F} = (e^z, \sin(xy), 2z)$ ,  $V$ : cylinder  $x^2 + y^2 \leq 1$ ,  $0 \leq z \leq 5$ .

**Decision:** Use the **Divergence Theorem** (volume integral), because computing the surface integral directly requires parameterizing three surfaces (curved side, top disk, bottom disk) with complicated integrands like  $e^z$  and  $\sin(xy)$ .

**Divergence:**

$$\nabla \cdot \mathbf{F} = 0 + x \cos(xy) + 2.$$

By the Divergence Theorem:

$$\oiint_{\partial V} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V (x \cos(xy) + 2) \, dV.$$

**Steps:**

- (a) The integral of 2 over  $V$ :  $2 \cdot \text{Vol}(V) = 2 \cdot \pi(1)^2(5) = 10\pi$ .
- (b) The integral of  $x \cos(xy)$  over  $V$ : Note  $x \cos(xy)$  is an *odd* function of  $x$  (replacing  $x \rightarrow -x$  gives  $-x \cos(-xy) = -x \cos(xy)$ ), and the domain is symmetric in  $x$ . Therefore  $\iiint_V x \cos(xy) \, dV = 0$ .

$$\boxed{\oiint_{\partial V} \mathbf{F} \cdot \mathbf{n} \, dS = 10\pi.}$$

72. **Solution.**  $\mathbf{F} = (x^2 + y^2, z, 3x)$ ,  $V$ : ball  $x^2 + y^2 + z^2 \leq 4$ .

$$\nabla \cdot \mathbf{F} = 2x + 0 + 0 = 2x.$$

In spherical coordinates:  $x = \rho \sin \phi \cos \theta$ ,  $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ .

$$\iiint_V 2x \, dV = \int_0^{2\pi} \int_0^\pi \int_0^2 2\rho \sin \phi \cos \theta \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

$$= 2 \int_0^{2\pi} \cos \theta \, d\theta \int_0^\pi \sin^2 \phi \, d\phi \int_0^2 \rho^3 \, d\rho.$$

Since  $\int_0^{2\pi} \cos \theta \, d\theta = 0$ , the entire integral equals  $\boxed{0}$ .

73. **Solution.** If a fluid is **incompressible**, then  $\nabla \cdot \mathbf{v} = 0$  everywhere. This means at every point, the rate of fluid creation (or destruction) per unit volume is zero—fluid is neither created nor destroyed anywhere.

By the Divergence Theorem, for any closed surface  $S = \partial V$  bounding a volume  $V$ :

$$\oiint_{\partial V} \mathbf{v} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{v} \, dV = \iiint_V 0 \, dV = 0.$$

The net outward flux through any closed surface is zero. Physically, this means whatever amount of fluid enters the volume through  $S$  must also leave through  $S$  (conservation of mass with constant density). No fluid accumulates inside, and none disappears. This is the mathematical statement of incompressibility: the volume of any fluid parcel is preserved over time.

# Chapter 6

## Chapter 6: Differential Equations

### 1. Solution.

- (a) The equation  $\frac{dy}{dx} = 3x^2$  involves a single independent variable  $x$  and ordinary derivatives. It is an **ODE**.
- (b) The equation  $\frac{\partial u}{\partial t} = 4\frac{\partial u}{\partial x}$  involves partial derivatives with respect to two independent variables  $t$  and  $x$ . It is a **PDE**.
- (c) The equation  $\frac{d^2y}{dx^2} + 5y = \sin(x)$  involves ordinary derivatives of  $y$  with respect to the single variable  $x$ . It is an **ODE**.

### 2. Solution.

- (a) The highest derivative is  $\frac{d^2y}{dx^2}$ , so the order is **2**.
- (b) The highest derivative is  $\frac{d^3y}{dx^3}$ , so the order is **3**.
- (c) The highest-order partial derivatives are  $\frac{\partial^2 u}{\partial x^2}$  and  $\frac{\partial^2 u}{\partial y^2}$ , so the order is **2**.

### 3. Solution.

- (a)  $\frac{dy}{dx} = y^2 - \sin(x)$ . The term  $y^2$  is nonlinear in the unknown  $y$ . This equation is **nonlinear**.
- (b)  $\frac{dy}{dx} + y \cos(x) = e^x$ . Every term involving  $y$  or its derivatives appears to the first power and is not multiplied by  $y$  or its derivatives. This equation is **linear**.
- (c)  $\frac{d^2y}{dx^2} + 3y\frac{dy}{dx} = x$ . The term  $3y\frac{dy}{dx}$  is a product of  $y$  and  $y'$ , which is nonlinear. This equation is **nonlinear**.

### 4. Solution.

- (a)  $y'' - 3y' + 2y = e^x$ . The right-hand side  $e^x \neq 0$ , so the equation is **non-homogeneous**.



- (b)  $y' + 2y = 0$ . The right-hand side is 0, so the equation is **homogeneous**.  
 (c)  $y'' + y = \sin(x)$ . The right-hand side  $\sin(x) \neq 0$ , so the equation is **non-homogeneous**.

5. **Solution.**

- (a)  $x y'' + 2y' - y = 0$ . The coefficient of  $y''$  is  $x$ , which is not constant. This ODE does **not** have constant coefficients.  
 (b)  $3y'' + 2y' - y = 0$ . All coefficients  $(3, 2, -1)$  are constants. This ODE **has constant coefficients**.  
 (c)  $y'' - 4y = 0$ . All coefficients  $(1, 0, -4)$  are constants. This ODE **has constant coefficients**.

6. **Solution.** Let  $y = e^{2x}$ . Then  $y' = 2e^{2x}$  and  $y'' = 4e^{2x}$ . Substituting into  $y'' - 4y$ :

$$y'' - 4y = 4e^{2x} - 4e^{2x} = 0.$$

Since the equation is satisfied,  $y = e^{2x}$  is indeed a solution.

7. **Solution.** Let  $y = 3e^{-x} + 2$ . Then  $y' = -3e^{-x}$ . Substituting into  $y' + y$ :

$$y' + y = -3e^{-x} + (3e^{-x} + 2) = 2.$$

The equation  $y' + y = 2$  is satisfied, confirming  $y = 3e^{-x} + 2$  is a solution.

8. **Solution.** Differentiate  $x^2 + y^2 = 1$  implicitly with respect to  $x$ :

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

This matches the given differential equation, so  $x^2 + y^2 = 1$  is a solution.

9. **Solution.** Starting from  $\ln |y| + y = x$ , differentiate both sides with respect to  $x$ :

$$\frac{1}{y} y' + y' = 1 \implies \left( \frac{1}{y} + 1 \right) y' = 1.$$

This confirms the implicit relation is a solution of the given ODE.

10. **Solution.** Separate variables:  $y dy = 2x dx$ . Integrate both sides:

$$\int y dy = \int 2x dx \implies \frac{y^2}{2} = x^2 + C.$$

Apply the initial condition  $y(1) = 2$ :  $\frac{4}{2} = 1 + C$ , so  $C = 1$ . Therefore  $\frac{y^2}{2} = x^2 + 1$ , giving

$$\boxed{y = \sqrt{2x^2 + 2}}.$$

(We take the positive root since  $y(1) = 2 > 0$ .)

11. **Solution.** Separate variables:  $e^{-y} dy = x dx$ . Integrate:

$$\int e^{-y} dy = \int x dx \implies -e^{-y} = \frac{x^2}{2} + C.$$

Solving for  $y$ :

$$e^{-y} = -\frac{x^2}{2} + C_1, \quad \text{where } C_1 = -C,$$

$$\boxed{y = -\ln\left(C_1 - \frac{x^2}{2}\right)},$$

valid where  $C_1 - \frac{x^2}{2} > 0$ .

12. **Solution.** The equation  $y' + 2y = \sin x$  is first-order linear. The integrating factor is  $\mu = e^{\int 2 dx} = e^{2x}$ . Multiply through:

$$\frac{d}{dx}(e^{2x}y) = e^{2x} \sin x.$$

Integrate the right side using integration by parts (or the standard formula  $\int e^{ax} \sin(bx) dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2}$ ):

$$\int e^{2x} \sin x dx = \frac{e^{2x}(2 \sin x - \cos x)}{5} + C.$$

Therefore

$$e^{2x}y = \frac{e^{2x}(2 \sin x - \cos x)}{5} + C,$$

$$\boxed{y = \frac{2 \sin x - \cos x}{5} + Ce^{-2x}}.$$

13. **Solution.** Write  $M = 2xy + y^2$  and  $N = x^2 + 2xy$ . Check exactness:

$$\frac{\partial M}{\partial y} = 2x + 2y, \quad \frac{\partial N}{\partial x} = 2x + 2y.$$

Since  $M_y = N_x$ , the equation is exact. Find  $F(x, y)$  such that  $F_x = M$  and  $F_y = N$ :

$$F = \int (2xy + y^2) dx = x^2y + xy^2 + g(y).$$

Then  $F_y = x^2 + 2xy + g'(y) = N = x^2 + 2xy$ , so  $g'(y) = 0$  and  $g(y) = C_0$ . The general solution is

$$\boxed{x^2y + xy^2 = C}.$$

14. **Solution.** The Bernoulli equation  $y' - y = e^x y^2$  has  $n = 2$ . Let  $v = y^{1-2} = y^{-1}$ , so  $y = v^{-1}$  and  $y' = -v^{-2}v'$ . Substituting:

$$-v^{-2}v' - v^{-1} = e^x v^{-2}.$$

Multiply by  $-v^2$ :

$$v' + v = -e^x.$$

This is first-order linear with integrating factor  $\mu = e^x$ :

$$\frac{d}{dx}(e^x v) = -e^{2x} \implies e^x v = -\frac{e^{2x}}{2} + C.$$

So  $v = -\frac{e^x}{2} + Ce^{-x}$ , and since  $v = 1/y$ :

$$y = \frac{1}{Ce^{-x} - \frac{1}{2}e^x} = \frac{2}{2Ce^{-x} - e^x}.$$

15. **Solution.** Rewrite as  $y' - \frac{y}{x} = x^2$ , a first-order linear ODE. The integrating factor is

$$\mu = e^{\int -1/x dx} = e^{-\ln x} = \frac{1}{x}.$$

Multiply:  $\frac{d}{dx}\left(\frac{y}{x}\right) = x$ . Integrate:

$$\frac{y}{x} = \frac{x^2}{2} + C \implies y = \frac{x^3}{2} + Cx.$$

Apply  $y(1) = 2$ :  $2 = \frac{1}{2} + C$ , so  $C = \frac{3}{2}$ . Therefore

$$y = \frac{x^3}{2} + \frac{3x}{2}.$$

16. **Solution.** Separate variables:  $\frac{dy}{\sqrt{y}} = 3 dx$ . Integrate:

$$2\sqrt{y} = 3x + C.$$

Apply  $y(4) = 9$ :  $2(3) = 3(4) + C$ , so  $6 = 12 + C$  and  $C = -6$ . Therefore  $2\sqrt{y} = 3x - 6$  and

$$y = \left(\frac{3x-6}{2}\right)^2 = \frac{9}{4}(x-2)^2.$$

17. **Solution.** The equation  $y' - \frac{y}{x} = x^2$  for  $x > 0$  has integrating factor  $\mu = e^{-\int dx/x} = \frac{1}{x}$ . Then

$$\frac{d}{dx}\left(\frac{y}{x}\right) = x \implies \frac{y}{x} = \frac{x^2}{2} + C,$$

so

$$y = \frac{x^3}{2} + Cx.$$

18. **Solution.** Separate variables:  $(1 + y^2) dy = 2x dx$ . Integrate:

$$y + \frac{y^3}{3} = x^2 + C.$$

The general solution in implicit form is

$$\boxed{y + \frac{y^3}{3} = x^2 + C}.$$

19. **Solution.** Let  $y = e^x - 1$ . Then  $y' = e^x$ . Substituting into  $y' - y$ :

$$y' - y = e^x - (e^x - 1) = 1.$$

The equation  $y' - y = 1$  is satisfied, confirming  $y = e^x - 1$  is a solution.

20. **Solution.** Write  $M = y \cos x - \sin x$  and  $N = \cos x$ . Check:

$$M_y = \cos x, \quad N_x = -\sin x.$$

Since  $M_y \neq N_x$ , the equation is **not exact** as stated. However, let us check for an integrating factor. We compute

$$\frac{M_y - N_x}{N} = \frac{\cos x - (-\sin x)}{\cos x} = 1 + \tan x.$$

This depends on  $x$  alone, so an integrating factor  $\mu$  depending on  $x$  satisfies:

$$\frac{1}{\mu} \frac{d\mu}{dx} = \frac{M_y - N_x}{N} = 1 + \tan x.$$

Then  $\ln \mu = x - \ln |\cos x|$ , so  $\mu = \frac{e^x}{\cos x} = e^x \sec x$ .

Multiply the equation by  $\mu$ :

$$\widetilde{M} = e^x \sec x (y \cos x - \sin x) = e^x y - e^x \tan x, \quad \widetilde{N} = e^x \sec x \cdot \cos x = e^x.$$

Now find  $F$  with  $F_y = \widetilde{N} = e^x$ , so  $F = ye^x + h(x)$ . Then  $F_x = ye^x + h'(x) = \widetilde{M} = ye^x - e^x \tan x$ , giving  $h'(x) = -e^x \tan x$ . Thus

$$F = ye^x - \int e^x \tan x dx.$$

Since  $\int e^x \tan x dx$  does not have a simple closed form, rewrite the original equation in standard linear form. Dividing  $(y \cos x - \sin x) dx + \cos x dy = 0$  by  $\cos x dx$  gives:

$$y' + y = \tan x.$$

This is first-order linear with integrating factor  $\mu = e^x$ :

$$\frac{d}{dx}(e^x y) = e^x \tan x, \quad \text{so} \quad e^x y = \int e^x \tan x dx + C.$$

Since  $\int e^x \tan x \, dx$  has no elementary closed form, the general solution is

$$y = e^{-x} \int e^x \tan x \, dx + Ce^{-x}.$$

The equation is **not exact** in its original form.

21. **Solution.** The Bernoulli equation  $y' + y = xy^2$  has  $n = 2$ . Set  $v = y^{-1}$ , so  $v' = -y^{-2}y'$ . From  $y' + y = xy^2$ , divide by  $y^2$ :

$$y^{-2}y' + y^{-1} = x.$$

Since  $v = y^{-1}$  gives  $v' = -y^{-2}y'$ , we have  $-v' + v = x$ , or

$$v' - v = -x.$$

Integrating factor:  $\mu = e^{-x}$ . Then

$$\frac{d}{dx}(e^{-x}v) = -xe^{-x}.$$

Integrate by parts:  $\int -xe^{-x} \, dx = xe^{-x} - e^{-x} + C = (x-1)e^{-x} + C$  (since  $\int xe^{-x} \, dx = -xe^{-x} + e^{-x}$ , so  $\int -xe^{-x} \, dx = xe^{-x} - e^{-x}$ ). Therefore

$$e^{-x}v = xe^{-x} - e^{-x} + C \implies v = x - 1 + Ce^x.$$

Since  $v = 1/y$ :

$$y = \frac{1}{x - 1 + Ce^x}.$$

22. **Solution.** Separate variables:  $\frac{dy}{y} = \frac{x \, dx}{1+x^2}$ . Integrate:

$$\ln |y| = \frac{1}{2} \ln(1+x^2) + C_1.$$

Exponentiate:

$$y = C\sqrt{1+x^2}.$$

23. **Solution.** Rewrite:  $y' - \frac{y}{x} = \ln x$ , first-order linear. Integrating factor  $\mu = \frac{1}{x}$ :

$$\frac{d}{dx}\left(\frac{y}{x}\right) = \frac{\ln x}{x}.$$

Integrate:  $\int \frac{\ln x}{x} \, dx = \frac{(\ln x)^2}{2} + C$ . So

$$\frac{y}{x} = \frac{(\ln x)^2}{2} + C \implies y = \frac{x(\ln x)^2}{2} + Cx.$$

Apply  $y(1) = 0$ :  $0 = 0 + C$ , so  $C = 0$ . Therefore

$$y = \frac{x(\ln x)^2}{2}.$$

24. **Solution.** The equation  $y' + \frac{2}{x}y = \sin x$  has integrating factor  $\mu = e^{\int 2/x dx} = x^2$ . Then

$$\frac{d}{dx}(x^2y) = x^2 \sin x.$$

Integrate by parts twice:

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C.$$

Therefore

$$x^2y = -x^2 \cos x + 2x \sin x + 2 \cos x + C,$$

$$y = -\cos x + \frac{2 \sin x}{x} + \frac{2 \cos x}{x^2} + \frac{C}{x^2}.$$

25. **Solution.** The Bernoulli equation  $y' - \frac{1}{x}y = xy^3$  has  $n = 3$ . Set  $v = y^{1-3} = y^{-2}$ , so  $v' = -2y^{-3}y'$ . From the ODE, divide by  $y^3$ :

$$y^{-3}y' - \frac{1}{x}y^{-2} = x.$$

Since  $v' = -2y^{-3}y'$ , we get  $y^{-3}y' = -\frac{v'}{2}$ :

$$-\frac{v'}{2} - \frac{v}{x} = x \implies v' + \frac{2}{x}v = -2x.$$

Integrating factor:  $\mu = x^2$ . Then

$$\frac{d}{dx}(x^2v) = -2x^3 \implies x^2v = -\frac{x^4}{2} + C.$$

So  $v = -\frac{x^2}{2} + \frac{C}{x^2}$ . Since  $v = y^{-2}$ :

$$\boxed{\frac{1}{y^2} = \frac{C}{x^2} - \frac{x^2}{2}}, \quad \text{i.e., } y^2 = \frac{x^2}{C - \frac{x^4}{2}} = \frac{2x^2}{2C - x^4}.$$

26. **Solution.** Separate variables:  $(1 + y) dy = e^x dx$ . Integrate:

$$y + \frac{y^2}{2} = e^x + C.$$

The general solution in implicit form is

$$\boxed{y + \frac{y^2}{2} = e^x + C}.$$

27. **Solution.** Characteristic equation:  $r^2 - 3r + 2 = 0 \implies (r-1)(r-2) = 0$ , so  $r = 1, 2$ . The general solution is

$$\boxed{y = C_1 e^x + C_2 e^{2x}}.$$

28. **Solution.** Characteristic equation:  $r^2 + 1 = 0 \implies r = \pm i$ . The general solution is

$$\boxed{y = C_1 \cos x + C_2 \sin x}.$$

29. **Solution.** Characteristic equation:  $r^2 + 4r + 4 = 0 \implies (r + 2)^2 = 0$ , so  $r = -2$  (repeated). The general solution is

$$\boxed{y = (C_1 + C_2 x)e^{-2x}}.$$

30. **Solution.** Characteristic equation:  $r^3 - 3r^2 + 3r - 1 = 0 \implies (r - 1)^3 = 0$ , so  $r = 1$  with multiplicity 3. The general solution is

$$\boxed{y = (C_1 + C_2 x + C_3 x^2)e^x}.$$

31. **Solution.** Characteristic equation:  $r^4 - 5r^3 + 8r^2 - 4r = 0$ . Factor out  $r$ :

$$r(r^3 - 5r^2 + 8r - 4) = 0.$$

Test  $r = 1$ :  $1 - 5 + 8 - 4 = 0$ , so  $(r - 1)$  is a factor. Divide:  $r^3 - 5r^2 + 8r - 4 = (r - 1)(r^2 - 4r + 4) = (r - 1)(r - 2)^2$ . Therefore the roots are  $r = 0, 1, 2, 2$ . The general solution is

$$\boxed{y = C_1 + C_2 e^x + (C_3 + C_4 x)e^{2x}}.$$

32. **Solution.** The homogeneous equation  $y'' - y = 0$  has roots  $r = \pm 1$ , so  $y_h = C_1 e^x + C_2 e^{-x}$ . Since  $e^x$  is already a solution of the homogeneous equation ( $r = 1$ ), we try  $y_p = A x e^x$  (multiply by  $x$  due to resonance).

Compute:  $y'_p = A e^x + A x e^x = A(1 + x)e^x$ ,  $y''_p = A(2 + x)e^x$ . Substitute:

$$y''_p - y_p = A(2 + x)e^x - A x e^x = 2A e^x = e^x.$$

So  $A = \frac{1}{2}$  and

$$\boxed{y_p = \frac{x}{2} e^x}.$$

33. **Solution.** The homogeneous solution has roots  $r = \pm 1$ . Since  $e^{-x}$  corresponds to  $r = -1$  (a root), we try  $y_p = A x e^{-x}$ .

Compute:  $y'_p = A(e^{-x} - x e^{-x}) = A(1 - x)e^{-x}$ ,  $y''_p = A(-2 + x)e^{-x}$ . Substitute:

$$y''_p - y_p = A(-2 + x)e^{-x} - A x e^{-x} = -2A e^{-x} = e^{-x}.$$

So  $A = -\frac{1}{2}$  and

$$\boxed{y_p = -\frac{x}{2} e^{-x}}.$$

34. **Solution.** The homogeneous equation  $y'' + 2y' + y = 0$  has characteristic root  $r = -1$  (double), so  $y_1 = e^{-x}$ ,  $y_2 = xe^{-x}$ . The Wronskian is

$$W = y_1 y_2' - y_2 y_1' = e^{-x}(e^{-x} - xe^{-x}) - xe^{-x}(-e^{-x}) = e^{-2x} - xe^{-2x} + xe^{-2x} = e^{-2x}.$$

For variation of parameters with  $g(x) = x$ :

$$u_1' = -\frac{y_2 g}{W} = -\frac{xe^{-x} \cdot x}{e^{-2x}} = -x^2 e^x,$$

$$u_2' = \frac{y_1 g}{W} = \frac{e^{-x} \cdot x}{e^{-2x}} = xe^x.$$

Integrate:

$$u_1 = -\int x^2 e^x dx = -(x^2 e^x - 2xe^x + 2e^x) = -(x^2 - 2x + 2)e^x,$$

$$u_2 = \int xe^x dx = (x - 1)e^x.$$

The particular solution is

$$\begin{aligned} y_p &= u_1 y_1 + u_2 y_2 = -(x^2 - 2x + 2)e^x \cdot e^{-x} + (x - 1)e^x \cdot xe^{-x} \\ &= -(x^2 - 2x + 2) + x(x - 1) = -x^2 + 2x - 2 + x^2 - x = x - 2. \end{aligned}$$

$y_p = x - 2.$

35. **Solution.** The homogeneous equation  $y'' - 2y' + y = 0$  has  $r = 1$  (double), so  $y_1 = e^x$ ,  $y_2 = xe^x$ . The Wronskian is

$$W = e^x(e^x + xe^x) - xe^x \cdot e^x = e^{2x} + xe^{2x} - xe^{2x} = e^{2x}.$$

With  $g(x) = xe^x$ :

$$u_1' = -\frac{xe^x \cdot xe^x}{e^{2x}} = -x^2, \quad u_2' = \frac{e^x \cdot xe^x}{e^{2x}} = x.$$

Integrate:  $u_1 = -\frac{x^3}{3}$ ,  $u_2 = \frac{x^2}{2}$ . The particular solution is

$$y_p = -\frac{x^3}{3}e^x + \frac{x^2}{2} \cdot xe^x = \left(-\frac{x^3}{3} + \frac{x^3}{2}\right)e^x = \frac{x^3}{6}e^x.$$

$y_p = \frac{x^3}{6}e^x.$



36. **Solution.** The homogeneous equation  $y'' + y = 0$  has  $y_h = C_1 \cos x + C_2 \sin x$ . Since  $\sin x$  is a homogeneous solution, the particular solution for  $y'' + y = \sin x$  requires the resonance form  $y_p = x(A \cos x + B \sin x)$ .

Compute  $y'_p = (A \cos x + B \sin x) + x(-A \sin x + B \cos x)$ , and

$$y''_p = -2A \sin x + 2B \cos x + x(-A \cos x - B \sin x).$$

Then  $y''_p + y_p = -2A \sin x + 2B \cos x = \sin x$ , giving  $A = -\frac{1}{2}$ ,  $B = 0$ .

So  $y_p = -\frac{x}{2} \cos x$ . The general solution is  $y = C_1 \cos x + C_2 \sin x - \frac{x}{2} \cos x$ .

Apply  $y(0) = 0$ :  $C_1 = 0$ . Then  $y = C_2 \sin x - \frac{x}{2} \cos x$ .

$y' = C_2 \cos x - \frac{1}{2} \cos x + \frac{x}{2} \sin x$ . Apply  $y'(0) = 1$ :  $C_2 - \frac{1}{2} = 1$ , so  $C_2 = \frac{3}{2}$ .

$$y = \frac{3}{2} \sin x - \frac{x}{2} \cos x.$$

37. **Solution.** Characteristic equation:  $r^2 + 2r + 5 = 0 \implies r = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i$ .  
General solution:

$$y = e^{-x}(C_1 \cos 2x + C_2 \sin 2x).$$

Apply  $y(0) = 2$ :  $C_1 = 2$ .

$y' = e^{-x}[(-C_1 + 2C_2) \cos 2x + (-C_2 - 2C_1) \sin 2x]$ . At  $x = 0$ :  $y'(0) = -C_1 + 2C_2 = -2 + 2C_2 = 0$ , so  $C_2 = 1$ .

$$y = e^{-x}(2 \cos 2x + \sin 2x).$$

38. **Solution.** The homogeneous equation  $y'' + 4y = 0$  has solutions  $\cos 2x$ ,  $\sin 2x$ . Since  $\cos 2x$  is a homogeneous solution, we have resonance. Try  $y_p = x(A \cos 2x + B \sin 2x)$ .

$$y'_p = (A \cos 2x + B \sin 2x) + x(-2A \sin 2x + 2B \cos 2x),$$

$$y''_p = -4A \sin 2x + 4B \cos 2x + x(-4A \cos 2x - 4B \sin 2x).$$

$$y''_p + 4y_p = -4A \sin 2x + 4B \cos 2x = \cos 2x.$$

So  $B = \frac{1}{4}$ ,  $A = 0$ :

$$y_p = \frac{x}{4} \sin 2x.$$

39. **Solution.** Characteristic equation:  $r^4 - 4r^2 + 4 = 0 \implies (r^2 - 2)^2 = 0$ , so  $r^2 = 2$  and  $r = \pm\sqrt{2}$ , each with multiplicity 2. The general solution is

$$y = (C_1 + C_2 x)e^{\sqrt{2}x} + (C_3 + C_4 x)e^{-\sqrt{2}x}.$$

40. **Solution.** Characteristic equation:  $r^3 - 2r^2 + r = 0 \implies r(r-1)^2 = 0$ , so  $r = 0, 1, 1$ .  
The homogeneous solution is  $y_h = C_1 + C_2 e^x + C_3 x e^x$ .

Since  $e^x$  corresponds to  $r = 1$  with multiplicity 2, we need  $y_p = Ax^2e^x$  (multiply by  $x^2$ ).

Compute:

$$\begin{aligned}y_p &= Ax^2e^x, & y'_p &= A(2x + x^2)e^x, & y''_p &= A(2 + 4x + x^2)e^x, \\y'''_p &= A(6 + 6x + x^2)e^x.\end{aligned}$$

Substitute into  $y''' - 2y'' + y'$ :

$$\begin{aligned}&A(6 + 6x + x^2)e^x - 2A(2 + 4x + x^2)e^x + A(2x + x^2)e^x \\&= Ae^x[(6 + 6x + x^2) - (4 + 8x + 2x^2) + (2x + x^2)] = Ae^x[2] = 2Ae^x.\end{aligned}$$

Set  $2A = 1$ , so  $A = \frac{1}{2}$ :

$$y_p = \frac{x^2}{2} e^x.$$

41. **Solution.** The homogeneous equation  $y'' - 2y' + y = 0$  has  $r = 1$  (double), so  $y_h = (C_1 + C_2x)e^x$ . Since  $e^x$  is a double root, try  $y_p = Ax^2e^x$ .

$$y_p = Ax^2e^x, \quad y'_p = A(2x + x^2)e^x, \quad y''_p = A(2 + 4x + x^2)e^x.$$

$$y''_p - 2y'_p + y_p = A[(2 + 4x + x^2) - 2(2x + x^2) + (x^2)]e^x = 2Ae^x = 3e^x.$$

So  $A = \frac{3}{2}$ . The general solution is

$$y = (C_1 + C_2x)e^x + \frac{3}{2}x^2e^x.$$

42. **Solution.** Characteristic equation:  $r^3 - 4r^2 + 5r - 2 = 0$ . Test  $r = 1$ :  $1 - 4 + 5 - 2 = 0$ . Factor:  $(r - 1)(r^2 - 3r + 2) = (r - 1)(r - 1)(r - 2) = (r - 1)^2(r - 2) = 0$ .

Roots:  $r = 1$  (double),  $r = 2$ . The general solution is

$$y = (C_1 + C_2x)e^x + C_3e^{2x}.$$

43. **Solution.** The homogeneous equation  $y'' + 2y' + y = 0$  has  $r = -1$  (double). Since  $e^{-x}$  corresponds to the double root, try  $y_p = Ax^2e^{-x}$ .

$$y_p = Ax^2e^{-x}, \quad y'_p = A(2x - x^2)e^{-x}, \quad y''_p = A(2 - 4x + x^2)e^{-x}.$$

$$y''_p + 2y'_p + y_p = A[(2 - 4x + x^2) + 2(2x - x^2) + (x^2)]e^{-x} = 2Ae^{-x} = e^{-x}.$$

So  $A = \frac{1}{2}$ :

$$y_p = \frac{x^2}{2} e^{-x}.$$

44. **Solution.** The homogeneous equation  $y'' + y = 0$  has roots  $r = \pm i$ . The forcing  $e^{ix}$  has  $r = i$ , which is a root. So we try  $y_p = Axe^{ix}$ .

$$y'_p = Ae^{ix} + Aixe^{ix} = A(1 + ix)e^{ix}, \quad y''_p = A(2i - x)e^{ix}.$$

$$y''_p + y_p = A(2i - x)e^{ix} + Axe^{ix} = 2Aie^{ix} = e^{ix}.$$

So  $A = \frac{1}{2i} = -\frac{i}{2}$ . The particular solution is

$$y_p = -\frac{i}{2}xe^{ix} = -\frac{i}{2}x(\cos x + i\sin x) = \frac{x\sin x}{2} - \frac{ix\cos x}{2}.$$

The general solution is  $y = C_1 \cos x + C_2 \sin x - \frac{i}{2}xe^{ix}$ .

Extracting the real part (for the physical solution corresponding to the real part of  $e^{ix} = \cos x$ ):

$$\boxed{\operatorname{Re}(y_p) = \frac{x}{2} \sin x}.$$

This matches the particular solution for  $y'' + y = \cos x$ .

45. **Solution.** Characteristic equation:  $r^4 - 2r^2 + 1 = 0 \implies (r^2 - 1)^2 = 0 \implies (r - 1)^2(r + 1)^2 = 0$ .

Roots:  $r = 1$  (double),  $r = -1$  (double). The general solution is

$$\boxed{y = (C_1 + C_2x)e^x + (C_3 + C_4x)e^{-x}}.$$

46. **Solution.** Characteristic equation:  $r^3 + 3r^2 + 3r + 1 = 0 \implies (r + 1)^3 = 0$ , so  $r = -1$  with multiplicity 3. The homogeneous solution is  $y_h = (C_1 + C_2x + C_3x^2)e^{-x}$ .

Since  $e^{-x}$  corresponds to a triple root, try  $y_p = Ax^3e^{-x}$ .

$y_p = Ax^3e^{-x}$ . Using the annihilator approach: when we substitute, the operator  $(D + 1)^3$  kills the homogeneous part, and  $(D + 1)^3(Ax^3e^{-x}) = A \cdot 6 \cdot e^{-x}$  (since  $\frac{d^3}{dx^3}[x^3] = 6$ ). So

$$(D + 1)^3y_p = 6Ae^{-x} = e^{-x} \implies A = \frac{1}{6}.$$

The general solution is

$$\boxed{y = (C_1 + C_2x + C_3x^2)e^{-x} + \frac{x^3}{6}e^{-x}}.$$

47. **Solution.** We solve  $y'' + y = \sin t$ ,  $y(0) = 0$ ,  $y'(0) = 1$  using Laplace transforms. Let  $Y = \mathcal{L}\{y\}$ .

Taking the Laplace transform:

$$s^2Y - sy(0) - y'(0) + Y = \frac{1}{s^2 + 1}.$$

With  $y(0) = 0$ ,  $y'(0) = 1$ :

$$(s^2 + 1)Y - 1 = \frac{1}{s^2 + 1} \implies Y = \frac{1}{s^2 + 1} + \frac{1}{(s^2 + 1)^2}.$$

We know  $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$  and  $\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^2}\right\} = \frac{\sin t - t \cos t}{2}$ .

Therefore

$$y = \sin t + \frac{\sin t - t \cos t}{2} = \frac{3}{2} \sin t - \frac{t}{2} \cos t.$$

$$\boxed{y = \frac{3}{2} \sin t - \frac{t}{2} \cos t}.$$

This agrees with the answer to Problem 36.

48. **Solution.** Take the Laplace transform of  $y'' - 3y' + 2y = e^{3t}$  with  $y(0) = y'(0) = 0$ :

$$(s^2 - 3s + 2)Y = \frac{1}{s - 3} \implies Y = \frac{1}{(s - 1)(s - 2)(s - 3)}.$$

Partial fractions:  $\frac{1}{(s-1)(s-2)(s-3)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$ .

$$A = \frac{1}{(1-2)(1-3)} = \frac{1}{2}, \quad B = \frac{1}{(2-1)(2-3)} = -1, \quad C = \frac{1}{(3-1)(3-2)} = \frac{1}{2}.$$

Inverse transform:

$$\boxed{y = \frac{1}{2}e^t - e^{2t} + \frac{1}{2}e^{3t}}.$$

49. **Solution.** The homogeneous solutions are  $y_1 = \cos x$ ,  $y_2 = \sin x$ ,  $W = 1$ . With  $g(x) = \tan x$ :

$$u_1' = -\frac{\sin x \tan x}{1} = -\frac{\sin^2 x}{\cos x} = -\frac{1 - \cos^2 x}{\cos x} = \cos x - \sec x,$$

$$u_2' = \frac{\cos x \tan x}{1} = \sin x.$$

Integrate:

$$u_1 = \sin x - \ln |\sec x + \tan x|, \quad u_2 = -\cos x.$$

The particular solution is

$$y_p = u_1 \cos x + u_2 \sin x = (\sin x - \ln |\sec x + \tan x|) \cos x - \cos x \sin x$$

$$= -\cos x \ln |\sec x + \tan x|.$$

$$\boxed{y_p = -\cos x \ln |\sec x + \tan x|}.$$

50. **Solution.** With  $y_1 = \cos x$ ,  $y_2 = \sin x$ ,  $W = 1$ , and  $g(x) = \sec^2 x$ :

$$u'_1 = -\sin x \sec^2 x = -\frac{\sin x}{\cos^2 x}, \quad u'_2 = \cos x \sec^2 x = \sec x.$$

Integrate:

$$u_1 = \int -\frac{\sin x}{\cos^2 x} dx = -\frac{1}{\cos x} = -\sec x,$$

$$u_2 = \int \sec x dx = \ln |\sec x + \tan x|.$$

The particular solution is

$$y_p = -\sec x \cos x + \sin x \ln |\sec x + \tan x| = -1 + \sin x \ln |\sec x + \tan x|.$$

$$\boxed{y_p = -1 + \sin x \ln |\sec x + \tan x|}.$$

51. **Solution.** Take the Laplace transform. Let  $Y = \mathcal{L}\{y\}$ . Note  $\mathcal{L}\left\{\int_0^t y(s) ds\right\} = \frac{Y}{s}$ .

$$sY - y(0) + \frac{Y}{s} = \frac{1}{s}.$$

With  $y(0) = 0$ :

$$sY + \frac{Y}{s} = \frac{1}{s} \implies Y\left(s + \frac{1}{s}\right) = \frac{1}{s} \implies Y \cdot \frac{s^2 + 1}{s} = \frac{1}{s},$$

$$Y = \frac{1}{s^2 + 1}.$$

Inverse transform:

$$\boxed{y(t) = \sin t}.$$

52. **Solution.** Take the Laplace transform of  $y'' + 2y' + 5y = u(t - 3)$  with  $y(0) = 0$ ,  $y'(0) = 1$ :

$$s^2Y - 1 + 2sY + 5Y = \frac{e^{-3s}}{s}.$$

$$(s^2 + 2s + 5)Y = 1 + \frac{e^{-3s}}{s}.$$

Note  $s^2 + 2s + 5 = (s + 1)^2 + 4$ . So

$$Y = \frac{1}{(s + 1)^2 + 4} + \frac{e^{-3s}}{s[(s + 1)^2 + 4]}.$$

The first term inverts to  $\frac{1}{2}e^{-t} \sin 2t$ .

For the second term, use partial fractions on  $\frac{1}{s(s^2 + 2s + 5)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2s + 5}$ .

$A = \frac{1}{5}$ . Then  $1 = \frac{1}{5}(s^2 + 2s + 5) + (Bs + C)s$ . Expanding:  $1 = \frac{s^2}{5} + \frac{2s}{5} + 1 + Bs^2 + Cs$ . Comparing  $s^2$ :  $0 = \frac{1}{5} + B$ , so  $B = -\frac{1}{5}$ . Comparing  $s$ :  $0 = \frac{2}{5} + C$ , so  $C = -\frac{2}{5}$ .

$$\frac{1}{s(s^2 + 2s + 5)} = \frac{1}{5} \cdot \frac{1}{s} - \frac{1}{5} \cdot \frac{s+2}{(s+1)^2 + 4} = \frac{1}{5} \cdot \frac{1}{s} - \frac{1}{5} \cdot \frac{(s+1)+1}{(s+1)^2 + 4}.$$

Inverse:  $\frac{1}{5} - \frac{1}{5}e^{-t} \cos 2t - \frac{1}{10}e^{-t} \sin 2t$ .

Applying the shift by 3 (second shifting theorem):

$$y(t) = \frac{1}{2}e^{-t} \sin 2t + u(t-3) \left[ \frac{1}{5} - \frac{1}{5}e^{-(t-3)} \cos 2(t-3) - \frac{1}{10}e^{-(t-3)} \sin 2(t-3) \right].$$

53. **Solution.** Take the Laplace transform of  $y'' - y = \delta(t-2)$  with  $y(0) = y'(0) = 0$ :

$$(s^2 - 1)Y = e^{-2s} \implies Y = \frac{e^{-2s}}{s^2 - 1} = \frac{e^{-2s}}{(s-1)(s+1)}.$$

Partial fractions:  $\frac{1}{(s-1)(s+1)} = \frac{1/2}{s-1} - \frac{1/2}{s+1}$ . Inverse of  $\frac{1}{s^2-1}$  is  $\sinh t$ . By the second shifting theorem:

$$y(t) = u(t-2) \sinh(t-2) = \frac{u(t-2)}{2} [e^{t-2} - e^{-(t-2)}].$$

54. **Solution.** Take the Laplace transform of  $y'' + 3y' + 2y = te^{-t}$  with  $y(0) = 1$ ,  $y'(0) = 0$ :

$$s^2Y - s + 3(sY - 1) + 2Y = \frac{1}{(s+1)^2}.$$

$$(s^2 + 3s + 2)Y = s + 3 + \frac{1}{(s+1)^2}.$$

$(s^2 + 3s + 2) = (s+1)(s+2)$ . So

$$Y = \frac{s+3}{(s+1)(s+2)} + \frac{1}{(s+1)^3(s+2)}.$$

First term:  $\frac{s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$ .  $A = \frac{1+3}{1} = 2$ ,  $B = \frac{-2+3}{-1} = -1$ . So  $\frac{2}{s+1} - \frac{1}{s+2}$ , giving  $2e^{-t} - e^{-2t}$ .

Second term:  $\frac{1}{(s+1)^3(s+2)}$ . Let  $w = s+1$ :

$$\frac{1}{w^3(w+1)} = \frac{A}{w} + \frac{B}{w^2} + \frac{C}{w^3} + \frac{D}{w+1}.$$

$C = \frac{1}{1} = 1$ .  $D = \frac{1}{(-1)^3} = -1$ . Expand:  $1 = Aw^2(w+1) + Bw(w+1) + C(w+1) + Dw^3$ . Setting  $w = 0$ :  $1 = C$ . Setting  $w = -1$ :  $1 = -D$ , so  $D = -1$ .

Expand fully:  $Aw^3 + Aw^2 + Bw^2 + Bw + Cw + C + Dw^3 = (A+D)w^3 + (A+B)w^2 + (B+C)w + C$ .

$C = 1$ . Coefficient of  $w^3$ :  $A+D = 0$ , so  $A = 1$ . Coefficient of  $w^2$ :  $A+B = 0$ , so  $B = -1$ . Check  $w$ :  $B+C = -1+1 = 0$ . So  $\frac{1}{(s+1)^3(s+2)} = \frac{1}{s+1} - \frac{1}{(s+1)^2} + \frac{1}{(s+1)^3} - \frac{1}{s+2}$ .

Inverse:  $e^{-t} - te^{-t} + \frac{t^2}{2}e^{-t} - e^{-2t}$ .

Combining both terms:

$$y = 2e^{-t} - e^{-2t} + e^{-t} - te^{-t} + \frac{t^2}{2}e^{-t} - e^{-2t} = 3e^{-t} - te^{-t} + \frac{t^2}{2}e^{-t} - 2e^{-2t}.$$

$$\boxed{y(t) = \left(3 - t + \frac{t^2}{2}\right)e^{-t} - 2e^{-2t}}.$$

55. **Solution.** Take the Laplace transform of  $y'' + y = u(t - \pi)$  with  $y(0) = y'(0) = 0$ :

$$(s^2 + 1)Y = \frac{e^{-\pi s}}{s} \implies Y = \frac{e^{-\pi s}}{s(s^2 + 1)}.$$

Partial fractions:  $\frac{1}{s(s^2+1)} = \frac{1}{s} - \frac{s}{s^2+1}$ . Inverse:  $1 - \cos t$ .

By the second shifting theorem:

$$y(t) = u(t - \pi)[1 - \cos(t - \pi)] = u(t - \pi)[1 + \cos t],$$

since  $\cos(t - \pi) = -\cos t$ .

$$\boxed{y(t) = u(t - \pi)(1 + \cos t)}.$$

56. **Solution.** Take the Laplace transform with  $y(0) = 0$ :

$$sY + \frac{Y}{s} = \frac{1}{s+1}.$$

$$Y \cdot \frac{s^2 + 1}{s} = \frac{1}{s+1} \implies Y = \frac{s}{(s+1)(s^2+1)}.$$

Partial fractions:  $\frac{s}{(s+1)(s^2+1)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+1}$ .

$A = \frac{-1}{1+1} = -\frac{1}{2}$ . Then  $s = -\frac{1}{2}(s^2+1) + (Bs+C)(s+1)$ . Expand:  $s = -\frac{s^2}{2} - \frac{1}{2} + Bs^2 + Bs + Cs + C$ .

$s^2$ :  $0 = -\frac{1}{2} + B$ , so  $B = \frac{1}{2}$ . Constant:  $0 = -\frac{1}{2} + C$ , so  $C = \frac{1}{2}$ . Check  $s$ :  $1 = B + C = 1$ .

$$Y = -\frac{1/2}{s+1} + \frac{s/2 + 1/2}{s^2+1} = -\frac{1}{2} \cdot \frac{1}{s+1} + \frac{1}{2} \cdot \frac{s}{s^2+1} + \frac{1}{2} \cdot \frac{1}{s^2+1}.$$

Inverse:

$$\boxed{y(t) = -\frac{1}{2}e^{-t} + \frac{1}{2}\cos t + \frac{1}{2}\sin t}.$$

57. **Solution.** Take the Laplace transform of  $y'' + y = e^t \sin 2t$  with  $y(0) = y'(0) = 0$ . Note  $\mathcal{L}\{e^t \sin 2t\} = \frac{2}{(s-1)^2+4}$ .

$$(s^2 + 1)Y = \frac{2}{(s-1)^2 + 4} = \frac{2}{s^2 - 2s + 5}.$$

$$Y = \frac{2}{(s^2 + 1)(s^2 - 2s + 5)}.$$

Partial fractions:  $\frac{2}{(s^2+1)(s^2-2s+5)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2-2s+5}.$

$$2 = (As + B)(s^2 - 2s + 5) + (Cs + D)(s^2 + 1).$$

$$\text{Set } s = i: 2 = (Ai + B)(4 - 2i) + (Ci + D)(0) = (Ai + B)(4 - 2i).$$

$$(Ai + B)(4 - 2i) = 4Ai - 2Ai^2 + 4B - 2Bi = 2A + 4B + (4A - 2B)i = 2.$$

So  $4A - 2B = 0$  and  $2A + 4B = 2$ . From the first:  $A = B/2$ . Then  $B + 4B = 2$ ,  $B = \frac{2}{5}$ ,  $A = \frac{1}{5}$ .

Set  $s = 1 + 2i$ . Since  $s^2 - 2s + 5 = (s - 1)^2 + 4$ , at  $s = 1 + 2i$  we get  $(2i)^2 + 4 = 0$ . Evaluating  $s^2 + 1$  at  $s = 1 + 2i$ :  $(1 + 2i)^2 + 1 = 1 + 4i - 4 + 1 = -2 + 4i$ .

$$2 = (C(1 + 2i) + D)(-2 + 4i).$$

$$C(1 + 2i) + D = (C + D) + 2Ci. \text{ Multiply by } (-2 + 4i): (-2(C + D) - 4C \cdot 2) + (4(C + D) - 2 \cdot 2C)i = (-2C - 2D - 8C) + (4C + 4D - 4C)i = (-10C - 2D) + 4Di.$$

So  $-10C - 2D = 2$  and  $4D = 0$ , giving  $D = 0$  and  $C = -\frac{1}{5}$ .

$$\begin{aligned} Y &= \frac{1}{5} \cdot \frac{s+2}{s^2+1} - \frac{1}{5} \cdot \frac{s}{s^2-2s+5} = \frac{1}{5} \cdot \frac{s}{s^2+1} + \frac{2}{5} \cdot \frac{1}{s^2+1} - \frac{1}{5} \cdot \frac{(s-1)+1}{(s-1)^2+4} \\ &= \frac{1}{5} \cdot \frac{s}{s^2+1} + \frac{2}{5} \cdot \frac{1}{s^2+1} - \frac{1}{5} \cdot \frac{s-1}{(s-1)^2+4} - \frac{1}{10} \cdot \frac{2}{(s-1)^2+4}. \end{aligned}$$

Inverse:

$$y(t) = \frac{1}{5} \cos t + \frac{2}{5} \sin t - \frac{1}{5} e^t \cos 2t - \frac{1}{10} e^t \sin 2t.$$

58. **Solution.** Take the Laplace transform of  $y'' + 2y' + y = \delta(t - 1) + \delta(t - 2)$  with  $y(0) = y'(0) = 0$ :

$$(s + 1)^2 Y = e^{-s} + e^{-2s} \implies Y = \frac{e^{-s}}{(s + 1)^2} + \frac{e^{-2s}}{(s + 1)^2}.$$

Since  $\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = te^{-t}$ , by the second shifting theorem:

$$y(t) = u(t - 1)(t - 1)e^{-(t-1)} + u(t - 2)(t - 2)e^{-(t-2)}.$$

59. **Solution.** The homogeneous equation  $y'' + 4y = 0$  has  $y_1 = \cos 2x$ ,  $y_2 = \sin 2x$ ,  $W = 2$ . With  $g(x) = e^{2x} \cos 3x$ :

$$u'_1 = -\frac{\sin 2x \cdot e^{2x} \cos 3x}{2}, \quad u'_2 = \frac{\cos 2x \cdot e^{2x} \cos 3x}{2}.$$



Use the product-to-sum identities:  $\sin 2x \cos 3x = \frac{1}{2}[\sin 5x - \sin x]$  and  $\cos 2x \cos 3x = \frac{1}{2}[\cos 5x + \cos x]$ .

$$u_1' = -\frac{e^{2x}}{4}(\sin 5x - \sin x), \quad u_2' = \frac{e^{2x}}{4}(\cos 5x + \cos x).$$

Using  $\int e^{ax} \sin bx \, dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2}$  and  $\int e^{ax} \cos bx \, dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2}$ :

$$u_1 = -\frac{1}{4} \left[ \frac{e^{2x}(2 \sin 5x - 5 \cos 5x)}{29} - \frac{e^{2x}(2 \sin x - \cos x)}{5} \right],$$

$$u_2 = \frac{1}{4} \left[ \frac{e^{2x}(2 \cos 5x + 5 \sin 5x)}{29} + \frac{e^{2x}(2 \cos x + \sin x)}{5} \right].$$

The particular solution  $y_p = u_1 \cos 2x + u_2 \sin 2x$ . After substantial algebra (combining the terms and using product-to-sum identities to simplify  $\cos 2x \sin 5x$ , etc.), one obtains:

$$\boxed{y_p} = \frac{e^{2x}(2 \cos 3x + 3 \sin 3x)}{4 + 9 + 4 \cdot 4}$$

$$= \frac{e^{2x}}{4} \left[ \frac{\cos x + 2 \sin x}{5} + \frac{-\cos 5x + 10 \sin 5x - 3 \cos 3x + 2 \sin 3x}{29} \right].$$

A cleaner form is obtained by the method of undetermined coefficients. Try  $y_p = e^{2x}(A \cos 3x + B \sin 3x)$ . Substituting into  $y'' + 4y$ :

$$y_p' = e^{2x}[(2A - 3B) \cos 3x + (2B + 3A) \sin 3x].$$

$$y_p'' = e^{2x}[(-5A - 12B) \cos 3x + (-5B + 12A) \sin 3x].$$

$$y_p'' + 4y_p = e^{2x}[(-5A - 12B + 4A) \cos 3x + (-5B + 12A + 4B) \sin 3x]$$

$$= e^{2x}[(-A - 12B) \cos 3x + (12A - B) \sin 3x] = e^{2x} \cos 3x.$$

So  $-A - 12B = 1$  and  $12A - B = 0$ , i.e.,  $B = 12A$  and  $-A - 144A = 1$ ,  $A = -\frac{1}{145}$ ,  $B = -\frac{12}{145}$ .

$$\boxed{y_p = -\frac{e^{2x}}{145}(\cos 3x + 12 \sin 3x)}.$$

60. **Solution.** Assume  $u(x, t) = X(x)T(t)$ . Substituting into  $u_t = \alpha u_{xx}$ :

$$XT' = \alpha X''T \implies \frac{T'}{\alpha T} = \frac{X''}{X} = -\lambda.$$

**Spatial problem:**  $X'' + \lambda X = 0$ ,  $X(0) = X(L) = 0$ . For nontrivial solutions,  $\lambda_n = \left(\frac{n\pi}{L}\right)^2$  with  $X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$ ,  $n = 1, 2, 3, \dots$

**Temporal problem:**  $T' + \alpha \lambda_n T = 0 \implies T_n(t) = e^{-\alpha(n\pi/L)^2 t}$ .

By superposition:

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha(n\pi/L)^2 t},$$

where the coefficients are determined by the initial condition  $u(x, 0) = f(x)$ :

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

61. **Solution.** Assume  $u(x, y) = X(x)Y(y)$ . Substituting into  $u_{xx} + u_{yy} = 0$ :

$$X''Y + XY'' = 0 \implies \frac{X''}{X} = -\frac{Y''}{Y} = -\lambda.$$

**X-problem:**  $X'' + \lambda X = 0$ ,  $X(0) = X(L) = 0$ . Eigenvalues:  $\lambda_n = (n\pi/L)^2$ , eigenfunctions  $X_n = \sin(n\pi x/L)$ .

**Y-problem:**  $Y'' - \lambda_n Y = 0$ ,  $Y(0) = 0$ . General solution:  $Y = A \cosh(n\pi y/L) + B \sinh(n\pi y/L)$ . From  $Y(0) = 0$ :  $A = 0$ , so  $Y_n = \sinh(n\pi y/L)$ .

Superposition:

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi y}{L}\right).$$

Apply  $u(x, H) = g(x)$ :

$$g(x) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi H}{L}\right) \sin\left(\frac{n\pi x}{L}\right).$$

$$B_n = \frac{2}{L \sinh(n\pi H/L)} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

62. **Solution.** Assume  $u(x, t) = X(x)T(t)$ . From  $u_{tt} = c^2 u_{xx}$ :

$$\frac{T''}{c^2 T} = \frac{X''}{X} = -\lambda.$$

**Spatial:**  $X'' + \lambda X = 0$ ,  $X(0) = X(L) = 0$ . Eigenvalues  $\lambda_n = (n\pi/L)^2$ ,  $X_n = \sin(n\pi x/L)$ .

**Temporal:**  $T'' + c^2 \lambda_n T = 0 \implies T_n = A_n \cos(cn\pi t/L) + B_n \sin(cn\pi t/L)$ .

Superposition:

$$u(x, t) = \sum_{n=1}^{\infty} \left[ A_n \cos\left(\frac{cn\pi t}{L}\right) + B_n \sin\left(\frac{cn\pi t}{L}\right) \right] \sin\left(\frac{n\pi x}{L}\right).$$

From  $u(x, 0) = f(x)$ :  $A_n = \frac{2}{L} \int_0^L f(x) \sin(n\pi x/L) dx$ .

From  $u_t(x, 0) = g(x)$ :

$$B_n = \frac{2}{cn\pi} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

63. **Solution.** The PDE  $u_x + 2u_y = 3$  with  $u(x, 0) = \sin x$ .

Characteristic equations:  $\frac{dx}{1} = \frac{dy}{2} = \frac{du}{3}$ .

From  $\frac{dx}{1} = \frac{dy}{2}$ :  $y = 2x + c_1$ , so  $c_1 = y - 2x$  is constant along characteristics.

From  $\frac{dx}{1} = \frac{du}{3}$ :  $u = 3x + c_2$ , so  $c_2 = u - 3x$  is constant.

General solution:  $u - 3x = \phi(y - 2x)$ , i.e.,  $u = 3x + \phi(y - 2x)$ .

Apply  $u(x, 0) = \sin x$ :  $\sin x = 3x + \phi(-2x)$ , so  $\phi(-2x) = \sin x - 3x$ . Let  $s = -2x$ , then  $x = -s/2$  and  $\phi(s) = \sin(-s/2) - 3(-s/2) = -\sin(s/2) + \frac{3s}{2}$ .

Therefore  $\phi(y - 2x) = -\sin\left(\frac{y-2x}{2}\right) + \frac{3(y-2x)}{2}$  and

$$u = 3x - \sin\left(\frac{y-2x}{2}\right) + \frac{3y-6x}{2} = \frac{3y}{2} - \sin\left(\frac{y-2x}{2}\right).$$

$$\boxed{u(x, y) = \frac{3y}{2} - \sin\left(\frac{y-2x}{2}\right)}.$$

64. **Solution.** The PDE  $u u_x + u_y = 0$  with  $u(x, 0) = x$ .

Parametrize characteristics by  $s$  with initial data at  $y = 0$ :  $x(0) = x_0$ ,  $y(0) = 0$ ,  $u(0) = x_0$ .

Characteristic equations:  $\frac{dx}{ds} = u$ ,  $\frac{dy}{ds} = 1$ ,  $\frac{du}{ds} = 0$ .

From  $du/ds = 0$ :  $u = x_0$  (constant along characteristics).

From  $dy/ds = 1$ :  $y = s$ .

From  $dx/ds = u = x_0$ :  $x = x_0 s + x_0 = x_0(1 + s) = x_0(1 + y)$ .

So  $x_0 = \frac{x}{1+y}$  and  $u = x_0$ :

$$\boxed{u(x, y) = \frac{x}{1+y}},$$

valid for  $y > -1$  (before characteristics cross).

65. **Solution.** We seek the solution of  $u_t = u_{xx} + H(x - L/2)$  with  $u(0, t) = u(L, t) = 0$ ,  $u(x, 0) = 0$ .

First find the steady-state solution  $v(x)$  satisfying  $v'' + H(x - L/2) = 0$ ,  $v(0) = v(L) = 0$ , then write  $u = v(x) + w(x, t)$  where  $w$  satisfies the homogeneous heat equation with adjusted initial condition.

For  $0 < x < L/2$ :  $v'' = 0$ , so  $v = Ax + B$ . With  $v(0) = 0$ :  $B = 0$ , so  $v = Ax$ .

For  $L/2 < x < L$ :  $v'' = -1$ , so  $v = -\frac{x^2}{2} + Cx + D$ .

Continuity at  $x = L/2$ :  $\frac{AL}{2} = -\frac{L^2}{8} + \frac{CL}{2} + D$ .

Continuity of  $v'$  at  $x = L/2$ :  $A = -\frac{L}{2} + C$ .

At  $x = L$ :  $v(L) = 0$ :  $-\frac{L^2}{2} + CL + D = 0$ .

From the last:  $D = \frac{L^2}{2} - CL$ . Substituting into continuity:  $\frac{AL}{2} = -\frac{L^2}{8} + \frac{CL}{2} + \frac{L^2}{2} - CL = \frac{3L^2}{8} - \frac{CL}{2}$ .

From  $A = C - L/2$ :  $(C - L/2)\frac{L}{2} = \frac{3L^2}{8} - \frac{CL}{2}$ , i.e.,  $\frac{CL}{2} - \frac{L^2}{4} = \frac{3L^2}{8} - \frac{CL}{2}$ , so  $CL = \frac{3L^2}{8} + \frac{L^2}{4} = \frac{5L^2}{8}$ ,  $C = \frac{5L}{8}$ .

Then  $A = \frac{5L}{8} - \frac{L}{2} = \frac{L}{8}$  and  $D = \frac{L^2}{2} - \frac{5L^2}{8} = -\frac{L^2}{8}$ .

Steady state:

$$v(x) = \begin{cases} \frac{Lx}{8}, & 0 < x < L/2, \\ -\frac{x^2}{2} + \frac{5Lx}{8} - \frac{L^2}{8}, & L/2 < x < L. \end{cases}$$

Now  $w = u - v$  satisfies  $w_t = w_{xx}$ ,  $w(0, t) = w(L, t) = 0$ ,  $w(x, 0) = -v(x)$ .

$$w(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 t},$$

where  $B_n = \frac{2}{L} \int_0^L [-v(x)] \sin(n\pi x/L) dx = -\frac{2}{L} \int_0^L v(x) \sin(n\pi x/L) dx$ .

$$u(x, t) = v(x) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 t},$$

with  $v(x)$  as above and  $B_n = -\frac{2}{L} \int_0^L v(x) \sin(n\pi x/L) dx$ .

66. **Solution.** Seek a steady-state particular solution:  $u_p(x)$  satisfying  $u_p'' + \sin(\pi x) = 0$ ,  $u_p(0) = u_p(1) = 0$ :

$$u_p'' = -\sin(\pi x) \implies u_p = \frac{\sin(\pi x)}{\pi^2}.$$

(Check:  $u_p'' = -\sin(\pi x)$ . ✓;  $u_p(0) = u_p(1) = 0$ . ✓.)

Write  $u = u_p + w$  where  $w_t = w_{xx}$ ,  $w(0, t) = w(1, t) = 0$ ,  $w(x, 0) = -u_p(x) = -\frac{\sin(\pi x)}{\pi^2}$ .

The solution is  $w(x, t) = -\frac{1}{\pi^2} \sin(\pi x) e^{-\pi^2 t}$  (only the  $n = 1$  Fourier mode is present). Therefore:

$$u(x, t) = \frac{\sin(\pi x)}{\pi^2} (1 - e^{-\pi^2 t}).$$

67. **Solution.** Assume  $u(r, \theta) = R(r)\Theta(\theta)$ . Substituting into Laplace's equation in polar coordinates:

$$r^2 \frac{R''}{R} + r \frac{R'}{R} = -\frac{\Theta''}{\Theta} = \lambda.$$

**Angular problem:**  $\Theta'' + \lambda\Theta = 0$  with  $\Theta(\theta + 2\pi) = \Theta(\theta)$  (periodicity). Eigenvalues:  $\lambda_n = n^2$ ,  $n = 0, 1, 2, \dots$  with  $\Theta_0 = 1$ ,  $\Theta_n = A_n \cos n\theta + B_n \sin n\theta$ .

**Radial problem:**  $r^2 R'' + rR' - n^2 R = 0$  (Euler equation). Solutions:  $R = r^n$  and  $R = r^{-n}$ . For boundedness at  $r = 0$ , take  $R_n = r^n$ .

Superposition:

$$u(r, \theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} r^n (A_n \cos n\theta + B_n \sin n\theta).$$

Apply  $u(a, \theta) = f(\theta)$ :

$$f(\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} a^n (A_n \cos n\theta + B_n \sin n\theta).$$

The Fourier coefficients are:

$$\boxed{A_n = \frac{1}{\pi a^n} \int_{-\pi}^{\pi} f(\theta) \cos n\theta \, d\theta, \quad B_n = \frac{1}{\pi a^n} \int_{-\pi}^{\pi} f(\theta) \sin n\theta \, d\theta.}$$

This is the Poisson integral representation of the solution.

68. **Solution.** Take the Fourier transform in  $x$ : let  $\hat{u}(k, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ikx} dx$ .

The wave equation becomes  $\hat{u}_{tt} = -c^2 k^2 \hat{u}$ , with  $\hat{u}(k, 0) = 1$  (since  $\mathcal{F}\{\delta(x)\} = 1$ ) and  $\hat{u}_t(k, 0) = 0$ .

Solution:  $\hat{u}(k, t) = \cos(ckt)$ .

Invert:

$$\begin{aligned} u(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos(ckt) e^{ikx} dk = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{i(x+ct)k} + e^{i(x-ct)k}}{2} dk \\ &= \frac{1}{2} [\delta(x+ct) + \delta(x-ct)]. \end{aligned}$$

$$\boxed{u(x, t) = \frac{1}{2} [\delta(x-ct) + \delta(x+ct)]}.$$

This is d'Alembert's solution for impulsive initial displacement.

69. **Solution.** The Green's function  $G(x, \xi)$  satisfies  $-G_{xx} = \delta(x - \xi)$  with  $G(0, \xi) = G(L, \xi) = 0$ .

For  $x < \xi$ :  $G'' = 0$ , so  $G = Ax$ , using  $G(0) = 0$ .

For  $x > \xi$ :  $G'' = 0$ , so  $G = B(L - x)$ , using  $G(L) = 0$ .

Continuity at  $x = \xi$ :  $A\xi = B(L - \xi)$ .

Jump condition (integrate  $-G'' = \delta$  across  $\xi$ ): since  $G'(\xi^-) = A$  and  $G'(\xi^+) = -B$ , we obtain  $-G'(\xi^+) + G'(\xi^-) = B + A = 1$ . Combined with  $A\xi = B(L - \xi)$ :

From  $B = 1 - A$ :  $A\xi = (1 - A)(L - \xi) = L - \xi - AL + A\xi$ , so  $0 = L - \xi - AL$ , hence  $A = \frac{L-\xi}{L}$  and  $B = \frac{\xi}{L}$ .

$$G(x, \xi) = \begin{cases} \frac{(L-\xi)x}{L}, & 0 \leq x \leq \xi, \\ \frac{\xi(L-x)}{L}, & \xi \leq x \leq L. \end{cases}$$

The solution is:

$$u(x) = \int_0^L G(x, \xi) f(\xi) d\xi = \frac{1}{L} \left[ (L-x) \int_0^x \xi f(\xi) d\xi + x \int_x^L (L-\xi) f(\xi) d\xi \right].$$

70. **Solution.** The Fourier sine coefficients are

$$B_n = \frac{2}{L} \int_0^L x(L-x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

Expand  $x(L-x) = Lx - x^2$ . Using standard integrals:

$$\int_0^L x \sin\left(\frac{n\pi x}{L}\right) dx = \frac{L^2(-1)^{n+1}}{n\pi},$$

Using integration by parts:

$$\int_0^L x^2 \sin\left(\frac{n\pi x}{L}\right) dx = \left[ -\frac{L}{n\pi} x^2 \cos\left(\frac{n\pi x}{L}\right) \right]_0^L + \frac{2L}{n\pi} \int_0^L x \cos\left(\frac{n\pi x}{L}\right) dx.$$

The first term is  $-\frac{L^3}{n\pi}(-1)^n$ . For the remaining integral:

$$\int_0^L x \cos\left(\frac{n\pi x}{L}\right) dx = \frac{L^2}{(n\pi)^2} [(-1)^n - 1].$$

So  $\int_0^L x^2 \sin(n\pi x/L) dx = -\frac{L^3(-1)^n}{n\pi} + \frac{2L^3}{(n\pi)^3} [(-1)^n - 1]$ .

Therefore:

$$\begin{aligned} B_n &= \frac{2}{L} \left[ \frac{L^3(-1)^{n+1}}{n\pi} + \frac{L^3(-1)^n}{n\pi} - \frac{2L^3}{(n\pi)^3} [(-1)^n - 1] \right] \\ &= \frac{2}{L} \cdot \left[ -\frac{2L^3}{(n\pi)^3} [(-1)^n - 1] \right] = \frac{-4L^2}{(n\pi)^3} [(-1)^n - 1]. \end{aligned}$$

When  $n$  is even,  $(-1)^n - 1 = 0$ , so  $B_n = 0$ . When  $n$  is odd,  $(-1)^n - 1 = -2$ :

$$B_n = \frac{-4L^2}{(n\pi)^3}(-2) = \frac{8L^2}{(n\pi)^3}.$$

$$f(x) = x(L-x) = \frac{8L^2}{\pi^3} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^3} \sin\left(\frac{n\pi x}{L}\right).$$

71. **Solution.** The PDE  $u_t + u_x = 0$  with  $u(x, 0) = e^{-x^2}$ .

Characteristic equations:  $\frac{dt}{1} = \frac{dx}{1}$ , so  $x - t = c$  (constant). Along characteristics,  $u$  is constant:  $\frac{du}{dt} = 0$  following  $dx/dt = 1$ .

So  $u(x, t) = \phi(x - t)$  where  $\phi$  is determined by the initial condition:  $u(x, 0) = \phi(x) = e^{-x^2}$ .

$$u(x, t) = e^{-(x-t)^2}.$$

72. **Solution.** The PDE  $xu_x + u_y = 0$  with  $u(x, 0) = \ln(1 + x^2)$ .

Characteristic equations:  $\frac{dx}{x} = \frac{dy}{1} = \frac{du}{0}$ .

From  $\frac{dx}{x} = dy$ :  $\ln|x| = y + c_1$ , so  $xe^{-y} = c_1$  (constant along characteristics).

From  $du = 0$ :  $u$  is constant along characteristics.

General solution:  $u = \phi(xe^{-y})$ .

Apply  $u(x, 0) = \ln(1 + x^2)$ :  $\phi(x) = \ln(1 + x^2)$ .

$$\boxed{u(x, y) = \ln(1 + x^2 e^{-2y})}.$$

73. **Solution.** The heat equation  $u_t = u_{xx}$  on  $0 < x < 1$  with  $u(0, t) = u(1, t) = 0$  and  $u(x, 0) = x(1 - x)$ .

The solution is  $u(x, t) = \sum_{n=1}^{\infty} B_n \sin(n\pi x) e^{-n^2\pi^2 t}$ , where

$$B_n = 2 \int_0^1 x(1 - x) \sin(n\pi x) dx.$$

From Problem 70 with  $L = 1$ :  $B_n = 0$  for  $n$  even, and  $B_n = \frac{8}{(n\pi)^3}$  for  $n$  odd.

The first three nonzero terms ( $n = 1, 3, 5$ ):

$$\boxed{u(x, t) \approx \frac{8}{\pi^3} \sin(\pi x) e^{-\pi^2 t} + \frac{8}{27\pi^3} \sin(3\pi x) e^{-9\pi^2 t} + \frac{8}{125\pi^3} \sin(5\pi x) e^{-25\pi^2 t}}.$$

74. **Solution.** With Neumann conditions  $u_x(0, t) = u_x(L, t) = 0$ , separation of variables  $u = X(x)T(t)$  gives:

$X'' + \lambda X = 0$ ,  $X'(0) = X'(L) = 0$ . Eigenvalues:  $\lambda_0 = 0$  with  $X_0 = 1$ , and  $\lambda_n = (n\pi/L)^2$  with  $X_n = \cos(n\pi x/L)$ ,  $n = 1, 2, \dots$

$T' + \lambda_n T = 0 \implies T_n = e^{-(n\pi/L)^2 t}$ , and  $T_0 = 1$ .

Superposition:

$$u(x, t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 t},$$

where the Fourier cosine coefficients are:

$$\boxed{A_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad n = 0, 1, 2, \dots}$$

75. **Solution.** Take the Fourier transform  $\hat{u}(k, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ikx} dx$ :

$$\hat{u}_t + ick\hat{u} = -Dk^2\hat{u} \implies \hat{u}_t = -(Dk^2 + ick)\hat{u}.$$

Solution:  $\hat{u}(k, t) = \hat{\phi}(k) e^{-(Dk^2 + ick)t}$ , where  $\hat{\phi} = \mathcal{F}\{\phi\}$ .

Inverse transform:

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\phi}(k) e^{ikx - (Dk^2 + ick)t} dk = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\phi}(k) e^{ik(x-ct) - Dk^2t} dk.$$

This is a convolution:

$$u(x, t) = \frac{1}{\sqrt{4\pi Dt}} \int_{-\infty}^{\infty} \phi(\xi) \exp\left(-\frac{(x-ct-\xi)^2}{4Dt}\right) d\xi.$$

The solution is the initial profile  $\phi$  advected at speed  $c$  and diffused with diffusivity  $D$ .

76. **Solution.** Take the Laplace transform in  $t$ : let  $U(x, s) = \mathcal{L}\{u(x, t)\}$ . With  $u(x, 0) = 0$ :

$$sU = U_{xx} \implies U_{xx} - sU = 0.$$

For  $x > 0$  with boundedness as  $x \rightarrow \infty$ :  $U(x, s) = A(s) e^{-\sqrt{s}x}$  (taking  $\text{Re}(\sqrt{s}) > 0$ ).

Boundary condition:  $U(0, s) = G(s) = \mathcal{L}\{g(t)\}$ , so  $A(s) = G(s)$ :

$$U(x, s) = G(s) e^{-x\sqrt{s}}.$$

Using the known inverse  $\mathcal{L}^{-1}\{e^{-x\sqrt{s}}\} = \frac{x}{2\sqrt{\pi}t^{3/2}} e^{-x^2/(4t)}$ , and applying the convolution theorem:

$$u(x, t) = \frac{x}{2\sqrt{\pi}} \int_0^t \frac{g(\tau)}{(t-\tau)^{3/2}} e^{-x^2/[4(t-\tau)]} d\tau.$$

77. **Solution.** The PDE  $(1+x^2)u_x + u_y = 0$  with  $u(x, 0) = \frac{1}{1+x^2}$ .

Characteristic equations:  $\frac{dx}{1+x^2} = \frac{dy}{1} = \frac{du}{0}$ .

From  $\frac{dx}{1+x^2} = dy$ :  $\arctan x = y + c$ , so  $c = \arctan x - y$ .

Since  $u$  is constant along characteristics:  $u = \phi(\arctan x - y)$ .

Apply  $u(x, 0) = \frac{1}{1+x^2}$ :  $\phi(\arctan x) = \frac{1}{1+x^2}$ . Let  $\theta = \arctan x$ , then  $x = \tan \theta$  and  $\frac{1}{1+x^2} = \cos^2 \theta$ . So  $\phi(\theta) = \cos^2 \theta$ .

Therefore:

$$u(x, y) = \cos^2(\arctan x - y).$$

78. **Solution.** Assume  $u(x, y, t) = X(x)Y(y)T(t)$ . From  $u_{tt} = c^2(u_{xx} + u_{yy})$ :

$$\frac{T''}{c^2T} = \frac{X''}{X} + \frac{Y''}{Y}.$$

Separate:  $\frac{X''}{X} = -\mu$ ,  $\frac{Y''}{Y} = -\nu$ ,  $\frac{T''}{c^2T} = -(\mu + \nu)$ .

**X-problem:**  $X'' + \mu X = 0$ ,  $X(0) = X(L) = 0$ . Eigenvalues  $\mu_m = (m\pi/L)^2$ ,  $X_m = \sin(m\pi x/L)$ .

**Y-problem:**  $Y'' + \nu Y = 0$ ,  $Y(0) = Y(H) = 0$ . Eigenvalues  $\nu_n = (n\pi/H)^2$ ,  $Y_n = \sin(n\pi y/H)$ .



**T-problem:**  $T'' + c^2\omega_{mn}^2 T = 0$  where  $\omega_{mn}^2 = \mu_m + \nu_n = (m\pi/L)^2 + (n\pi/H)^2$ .

$$T_{mn} = A_{mn} \cos(c\omega_{mn}t) + B_{mn} \sin(c\omega_{mn}t).$$

Superposition:

$$u = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} [A_{mn} \cos(c\omega_{mn}t) + B_{mn} \sin(c\omega_{mn}t)] \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi y}{H}\right).$$

From  $u(x, y, 0) = f(x, y)$ :

$$A_{mn} = \frac{4}{LH} \int_0^L \int_0^H f(x, y) \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi y}{H}\right) dy dx.$$

From  $u_t(x, y, 0) = g(x, y)$ :

$$\boxed{B_{mn} = \frac{4}{c\omega_{mn}LH} \int_0^L \int_0^H g(x, y) \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi y}{H}\right) dy dx},$$

where  $\omega_{mn} = \pi\sqrt{(m/L)^2 + (n/H)^2}$ .

# Chapter 7

## Chapter 7: Complex Variables

1. **Solution.** Let  $z = 3 - 4i$ . The modulus is

$$|z| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

The complex conjugate is  $\bar{z} = 3 + 4i$ . We verify:

$$z\bar{z} = (3 - 4i)(3 + 4i) = 9 + 12i - 12i - 16i^2 = 9 + 16 = 25 = 5^2 = |z|^2. \quad \square$$

2. **Solution.** Let  $z_1 = 2 + 3i$  and  $z_2 = -1 + 4i$ .

(a)  $z_1 + z_2 = (2 + (-1)) + (3 + 4)i = 1 + 7i$ .

(b)  $z_1 - z_2 = (2 - (-1)) + (3 - 4)i = 3 - i$ .

(c)  $z_1 \cdot z_2 = (2 + 3i)(-1 + 4i) = -2 + 8i - 3i + 12i^2 = -2 + 5i - 12 = -14 + 5i$ .

(d) We rationalize:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{2 + 3i}{-1 + 4i} \cdot \frac{-1 - 4i}{-1 - 4i} = \frac{(2 + 3i)(-1 - 4i)}{(-1)^2 + 4^2} \\ &= \frac{-2 - 8i - 3i - 12i^2}{1 + 16} = \frac{-2 - 11i + 12}{17} = \frac{10 - 11i}{17} = \frac{10}{17} - \frac{11}{17}i. \end{aligned}$$

3. **Solution.** Let  $z = -1 - i$ . The modulus is

$$|z| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}.$$

Since  $z$  lies in the third quadrant ( $\operatorname{Re}(z) < 0$ ,  $\operatorname{Im}(z) < 0$ ), the argument is

$$\arg(z) = \pi + \arctan\left(\frac{1}{1}\right) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}.$$

(Equivalently,  $\arg(z) = -\frac{3\pi}{4}$  if we use the range  $(-\pi, \pi]$ .) The polar form is

$$z = \sqrt{2} e^{i5\pi/4} = \sqrt{2} \left( \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right).$$

4. **Solution.** Euler's formula states  $e^{i\theta} = \cos \theta + i \sin \theta$ . We verify by comparing Taylor series. The Taylor series of  $e^{i\theta}$  is

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots$$

Using  $i^2 = -1$ ,  $i^3 = -i$ ,  $i^4 = 1$ :

$$= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \cdots$$

Separating real and imaginary parts:

$$= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right) = \cos \theta + i \sin \theta.$$

This confirms Euler's formula by comparing the real part with the Taylor series of  $\cos \theta$  and the imaginary part with that of  $\sin \theta$ .

5. **Solution.** First, note  $\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = e^{i\pi/4}$ , which has modulus 1 and argument  $\pi/4$ .

By De Moivre's theorem:

$$(e^{i\pi/4})^5 = e^{i5\pi/4} = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}.$$

6. **Solution.** We solve  $z^3 = 8i$ . Write  $8i = 8e^{i\pi/2}$ . The cube roots are

$$z_k = 8^{1/3} e^{i(\pi/2+2\pi k)/3} = 2 e^{i(\pi/6+2\pi k/3)}, \quad k = 0, 1, 2.$$

$$k = 0: z_0 = 2e^{i\pi/6} = 2 \left( \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 2 \left( \frac{\sqrt{3}}{2} + \frac{i}{2} \right) = \sqrt{3} + i.$$

$$k = 1: z_1 = 2e^{i5\pi/6} = 2 \left( \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = 2 \left( -\frac{\sqrt{3}}{2} + \frac{i}{2} \right) = -\sqrt{3} + i.$$

$$k = 2: z_2 = 2e^{i3\pi/2} = 2 \left( \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = 2(0 - i) = -2i.$$

7. **Solution.**  $z^2 + 4 = 0 \implies z^2 = -4$ . Write  $-4 = 4e^{i\pi}$ . The square roots are

$$z_k = 2e^{i(\pi+2\pi k)/2} = 2e^{i(\pi/2+\pi k)}, \quad k = 0, 1.$$

$$k = 0: z_0 = 2e^{i\pi/2} = 2i. \quad k = 1: z_1 = 2e^{i3\pi/2} = -2i.$$

The solutions are  $z = \pm 2i$ .

8. **Solution.** Multiplication by  $-i = e^{-i\pi/2}$  rotates a complex number by  $-\pi/2$  (i.e.,  $90^\circ$  clockwise) while preserving its modulus.

**Illustration:** Take  $z = 1 + i$ . Then

$$(-i)(1 + i) = -i - i^2 = -i + 1 = 1 - i.$$

The original point  $(1, 1)$  is mapped to  $(1, -1)$ , which is indeed a  $90^\circ$  clockwise rotation about the origin. Both have modulus  $\sqrt{2}$ ; the argument changes from  $\pi/4$  to  $-\pi/4$ , a decrease of  $\pi/2$ .

9. **Solution.** The equation  $|z - (1 + 2i)| = 3$  describes the set of all points in the complex plane at distance 3 from the point  $1 + 2i$ . Writing  $z = x + iy$ :

$$|(x - 1) + i(y - 2)| = 3 \implies (x - 1)^2 + (y - 2)^2 = 9.$$

This is a **circle** centered at  $(1, 2)$  with radius 3.

10. **Solution.** The 5th roots of unity are the solutions of  $z^5 = 1$ . They are

$$z_k = e^{2\pi i k/5}, \quad k = 0, 1, 2, 3, 4.$$

Explicitly:

$$\begin{aligned} z_0 &= 1, \\ z_1 &= e^{2\pi i/5} = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \approx 0.309 + 0.951i, \\ z_2 &= e^{4\pi i/5} = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} \approx -0.809 + 0.588i, \\ z_3 &= e^{6\pi i/5} = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} \approx -0.809 - 0.588i, \\ z_4 &= e^{8\pi i/5} = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} \approx 0.309 - 0.951i. \end{aligned}$$

These five points are equally spaced on the unit circle, forming the vertices of a regular pentagon.

11. **Solution.** Let  $z = 5 - 2i$ . Then  $\bar{z} = 5 + 2i$  and

$$z\bar{z} = (5 - 2i)(5 + 2i) = 25 + 10i - 10i - 4i^2 = 25 + 4 = 29.$$

Also,  $|z|^2 = 5^2 + (-2)^2 = 25 + 4 = 29$ . Hence  $z\bar{z} = |z|^2 = 29$ .  $\square$

12. **Solution.** Write  $z = x + iy$ . Then  $|z - 2| = |z + 2|$  becomes

$$\sqrt{(x - 2)^2 + y^2} = \sqrt{(x + 2)^2 + y^2}.$$

Squaring both sides:

$$(x - 2)^2 + y^2 = (x + 2)^2 + y^2 \implies x^2 - 4x + 4 = x^2 + 4x + 4 \implies -8x = 0 \implies x = 0.$$

The locus is the **imaginary axis** (the line  $x = 0$ ). Geometrically, this is the set of all points equidistant from 2 and  $-2$ , which is the perpendicular bisector of the segment joining  $-2$  and  $2$  on the real axis.

13. **Solution.** Using the difference of cubes factorization:

$$z^3 - 8 = (z - 2)(z^2 + 2z + 4).$$

The quadratic  $z^2 + 2z + 4 = 0$  has roots

$$z = \frac{-2 \pm \sqrt{4 - 16}}{2} = \frac{-2 \pm \sqrt{-12}}{2} = \frac{-2 \pm 2i\sqrt{3}}{2} = -1 \pm i\sqrt{3}.$$

Therefore the complete factorization over  $\mathbb{C}$  is

$$z^3 - 8 = (z - 2)(z - (-1 + i\sqrt{3}))(z - (-1 - i\sqrt{3})),$$

and the roots are  $z = 2, -1 + i\sqrt{3}, -1 - i\sqrt{3}$ .

14. **Solution.**  $z^4 = -16 = 16e^{i\pi}$ . The fourth roots are

$$z_k = 16^{1/4} e^{i(\pi+2\pi k)/4} = 2 e^{i(\pi/4+\pi k/2)}, \quad k = 0, 1, 2, 3.$$

$$k = 0: z_0 = 2e^{i\pi/4} = 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = \sqrt{2} + i\sqrt{2}.$$

$$k = 1: z_1 = 2e^{i3\pi/4} = 2\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} + i\sqrt{2}.$$

$$k = 2: z_2 = 2e^{i5\pi/4} = 2\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = -\sqrt{2} - i\sqrt{2}.$$

$$k = 3: z_3 = 2e^{i7\pi/4} = 2\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = \sqrt{2} - i\sqrt{2}.$$

15. **Solution.** We solve  $z^2 + (3 - 2i)z + (5 - i) = 0$  using the quadratic formula. The discriminant is

$$\Delta = (3 - 2i)^2 - 4(5 - i) = 9 - 12i + 4i^2 - 20 + 4i = 9 - 12i - 4 - 20 + 4i = -15 - 8i.$$

We need  $\sqrt{-15 - 8i}$ . Let  $\sqrt{-15 - 8i} = a + bi$ . Then  $a^2 - b^2 = -15$  and  $2ab = -8$ , so  $b = -4/a$ . Substituting:

$$a^2 - \frac{16}{a^2} = -15 \implies a^4 + 15a^2 - 16 = 0 \implies (a^2 + 16)(a^2 - 1) = 0.$$

So  $a^2 = 1$ , giving  $a = 1, b = -4$  (or  $a = -1, b = 4$ ). Thus  $\sqrt{-15 - 8i} = \pm(1 - 4i)$ .

The solutions are

$$z = \frac{-(3 - 2i) \pm (1 - 4i)}{2} = \frac{-3 + 2i \pm (1 - 4i)}{2}.$$

$$z_1 = \frac{-3 + 2i + 1 - 4i}{2} = \frac{-2 - 2i}{2} = -1 - i.$$

$$z_2 = \frac{-3 + 2i - 1 + 4i}{2} = \frac{-4 + 6i}{2} = -2 + 3i.$$

**Verification:** For  $z_1 = -1 - i$ :  $(-1 - i)^2 + (3 - 2i)(-1 - i) + (5 - i) = (2i) + (-3 - 3i + 2i + 2i^2) + (5 - i) = 2i + (-5 - i) + (5 - i) = 0$ .

16. **Solution.** Let  $z = x + iy$ . Then

$$f(z) = z^2 = (x + iy)^2 = x^2 + 2ixy + i^2y^2 = (x^2 - y^2) + i(2xy).$$

So  $u(x, y) = x^2 - y^2$  and  $v(x, y) = 2xy$ .

If  $z$  lies on the real axis, then  $y = 0$ , so  $f(z) = (x^2 - 0) + i(0) = x^2$ , which is real.  $\square$

17. **Solution.** For  $z = x + iy \neq 0$ :

$$f(z) = \frac{1}{z} = \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy} = \frac{x - iy}{x^2 + y^2}.$$

Therefore:

$$u(x, y) = \frac{x}{x^2 + y^2}, \quad v(x, y) = \frac{-y}{x^2 + y^2}.$$

The function  $f(z) = 1/z$  is undefined at  $z = 0$  (i.e., at the origin  $(0, 0)$ ).

18. **Solution.** For  $z = x + iy$ :

$$f(z) = e^z = e^{x+iy} = e^x(\cos y + i \sin y).$$

On the vertical line  $x = 0$  (so  $z = iy$ ):

$$f(iy) = e^0(\cos y + i \sin y) = \cos y + i \sin y.$$

As  $y$  varies over  $\mathbb{R}$ ,  $w = \cos y + i \sin y$  traces the **unit circle**  $|w| = 1$  in the  $w$ -plane. The image is traversed repeatedly, with the full circle traced once for each interval of length  $2\pi$ .

19. **Solution.** The complex logarithm is defined for  $z \neq 0$  as

$$\ln(z) = \ln |z| + i \arg(z).$$

Since  $\arg(z)$  is multi-valued (defined up to multiples of  $2\pi$ ),  $\ln(z)$  is multi-valued. Its natural domain is  $\mathbb{C} \setminus \{0\}$ .

To obtain a single-valued function, we introduce a **branch cut**, typically along the negative real axis  $(-\infty, 0]$ . This restricts the argument to the interval  $(-\pi, \pi]$ , giving the **principal branch**:

$$\text{Log}(z) = \ln |z| + i \text{Arg}(z), \quad \text{Arg}(z) \in (-\pi, \pi].$$

The branch cut is needed because as one traverses a closed loop around the origin, the argument increases by  $2\pi$ , so  $\ln(z)$  would not return to its original value. Removing the branch cut prevents such loops, making the function single-valued and continuous on  $\mathbb{C} \setminus (-\infty, 0]$ .

20. **Solution.** Let  $z = x + iy$ . Then  $iz = ix - y$  and  $-iz = -ix + y$ . So:

$$e^{iz} = e^{i(x+iy)} = e^{ix-y} = e^{-y}(\cos x + i \sin x),$$

$$e^{-iz} = e^{-i(x+iy)} = e^{-ix+y} = e^y(\cos x - i \sin x).$$

Therefore:

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = \frac{e^{-y}(\cos x + i \sin x) - e^y(\cos x - i \sin x)}{2i}$$

$$= \frac{(e^{-y} - e^y) \cos x + i(e^{-y} + e^y) \sin x}{2i}.$$

Dividing numerator by  $2i$  (multiply by  $\frac{1}{2i} = \frac{-i}{2}$ ):

$$\begin{aligned} \sin z &= \frac{(e^{-y} - e^y) \cos x}{2i} + \frac{i(e^{-y} + e^y) \sin x}{2i} \\ &= \frac{(e^{-y} - e^y) \cos x \cdot (-i)}{2} + \frac{(e^{-y} + e^y) \sin x}{2}. \end{aligned}$$

Using  $\cosh y = \frac{e^y + e^{-y}}{2}$  and  $\sinh y = \frac{e^y - e^{-y}}{2}$ :

$$\sin z = \sin x \cosh y + i \cos x \sinh y. \quad \square$$

21. **Solution.** Write  $z = x + iy$ . Then  $e^z = e^x e^{iy} = e^x(\cos y + i \sin y) = -2$ . Comparing:

$$e^x \cos y = -2, \quad e^x \sin y = 0.$$

From the second equation,  $\sin y = 0$ , so  $y = n\pi$  for  $n \in \mathbb{Z}$ . From the first equation,  $e^x \cos(n\pi) = -2$ . Since  $\cos(n\pi) = (-1)^n$ , we need  $(-1)^n e^x = -2$ , so  $(-1)^n$  must be negative, meaning  $n$  is odd. Then  $e^x = 2$ , so  $x = \ln 2$ .

All solutions:

$$z = \ln 2 + i(2k + 1)\pi, \quad k \in \mathbb{Z}.$$

22. **Solution.** Let  $|z| = 1$ , so  $z = e^{i\theta}$ . Then  $f(z) = z^3 = e^{3i\theta}$ , which has modulus  $|e^{3i\theta}| = 1$  (on the unit circle) and argument  $3\theta$  (tripled).

**Verification with  $z = e^{i\pi/3}$ :**

$$f(z) = e^{3i\pi/3} = e^{i\pi} = -1.$$

Indeed  $|f(z)| = 1$  (on the unit circle), and  $\arg(f(z)) = \pi = 3 \cdot \frac{\pi}{3}$ , confirming the argument is tripled.  $\square$

23. **Solution.**  $f(z) = \frac{z^2 + 1}{z - i}$ . The denominator vanishes at  $z = i$ , so  $f$  is undefined there. However, we can factor the numerator:

$$z^2 + 1 = (z - i)(z + i).$$

Therefore, for  $z \neq i$ :

$$f(z) = \frac{(z - i)(z + i)}{z - i} = z + i.$$

The singularity at  $z = i$  is a **removable singularity**. If we define  $f(i) = 2i$ , the function extends to an entire function. The domain is  $\mathbb{C} \setminus \{i\}$ .

24. **Solution.** Using the principal branch:  $\sqrt{z} = |z|^{1/2} e^{i \operatorname{Arg}(z)/2}$ .

For  $z = 4e^{i\pi}$ : we have  $|z| = 4$  and  $\operatorname{Arg}(z) = \pi$  (since  $4e^{i\pi} = -4$ , which lies on the negative real axis; the principal argument is  $\pi$ ). Therefore:

$$f(4e^{i\pi}) = \sqrt{4} e^{i\pi/2} = 2(\cos(\pi/2) + i \sin(\pi/2)) = 2(0 + i) = 2i.$$

25. **Solution.** Let  $z = -1 - i$ . The modulus is  $|z| = \sqrt{1+1} = \sqrt{2}$ .

The argument: since both real and imaginary parts are negative (third quadrant),  $\arg(z) = -\frac{3\pi}{4} + 2k\pi$  for  $k \in \mathbb{Z}$  (using the convention that the principal argument is  $-\frac{3\pi}{4}$ ).

All values of  $\ln(-1 - i)$ :

$$\ln(-1 - i) = \ln \sqrt{2} + i \left( -\frac{3\pi}{4} + 2k\pi \right) = \frac{1}{2} \ln 2 + i \left( -\frac{3\pi}{4} + 2k\pi \right), \quad k \in \mathbb{Z}.$$

The **principal value** (with  $\text{Arg}(z) \in (-\pi, \pi]$ ) is:

$$\text{Log}(-1 - i) = \frac{1}{2} \ln 2 - \frac{3\pi}{4} i.$$

26. **Solution.** On the line  $y = 1$ , we have  $z = x + i$ , so:

$$w = e^z = e^{x+i} = e^x \cdot e^i = e^x (\cos 1 + i \sin 1).$$

Writing  $w = u + iv$ :  $u = e^x \cos 1$ ,  $v = e^x \sin 1$ . Since  $\cos 1$  and  $\sin 1$  are constants (both positive), as  $x$  ranges over  $\mathbb{R}$ ,  $e^x$  ranges over  $(0, \infty)$ .

Therefore  $u/\cos 1 = v/\sin 1 = e^x > 0$ , which gives the ray  $v = u \tan 1$  with  $u > 0$ .

The image is a **ray** emanating from the origin at angle 1 radian (approximately  $57.3^\circ$ ) from the positive real axis, not including the origin.

27. **Solution.** Using  $\cos z = \frac{e^{iz} + e^{-iz}}{2}$ , set  $\cos z = 0$ :

$$e^{iz} + e^{-iz} = 0 \implies e^{iz} = -e^{-iz} \implies e^{2iz} = -1.$$

Write  $-1 = e^{i(\pi+2k\pi)}$ , so  $2iz = i(\pi + 2k\pi)$ , giving

$$z = \frac{\pi + 2k\pi}{2} = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

These are exactly the same zeros as in the real case:  $z = (2k+1)\frac{\pi}{2}$  for  $k \in \mathbb{Z}$ . All zeros are real.

28. **Solution.** Let  $z = x + iy$  with  $z \neq 0$ . Then:

$$f(z) = z + \frac{1}{z} = (x + iy) + \frac{x - iy}{x^2 + y^2} = \left( x + \frac{x}{x^2 + y^2} \right) + i \left( y - \frac{y}{x^2 + y^2} \right).$$

For  $f(z)$  to be real, the imaginary part must vanish:

$$y - \frac{y}{x^2 + y^2} = 0 \implies y \left( 1 - \frac{1}{x^2 + y^2} \right) = 0.$$

This holds when either:



- (i)  $y = 0$  (i.e.,  $z$  is real and nonzero), or  
 (ii)  $x^2 + y^2 = 1$  (i.e.,  $|z| = 1$ , so  $z$  lies on the unit circle).

29. **Solution.** Let  $z = iy$  with  $y \in \mathbb{R}$ . Then:

$$f(iy) = \frac{iy - 1}{iy + 1}.$$

We compute the modulus:

$$|f(iy)| = \left| \frac{iy - 1}{iy + 1} \right| = \frac{|iy - 1|}{|iy + 1|} = \frac{\sqrt{1 + y^2}}{\sqrt{1 + y^2}} = 1.$$

Since  $|f(iy)| = 1$  for all  $y \in \mathbb{R}$ , the image lies on the unit circle.  $\square$

30. **Solution.** Let  $z = x + iy$ . Then  $z^2 = (x^2 - y^2) + 2ixy$ , so:

$$f(z) = e^{z^2} = e^{(x^2 - y^2) + 2ixy} = e^{x^2 - y^2} (\cos(2xy) + i \sin(2xy)).$$

Therefore:  $u(x, y) = e^{x^2 - y^2} \cos(2xy)$  and  $v(x, y) = e^{x^2 - y^2} \sin(2xy)$ .

31. **Solution.**

(a)  $\lim_{z \rightarrow 0} z^3 = 0^3 = 0$ . (Direct substitution;  $z^3$  is continuous.)

(b)  $\lim_{z \rightarrow 0} \frac{\sin z}{z}$ . Using the Taylor series  $\sin z = z - \frac{z^3}{3!} + \dots$ :

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \dots \rightarrow 1 \quad \text{as } z \rightarrow 0.$$

(c)  $\lim_{z \rightarrow 0} \frac{e^z - 1}{z}$ . Using  $e^z = 1 + z + \frac{z^2}{2!} + \dots$ :

$$\frac{e^z - 1}{z} = 1 + \frac{z}{2!} + \dots \rightarrow 1 \quad \text{as } z \rightarrow 0.$$

(d)  $\lim_{z \rightarrow 0} \frac{1 - \cos z}{z^2}$ . Using  $\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots$ :

$$\frac{1 - \cos z}{z^2} = \frac{1}{2} - \frac{z^2}{4!} + \dots \rightarrow \frac{1}{2} \quad \text{as } z \rightarrow 0.$$

32. **Solution.** Using  $\sin z = z - \frac{z^3}{6} + \frac{z^5}{120} - \dots$ :

$$\frac{\sin z - z}{z^3} = \frac{-\frac{z^3}{6} + \frac{z^5}{120} - \dots}{z^3} = -\frac{1}{6} + \frac{z^2}{120} - \dots \rightarrow -\frac{1}{6} \quad \text{as } z \rightarrow 0.$$

33. **Solution.** The limit  $\lim_{z \rightarrow 1+i} \frac{z^3 - (1+i)^3}{z - (1+i)}$  is by definition the derivative of  $g(z) = z^3$  evaluated at  $z_0 = 1 + i$ :

$$g'(z) = 3z^2, \quad g'(1+i) = 3(1+i)^2 = 3(1+2i+i^2) = 3(1+2i-1) = 3(2i) = 6i.$$

34. **Solution.** We must show: for every  $\epsilon > 0$ , there exists  $\delta > 0$  such that  $|z^n - 0| < \epsilon$  whenever  $0 < |z - 0| < \delta$ .

Choose  $\delta = \epsilon^{1/n}$ . If  $|z| < \delta$ , then

$$|z^n| = |z|^n < \delta^n = (\epsilon^{1/n})^n = \epsilon.$$

Therefore  $\lim_{z \rightarrow 0} z^n = 0$  for all  $n \geq 1$ .  $\square$

35. **Solution.** Using the Taylor series  $\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$  (valid for all  $z \in \mathbb{C}$ ):

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \dots \rightarrow 1 \quad \text{as } z \rightarrow 0.$$

The result is the same as the real-variable limit because the Taylor series for  $\sin z$  in  $\mathbb{C}$  is the same power series as for the real  $\sin x$ . Since the complex limit requires convergence along *every* path to 0, and the series converges absolutely for all  $z$ , the limit equals 1 regardless of the direction of approach.

36. **Solution.** Let  $z = x + iy$  with  $y \neq 0$ . Then  $\bar{z} = x - iy$ , so:

$$z^2 - \bar{z}^2 = (x + iy)^2 - (x - iy)^2 = (x^2 + 2ixy - y^2) - (x^2 - 2ixy - y^2) = 4ixy.$$

$$z - \bar{z} = (x + iy) - (x - iy) = 2iy.$$

Therefore:

$$f(z) = \frac{4ixy}{2iy} = 2x.$$

As  $z \rightarrow 2 + i$ , we have  $x \rightarrow 2$ , so:

$$\lim_{z \rightarrow 2+i} f(z) = 2(2) = 4.$$

37. **Solution.** Write  $z = re^{i\theta}$ . Then  $z \cdot e^{i\theta} = re^{2i\theta}$ —but the expression in question is simply  $re^{2i\theta}$ . We have:

$$|re^{2i\theta}| = r|e^{2i\theta}| = r \cdot 1 = r.$$

Therefore, as  $r \rightarrow 0$ :

$$|re^{2i\theta} - 0| = r \rightarrow 0,$$

regardless of  $\theta$ . The limit does not depend on  $\theta$  because the modulus of  $e^{2i\theta}$  is always 1, so the magnitude is controlled entirely by  $r$ , which tends to 0.  $\square$

38. **Solution.** Write  $z = re^{i\theta}$  with  $r > 0$ . Then  $\bar{z} = re^{-i\theta}$ , so:

$$\frac{\bar{z}}{z} = \frac{re^{-i\theta}}{re^{i\theta}} = e^{-2i\theta}.$$

This depends on  $\theta$  and is independent of  $r$ . Different directions of approach give different limits:

- Along the real axis ( $\theta = 0$ ):  $\frac{\bar{z}}{z} = e^0 = 1$ .
- Along the imaginary axis ( $\theta = \pi/2$ ):  $\frac{\bar{z}}{z} = e^{-i\pi} = -1$ .
- Along  $\theta = \pi/4$ :  $\frac{\bar{z}}{z} = e^{-i\pi/2} = -i$ .

Since the value depends on the direction of approach, the limit does not exist.  $\square$

39. **Solution.**

(a) For  $|z| \rightarrow \infty$ :  $\left|\frac{1}{z}\right| = \frac{1}{|z|} \rightarrow 0$ . Therefore  $\lim_{|z| \rightarrow \infty} \frac{1}{z} = 0$ .

(b) Write  $z = re^{i\theta}$ . Then  $\frac{z}{|z|} = \frac{re^{i\theta}}{r} = e^{i\theta}$ . As  $|z| \rightarrow \infty$ , the value  $e^{i\theta}$  depends on the direction of approach. For instance:

- Along the positive real axis ( $\theta = 0$ ):  $e^{i\theta} = 1$ .
- Along the positive imaginary axis ( $\theta = \pi/2$ ):  $e^{i\theta} = i$ .

Since the limiting value depends on  $\theta$ , the limit **does not exist**.

40. **Solution.** Let  $z = x + iy$ , so  $\bar{z} = x - iy$  and  $z - \bar{z} = 2iy$ . Then:

$$\frac{e^z - e^{\bar{z}}}{z - \bar{z}} = \frac{e^{x+iy} - e^{x-iy}}{2iy} = \frac{e^x(e^{iy} - e^{-iy})}{2iy} = \frac{e^x \cdot 2i \sin y}{2iy} = e^x \cdot \frac{\sin y}{y}.$$

As  $z \rightarrow 0$ , we have  $x \rightarrow 0$  and  $y \rightarrow 0$ , so:

$$\lim_{z \rightarrow 0} e^x \cdot \frac{\sin y}{y} = e^0 \cdot 1 = 1.$$

41. **Solution. Definition:** We say  $\lim_{z \rightarrow z_0} f(z) = L$  if for every  $\epsilon > 0$ , there exists a  $\delta > 0$  such that

$$0 < |z - z_0| < \delta \implies |f(z) - L| < \epsilon.$$

**Why path independence is crucial:** In  $\mathbb{R}$ , a variable can approach a point from only two directions (left and right). In  $\mathbb{C}$ , a variable  $z$  can approach  $z_0$  from infinitely many directions—along straight lines at any angle, along spirals, along curves of any shape. The  $\epsilon$ - $\delta$  definition requires  $|f(z) - L| < \epsilon$  for *all*  $z$  in the punctured disk  $0 < |z - z_0| < \delta$ , regardless of the path. If the limit depended on the path, the  $\epsilon$ - $\delta$  condition would fail. This makes the existence of a complex limit a much stronger condition than in real analysis, and it is the reason why complex differentiability implies so much more than real differentiability (e.g., analyticity, harmonicity of components, etc.).

42. **Solution.** The **extended complex plane** (or Riemann sphere) is  $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ . It is obtained by adding a single point at infinity to  $\mathbb{C}$ . A neighborhood of  $\infty$  is any set of the form  $\{z \in \mathbb{C} : |z| > R\} \cup \{\infty\}$  for some  $R > 0$ .

We say  $\lim_{z \rightarrow z_0} f(z) = \infty$  if for every  $M > 0$ , there exists  $\delta > 0$  such that

$$0 < |z - z_0| < \delta \implies |f(z)| > M.$$

**Proof that  $\lim_{z \rightarrow 0} 1/z = \infty$ :** Given  $M > 0$ , choose  $\delta = 1/M$ . If  $0 < |z| < \delta = 1/M$ , then

$$\left| \frac{1}{z} \right| = \frac{1}{|z|} > \frac{1}{\delta} = M.$$

Therefore  $|1/z| > M$  whenever  $0 < |z| < \delta$ , which proves  $\lim_{z \rightarrow 0} 1/z = \infty$ .  $\square$

43. **Solution.**

$$\begin{aligned} f'(z) &= \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = \lim_{h \rightarrow 0} \frac{(z+h)^2 - z^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{z^2 + 2zh + h^2 - z^2}{h} = \lim_{h \rightarrow 0} \frac{2zh + h^2}{h} = \lim_{h \rightarrow 0} (2z + h) = 2z. \end{aligned}$$

44. **Solution.**

$$(e^z)' = \lim_{h \rightarrow 0} \frac{e^{z+h} - e^z}{h} = \lim_{h \rightarrow 0} \frac{e^z \cdot e^h - e^z}{h} = e^z \lim_{h \rightarrow 0} \frac{e^h - 1}{h}.$$

Using the series  $e^h = 1 + h + \frac{h^2}{2!} + \cdots$ :

$$\frac{e^h - 1}{h} = 1 + \frac{h}{2!} + \frac{h^2}{3!} + \cdots \rightarrow 1 \quad \text{as } h \rightarrow 0.$$

Therefore  $(e^z)' = e^z \cdot 1 = e^z$ .  $\square$

45. **Solution.** On the principal branch,  $\text{Log}(z) = \ln|z| + i \text{Arg}(z)$  for  $z \notin (-\infty, 0]$ . We compute:

$$(\text{Log } z)' = \lim_{h \rightarrow 0} \frac{\text{Log}(z+h) - \text{Log}(z)}{h}.$$

Since  $e^{\text{Log } z} = z$ , we use the inverse function approach. Let  $w = \text{Log } z$ , so  $z = e^w$ . The exponential is holomorphic with  $(e^w)' = e^w \neq 0$ . By the inverse function theorem for holomorphic functions:

$$(\text{Log } z)' = \frac{1}{(e^w)'} \bigg|_{w=\text{Log } z} = \frac{1}{e^{\text{Log } z}} = \frac{1}{z}.$$

Alternatively, using power series: for  $|h/z| < 1$ ,

$$\text{Log}(z+h) - \text{Log}(z) = \text{Log}\left(1 + \frac{h}{z}\right) = \frac{h}{z} - \frac{1}{2} \left(\frac{h}{z}\right)^2 + \cdots$$

Dividing by  $h$ :

$$\frac{\text{Log}(z+h) - \text{Log}(z)}{h} = \frac{1}{z} - \frac{h}{2z^2} + \cdots \rightarrow \frac{1}{z} \quad \text{as } h \rightarrow 0. \quad \square$$

46. **Solution.**

$$(\sin z)' = \lim_{h \rightarrow 0} \frac{\sin(z+h) - \sin z}{h}.$$

Using the addition formula  $\sin(z+h) = \sin z \cos h + \cos z \sin h$ :

$$= \lim_{h \rightarrow 0} \frac{\sin z \cos h + \cos z \sin h - \sin z}{h} = \lim_{h \rightarrow 0} \left[ \sin z \cdot \frac{\cos h - 1}{h} + \cos z \cdot \frac{\sin h}{h} \right].$$

Using the known limits  $\frac{\cos h - 1}{h} \rightarrow 0$  and  $\frac{\sin h}{h} \rightarrow 1$  as  $h \rightarrow 0$  (in  $\mathbb{C}$ ):

$$(\sin z)' = \sin z \cdot 0 + \cos z \cdot 1 = \cos z.$$

47. **Solution.** For  $z \neq 0$ :

$$\left(\frac{1}{z}\right)' = \lim_{h \rightarrow 0} \frac{\frac{1}{z+h} - \frac{1}{z}}{h} = \lim_{h \rightarrow 0} \frac{z - (z+h)}{h \cdot z(z+h)} = \lim_{h \rightarrow 0} \frac{-h}{h \cdot z(z+h)} = \lim_{h \rightarrow 0} \frac{-1}{z(z+h)} = -\frac{1}{z^2}.$$

48. **Solution.** Let  $f(z) = |z|^2 = z\bar{z}$ . Consider the difference quotient at  $z_0 \neq 0$ :

$$\frac{f(z_0+h) - f(z_0)}{h} = \frac{|z_0+h|^2 - |z_0|^2}{h} = \frac{(z_0+h)\overline{(z_0+h)} - z_0\bar{z}_0}{h}.$$

Expanding:

$$= \frac{z_0\bar{z}_0 + z_0\bar{h} + h\bar{z}_0 + h\bar{h} - z_0\bar{z}_0}{h} = \frac{z_0\bar{h} + h\bar{z}_0 + |h|^2}{h} = z_0 \cdot \frac{\bar{h}}{h} + \bar{z}_0 + \bar{h}.$$

As  $h \rightarrow 0$ ,  $\bar{h} \rightarrow 0$ , so this becomes  $z_0 \cdot \frac{\bar{h}}{h} + \bar{z}_0$ .

The term  $\bar{h}/h$  depends on the direction: if  $h = re^{i\theta}$ , then  $\bar{h}/h = e^{-2i\theta}$ . For  $z_0 \neq 0$ , the term  $z_0 e^{-2i\theta}$  takes different values for different  $\theta$ , so the limit does not exist. Therefore  $f$  is not complex differentiable at any  $z_0 \neq 0$ .  $\square$

49. **Solution.** Let  $f(z) = \operatorname{Re}(z) = x$  where  $z = x + iy$ . So  $u(x, y) = x$ ,  $v(x, y) = 0$ . The Cauchy–Riemann equations require:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x.$$

We compute:  $u_x = 1$ ,  $v_y = 0$ ,  $u_y = 0$ ,  $v_x = 0$ .

The first equation gives  $1 = 0$ , which is false everywhere. Therefore the Cauchy–Riemann equations are never satisfied, and  $f(z) = \operatorname{Re}(z)$  is **nowhere differentiable** in  $\mathbb{C}$ .  $\square$

50. **Solution.** Let  $z = x + iy$ . Then:

$$z^3 = (x + iy)^3 = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 = x^3 + 3ix^2y - 3xy^2 - iy^3.$$

So  $u(x, y) = x^3 - 3xy^2$  and  $v(x, y) = 3x^2y - y^3$ .

**Cauchy–Riemann equations:**

$$u_x = 3x^2 - 3y^2, \quad v_y = 3x^2 - 3y^2 \implies u_x = v_y.$$

$$u_y = -6xy, \quad v_x = 6xy \implies u_y = -v_x.$$

Both Cauchy–Riemann equations hold everywhere, confirming  $z^3$  is entire.  $\square$

51. **Solution.** Let  $f(z) = \bar{z} = x - iy$ . So  $u(x, y) = x$  and  $v(x, y) = -y$ .

Cauchy–Riemann equations:

$$u_x = 1, \quad v_y = -1 \implies u_x \neq v_y \quad (1 \neq -1).$$

The first Cauchy–Riemann equation fails everywhere. Therefore  $f(z) = \bar{z}$  is not holomorphic at any point.  $\square$

52. **Solution. Proof outline.** Let  $g$  be holomorphic at  $z_0$  and  $f$  holomorphic at  $w_0 = g(z_0)$ . We want to show  $(f \circ g)'(z_0) = f'(w_0) g'(z_0)$ .

Consider:

$$\frac{f(g(z)) - f(g(z_0))}{z - z_0} = \frac{f(g(z)) - f(g(z_0))}{g(z) - g(z_0)} \cdot \frac{g(z) - g(z_0)}{z - z_0},$$

valid when  $g(z) \neq g(z_0)$ . As  $z \rightarrow z_0$ , continuity of  $g$  gives  $g(z) \rightarrow g(z_0) = w_0$ , so:

$$\frac{f(g(z)) - f(w_0)}{g(z) - w_0} \rightarrow f'(w_0), \quad \frac{g(z) - g(z_0)}{z - z_0} \rightarrow g'(z_0).$$

To handle the case when  $g(z) = g(z_0)$  for some  $z$  near  $z_0$ , define the auxiliary function

$$\phi(w) = \begin{cases} \frac{f(w) - f(w_0)}{w - w_0} & w \neq w_0, \\ f'(w_0) & w = w_0. \end{cases}$$

Then  $\phi$  is continuous at  $w_0$  (since  $f$  is holomorphic there), and  $f(w) - f(w_0) = \phi(w)(w - w_0)$  for all  $w$  near  $w_0$ . Substituting  $w = g(z)$ :

$$\frac{f(g(z)) - f(g(z_0))}{z - z_0} = \phi(g(z)) \cdot \frac{g(z) - g(z_0)}{z - z_0} \rightarrow \phi(w_0) \cdot g'(z_0) = f'(w_0) g'(z_0). \quad \square$$

53. **Solution.**

(a)  $f(z) = z^2 e^z$ . By the product rule:

$$f'(z) = 2z \cdot e^z + z^2 \cdot e^z = e^z(2z + z^2) = ze^z(z + 2).$$

(b)  $f(z) = \sin(e^z)$ . By the chain rule:

$$f'(z) = \cos(e^z) \cdot e^z.$$

(c)  $f(z) = z^3 + 2z^2 - 3z + 1$ . By the power rule:

$$f'(z) = 3z^2 + 4z - 3.$$

54. **Solution.** Write  $z = re^{i\theta}$  with  $r > 0$  and  $\theta = \text{Arg}(z) \in (-\pi, \pi)$ . Then:

$$f(z) = z^{3/2} = r^{3/2}e^{3i\theta/2}.$$

Since  $z^{3/2} = e^{(3/2)\text{Log } z}$ , by the chain rule:

$$f'(z) = \frac{3}{2} \cdot e^{(3/2)\text{Log } z} \cdot \frac{1}{z} = \frac{3}{2} \cdot \frac{z^{3/2}}{z} = \frac{3}{2}z^{1/2} = \frac{3}{2}\sqrt{z}.$$

55. **Solution.** Suppose  $f'(z) = 0$  for all  $z$  in a connected domain  $D$ . Write  $f(z) = u(x, y) + iv(x, y)$ . Since  $f$  is holomorphic with  $f'(z) = 0$ , we have  $f'(z) = u_x + iv_x = 0$ , so  $u_x = 0$  and  $v_x = 0$  everywhere in  $D$ .

By the Cauchy–Riemann equations,  $u_x = v_y$  and  $u_y = -v_x$ , so also  $v_y = 0$  and  $u_y = 0$  everywhere in  $D$ .

Since all partial derivatives of  $u$  and  $v$  vanish on the connected open set  $D$ , both  $u$  and  $v$  are constant on  $D$  (by a standard result from real analysis: a function with zero gradient on a connected open set is constant). Therefore  $f$  is constant.  $\square$

56. **Solution.** In polar coordinates,  $z = re^{i\theta}$ , so  $f(z) = z^2 = r^2e^{2i\theta}$ . Thus:

$$u(r, \theta) = r^2 \cos(2\theta), \quad v(r, \theta) = r^2 \sin(2\theta).$$

**Polar Cauchy–Riemann equations:** For  $f = u + iv$  in polar form, the Cauchy–Riemann equations become

$$u_r = \frac{1}{r}v_\theta, \quad v_r = -\frac{1}{r}u_\theta.$$

**Derivation:** Using  $x = r \cos \theta$ ,  $y = r \sin \theta$ , by the chain rule:

$$u_r = u_x \cos \theta + u_y \sin \theta, \quad u_\theta = -u_x r \sin \theta + u_y r \cos \theta.$$

Applying  $u_x = v_y$  and  $u_y = -v_x$ , and similarly computing  $v_r$  and  $v_\theta$ , one obtains the polar C–R equations.

**Verification for  $f(z) = z^2$ :**

$$\begin{aligned} u_r &= 2r \cos(2\theta), & \frac{1}{r}v_\theta &= \frac{1}{r} \cdot 2r \cos(2\theta) = 2 \cos(2\theta). \\ v_r &= 2r \sin(2\theta), & -\frac{1}{r}u_\theta &= -\frac{1}{r} \cdot (-2r \sin(2\theta)) = 2 \sin(2\theta). \end{aligned}$$

57. **Solution.** Let  $u(x, y) = \frac{1}{2} \ln(x^2 + y^2)$ . Compute the partial derivatives for  $(x, y) \neq (0, 0)$ :

$$\begin{aligned} u_x &= \frac{x}{x^2 + y^2}, & u_{xx} &= \frac{(x^2 + y^2) - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}. \\ u_y &= \frac{y}{x^2 + y^2}, & u_{yy} &= \frac{(x^2 + y^2) - y \cdot 2y}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}. \end{aligned}$$

Therefore:

$$u_{xx} + u_{yy} = \frac{y^2 - x^2}{(x^2 + y^2)^2} + \frac{x^2 - y^2}{(x^2 + y^2)^2} = 0.$$

So  $u$  is harmonic on  $\mathbb{R}^2 \setminus \{(0, 0)\}$ .  $\square$

58. **Solution.** Let  $u(x, y) = e^x \cos y$ . Then:

$$u_x = e^x \cos y, \quad u_{xx} = e^x \cos y.$$

$$u_y = -e^x \sin y, \quad u_{yy} = -e^x \cos y.$$

Therefore:

$$u_{xx} + u_{yy} = e^x \cos y - e^x \cos y = 0.$$

Laplace's equation is satisfied, so  $u$  is harmonic on  $\mathbb{R}^2$ .  $\square$

(Note:  $u = e^x \cos y = \operatorname{Re}(e^z)$ , so this also follows from the fact that  $e^z$  is entire.)

59. **Solution.** Let  $u(x, y) = x^3 - 3xy^2$ . Then:

$$u_x = 3x^2 - 3y^2, \quad u_{xx} = 6x.$$

$$u_y = -6xy, \quad u_{yy} = -6x.$$

Therefore:

$$u_{xx} + u_{yy} = 6x + (-6x) = 0. \quad \square$$

60. **Solution.** The geometric series gives:

$$f(z) = \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \cdots$$

This series converges when  $|z| < 1$  and diverges when  $|z| > 1$ . Since the nearest singularity of  $f$  to  $z = 0$  is at  $z = 1$  (distance 1 from the center), the **radius of convergence is**  $R = 1$ .

61. **Solution.** The Taylor series of  $\operatorname{Log}(1+z)$  about  $z = 0$  is:

$$\operatorname{Log}(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} z^n = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \cdots$$

This can be derived by integrating the geometric series  $\frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n$  term by term.

The nearest singularity of  $\operatorname{Log}(1+z)$  to  $z = 0$  is the branch point at  $z = -1$  (distance 1). Therefore the **radius of convergence is**  $R = 1$ .



62. **Solution.** The path  $\gamma$  is the vertical segment from 1 to  $1 + 2i$ . Parametrize:  $z(t) = 1 + 2it$ ,  $t \in [0, 1]$ , so  $dz = 2i dt$ .

$$\int_{\gamma} dz = \int_0^1 2i dt = 2i.$$

Alternatively, since  $f(z) = 1$  is entire with antiderivative  $F(z) = z$ :

$$\int_{\gamma} dz = F(1 + 2i) - F(1) = (1 + 2i) - 1 = 2i.$$

63. **Solution.** Parametrize  $\gamma$ :  $z(t) = (1 + i)t$ ,  $t \in [0, 1]$ , so  $dz = (1 + i) dt$ . Then:

$$\int_{\gamma} e^z dz = \int_0^1 e^{(1+i)t} (1 + i) dt = (1 + i) \left[ \frac{e^{(1+i)t}}{1 + i} \right]_0^1 = e^{(1+i)} - e^0 = e^{1+i} - 1.$$

Alternatively, since  $e^z$  has antiderivative  $e^z$ :

$$\int_{\gamma} e^z dz = e^{1+i} - e^0 = e(\cos 1 + i \sin 1) - 1.$$

64. **Solution.** Since  $f(z) = z^2$  is entire (holomorphic on all of  $\mathbb{C}$ ) and  $\gamma$  is a closed contour, by **Cauchy's theorem**:

$$\oint_{\gamma} z^2 dz = 0.$$

One can also verify directly:  $z^2$  has antiderivative  $F(z) = z^3/3$ , so the integral over any closed curve vanishes.

65. **Solution.** On  $\gamma_R$ :  $z = Re^{it}$ ,  $t \in [0, \pi]$ . The length of  $\gamma_R$  is  $L = \pi R$ . For  $k > 0$  and  $z = Re^{it} = R \cos t + iR \sin t$ :

$$|e^{ikz}| = |e^{ik(R \cos t + iR \sin t)}| = |e^{ikR \cos t - kR \sin t}| = e^{-kR \sin t}.$$

For  $t \in [0, \pi]$ ,  $\sin t \geq 0$ , so  $|e^{ikz}| \leq 1$ , and  $M = \max_{z \in \gamma_R} |e^{ikz}| \leq 1$ .

By the **ML inequality**:

$$\left| \int_{\gamma_R} e^{ikz} dz \right| \leq M \cdot L \leq 1 \cdot \pi R = \pi R.$$

However, this bound does not tend to zero. For the integrand  $e^{ikz}$  alone on the semi-circle, a sharper estimate is needed; in typical applications, additional decay from a factor such as  $1/p(z)$  for some polynomial  $p$  is present.

More precisely, using **Jordan's lemma**: For  $k > 0$ ,

$$\left| \int_{\gamma_R} e^{ikz} dz \right| \leq \frac{\pi}{k} (1 - e^{-kR}) < \frac{\pi}{k}.$$

This follows from the estimate  $\int_0^\pi e^{-kR \sin t} dt \leq \frac{\pi}{kR}$  (using  $\sin t \geq \frac{2t}{\pi}$  for  $t \in [0, \pi/2]$ ). With the factor  $|dz| = R dt$ :

$$\left| \int_{\gamma_R} e^{ikz} dz \right| \leq R \int_0^\pi e^{-kR \sin t} dt \leq R \cdot \frac{\pi}{kR} = \frac{\pi}{k}.$$

If the problem intends for an additional factor such as  $e^{ikz}/z$  or similar (as is standard in applications), the bound  $R \cdot \frac{\pi}{kR} \rightarrow 0$  does require the  $1/R$  decay. In the standard application of Jordan's lemma,  $\int_{\gamma_R} \frac{e^{ikz}}{p(z)} dz \rightarrow 0$  when  $\deg p \geq 1$ . For  $e^{ikz}$  alone, the integral remains bounded but need not tend to zero.

**Corrected interpretation:** If the integrand is  $e^{ikz}/z^n$  for  $n \geq 1$ , then  $|e^{ikz}/z^n| \leq 1/R^n$  on  $\gamma_R$ , and

$$\left| \int_{\gamma_R} \frac{e^{ikz}}{z^n} dz \right| \leq \frac{1}{R^n} \cdot \pi R = \frac{\pi}{R^{n-1}} \rightarrow 0 \text{ as } R \rightarrow \infty. \quad \square$$

66. **Solution.** Parametrize  $\gamma$ :  $z = e^{it}$ ,  $t \in [0, 2\pi]$ , so  $dz = ie^{it} dt$ . Then:

$$\oint_{\gamma} \frac{dz}{z} = \int_0^{2\pi} \frac{ie^{it}}{e^{it}} dt = \int_0^{2\pi} i dt = 2\pi i.$$

The answer  $2\pi i$  equals  $2\pi i \cdot n(\gamma, 0)$ , where  $n(\gamma, 0) = 1$  is the winding number of  $\gamma$  about the origin. In general, for any closed contour  $\gamma$  not passing through the origin,  $\oint_{\gamma} \frac{dz}{z} = 2\pi i \cdot n(\gamma, 0)$ . The unit circle traversed once counterclockwise has winding number 1.

67. **Solution.** An antiderivative of  $f(z) = z^3$  is  $F(z) = \frac{z^4}{4}$ . By the Fundamental Theorem for contour integrals:

$$\int_{\gamma} z^3 dz = F(2) - F(1) = \frac{2^4}{4} - \frac{1^4}{4} = \frac{16}{4} - \frac{1}{4} = \frac{15}{4}.$$

68. **Solution.** With  $\gamma(t) = t + it$ ,  $t \in [0, 1]$ , so  $dz = (1+i) dt$ . Since  $\sin z$  has antiderivative  $-\cos z$ :

$$\int_{\gamma} \sin(z) dz = [-\cos z]_{z=0}^{z=1+i} = -\cos(1+i) + \cos(0) = 1 - \cos(1+i).$$

Now,  $\cos(1+i) = \cos 1 \cosh 1 - i \sin 1 \sinh 1$ . Therefore:

$$\int_{\gamma} \sin(z) dz = 1 - \cos 1 \cosh 1 + i \sin 1 \sinh 1.$$

69. **Solution.** The function  $\frac{1}{z-1}$  has a singularity at  $z = 1$ , which lies inside the circle  $|z| = 2$ . By **Cauchy's Integral Formula** (with  $f(z) = 1$ , holomorphic inside  $\gamma$ ):

$$\oint_{|z|=2} \frac{dz}{z-1} = 2\pi i \cdot f(1) = 2\pi i \cdot 1 = 2\pi i.$$

70. **Solution. Proof.** Let  $\gamma : [a, b] \rightarrow D$  be a piecewise smooth curve from  $z_1 = \gamma(a)$  to  $z_2 = \gamma(b)$ . Since  $F' = f$  in  $D$ :

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt = \int_a^b F'(\gamma(t)) \gamma'(t) dt = \int_a^b \frac{d}{dt}[F(\gamma(t))] dt,$$

where the last equality uses the chain rule. By the Fundamental Theorem of Calculus (for real integrals applied component-wise):

$$= F(\gamma(b)) - F(\gamma(a)) = F(z_2) - F(z_1).$$

This depends only on the endpoints  $z_1$  and  $z_2$ , *not* on the path  $\gamma$ . Therefore the integral is path-independent.  $\square$

71. **Solution.** We find an antiderivative of  $\frac{z}{z^2+1}$ . Let  $F(z) = \frac{1}{2} \text{Log}(z^2 + 1)$ . Then  $F'(z) = \frac{z}{z^2+1}$ .

The singularities of the integrand are at  $z = \pm i$ , which do not lie on the segment from 0 to 2. On this segment,  $z^2 + 1 > 0$  (since  $z$  is real and  $z^2 + 1 \geq 1$ ), so  $\text{Log}(z^2 + 1)$  is well-defined and holomorphic.

By the Fundamental Theorem:

$$\int_{\gamma} \frac{z}{z^2+1} dz = F(2) - F(0) = \frac{1}{2} \ln(4+1) - \frac{1}{2} \ln(0+1) = \frac{1}{2} \ln 5 - 0 = \frac{\ln 5}{2}.$$

72. **Solution.** Let  $\gamma : [a, b] \rightarrow \mathbb{C}$  be a piecewise smooth curve and let  $\phi : [c, d] \rightarrow [a, b]$  be a smooth, orientation-preserving reparametrization (i.e.,  $\phi' > 0$ ,  $\phi(c) = a$ ,  $\phi(d) = b$ ). Define  $\tilde{\gamma}(s) = \gamma(\phi(s))$ .

$$\int_{\tilde{\gamma}} f(z) dz = \int_c^d f(\tilde{\gamma}(s)) \tilde{\gamma}'(s) ds = \int_c^d f(\gamma(\phi(s))) \gamma'(\phi(s)) \phi'(s) ds.$$

By the substitution  $t = \phi(s)$ ,  $dt = \phi'(s) ds$ :

$$= \int_a^b f(\gamma(t)) \gamma'(t) dt = \int_{\gamma} f(z) dz.$$

The value is unchanged. This **invariance property** means that the complex line integral depends only on the oriented curve as a geometric object, not on the particular parametrization chosen.  $\square$

73. **Solution.** Let  $\gamma$  wind  $n$  times counterclockwise around the origin. We can decompose  $\gamma$  (by homotopy or deformation) into  $n$  copies of a simple closed loop around the origin. For a single counterclockwise loop  $C$  around 0:

$$\oint_C \frac{dz}{z} = 2\pi i$$

(proved by direct parametrization:  $z = re^{it}$ ,  $t \in [0, 2\pi]$ ).

Since the integral is additive under concatenation of paths:

$$\oint_{\gamma} \frac{dz}{z} = n \cdot 2\pi i = 2\pi i n.$$

This is precisely  $2\pi i$  times the winding number  $n(\gamma, 0) = n$ .  $\square$

74. **Solution.** The function  $e^z$  is entire (holomorphic on all of  $\mathbb{C}$ ), and it has an antiderivative, namely  $F(z) = e^z$  (since  $F'(z) = e^z$ ). For any closed piecewise smooth curve  $\gamma$  from  $z_1$  back to  $z_1$ :

$$\oint_{\gamma} e^z dz = F(z_1) - F(z_1) = 0.$$

Alternatively, by **Cauchy's theorem**, since  $e^z$  is entire (and hence holomorphic inside and on  $\gamma$ ), the integral over any closed contour is zero.  $\square$

75. **Solution.** Parametrize  $\gamma$ :  $z = e^{it}$ ,  $t \in [0, 2\pi]$ ,  $dz = ie^{it} dt$ :

$$\oint_{\gamma} \frac{dz}{z} = \int_0^{2\pi} \frac{ie^{it}}{e^{it}} dt = \int_0^{2\pi} i dt = i \cdot 2\pi = 2\pi i. \quad \square$$

76. **Solution.** By **Cauchy's Integral Formula**: if  $f$  is holomorphic inside and on  $\gamma$  and  $z_0$  is inside  $\gamma$ , then

$$\oint_{\gamma} \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

Here  $f(z) = e^z$  (entire),  $z_0 = 2$ , and  $|z| = 3$  encloses  $z_0 = 2$ . Therefore:

$$\oint_{|z|=3} \frac{e^z}{z - 2} dz = 2\pi i \cdot e^2.$$

77. **Solution.** The generalized Cauchy Integral Formula gives:

$$f''(z_0) = \frac{2!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^3} dz.$$

With  $f(z) = \sin z$  and  $z_0 = 0$ :

$$f''(0) = \frac{2}{2\pi i} \oint_{|z|=R} \frac{\sin z}{z^3} dz.$$

Now,  $f(z) = \sin z$  gives  $f'(z) = \cos z$  and  $f''(z) = -\sin z$ , so  $f''(0) = -\sin 0 = 0$ .

Therefore:

$$\oint_{|z|=R} \frac{\sin z}{z^3} dz = \frac{2\pi i}{2} \cdot f''(0) = \pi i \cdot 0 = 0.$$

78. **Solution.** By Cauchy's Integral Formula with  $f(z) = 1$  (entire) and  $z_0 = 1$  (inside  $|z| = 2$ ):

$$\oint_{|z|=2} \frac{dz}{z-1} = 2\pi i \cdot f(1) = 2\pi i.$$

The answer depends on  $z = 1$  lying inside the contour because  $\frac{1}{z-1}$  has a singularity at  $z = 1$ . If  $z = 1$  were outside the contour, the integrand would be holomorphic inside and on the contour, and by Cauchy's theorem, the integral would be 0.

79. **Solution.** Since  $\frac{1}{(z-1)^2}$  has a pole of order 2 at  $z = 1$ , we apply the generalized Cauchy Integral Formula with  $f(z) = 1$  and  $n = 1$ :

$$\frac{f'(z_0)}{1!} = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z-z_0)^2} dz \implies \oint_{\gamma} \frac{dz}{(z-1)^2} = 2\pi i \cdot f'(1).$$

Since  $f(z) = 1$ ,  $f'(z) = 0$ , so:

$$\oint_{\gamma} \frac{dz}{(z-1)^2} = 2\pi i \cdot 0 = 0.$$

**Explanation:** The integrand  $\frac{1}{(z-1)^2}$  has a pole of order 2 at  $z = 1$ . Its Laurent series about  $z = 1$  is simply  $(z-1)^{-2}$ , so the residue (the coefficient of  $(z-1)^{-1}$ ) is 0. By the Residue Theorem, the integral equals  $2\pi i$  times the residue, which is 0. In general, for a pole of order  $n \geq 2$  with no  $(z-z_0)^{-1}$  term, the contour integral vanishes.

80. **Solution.** Partial fractions:  $\frac{1}{z(z-1)} = \frac{A}{z} + \frac{B}{z-1}$ . Setting  $z = 0$ :  $A = -1$ . Setting  $z = 1$ :  $B = 1$ . So:

$$\frac{1}{z(z-1)} = \frac{-1}{z} + \frac{1}{z-1}.$$

Both singularities  $z = 0$  and  $z = 1$  lie inside  $|z| = 2$ . Therefore:

$$\oint_{|z|=2} \frac{dz}{z(z-1)} = -\oint_{|z|=2} \frac{dz}{z} + \oint_{|z|=2} \frac{dz}{z-1} = -2\pi i + 2\pi i = 0.$$

81. **Solution.** Use partial fractions:  $\frac{1}{z(z-2)} = \frac{A}{z} + \frac{B}{z-2}$ , giving  $A = -1/2$  and  $B = 1/2$ :

$$\frac{1}{z(z-2)} = -\frac{1}{2z} + \frac{1}{2(z-2)}.$$

For the region  $0 < |z| < 2$ , expand  $\frac{1}{z-2} = \frac{-1}{2} \cdot \frac{1}{1-z/2}$  as a geometric series (valid for  $|z| < 2$ ):

$$\frac{1}{z-2} = -\frac{1}{2} \sum_{n=0}^{\infty} \frac{z^n}{2^n} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}.$$

Therefore:

$$f(z) = -\frac{1}{2z} + \frac{1}{2} \left( -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} \right) = -\frac{1}{2z} - \sum_{n=0}^{\infty} \frac{z^n}{2^{n+2}}.$$

$$= -\frac{1}{2} \cdot z^{-1} - \frac{1}{4} - \frac{z}{8} - \frac{z^2}{16} - \dots$$

The Laurent series has a single term with negative power ( $z^{-1}$  with coefficient  $-1/2$ ), so  $z = 0$  is a **simple pole** (pole of order 1) with residue  $-1/2$ .

82. **Solution.** We need the residue of  $f(z) = \frac{e^{1/z}}{1+z}$  at  $z = 0$ , i.e., the coefficient  $a_{-1}$  in the Laurent expansion about  $z = 0$ .

Expand  $e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$  and  $\frac{1}{1+z} = \sum_{m=0}^{\infty} (-1)^m z^m$  (for  $|z| < 1$ ).

$$f(z) = \left( \sum_{n=0}^{\infty} \frac{z^{-n}}{n!} \right) \left( \sum_{m=0}^{\infty} (-1)^m z^m \right).$$

The coefficient of  $z^{-1}$  comes from pairing terms where  $m - n = -1$ , i.e.,  $m = n - 1$ :

$$a_{-1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n!} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)!} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)!}.$$

This can be evaluated:  $\sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)!} = \frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \dots$

Recall that  $e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \dots$ , so  $e^{-1} - 1 = \sum_{k=1}^{\infty} \frac{(-1)^k}{k!} = -\left(\frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \dots\right)$ .

Therefore  $a_{-1} = 1 - e^{-1}$ .

The residue of  $f(z)$  at  $z = 0$  is  $\boxed{1 - e^{-1}}$ .

83. **Solution.** First, partial fractions. We write:

$$\frac{z^2 + 1}{z(z-1)(z-2)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z-2}.$$

Setting  $z = 0$ :  $\frac{1}{(-1)(-2)} = \frac{1}{2}$ , so  $A = 1/2$ .

Setting  $z = 1$ :  $\frac{2}{(1)(-1)} = -2$ , so  $B = -2$ .

Setting  $z = 2$ :  $\frac{5}{(2)(1)} = 5/2$ , so  $C = 5/2$ .

For  $0 < |z| < 1$ , expand  $\frac{1}{z-1} = \frac{-1}{1-z} = -\sum_{n=0}^{\infty} z^n$  and  $\frac{1}{z-2} = \frac{-1}{2} \cdot \frac{1}{1-z/2} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}$ .

$$\begin{aligned} f(z) &= \frac{1}{2z} + (-2) \left( -\sum_{n=0}^{\infty} z^n \right) + \frac{5}{2} \left( -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} \right) \\ &= \frac{1}{2z} + 2 \sum_{n=0}^{\infty} z^n - \frac{5}{2} \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} \\ &= \frac{1}{2z} + \sum_{n=0}^{\infty} \left( 2 - \frac{5}{2^{n+2}} \right) z^n. \end{aligned}$$

The only negative power is  $z^{-1}$  with coefficient  $1/2$ . Therefore  $z = 0$  is a **simple pole** with residue  $1/2$ .

84. **Solution.** The singularities of  $\frac{e^z}{z^2+1} = \frac{e^z}{(z-i)(z+i)}$  are at  $z = i$  and  $z = -i$ . Both lie inside  $|z| = 3$ .

By the Residue Theorem:

$$\oint_{|z|=3} \frac{e^z}{z^2+1} dz = 2\pi i [\text{Res}(f, i) + \text{Res}(f, -i)].$$

At  $z = i$  (simple pole):  $\text{Res}(f, i) = \lim_{z \rightarrow i} (z - i) \frac{e^z}{(z-i)(z+i)} = \frac{e^i}{2i}$ .

At  $z = -i$  (simple pole):  $\text{Res}(f, -i) = \lim_{z \rightarrow -i} (z + i) \frac{e^z}{(z-i)(z+i)} = \frac{e^{-i}}{-2i}$ .

Sum of residues:

$$\frac{e^i}{2i} + \frac{e^{-i}}{-2i} = \frac{e^i - e^{-i}}{2i} = \sin 1.$$

Therefore:

$$\oint_{|z|=3} \frac{e^z}{z^2+1} dz = 2\pi i \sin 1.$$

85. **Solution.** Consider  $\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$ . Let  $f(z) = \frac{1}{z^2+1} = \frac{1}{(z-i)(z+i)}$ . Close the contour with a semicircle  $\gamma_R$  in the upper half-plane.

The contour  $C_R$  consists of  $[-R, R]$  on the real axis plus  $\gamma_R$ . The only pole in the upper half-plane is  $z = i$ .

$$\text{Res}(f, i) = \lim_{z \rightarrow i} (z - i) \frac{1}{(z-i)(z+i)} = \frac{1}{2i}.$$

By the Residue Theorem:

$$\oint_{C_R} f(z) dz = 2\pi i \cdot \frac{1}{2i} = \pi.$$

On  $\gamma_R$ :  $|f(z)| \leq \frac{1}{R^2-1}$ , so by the ML inequality,  $\left| \int_{\gamma_R} f(z) dz \right| \leq \frac{\pi R}{R^2-1} \rightarrow 0$  as  $R \rightarrow \infty$ .

Therefore:

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \pi.$$

86. **Solution. Proof of Liouville's Theorem.** Let  $f$  be entire and bounded:  $|f(z)| \leq M$  for all  $z \in \mathbb{C}$ . Since  $f$  is entire, it has a power series  $f(z) = \sum_{n=0}^{\infty} a_n z^n$  converging for all  $z$ .

By **Cauchy's estimate**, for any  $R > 0$  and any  $n \geq 1$ :

$$|a_n| \leq \frac{\max_{|z|=R} |f(z)|}{R^n} \leq \frac{M}{R^n}.$$

Since this holds for all  $R > 0$ , we let  $R \rightarrow \infty$ :

$$|a_n| \leq \lim_{R \rightarrow \infty} \frac{M}{R^n} = 0 \quad \text{for all } n \geq 1.$$

Therefore  $a_n = 0$  for all  $n \geq 1$ , and  $f(z) = a_0$ , a constant.  $\square$

87. **Solution.** By the generalized Cauchy Integral Formula with  $f(z) = e^{z^2}$ ,  $z_0 = i$ , and  $n = 2$ :

$$\oint_{|z-i|=R} \frac{e^{z^2}}{(z-i)^3} dz = \frac{2\pi i}{2!} f''(i) = \pi i \cdot f''(i).$$

Compute  $f''(i)$  for  $f(z) = e^{z^2}$ :

$$f'(z) = 2ze^{z^2}, \quad f''(z) = 2e^{z^2} + 4z^2e^{z^2} = (2 + 4z^2)e^{z^2}.$$

At  $z = i$ :  $i^2 = -1$ , so:

$$f''(i) = (2 + 4(-1))e^{i^2} = (2 - 4)e^{-1} = -\frac{2}{e}.$$

Therefore:

$$\oint_{|z-i|=R} \frac{e^{z^2}}{(z-i)^3} dz = \pi i \cdot \left(-\frac{2}{e}\right) = -\frac{2\pi i}{e}.$$

88. **Solution.** We evaluate  $\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$  by contour integration. Let  $f(z) = \frac{1}{z^2+1} = \frac{1}{(z-i)(z+i)}$ .

Close the contour in the upper half-plane with a semicircular arc  $\gamma_R$  of radius  $R$ . The closed contour  $C_R = [-R, R] \cup \gamma_R$  encloses the pole at  $z = i$  (for  $R > 1$ ).

**Residue at  $z = i$ :**

$$\text{Res}(f, i) = \lim_{z \rightarrow i} (z - i) \cdot \frac{1}{(z - i)(z + i)} = \frac{1}{i + i} = \frac{1}{2i}.$$

**Residue Theorem:**

$$\oint_{C_R} f(z) dz = 2\pi i \cdot \frac{1}{2i} = \pi.$$

**Semicircular contribution:** On  $\gamma_R$  ( $z = Re^{i\theta}$ ,  $\theta \in [0, \pi]$ ):

$$|f(z)| = \frac{1}{|z^2+1|} \leq \frac{1}{R^2-1}, \quad \left| \int_{\gamma_R} f(z) dz \right| \leq \frac{\pi R}{R^2-1} \xrightarrow{R \rightarrow \infty} 0.$$

**Conclusion:**

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \lim_{R \rightarrow \infty} \left[ \oint_{C_R} f(z) dz - \int_{\gamma_R} f(z) dz \right] = \pi - 0 = \pi.$$



89. **Solution.**

(a)  $f(z) = \frac{\sin z}{z}$  at  $z = 0$ .

The Taylor series of  $\sin z$  is  $z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$ , so:

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots$$

This is a power series with no negative powers of  $z$ , so the singularity at  $z = 0$  is **removable**. Defining  $f(0) = 1$  extends  $f$  to an entire function.

(b)  $f(z) = \frac{1}{(z-1)^3}$  at  $z = 1$ .

The Laurent series about  $z = 1$  is simply  $(z - 1)^{-3}$ . The most negative power is  $(z - 1)^{-3}$ , so  $z = 1$  is a **pole of order 3**.

(c)  $f(z) = e^{1/z}$  at  $z = 0$ .

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n! z^n} = 1 + \frac{1}{z} + \frac{1}{2! z^2} + \frac{1}{3! z^3} + \cdots$$

The Laurent series has infinitely many negative powers of  $z$ . Therefore  $z = 0$  is an **essential singularity**.